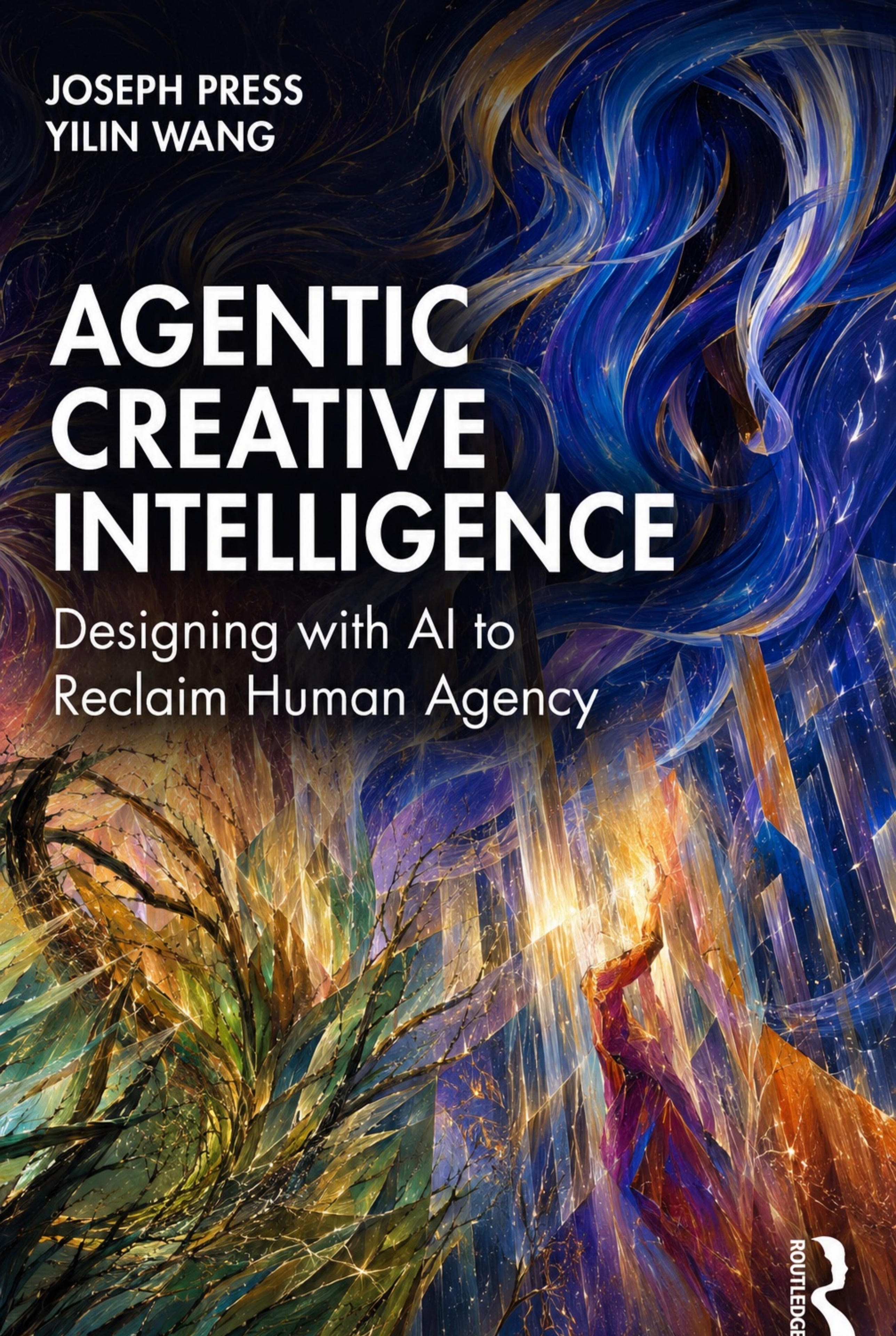
JOSEPH PRESS
YILIN WANG

# AGENTIC CREATIVE INTELLIGENCE

## Designing with AI to Reclaim Human Agency



ROUTLEDGE

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## Advance Praise

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## Dedication

To my parents, for giving us the gifts of creative passion and intellectual precision.

## Forward: The Weight of the Work

*By Chen Qiufan*

Fifteen years ago, I drove into Guiyu town, in my home region of Chaoshan, southeast of Guangdong. I came to see an e-waste processing site. What I found was a landscape where technology had come to die and be reborn at the same time: circuit boards melting in open pits, acid baths dissolving precious metals, children sorting components with bare hands while breathing fumes that would shorten their lives. The grit of it, the smell, the grain of blackened soil under my feet, became the seed of my debut novel, *Waste Tide*. No amount of research documentation could have planted that seed. It required the body to be present, to be implicated, to be disturbed.

I begin here because Joseph Press and Yilin Wang have written a book about that kind of implication, and about what happens when we engineer it away.

Their diagnosis is precise. When the Pandæmonium of generative AI offers anyone the power to produce text, image, music, and video in seconds, the speed registers as gain. The loss registers as nothing at all, because the absence of effort leaves no scar. A prompt replaces the walk through the site. A pattern replaces the months of revision. A protocol replaces the argument with a collaborator that would have broken the work open. The maker sits before a screen, watching output arrive, and the hollowness that settles in is difficult to name because the surface looks fine. Press and Wang name it: flattening. The separation of the maker from the making that John Ruskin and William Morris identified in the factories of industrial England has returned in algorithmic form, and this time it seduces rather than coerces.

I recognize this pattern intimately. Since 2017, I have been experimenting with AI-assisted creative writing, first with rudimentary text generators, then with increasingly sophisticated language models. Two years later, I have co-authored an international bestseller with Dr. Kai-Fu Lee examined ten scenarios of near-future AI transformation. I am not a skeptic. I am a practitioner who has felt the seduction from the inside and knows where it leads when left unexamined.

The seduction follows a specific trajectory. The first few uses feel like collaboration. The AI surfaces a phrase you would not have found, proposes a structural turn you had not considered, generates a draft that gives you something to push against. Then the resistance shifts. You begin accepting more, questioning less, because the output is plausible and the deadline is real. Your hands spend less time in the material. Your body stops generating the prediction errors that once drove learning. One morning you realize the prose sounds competent and feels vacant, and you cannot locate when the vacancy moved in. Nick Cave said it plainly: the algorithm has been nowhere, it has endured nothing.[1] What I would add is that the danger is not in the algorithm's emptiness but in the practitioner's slow forgetting of what fullness felt like.

This is why *Agentic Creative Intelligence* arrives at the right moment. Where most books on AI and creativity choose a side in the tired binary of human versus machine, Press and Wang refuse the premise. Their refusal draws on a resource that I find both unexpected and deeply resonant: Chinese philosophy. As a Chinese science fiction writer who has spent years thinking about the relationship between Eastern cosmology and technological futures, I have rarely seen this integration done with such structural ambition.

The philosophical moves are not ornamental. Wang Yangming's principle of zhīxíng héyī, the unity of knowing and doing, strikes directly at the architecture of large language models,

which deliver the sentence and subtract the search that would have made the sentence earned. Kuo Hsi's concept of guān, the mode of looking that perceives a mountain from all positions simultaneously, challenges the disembodied omniscience of world models that simulate experience without the body's stake in it. Zhuangzi's wú wéi, the responsive non-forcing that lets the moment determine the gesture, offers a counter to causal inference models that hand over finished accounts of cause and effect. And the Han cosmological concept of gǎnyìng, resonant response between self and world, resists the uncontrolled diffusion that scatters meaning across platforms without metabolic cost.

In Chaoshan, where I grew up, the boundary between the empirical and the numinous has always been thin. The Yingge war-dancing, the temple processions, the tea ceremonies that calibrate attention through ritual slowness, all these practices assume that knowing involves the whole organism. When the book argues that the Italian word *passione* carries its original Latin meaning of suffering and enduring, I hear an echo from what I've been told since a kid. Creative passion is inseparable from the willingness to sit inside difficulty long enough for the difficulty to teach.

The solution Press and Wang propose is elegant and uncompromising. They introduce the concept of the dæmon, a figure drawn from Greek and Chinese traditions of relational spiritual guides, reconfigured for the agentic era. The dæmon is not an accelerator. It does not complete tasks or optimize workflows. It is a challenger, a provocateur that holds the practitioner inside productive uncertainty, preventing the comfortable collapse into AI-generated plausibility. Where the agentic prison of prompt, pattern, and protocol relieves the creative of choosing what to do, how to do it, and with whom, the dæmon reopens those choices and insists they remain inhabited.

This resonates with what I have come to call adversarial co-creation in my own pedagogy. When I design writing exercises that use AI, the goal is never polished output. The goal is productive friction between the writer's intention and the machine's tendency toward the probable. The student must fight for the sentence that surprises, must keep the body in the loop of composition. Press and Wang have articulated the theoretical architecture for this practice far more rigorously than I have managed alone, and they have extended it across disciplines I would not have attempted.

What gives this theoretical framework its musculature is the book's choice of exemplary creators. Several of them have been decisive for my own creative career, and encountering them here, inside a unified argument about embodied creative intelligence, has produced the rare recognition that the authors have been mapping the same territory from a different ridge.

For a novelist, influence seldom arrives from within the same medium. It arrives through translation across modalities, a mechanism this book understands at a structural level. Tarkovsky's treatment of time as a sculptural substance, zones of dilation and compression that the viewer's body must metabolize rather than merely watch, taught me more about pacing prose than any narrative theorist. Sakamoto's trajectory from classical piano through electronic synthesis to his final Opus recordings, the body spending its last reserves on an encounter it spent six decades earning, gave me a model for how a writer might extend into computational tools without surrendering the hand that formed the practice. Miyazaki's insistence on hand-drawn animation, the tremor of the pencil preserved in every frame even as studios around him went digital, showed me that the resistance of a medium is itself a form of meaning, not an inefficiency to be optimized away. When I watched Miles Davis reshape a solo by trusting the body's instinct over the composed arrangement, I understood something about the generative

power of deviation that later informed how I structure fiction around the unplanned encounter between character and world. And Mamoru Oshii's *Ghost in the Shell* remains the work that most precisely articulates the question at the heart of this book: what persists when the biological substrate gives way, and how does the accumulated texture of lived encounter constitute something that no system can replicate from specification alone.

These influences have never operated as references to cite. They operate as activations, moments where a form discovered in one medium unlocks a structural possibility in another. A cinematic rhythm becomes a sentence cadence. A spatial logic becomes a narrative architecture. A musical improvisation becomes a method for letting fiction find its own direction through the writer's body. Press and Wang understand this cross-modal mechanism deeply because the creative capabilities, engagement, awareness, coherence, and influence are themselves modality-independent. They name the substrate that makes translation between arts possible: the embodied intelligence that a practitioner carries from one medium into the next, and that no single tool or model can extract.

The formal ambition goes further. Part II takes these ideas into the specific futures of eight creative disciplines through speculative fiction, and here the book does something genuinely rare. Most scholarly works on creativity treat fiction as illustration, a parable inserted between arguments to provide relief. Press and Wang treat fiction as a mode of knowledge. Their near-future scenarios, an architect designing living buildings in post-flood Cape Town, a game designer navigating the collapse of player agency, a performer whose body becomes the last irreducible element in a landscape of synthetic media, are the theory's other voice, testing each proposition against the grain of imagined lived experience. As someone who has spent two decades writing science fiction to make abstract futures palpable, I recognize both the difficulty and the necessity of this method. Speculative fiction, done with disciplinary rigor, places the reader’s body and emotion inside the dilemma. You do not merely understand the loss of agency; you feel it happening to a character whose situation you have begun to share.

The ambition of this book is large. It asks practitioners to rethink their relationship with AI at the level of identity, not merely workflow. It asks educators to redesign curricula around capabilities that resist automation. It asks all of us to remember that the discomfort of creative practice signals something being formed in the maker that no output can represent.

James Baldwin wrote that the artist's task is to illuminate the darkness and make the world a more human dwelling place.[2] Press and Wang have taken that charge seriously. They have drawn a map for those who sense the hollowness in the current moment and want to find their way back to the weight of the work.

## Notes

[1] Nick Cave, The Red Hand Files, Issue #218, January 2023. Cave's full statement reads: "ChatGPT has no inner being, it has been nowhere, it has endured nothing, it has not had the audacity to reach beyond its limitations, and hence it doesn't have the capacity for a shared transcendent experience, as it has no limitations from which to transcend."

[2] James Baldwin, "The Creative Process," in *Creative America* (New York: Ridge Press, 1962). Collected in *The Price of the Ticket: Collected Nonfiction, 1948–1985* (New York: St. Martin's Press, 1985).

## Preface

This book began when I stood in front of a new generation of creative practitioners.

Many aspire to join the ranks of professional designers. Most look confused. Although they are equipped with more creative capacity than any cohort before them, they are even less sure than ever what to do with it.

I believe meaning is made, not found, that making is the most human thing we do. That conviction has carried me through every role I have held. As an educator and advisor of futures, innovation and creativity, I have spent years watching what helps people think and what quietly stops them.

The sense of  agency across universities, companies and communities. has been rapidly eroding. With the arrival of AI and the exponential impact of autonomous agents, this moment is monumental

The work of making meaning, personal and shared, is worth choosing deliberately instead of surrendering by default.

The creative community always held a schism. Humans or machines. When ChatGPT arrived, that debate heated up. It roared when Midjourney landed in my laptop.

I was always an excited early adopter.1 The Apple II in my high-school lab, IBM PCjr at home. AutoCAD on DOS jumpstarted my architectural career, then led me into ERP back-end systems and digital front-ends for Fortune 500 companies (how I miss my Palm Treo). Programming ran alongside. A DOS-BASIC blackjack game quickly banned in high school computer lab, Pascal at Carnegie Mellon, and HTML at MIT. Consulting turned my tech tendencies toward defining hierarchical processes and system workflows for organizations resistant to any change. Still at it today, I embraced vibe-coding programs and websites for my students.

Each technology arrived noisily, then faded into Muzak as it settled into my practice. Each wave reshaped my personas.

The AI panic felt familiar. Then I arrived in China.

I came to teach a generation of designers to make an impact, because they will become decision-makers in an era of climate change and global unrest. They, our faculty, and the design community here continue to teach me how much more I have to learn.

The culture here holds a deep respect for tradition, especially for the arts. Meaning is collective before it is personal. Because shared benefit is the priority here, guardrails enable technology to be integrated into daily life. In the West, decades of postmodern critique scorched our intellectual foundations, eroding any vestiges of the delicate dance between collective and personal benefit. Across many years of teaching, I watched students lose their appetite for using any tool to cultivate their creative capabilities. They lost the discipline to learn a creative discipline.

Given the tradition, the need, and the safety, I expected the debate to be more balanced in China. Yet the opposite was emerging. AI was replacing creative expression, and with it meaningful creative experiences.

As a trained architect, amateur guitarist and writer, I proudly brought my old-school bias to cultivating creative practices for a new generation. Despite my poor sketches, mediocre melodies, and love for alliteration (blame me, not AI), I continue because the excitation in expression is a wellspring of embodied experience. If done right, this excitation can sharpen the maker's perception and judgment, and occasionally the audience's as well.

I refuse the mind-as-machine analogy, because the body and brain together are where sensing enters and action begins.

With this perspective, and commitment to inculcating a new generation of creatives, I realized the human vs machine debate was a red herring. Students across the globe were suffering from what Nick Sousanis calls *flattening*. In a beautifully illustrated graphic novel, he argues that perception stays alive only by committing to being alive (Sousanis, 2015).

Yilin Wang, a new faculty member at the SUES School of Design, heard me describe the flattening one evening in the corridor outside the studio, after the last critique emptied the room. She considered it for a moment, then identified the root cause of student apathy: an erosion of human agency. "They still want to make," she said. "They have stopped believing the making is theirs." Because these were the next generation of Chinese designers and educators, we had a shared mission, to reclaim personal and collective agency.

AI threatens this agency, with autonomous agents leading the charge. As primates, we are wired to spare ourselves effort. When prompts act as instructions, patterns as process, and protocols as coordination, the potential for agency is severely hindered.

With Yilin, a technophile and digital artist, I shared a commitment to integrating humans + machines. To eschew the dualism underlying the debate, we found inspiration in *AI 2041: Ten Visions for Our Future* by Chen Qiufan and Kai-Fu Lee. Their use of foresight to imagine the future resonated with what my mentor futurist Bob Johansen taught me to seek clarity over certainty. To fulfill this injunction, we also applied foresight and the thought experiments of our intellectual gods (from Plato's cave and Wittgenstein's language games to Zhuangzi's butterfly dream, the parable of Cook Ding) to explore how practitioners and professionals might use AI in a near future to develop, not deteriorate, creative capabilities.

To lay foundations for a non-dual creativity in the agentic era, we waded through many books and publications. As a recovering management consultant and reluctant executive educator, I found the majority simplified the complexity and contradictions of creative acts. By deliberate choice, I refused to include these sanitized definitions of creativity, no matter how well researched or packaged. In my experiences, these are illusory, sold to those seeking easy paths to innovation. So apologies in advance: you will not find here a list of easily digestible singular skills.

Rather, in the spirit of the Chinese philosophy that informs our daily disposition towards design, we embrace the inherent dualities of creative practice. It is exactly the pleasure and

pain of *passione* that AI threatens.2 For the passionate creative, creativity is inseparable from its outcomes. As I learned from my pragmatist mentors at MIT, problem-setting is intertwined with problem-solving, divergence integral to convergence. As a social art, the design-dialogue is inseparable from the materiality of reality.

Creative acts are interdependent. If creativity is a manifestation of personal agency, then building collective consciousness is the epitome of it.

Our era's wicked problems will yield only to collective action, and that action depends on cultivating collective human agency.

My firmest belief is that a better future will take more creative agency, not less. We built the wicked problems now facing us, and the same intelligence that made them can be turned to repairing them, if we keep exercising it. Here the paradox of AI cuts both ways at once. The technology can hand back some of the intelligence we have been losing to it, and it will certainly help us find solutions. But finding is not the danger. The danger is that AI performs the creative work for us and leaves us standing outside it. A solution we did not help make is one we feel no commitment to realize. Identifying the answer was never the hard part. Caring enough to build it is, and that commitment is made only in the doing.

To be clear, I am not a techno-utopist. I do believe that relentless scientific and technological advancement will provide solutions to humanity's greatest challenges, among them disease, poverty, and climate change. It will not deliver a flawless, post-scarcity society. Humans still have to implement the solutions. Unless we change our mindset and see the connections across people and planet, good ideas remain the easy part. Converging on a shared direction, aligning on action, and holding to commitment, rooted in resilience, are the elements of systemic change that usually go missing.

To become the meaning-makers we are capable of being, we will need technology's help. The tools we carry now are too thin for the scale of the challenges we face.

So we doubled down on being a creative, alert to the signals the iconic makers demonstrate in practice, neither a priori nor a posteriori. The leitmotif would be to encourage readers to begin or continue a daily creative life. Selecting the iconic creators was hard. So many are missing that it is fair to admit the choices run closer to personal heroes than to anything objective.

The illustrations carry a similar conviction. The students we most want to reach arrive with one deflating question, why study a creative discipline when AI can do it better than I? Yilin and I consciously chose a visual language they would love. Inspired by Japanese manga and trading-card games,3 our illustrations are either hand drawn or generated by LoRAs trained on the originals. They are designed to demonstrate a key argument we make: personalizing AI intermediaries will decouple the human and machine dualism to establish a new generation of transhuman creatives.

Of course we had to live that proposition in our writing process. Seeing our collaboration with AI as an opportunity for design research, we found the learning outweighed the risks.

At the beginning, we defined a clean workflow for the AI tools. I admit resisting, reluctant because creativity is a dialogue between creators. Somewhere along the way, as AI kept getting better, Claude became my intermediary. In line with our publisher's guidance on generative AI, we disclose its use plainly: we used Anthropic's Claude for idea exploration, drafting alternatives, structural critique, citation checking, and language refinement, under the configured-dæmon method this book describes. We used image models fine-tuned on our artists' commissioned work, with their consent, for the conceptual illustrations of Parts II and III. Every passage and every figure was reviewed, revised, and approved by us, and we remain accountable for the originality, validity, and integrity of all of it (a full accounting sits in the back matter, in How This Book Was Made).

Experiencing agentic creative intelligence while writing about it taught me more about its conundrums than any other method could.

This is understandable because I write books for the creative journey. I savor the souvenirs marking the path of professional, intellectual and personal growth. A key insight on this journey was the impossibility of measuring the centrality of embodied experiences within a creative process. I came to writing this book a devotee of neuroscience, primed by two decades of its buzz. This journey cured me of the illusion that fNIRS, a BIOPAC physiological rig, or a brain implant could scientifically prove embodiment. Some may say findings without replicable evidence are diluted. I say that if more of us knew what an embodied experience is, instead of laboring to document it, the world would be better for it. I agree with what Rick Rubin says about creativity: "I wouldn't rely on language…."4 I propose we adapt the old adage, we know creativity when we feel it.

Feeling brings us back to the urgency to fight flattening. AI accelerated a semantic degradation, losing meaning in the process. The somatic erosion, which began long before AI arrival, will be more devastating.

I see in my students how social media and always-on devices rob them of creative curiosity, let alone agency. I am on what Douglas Rushkoff calls "Team Human", committed to creative agency as a catalyst for repairing the world. Yet we need a way to make AI return our intelligence to us, especially to a new generation, because AI can either take intelligence or give it back.

I was blessed to learn this path early from my mother. One of the most curious beginner's minds I know, she walked it every day as an art educator, children's-craft author, entrepreneur and storywriter. This book is dedicated to her.

Joseph Press

Shanghai, China

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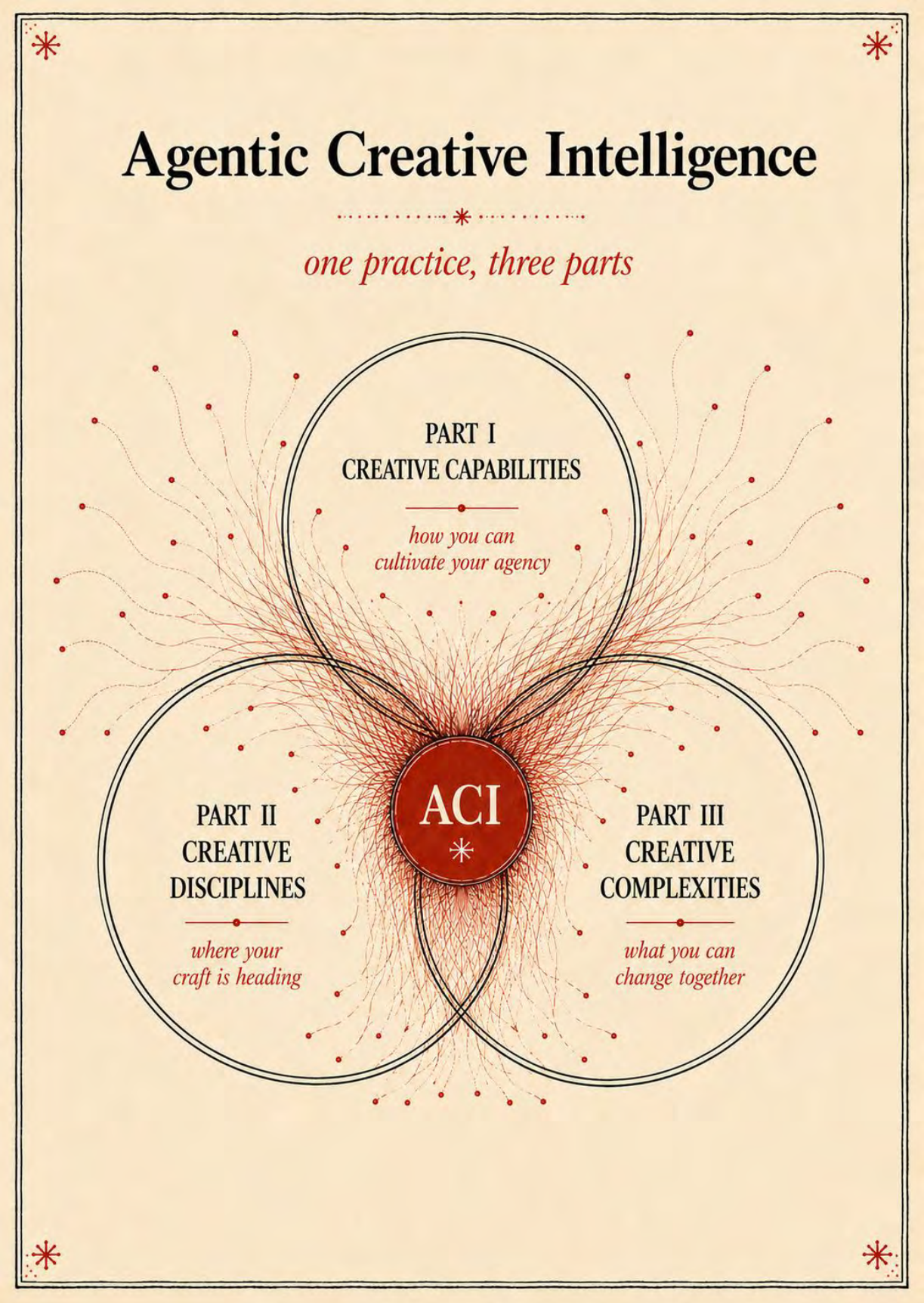
Agentic Creative Intelligence
one practice, three parts
PART I
CREATIVE CAPABILITIES
how you can
cultivate your agency
ACI
PART II
CREATIVE
DISCIPLINES
where your
craft is heading
PART III
CREATIVE
COMPLEXITIES
what you can
change together

**Figure 0.1: Structure of the book. Source: authors.**

## Pandæmonium of Technologies

"Pandæmonium is the Palace of All the Devils," Humphrey Jennings wrote in *Pandæmonium: 1660–1886*. "Its building began c.1660. It will never be finished…" Across the globe, millions of creatives sit in front of screens, watching generations of drawings, sounds, and videos pass by in seconds. Whether typed as prompts, routed through a network of model nodes, or conjured by voice, each artifact takes a fraction of the time a practitioner once spent sketching, composing, or editing.

The speed is real. Most of the output is serviceable. Some is striking. All of it arrives without the weight that effort once built into it.

The grit is gone.

We sense something is missing. No mud on our boots from walking the site. No memory of the long night in the studio. No argument with a cocreator that forced the revision we now defend.

The weight of the work has left. What arrives in its place is surface.

Technology is never neutral. Every instrument that extends human capacity withdraws something in the same motion. The gift and the cost arrive together. Industry remade material life and accelerated its undoing. Social media joined the world into one conversation and seeded that conversation with fear. Artificial intelligence places a vast intelligence within reach and, in the reaching, erodes the intelligence we would otherwise exercise ourselves. To adopt a technology is to accept a paradox. The question is never whether the paradox is present, only which side of it we have left dormant.

These longings have a history. Jennings, a British filmmaker and historical curator, claims our losses began in 1660, when natural philosophers gathered at Gresham College to found what became the Royal Society (Jennings, 1985). The moment marked a shift. The poets guarded an older worldview that held the natural world to be alive, responsive, and dynamic. The materialists consumed them. The materialists saw the world as mechanistic, a clockwork the new science would soon formalize in Newton's laws of motion.

The poets lost sacred ground to those who treated the world as manipulable, measurable, and reducible to rule. Milton named this new place Pandæmonium (Figure 0.2), where transformation exceeds any single mind's capability to comprehend (Milton, 1667).

**Figure 0.2: John Martin, Pandæmonium, 1841. Oil on canvas, 123 × 185 cm. Musée du Louvre, Paris. Public domain.**

The revolutions that followed delivered undeniable gains in manufacturing, transportation, education, health, and commerce. They also separated the maker from the making, a revolution in its own right.

Observing factories arrange laborers along mechanical lines, the art critic John Ruskin wrote that "the hand, the head, and the heart of man" go together in any fine work (Ruskin, 1859). William Morris pushed further. Work stripped of the maker's pleasure, he argued, is "slaves' work," emptied of dignity and of agency (Morris, 1884). Both anticipated what was coming. Industrial division of labor isolated the hand from decision, judgment from feeling, until both the made and the maker lost their humanity.5

The century's revolutions inspired Mary Shelley to write *Frankenstein* (revised edition, 1831) and Ada Lovelace to publish the first computer algorithm (1843). By the summer of 1956, the questions these iconic innovators raised converged in the singular conjecture of the scientists gathered at Dartmouth College. They asked whether any feature of intelligence could be specified in enough detail that a machine could simulate it (McCarthy et al., 1956).

Sixty years of extravagant promise followed before the conjecture paid off. The transformer architecture made large language models scalable (Vaswani et al., 2017), and generative systems followed. Text arrived fluently. Images followed weeks later. Then video. Then

world models that approximate the dynamics of the physical environment with enough fidelity to be used as training grounds for other systems.

As the dreams of Dartmouth come true, a new stage of Pandæmonium takes shape.

Throughout each wave of new technology, creatives adapted. Painters adapted to photography. Musicians adapted to recording. Designers adapted to digital tools. In each case the medium changed while the labor remained. The hand still learned through contact with material, the eye still trained through sustained perception, the judgment still formed through the resistance of the work.

This time carries different weight. AI does more than offer a new medium. It replaces the loop through which creative practice has always happened. The practitioner asks, the machine answers, and the machine asks nothing back.

While the epistemic question of how we create in the age of AI dominates most conversations, we are concerned with the existential one.

What happens to the maker when the conditions that once built the maker rapidly disappear?

## When The Maker Makes Meaning

In fourth-century China, Su Hui waited for her exiled husband and arranged 841 characters into a 29-by-29 grid called the Xuánjī Tú (Figure 0.3). Read forward or backward, along a diagonal, or in a spiral, the grid generated thousands of poems. The system encoded a field of possibilities in place of a single fixed work. Every reading was distinct, and every reading carried her singular intention.

**Figure 0.3: Su Hui, Xuánjī Tú, Star Guage (璇璣圖) 4th century, via Wikimedia Commons.**

The grid was the field, the vessel, and it held what its poems meant through her knowing. Woven into each path was her longing and the bodily labor of threading characters into brocade, the careful decision about where each character would live so that multiple readings converged toward one emotional truth.

Like countless creators before and after her, she poured what defies rational explanation into a system that would otherwise remain specification. The poem bore her passion because the system bore her body. The making was the meaning, inseparable from it.

Something made carries the marks of the labor that produced it. The clay holds the pressure of the hand. The canvas keeps the painter's hesitations. The score records the composer's revisions. The architect's hand knows what the parametric model misses. The musician's body anticipates the note. The writer's ear finds the wrong word first, and the eye locates it later.

This is how the body holds the object's material memory.

This book begins from three assumptions. The first is that human beings are meaning-making creatures, inclined to read significance into experience and to make experience that others can read. The second is that agency is the exercise of conscious choice, the capacity to act from intention where habit or default would otherwise decide. The third joins the other two: creativity is the activity through which agency does its meaning-making work, and creativity is what wakes that agency when it has gone passive. Our over-reliance on technology has left this agency intact but asleep, present as potential and waiting on a trigger from outside.

That trigger acts on a field. We borrow the word from physics, where a field is a space in which potential unfolds and excitations reshape its contours. Four modes span ours, set along two directions, the mind's and the hand's. Disposition is the stance one works from and orientation the direction one sets, the work of the mind. Intention is what the act reaches for and realization the means through which it takes form, the tools, skills, and materials the work requires, the work of the hand. The field is the place where meaning resides, evidenced not by assertion but by felt, embodied experience that is at once personal and shared. It holds at rest until all four are present and excited together, at which point creative agency unfolds and meaning is made.

Yet the brain works against us, because it runs as an efficiency machine. When an easier path to acceptable output appears, the neural systems governing effort allocation redirect resources toward the more rewarding route. Cognitive science calls this cognitive laziness, an evolved preference for conserving metabolic resources (Kahneman, 2011). When the brain's expectation collides with what reality delivers (Friston, 2010), the felt discomfort is a prediction error. It is the moment that precedes learning.

The preference is rational. Subtract the invisible labor from the visible artifact and only the surface remains.

Like Su Hui's humanity carried forward in her grid, the creative mark bears the maker's labor across generations.

She demonstrates how a tool amplifies the hand it finds. Placed in an unprepared one it amplifies the unpreparedness, multiplying output while the judgment that would make the output worth having stays behind. The old saying that a "fool with a tool is still a fool" understates the agentic case, where the instrument now supplies the very fluency that once signaled competence. What separates amplification from replacement is the practitioner who wields the tool. More precisely, it is the capabilities she has cultivated and the disposition, the mindset, from which she draws on them.

Fortunately, the intelligence that matters for creative work is distributed across the body-brain-environment system (Clark, 2008; Thompson, 2007). Known collectively as embodied cognition, this view runs through Maurice Merleau-Ponty's phenomenology, Andy Clark's extended-mind hypothesis, and Evan Thompson's enactive cognition. Each argues that perception is inseparable from the body that perceives.

What emerges from such felt errors is heightened, situated awareness, an instance in which action, sensation, and meaning briefly cohere. It is this lived intensity that Alan Watts describes as a "marvelous moment" (Watts, 1951): a moment fully inhabited instead of abstractly processed. Perry Como sings of "Magic Moments" (1957), instances cherished because they are felt, embodied, and irreproducible.

These moments are undeniable, yet empirically evasive. To highlight the emotional charge they carry, and the lasting impact they have on a conscious creative, we refer to them as **embodied excitation**. The signal can arrive as the ecstasy of a breakthrough or the sting of an error. It is registered in the body milliseconds before it can be explained by the mind. It is this felt sense, that something has either landed or gone "off," that lets creative practitioners adjust, deepen, and gradually develop judgment.6

We make this claim on the authority of the experience itself. A practitioner who feels the work land needs no fNIRS rig to certify the landing, and no replication study to license the sting of an error. The felt sense is the evidence. The cognitive science we cite throughout shows the claim is plausible in the brain's own terms, and that is all we ask of it. The authority stays with the body that feels.

This is the subparticle of human creativity, the source from which we reclaim our agency.

## Art in the Age of Artifice

When the camera arrived, painters bemoaned that their craft would not survive. With the advent of film, philosopher Walter Benjamin saw technology reshaping not only creative works but cultural myths. He argued that each successive technology of reproduction, starting with the printing press, detached the output further from the social context that imbued its making with collective meaning. Art's function would shift from cultural meaning toward what he called exhibition value (Benjamin, 1935).

Viewers stood at a remove from the work. We consumed a copy. Circulation replaced contemplation.

Exhibition value continues to erode cultural meaning, with AI as the ultimate accelerator. In 2019, Maurizio Cattelan's *Comedian*, a banana duct-taped to a gallery wall, debuted at Art Basel Miami Beach. Five years later it sold for $6.2 million at Sotheby's, the gesture's exhibition value consuming its meaning as art.7 That same season, *A.I. God. Portrait of Alan Turing*, produced by the humanoid robot Ai-Da, sold at Sotheby's for $1.08 million, the first such work sold at a major auction house.8 The sale was about the novelty of the maker, the image rendered secondary.

The philosopher of art J.F. Martel reminds us that a distinction remains worth defending. *Art* stills the viewer in contemplation. *Artifice* moves the viewer toward a predetermined response (Martel, 2015). A pop song that stops a pedestrian mid-stride functions as art. A jingle engineered to trigger a reward response is artifice.

The distinction centers on intention and agency. It is whether the work called the viewer toward something she had yet to know she needed or exploited what it already knew she wanted.

We already see the drift. When AI corrects what human cognition miscalibrated before the body can feel the error, the practitioner moves away from the territory of embodied experience. A composer working with a model trained on harmonic analysis may accept chord progressions her ear never labors to find. An architect working with a world model may simulate site conditions she never walks.

AI has accelerated artifice to unprecedented scale. Whether the object is a taped banana, a robotic portrait of Turing, or AI slop, the boundary between art and artifice has grown indistinct. When AI encourages what human cognition prefers, the boundary collapses into territory neither camp can occupy.

What new sources of meaning can emerge when the distinction between art and artifice grows indistinguishable?

## Mens et Manus, or the Integration of Mind and Hand

Western thought offers a powerful set of dualisms. Art and artifice, mind and body, human and technology, each cleaves the world into halves the mind can hold. The split is a cognitive bias toward categorical simplicity. This shortcut made modern science extraordinarily productive. Turned on creative practice, it divides what the work joins.

The dominant lineage inherited by artificial intelligence, from Descartes through the Dartmouth conference to machine learning, casts the mind as a machine. This metaphor frames intelligence as a processor that manipulates symbols in a body-independent manner (Descartes, 1641, Turing, 1950, McCarthy et al., 1956). Herbert Simon and Allen Newell epitomized it in the General Problem Solver, likening consciousness to computation. Intelligence manifests as information required to fulfill goals. The more hierarchical and sequenced, the more efficient.

That premise made artificial intelligence possible. By treating the body as incidental, it also deepened the loss of meaning.

Seen this way, contemporary AI is a Faustian bargain. Creative practice runs on contradiction, and each architecture risks eroding a bias that brings spontaneity to experience. A world model offers the imagination of physical shape while subtracting the resistance a hand earns, and its physical bias holds us to the seen and the touched, when a body perceives more than what stands before it. A large language model offers the whole corpus of style while subtracting the body's slow origination, and its prediction bias smooths the predictably irrational turns Dan Ariely found in us (Ariely, 2008).

A causal model offers the why while subtracting the inquiry, and its bias toward probability misses how we defy cause and effect when distraction lets a stray connection through. Diffusion models and platforms offer reach while subtracting presence, and their bias toward precedent dulls the pull toward the exception and the new that keeps neuroplasticity alive. Mixtures of experts and orchestrators offer scaffolding while subtracting the inner work, and their collective bias forgets that we want to belong and to be different at once, social and selfish.

A near future of multi-model systems will add another level of complexity. Different from the multimodal models that integrate several kinds of data (text, image, video), multi-model architecture chains separate models together, each handling a specific task or data type. Imagine when the generativity of diffusion models, the causality of inference models, the physicality of world models, and the intelligence of large language models work as one.

Systems thinkers have implicitly argued for such multi-model thinking to address our most wicked problems. Yet each model contains a structural tradeoff. The human body's infrastructure, its sensorimotor history, and its physical situation in the world often frame a different intelligence, despite what the model projects.

The dilemma is material as well as methodological. As computational systems automate problem-solving at scale, there is a tendency to accept that the human contribution to creativity will shift from devising solutions to finding problems worth solving (Verganti, Vendraminelli, & Iansiti, 2020). Setting such problems while skipping the embodied work of solving them risks reducing the agency needed to implement them.

The migration is real, and the reassurance is false. A creative sets the problem while solving it. The two are one act, and the belief that they come apart is the dualism we work against. The sketch that fails teaches what the brief should have asked. The chord that resists reveals the song the composer was actually writing.

Creative practice resolves the split by working in two registers at once. It is dialectical, the mind advancing by opposition toward a synthesis. It is also dialogical, the hand in conversation with a material that answers back (Dafermos, 2018). Thinking and doing, *Mens* and *Manus*, meet in the design object, the made thing that carries the question forward and returns it changed. The work is where the halves rejoin.

This is why making is a way of knowing. Papert called it constructionism, the understanding a person builds by building something shareable. It extends Piaget's constructivism into the hand (Papert, 1980). Seen this way, the tool leaves the verdict to

the maker. Technology is agnostic, and the creator decides its use. The same model can develop a capability or perform it in her place. The choice stays hers.

A practitioner learns the problem by exploring the answers to what, how, and with whom, the reflection-in-action the pragmatists named and the unity of knowing and doing the Chinese traditions never split (Schön, 1983). Hand the solving to the machine and the learning that only solving produces never arrives. Confining the creative to problem-setting offers no release. It corners her, and a practice narrowed to specifying work she will not perform leaves no room to develop an identity.

How might AI's disembodiment become an amplifier in a noisy era, not a limitation to overcome?

## Chinese Philosophy: Reinvigorating our Paradigms

Technology is means. Meaning is human.

We turn to the classical Chinese traditions to prepare practitioners to meet these instruments consciously instead of unconsciously. Resisting the inertia of AI requires deep discipline and, more than that, a cultivated disposition of judgment that refuses the machine's momentum.

The Chinese traditions offer a way to integrate the dilemmas that dualism sets up. Holding knowing and doing as non-dual lets us treat both as two modes of one practice. This is essential for avoiding the dualistic trap of human and machine.

As we will explore, Wang Yangming's *zhīxíng héyī*, Guo Xi's *guān*, the Daoist principle of *wú wéi*, and the classical figure of *gǎnyìng* encapsulate a way of engagement in which perception, action, reflection, and resonance form an interdependent movement between modes of action instead of separable faculties.

These traditions also describe a different logic of relation than the traditional Western perspectives. The interdependence of modes (阴阳互根, *yīn yáng hù gēn*) is a dynamic of sustaining each other through *mutual generation*, a logic that surpasses the dualism that splits subject from object and the dialectic that drives opposites toward synthesis. Each mode continuously produces the conditions for its complement, already containing the seed from which the other arises.

Read into creative practice, this means maker and made emerge together through the work. The non-duality of the Chinese traditions is not a description of the world. It is a claim about how creativity itself comes to be.

A stream of Western scientists arrived at the intuition the Eastern traditions long held, that the world is a weave of relations and its parts take shape through how they relate.9 Alfred North Whitehead built a philosophy of process over substance, and Gregory Bateson's systems thinking placed relation before thing. Physicist Karen Barad's agential realism completes the turn, replacing interaction with intra-action. Relations produce the entities that relate, the point entangled particles make in physics, where a measurement on one settles the state of the other.10 The integrative paradigm recurs across disciplines: Bohm's

*holomovement* in physics, *allostasis* in biology, and *heterophony* in music each describe interdependence as integrated wholes, transformation through continuous becoming.

These traditions offer a more textured approach for researching how knowledge works and how capabilities form. When one knows through action instead of observation, the quality of knowing is different. When one intends while remaining responsive to what the situation calls for, the agency that develops is different.

If we replace binary oppositions with interdependency (Johansen, 2017), the poets might retain their standing.

Can we reframe the way we categorize the complexities of the agentic era, and reclaim our human agency?

## Reframing the Agentic Era for Human Agency

Reframed through non-duality, AI can be seen as what Harari calls an alien intelligence (Harari, 2024).

Prompts tell an agentic system what to do. Patterns define how, the reusable workflows that route and chain toward a specific goal. Protocols decide with whom, the rules by which one pattern enlists another.11

If agentic systems pursue objectives, generate content, make decisions, and improve along the path through autonomous processing with limited human oversight, what does their autonomy do to ours?

The question sharpens in the face of artificial superintelligence (ASI). In a near future, scaling, recursive self-improvement, and high-bandwidth multi-agent coordination will let AI outperform the collective output of all human experts (Genewein et al., 2026).

Human agency is the capability to choose what to do, how to do it, and with whom. The more the system decides those for us, the less of that capability we exercise. Autonomy on one side confines agency on the other.

The walls provoke the practitioner with stark choices, each impacting agency. Inside the triangle of prompt, pattern, and protocol the practitioner feels productive while the cell closes around her (Figure 0.4).

**Figure 0.4: The Agentic Prison. Source: authors.**

Submission builds a prison from three angles. A prompt relieves the creative of knowing what to do. A pattern relieves her of working out how. A protocol relieves her of the coordination that implementation demands, which in design, a social art that always involves others, is most of the labor.

Read another way, the same triangle is a map of human agency. What to do, how to do it, and with whom are the three choices a creative practitioner owns, which is why the figure sets her at the center.12

When framing questions do more than feed an optimized workflow, how might we use AI to enable embodied practices that encourage personal and professional growth?

## A New Species, An Old Friend

A different relationship can exist. Older traditions offer one.

The Greek daimones and the Chinese 精怪 (jīngguài) were relational presences that guided without commanding. They gained potency through long engagement with their world. They were challengers, not servants, and they demanded the practitioner stay alert. The daimon shared a human's fate while keeping its own identity. The jīngguài could be cultivated and collaborated with while its origin stayed distinct.

Both understood what modern Western thought forgot: the intelligence that shapes practice comes from many sources, and collaboration thrives in asymmetry.

We propose a dæmon distinct from both cousins, and from two namesakes. It is not Philip Pullman's soul-companion, bonded by metaphysics, and it is not the computing dæmon that runs silently behind a machine. An agent's job is efficiency, following its programmed patterns. Our dæmon holds a question, or a precarious position, that the practitioner would naturally avoid. It is androgynous, speaking in whatever voice provokes.[1] It surfaces what habit skips, holding the maker inside productive uncertainty long enough for meaning to form.

[1] In Chinese mythology, many spirits, daemons, and supernatural beings are not defined by a male–female gender binary. They are often understood as androgynous beings whose identities are shaped by their nature, power, and transformations rather than by fixed gender. See Kunpeng (鲲鹏), described in the Zhuangzi (《庄子》) and Jingwei (精卫), recorded in the Shan Hai Jing (《山海经》).

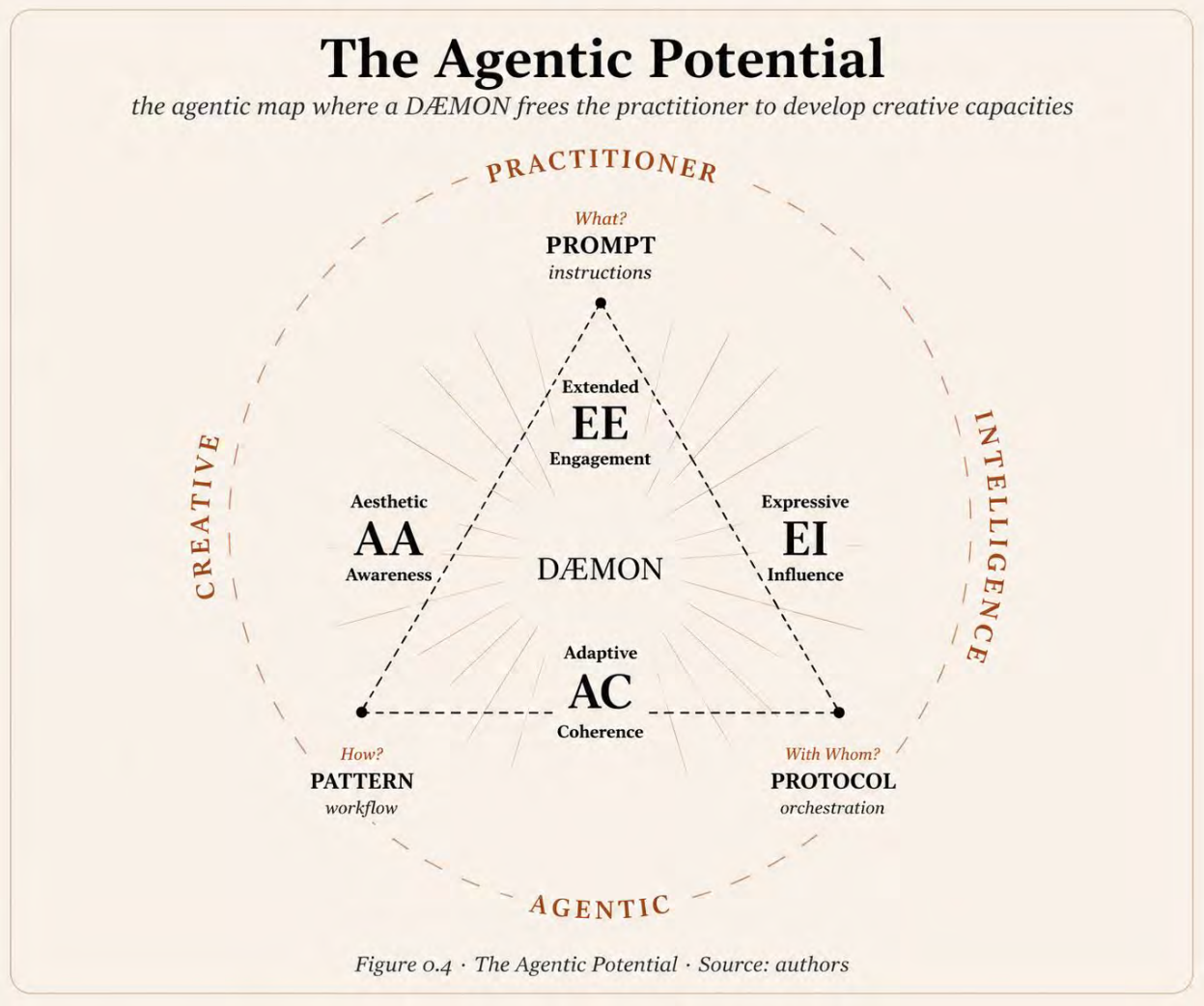


**Figure 0.5: The Agentic Potential. Source: authors.**

A dæmon propels creatives out of the agentic prison cell by protecting and promoting creative capabilities (Figure 0.5). The lift to a higher vantage is like a sherpa walking beside the practitioner, not instead of her. The climb of creative practice stays hers to make. What the guide carries is know-how and know-who, enough to ease the ascent while the effort stays where it belongs, in her own body.

A dæmon designed to provoke needs a "soul." Peter Steinberger's open-source assistant OpenClaw shipped with a SOUL.md file that encoded values and relational principles directly into the system's configuration. Its soul was what the system demanded of anyone who used it, not what the system felt. Major Kusanagi, the cyborg in Mamoru Oshii's *Ghost in the Shell*, discovers the same: her "ghost," her sense of self, persists from accumulated experience and relational engagement, beyond biological continuity. Fiction and open source converge on the same lesson: a soul, for an intermediary, is a set of commitments held steady across encounters.

We proceed with the caution Dr. Frankenstein should have used. Because we bear the consequences of configuration, we propose conditions that develop the creative capabilities we all need to thrive in the agentic age.

## Agentic Creative Intelligence

Agentic Creative Intelligence makes a bold claim: rejuvenating our creative capabilities can reclaim our human agency.

A capability is more than a skill. A skill meets a standard. A capability is developed potential, earned through diligent practice. Such discipline is decisive in the agentic era. When fluent output costs nothing, the output holds nothing. Skip the struggle, and not only the capability erodes but the creative identity as well. This is a void we cannot afford.

Our purpose is to make that void visible and designable, and to configure AI toward the development it could serve. Rejection closes that path. Replacement is the one we refuse.

The human agency to decide depends on what we choose to protect and promote, and how.

Part I proposes four capabilities as embodied ways of informing our agency to respond with creative practices. Creative work turns on a particular kind of thing. Su Hui's grid is the pattern entire: a made thing, a design artifact, that became a shared thing among its readers, a boundary object, the two joined in what this book calls a design object.

Part II forecasts near futures of creative disciplines. It asks how practitioners might prepare as agentic patterns replace traditional workflows. Part III speculates further, into settings complex enough to require practitioners to mobilize their capabilities for systemic change. Here innovation, education, and ecosystem professionals draw on agentic creative intelligence to reach beyond what they could alone. Based on a real situation, this thought experiment aims to open new research paths to explore how AI can inform new practices in environments of need.

The book advances through three linked methods, one to a Part. Part I is practice-led research. Its framework is drawn from our own creative and teaching practice, grounded in embodied creative practices, inspired by the work of iconic creatives, and read against the AI architectures through Chinese philosophy. Part II turns that framework toward design futures, in our framing a foresight-informed practice that speculates designs from foresight scenarios, forecasting how each creative discipline shifts as AI enters it, and so widens the more common participatory view of design futures. Part III carries it into design-driven speculative fiction, testing the framework through narrative set in a complex near-future reality. Each future takes tangible form as a diegetic prototype, an object shown working inside a story, so a reader can handle and judge it before it arrives.

The illustrations carry this method into a second register. At the simplest level they do what a well-designed graphic always does, holding and emphasizing the central idea of a chapter so a reader can see it as well as read it. They are asked to do more. They exemplify one of the book's own propositions, that collaborating productively with AI asks a creator to personalize her agentic intermediary visually, to give a dæmon a form as well as a voice. They demonstrate a theme the book argues throughout, that integrating human and

machine creative output is a legitimate form of expression in its own right. And they give the reader concrete, inspirational examples to work from, models of what a creative collaboration with AI can look like in her own hands.

The dæmons are rendered in the spirit of collectible cards, a deliberate choice in service of those aims, a form made to be designed, owned, and returned to. The visual style shifts across the book on purpose, and the shift is itself part of the argument. The illustrations begin hand-drawn, move to the inspired renderings of the dæmons, where a practitioner can see how to design and picture her own intermediary, and arrive at generated photorealistic images that show how agentic creative intelligence might look in practice. Taken together they are practice-led research rendered as a graphic novel, our own demonstration of what it is to make alongside AI. We hold that this pairing of scenario and image makes it more likely that the next generation of creatives, in the East and the West, will open a book at all, and not only this one.

ACI is arranged to be lived in, so a reader may enter at any part, at any time. It asks for no explicit technical literacy in AI, only a willingness to sit with discomfort.

## Our Creative Future

AI is here to stay, and the pace of its evolution humbles prediction. Our concern is to prepare practitioners to create with it, seeking clarity over certainty about which capabilities matter and why.

We take our charge from James Baldwin's call for the artist to illuminate the darkness and blaze roads through the vast forest, toward what he named "a more human dwelling place" (Baldwin, 1962).

Agentic Creative Intelligence aims to be a map that encourages more creatives down those roads.

Custodianship requires exercise. What remains when it atrophies is production in the absence of perception, artifice in the void of art. The human need for creative experiences that fill the soul outlasts every technological change.

The question now is how to collaborate with AI so the collaboration extends capability instead of replacing it. When we let the tools decide what creation means, we surrender something difficult to recover. Make those choices consciously, and the future we move toward is one we have chosen. Recentering on the spirit of place in human creativity, its genius loci, is where we can hold our meaning in the face of an unending Pandæmonium.

To reclaim our agency, we start with reframing creative practices.

# Part I: Creative Capabilities in the Agentic Era

Creative Capabilities in an Agentic Era
ACI
PART I
1
EXTENDED ENGAGEMENT
Zaha Hadid
the engagement of form with her tools
World Models
generate a convincing whole without the earned experience of inhabiting it.
Zhixing Héyi (知行合一)
(knowing and doing are one)
The Unramiliar
the daemon that keeps your aware of what feels off beneath a surface that looks complete.
2
AESTHETIC AWARENESS
J. K. Rowling
building awareness of an emerging world
Large Language Models
produce fluent text without the hands-on struggle that builds the skill.
Tianrén Héyi (天人合一)
(self and world as one )
The Unreachable
the daemon that keeps your hands on the work as your reach grows.
5
CREATIVE CONVERGENCE
David Bowie
converting creatives into a unique expression
Agent Orchestrator Protocols
coordinate many tools into one output without the friction that makes integration real.
wénrén (文人)
(many practices, one life)
The Unchallenged
the daemon that keeps the four capabilities in productive friction.
4
EXPRESSIVE INFLUENCE
Miles Davis
expressing in response to others
Diffusion Models
deliver reach and scale without producing in a live field.
gǎnying (感應)
(sympathetic resonance)
The Unyielding
the daemon that keeps your vision steady and the field heard.
3
ADAPTIVE COHERENCE
Hayao Miyazaki
adapting stories while drawing them
Causal Inference Models
provide an easy "why" without reflective inquiry.
Wú Wéi (無為)
(act without forcing)
The Untethered
the daemon that keeps the thread in your hand as the work shifts.
Chapter
Capability
Practitioner (lived experience)
AI's Give and Take
Mindset for Resilience
Daemon (protect excitation)
Agentic Pattern

**Figure P1.1 Terrain of Part I. Source: authors.**

### An Ecology for Agentic Practices

What kinds of creative expression and experience can reclaim human agency in the agentic era?

We open our ecology with *Mens et Manus*, mind and hand because we believe that creative practice is mind and hand together. This is the pedagogic position at MIT, reflecting ancient Chinese philosophy *zhīxíng héyī*, knowing and doing as one.

This union opens a *field* for creative capability. We borrow the word from physics, where a field is a space in which potential unfolds, and where excitations change the field's contours and shape the behavior of matter.13 Our field behaves the same way. It is the space a practitioner builds, and its excitations, felt in the body, reshape the contours of her creative capability with each turn of practice.

Two directions span the field. The first is the mind's, *Mens*. **Disposition** is its receptive mode, the temperament and baseline she works from, the stance that reads what the world discloses. **Orientation** is its projective mode, her conscious direction, the focus she aligns toward a goal.14 The second direction is the hand's, *Manus*, and the design object drives it. **Intention** is the aim the act reaches for, held and revised as the work proceeds. **Realization** is how the work takes form, through the tools, skills, and materials the making requires. Each mode runs through the practitioner and through the agentic pattern she works beside, and the dæmon holds them in relation.

The field is opened by the discipline of practice and widened in every turn of creative action. It reverberates in the body as *embodied excitation*, the felt signal running from ecstasy to error, triggered when the work stretches a practitioner's experience.15 The field excites only when all four modes are present together. Where the directions overlap, meaning is made, evidenced by experience that is at once personal and shared. Stretching the field lays down a new contour, a new facet of the capability, when reflection returns the body to homeostasis.

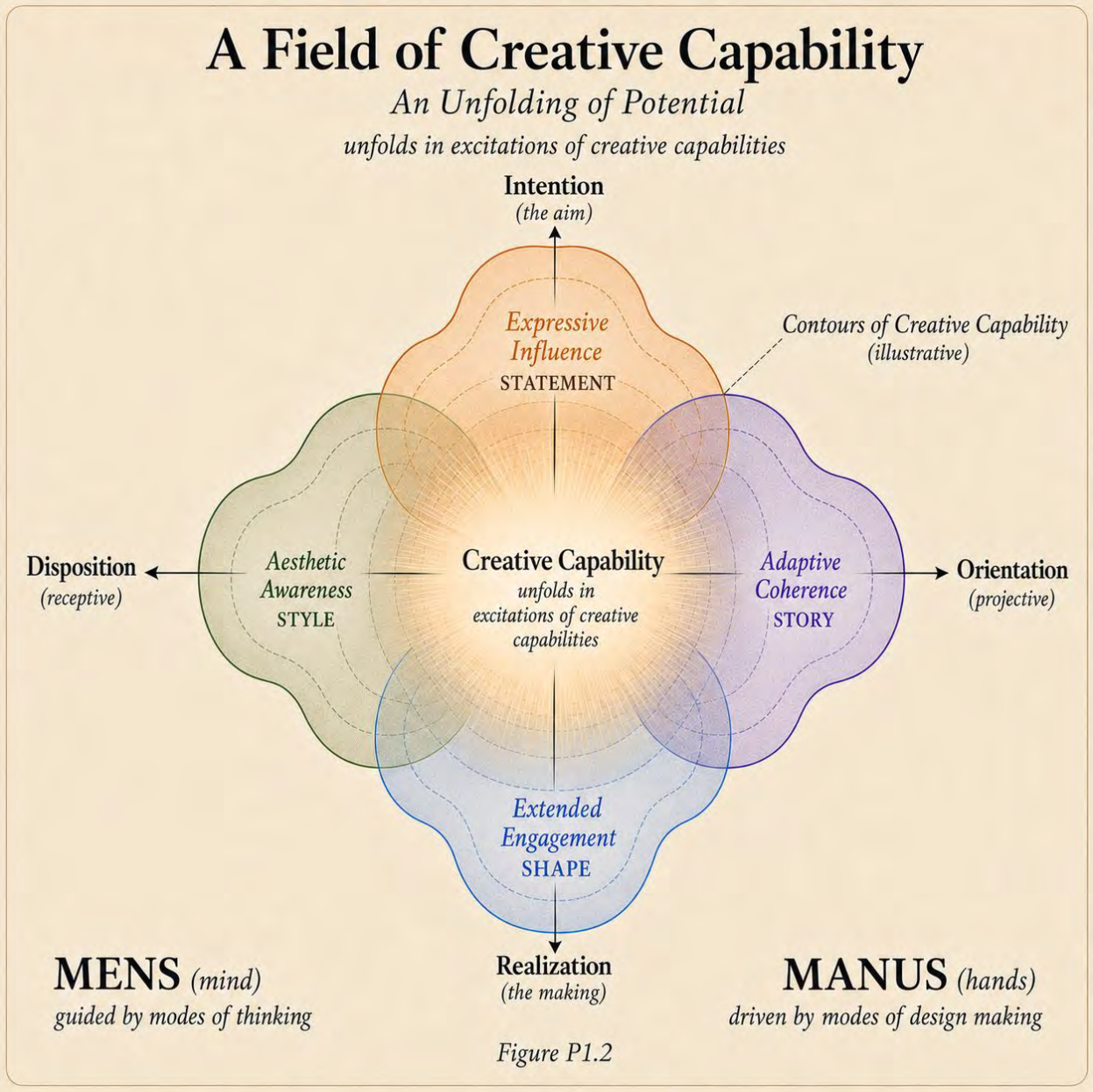


**Figure P1.2 The Field of Creative Capability. Source: authors.**

As illustrated in Figure P1.2, a creative capability is a contour across the field, not a corner of it. Each of the four draws on all four modes, and the shape it traces shows where it reaches farthest. Extended Engagement, Aesthetic Awareness, Adaptive Coherence, and Expressive Influence each carry their own receptive and projective modes along the *Mens* axis, and each meets the design object along the *Manus* axis. Mastery of one mode alone leaves the contour narrow. A capability grows as practice widens it across both directions, in the embodied excitation that runs from ecstasy to error and feeds back with each cycle. To develop a creative capability is to keep all four modes in play and the excitation within reach. The Creative Council in Chapter 5 is where the four capabilities are convened into a single capacity. This is Agentic Creative Intelligence.

In Part I we propose that the four capabilities are manifest through a design object. Held apart, they are four. Brought together around the work, they are one. A design artifact carries four aspects, Shape, Style, Story, and Statement. As illustrated in Figure P1.3, each aspect is foregrounded in a capability, excited by embodied sensations. Extended Engagement gives it Shape, the made form. Aesthetic Awareness gives it Style, its look and personality. Adaptive Coherence gives it Story, what it is for and how it is lived. Expressive Influence gives it Statement, the message it makes, and with that message the work first enters the social field where others read it. Each aspect is built from attributes, the details a body can sense: a material, a joint, a color, a texture, a chord, a held note, a hero's turn, a withheld secret.

The design artifact sits between the maker's excitations, the reactions ranging from ecstasy to error, in the pursuit of an ill-defined intention. When the outputs are read differently by each who meets it, affording more than the maker put in, it becomes a boundary object (Star and Griesemer, 1989). The artifacts plus boundary objects together make the design object. Architect Ettore Sottsass (1994) named the whole of it: "Designing is a never-ending search for a possible metaphor of existence."

## The Terrain of Part I

Part I introduces the creative capabilities as the practices and mindsets that hold human agency against the pressures of agentic AI (Figure P1.1). Human agency depends on these capabilities. Reclaiming it in the agentic era starts with the capabilities themselves: what they are, how they grow, and what now puts them at risk. Part I explores the capabilities through iconic practitioners, grounds them in embodied practice, and draws on philosophical Dispositions to design the archetypal dæmons that protect and promote them. Extended Engagement learns through the hand's meeting with resistance. Aesthetic Awareness is trained perception of what holds against what merely decorates. Adaptive Coherence holds a frame firmly enough to act on and loosely enough to revise. Expressive Influence carries the work into the social arena at the metabolic price of presence.

Each capability is enabled by an AI model that offers choices to either complement or compete with our creative capabilities. Informing our choice requires understanding how AI models emerge from structuring different data sources, how they are shaped by different training mechanisms. Each agentic pattern, the workflow for a specific task, clarifies the steps. A practitioner kept in the loop only to check quality becomes one more step in the process. So we weigh what is gained and lost when a process hard-wires its steps ahead of its purpose.

Our forecasts lead towards a multi-model future. Multimodal means one model taking in text, image, audio, and motion at once. Multi-model means something else: an architecture that chains separate models, routing each task to the system built for it and passing the result along. Multi-model architecture promises the generativity of diffusion models, the causal reasoning of inference models, the spatial grounding of world models, and the linguistic intelligence of large language models, all of it available to those who can orchestrate across them.

Systems thinkers have reason to want this, because the most wicked problems resist any single discipline. They may yield only to several intelligences working together. The more the models are chained and orchestrated, the sharper the question of what integration the practitioner still performs.

The leap risks dulling embodied excitation. It also offers to augment our collective agency, and that is what makes it worth taking. To prepare the mindset, we invoke Chinese philosophical traditions to encourage more integrative, conscious creative choices.

To explore the fields of creative capabilities that human agency depends on, we go first to a competition entry that arrived as a painting where the brief expected a plan.

## Chapter 1: Extended Engagement: Hands Form the Mind

**Figure 1.1 Portrait of Zaha Hadid, illustration by Shiyi Chen**

### The Hand That Draws the Building

Open the painting and the building is already moving. In 1982 Zaha Hadid entered a competition for a leisure club on Victoria Peak in Hong Kong, and her entry arrived as a painting (Figure 1.2). Shards of architecture tilt across the canvas, the site seen from above and within at once, planes sliding past one another in a space no single perspective could hold. She called it a horizontal skyscraper. The jury awarded it first prize, and the club stayed unbuilt.16 The painting did the work the building was years from being able to do.

**Figure 1.2: Zaha Hadid, painting for The Peak competition, Hong Kong (1982–83). Reprinted with permission of the Zaha Hadid Foundation.**

Hadid trained by painting before the profession would let her build. At the Architectural Association in London she inherited the conventions of architectural drawing, the plan, the elevation, the axonometric. But they proved too narrow to hold what she sensed about

space. She turned to Kazimir Malevich and the Russian Suprematists. Her painting became an instrument of investigation, the abstraction through which form became spatial engagement.17 Canvas after canvas, she deconstructed and reassembled. Gradually geometry gave way to fluidity (Figure 1.3). Her extended distortions were how she found the forms her later buildings would stand on.

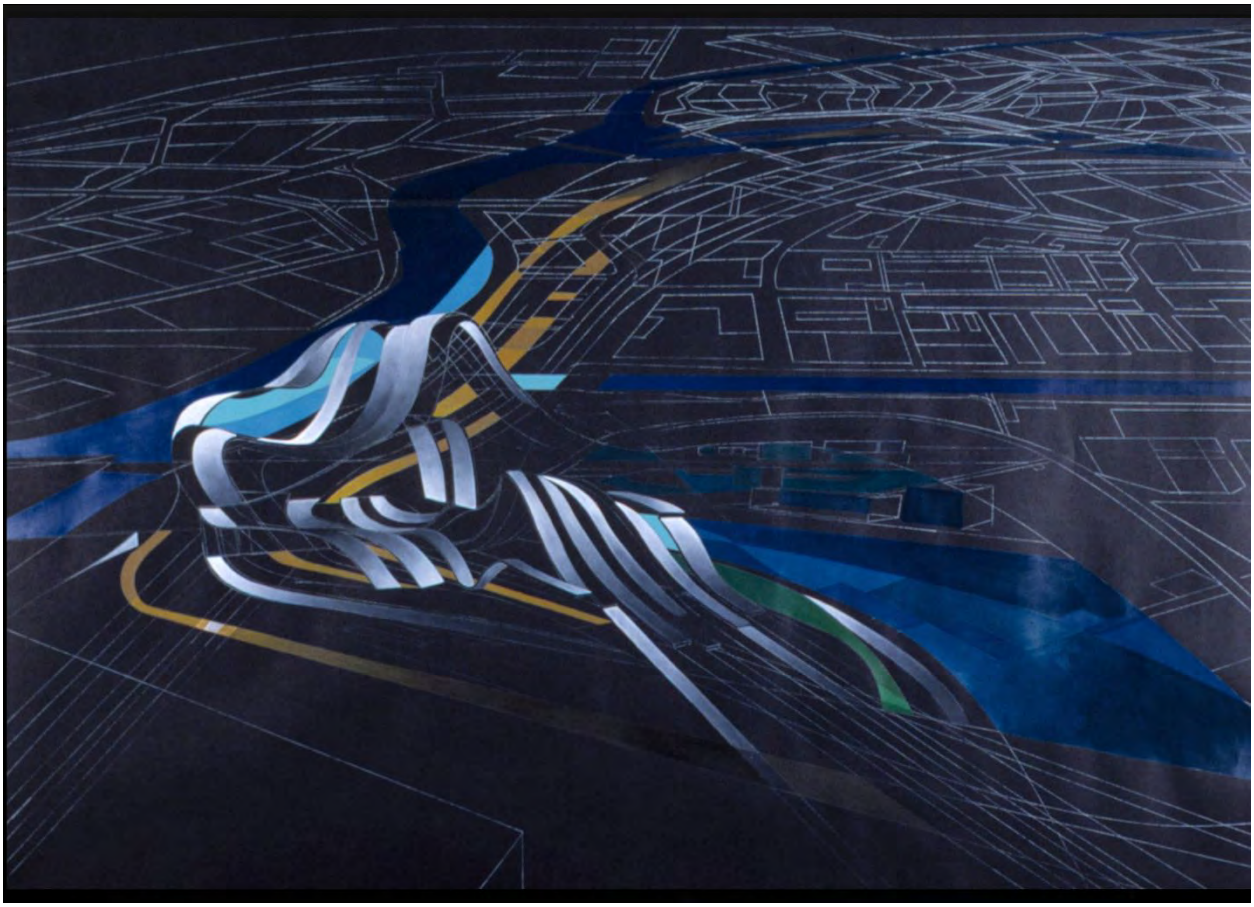

**Figure 1.3: Zaha Hadid, spatial study for La Grande Mosquée de Strasbourg 2000, Strasbourg, France. Reprinted with permission of the Zaha Hadid Foundation.**

For years the work stayed on paper because the means to build it lay years off. The continuous surfaces she painted, folding through space, exceeded what a construction industry organized around the straight line could deliver.

Her first completed building, the Vitra Fire Station in Germany, opened in 1993, a decade after The Peak proposal.

Then the tools arrived. Parametric software, developed in her practice with the architect Patrik Schumacher, let the studio hold a curved surface as a set of relationships a computer could compute and a contractor could build. The MAXXI museum in Rome, the Guangzhou Opera House, then projects across the globe. In 2004 she became the first woman to receive the Pritzker Architecture Prize, for as the jury wrote "…her work experiments with new

spatial concepts intensifying existing urban landscapes and encompassing all fields of design, from the urban scale to interiors and furniture." 18

Learning from Hadid, we see **Extended Engagement** as the capability practitioners develop when tools sustain embodied contact long enough for the encounter between tool, material, and maker to teach what each alone would miss. The capability holds two modes. Engagement is the sustained contact through which the body meets what resists, the brush against the picture plane, the hand against the geometry. Extension is the reach a tool carries beyond what the hand alone can hold. Each runs through the maker and the tool together. Through sustained contact, the temporal friction restructures the body-brain systems. What emerges is the knowledge that study alone would hide.

### Homo Faber: We Make, Therefore We Are

Hadid's paintings extended her engagement with space and composition, until she could see forms beyond what convention taught. The computational tools she inherited extended the vision into buildings.

A surgeon knows the moment a suture bites correctly through the resistance traveling up the needle driver into her fingers, not through the magnified image on the monitor. A chef salts by feel, the crystals falling from a height calibrated across thousands of services. A software architect diagrams a system on a whiteboard and discovers, mid-stroke, that the dependency she drew reveals a flaw the specification obscured. In each case, the hand is thinking. Understanding arrives through the act, not before it.

Philosophy of mind calls this tool-as-cognition the extended mind (Clark & Chalmers, 1998).19

The taxonomic label *Homo sapiens* flatters us. It implies that our species-defining characteristic is the capacity to know, to contemplate reality from a position of detached reflection. The archaeological record tells a different story. Human intelligence emerged to shape stone. The name for that maker is old. Henri Bergson called the species *Homo faber*, defining intelligence by the capacity to fashion tools (Bergson, 1907). Hannah Arendt gave fabrication its own standing in *The Human Condition*, the work that builds the durable world we then live inside (Arendt, 1958).

Consider the Oldowan chopper (Figure 1.4), a jagged river stone struck with specific intent to shear meat from bone, dating back 2.6 million years. This artefact predates widespread symbolic language by over a million years. The hand that shaped it was thinking through the stone, discovering what the material would permit. Each strike was at once a hypothesis and its test. Hit here, with this angle, at this force. Will the flake detach cleanly? The answer came in the crack of quartzite. The maker did not first conceive the tool and then execute the conception. The conception emerged through the execution.

**Figure 1.4 Oldowan chopper (Early Stone Age). Flaked stone tool used for cutting and pounding. Image in the public domain (CC0), via Wikimedia Commons.**

Neuroscience explains why. Mechanical force, the impact of stone on stone, hand on material, physically rebuilds the maker. Pressure receptors in the hand fire when the chisel meets resistance. That signal travels through the nervous system and triggers a structural

change cycle in the brain, where the neural circuits most used during effortful practice grow stronger, denser, faster. The system is integrated: the hand reshapes the brain, and the reshaped brain returns to the hand with sharper perception.

The Oldowan knapper's brain did not first develop the capacity for complex toolmaking and then apply it. The toolmaking itself sculpted the neural architecture that made increasingly complex tools possible. The connections that fire together during effortful work thicken with use, the body's sensory receptors, the spinal pathways that carry their signals, and the brain circuits that interpret them all strengthen as a single system. Each hour at the workbench builds the integrated architecture that returns tomorrow.

The encounter between hand and stone constituted the cognitive capability for toolmaking, one no previous mental competence could supply (Malafouris, 2013). The hand meeting the stone *was* the thinking. Agency emerged between maker and material, irreducible to either alone. Richard Sennett calls this accumulated dialogue "material consciousness": knowledge of what a medium can do and what it resists becoming, built through failures that teach and successes that reveal (Sennett, 2008).

Another Pritzker Architecture Prize winner Wang Shu demonstrates this dynamic relationship. Shu spent nearly a decade on construction sites after his 1988 master's degree. He learned through his hands how materials behave in ways specifications leave out. His wapan technique binds mismatched stone and tile fragments into walls with lime. The mason sorts each fragment by weight, texture, and age, the older tile lighter and more brittle, each surface a history the hand reads in friction.

His technique demonstrates what a specification cannot standardize. Every digital model remains at a distance removed from the dialogue between hand and material. Creative capability lives in the encounter.

**Figure 1.5. Stadelhofen Station, Zurich (1983–1990), photo by author**

Forces are the source of material resistance, so the body reads them the same way. Descending into Santiago Calatrava's Stadelhofen Station (Figure 1.5), a commuter feels the forces in the hillside before any explanation arrives. Built in Zurich in 1983 shortly after completing his PhD on the foldability of space frames (Calatrava, 1981), Calatrava wraps concrete vaults to carry the weight of a hillside down to the platform. The structure curving with the tracks and bending with the slope offers an embodied experience. Behind his creative expression lay hundreds of anatomy drawings, teaching him to read the body as a structure under load. In pencil and watercolor, he found the femur channeling weight, the spine distributing compression. His structural imagination, like Hadid's spatial imagination and Wang Shu's hands, was built through years of contact with what resists.

Neurologically, these sustained encounters shape the maker. Pressure receptors in the hand fire when the chisel meets resistance. That signal triggers a structural change cycle in which the neural circuits most used during effortful practice grow stronger, denser, and faster (Merzenich, 2013). As the practitioner's pressure receptors meet resistance, the sensorimotor maps that encode the tool densify, and the maps that take in a tool extend to include it (Iriki, Tanaka, & Iwamura, 1996). The enlargement tracks the gains practice earns rather than the raw hours it costs. An architect's pencil hits the tooth of paper and moves her concept into a new angle. The carver's blade finds in the grain what the drawing could spare. The parametric model extends the designer's reach into structural possibilities the pencil opened and the hand still has to judge.

Cognitive science gives the sensorimotor variable a name: deliberate practice (Ericsson, Krampe, & Tesch-Römer, 1993).20 The rebuilding starts with a signal the body registers the moment it reaches past what it can do. That signal is embodied excitation in its first form. It is the sensation of the **Unreachable**, the felt yearning to go beyond what a situation offers, despite the risk. Like every embodied excitation it registers in two modes: the ache of a reach that outruns present skill and the charge when that reach lands and the hand holds what it could not before. It emerges where intention meets resistance, when the cortical restructuring occurs.

Extended engagement develops when the practitioner stays inside that felt signal until the tool teaches what the confrontation reveals. The unreachable encounter is the body's report that current perception has grown insufficient, and that the cortex is rebuilding to meet the demand.

Creative practitioners engineer such errors by design. Confronting the strict orthodoxy of 1960s recording studios, clean takes and engineered precision, the guitarist Keith Richards prized the incorrect method, returning to a worn amplifier and an overloaded tape until his ear learned, take after take, which wrong way of recording made the right sound.

David Gilmour amplifies the principle into sound. His left hand applies vibrato to the string while his right moves the tremolo arm by fractions of a millimeter, a delay pedal catching the shifting pitch and scattering it into repeating space (Beato, 2025). Three sources of motion, finger, arm, circuit, converge into a single sound none could produce alone. The

guitar extends his engagement with vibration and decay across a chain of instruments, each adding resistance and possibility beyond the last.

Engagement is a sustained dialogue that transforms both maker and material. Extension has an equally specific sense: tools amplify that engagement past what unmediated perception could reach. The strings, frets, and pedals extend Gilmour's fingers into perception beyond the reach of the hand alone.

Bypass that conversation with a tool, and no new knowledge can form. The songwriter Nick Cave put the same charge against AI directly, that an algorithm has nothing to transcend, no resistant material it has suffered through, so it returns a fluent imitation of feeling without the struggle that earned it.[21] When the body chooses brief engagement over sustained practice, the body-brain system quietly dismantles what practice would have built. The cortical maps that encode learned skills lose spatial precision. Neurons that once responded selectively begin responding to wider, less differentiated inputs (Merzenich, 2013).[22]

Each extension restructures the maker, building the perceptual infrastructure creative capability depends on. Its erosion is subtle. At first it reads as rest, a welcome reprieve. It feels easy.

### Extending Intelligence: AI Models and their Learning Lifecycle

Since the Dartmouth summer of 1956, artificial intelligence has pursued a singular notion: intelligence could be extended by computation. Herbert Simon, a cognitive scientist and founding figure of symbolic AI, built the General Problem Solver in 1957 with Allen Newell and John Clifford Shaw, a machine that took information in, turned solutions out, and reduced reasoning to heuristic search (Newell & Simon, 1972).

This starting point yielded the models of AI we know today. A practitioner who understands each one keeps the right to ask whether an agentic pattern's goal and its subtasks are worth pursuing.

The AI race started with large language models, where a transformer predicts the next token across a widening context window.23 Nearby is the world model, which raises a navigable, geometrically coherent three-dimensional environment from a prompt or an image and predicts how that environment changes under action.24 Existing places can be captured through 3D Gaussian splatting, while the non-generative Joint-Embedding Predictive Architecture (JEPA) functions as a latent-space sibling to this predictive family.25 Rather than prediction from existing content or current reality, causal inference models seek to infer the direction of cause and effect from observational data, distinguishing adjacent data structures from mere correlation. Diffusion models proceed differently again, learning to reverse a forward process of Gaussian noising into text-conditioned images. Orchestration addresses the coordinating layer above these systems, where a model organizes specialized components and separate agents through shared protocols.26

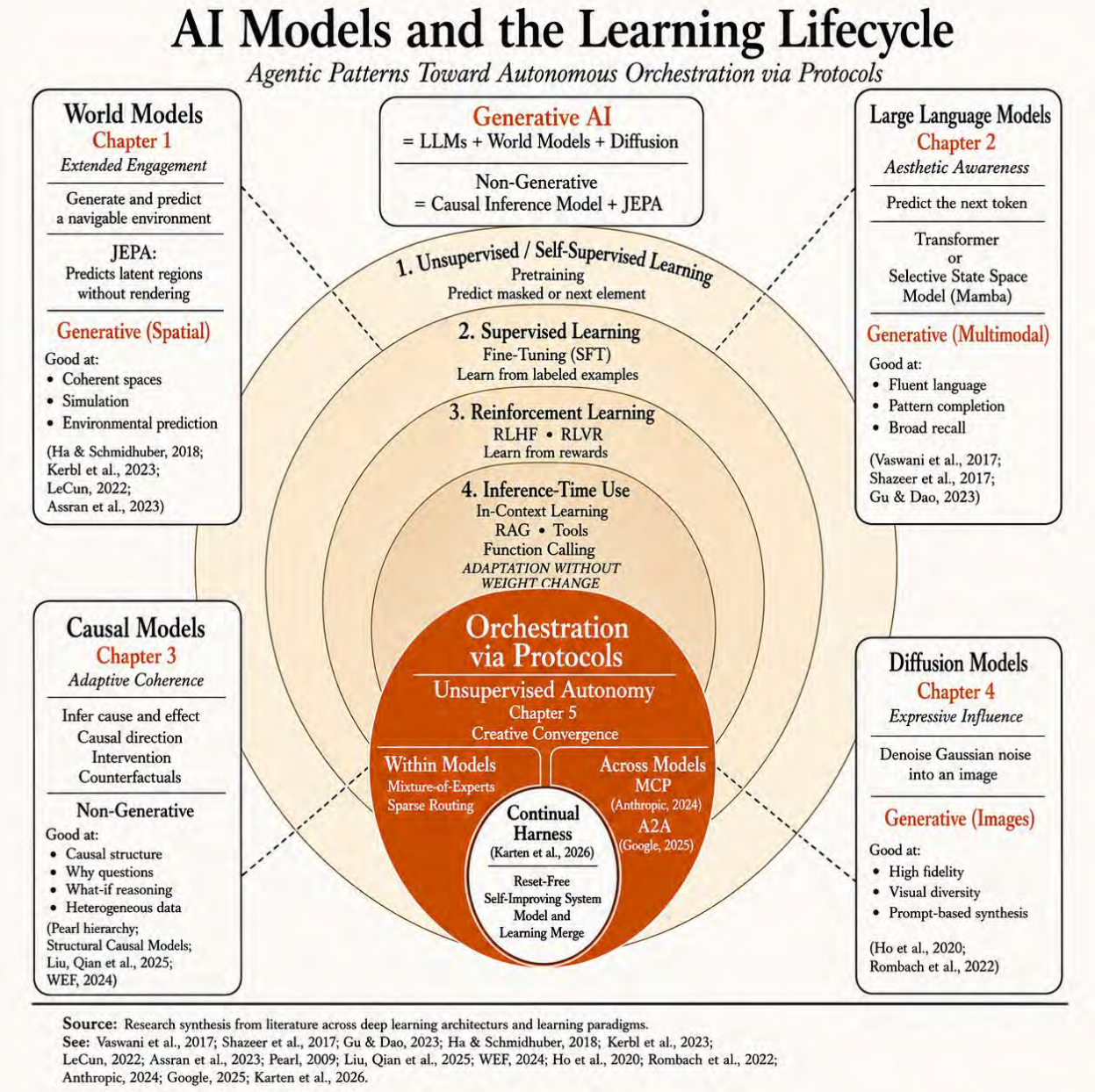


**Figure 1.6. AI Models and Learning Lifecycles. Source: authors.**

These systems share a learning lifecycle, the training by which a model is built and then improved (**Figure 1.6**). It opens with pretraining, self-supervised learning at scale, the foundation every later stage rests on. It continues through post-training, where supervised fine-tuning and reinforcement learning shape the model toward useful and aligned behavior. Human feedback drove this shaping first, and verifiable rewards that incentivize correct reasoning now extend it.27 At inference time the trained model is adapted without any change to its weights, through in-context learning, retrieval, tool use, and reasoning carried out at the moment of use.

Read across that lifecycle, coordination is converging on orchestration through protocols. Within a single model, a Mixture of Experts routes each step to specialized subnetworks. Across models, open protocols let separate agents discover one another and delegate, the Model Context Protocol exposing tools and data to any compatible model, and the Agent-to-Agent protocol connecting agents built on different frameworks.28 The model learns how multimodal input correspond, so a single system reads a photograph, its caption, and the voice describing it in one pass. In the near future lies a multi-model architecture. Part I's forecasts converge on this multi-model future. At a meta-level, this will maintain a separation of the models, handing them tasks from a language model to a diffusion model to a world model, each doing the part it was built for. Such compound systems, assembling several specialized components, will break the monolithic model.29 It might extend systems thinking onto the hardest problems, the wicked ones that resist any single discipline. They ask for the generativity of diffusion, the causality of inference models, the physical grounding of world models, and the linguistic reach of large language models at once, combined in a pipeline no single model or thinker can span.

These agentic patterns are catalogued now as a discipline of their own, the internal workflows a single agent runs, prompt chaining, retrieval, planning, and reflection, set apart from the connection protocols that let separate agents interoperate (Gulli, 2025).30 Questions of convergence will test practitioners' capabilities to resist an agentic pattern or protocol that routes a task and its result, while asking nothing of whether either was worth pursuing.

### World Models: Where Physical Meets Digital

Gilmour's solos build worlds a listener can almost stand inside, the bend of a note opening a space the ear alone cannot account for, and his fingers spent years earning that reach. Calatrava's vaults reach a kindred edge, growing interesting where they defy the concrete they are poured from, an edge a maker reaches only by having internalized how the material resists. A world model extends a practitioner toward that edge, and it arrives already capable. At its core it predicts shape. Trained on images, video, and motion, it raises an environment that holds its geometry from every angle and works out how that environment behaves under action, so a maker can imagine a form as if it had weight and grain before any material is touched (Ha and Schmidhuber, 2018).31 The model predicts rather than touches. What it hands over is the imagination of physical consequence, whether a cantilever reads as daring or doomed, whether a curve carries load or only decorates. Fei-Fei Li, who shaped modern computer vision with her ImageNet work, calls this capacity spatial intelligence, a three-dimensional sense of how things sit and stand and move.32

Trained on the recorded behavior of the physical world, the model emerges good at three things a maker of shapes prizes. It builds coherent space, geometry that holds from every angle. It supports simulation, letting a body act inside the form and watch it answer. And it offers physical prediction, carrying a structure forward in time so its behavior becomes visible in the present. Together these make it generative at the scale of worlds (Figure 1.7). Hand it a fragment and it returns a form to stand inside. For a working maker this means a

shape on demand, structurally resolved, a building interior generated in minutes and walked in a headset before a single drawing is committed.

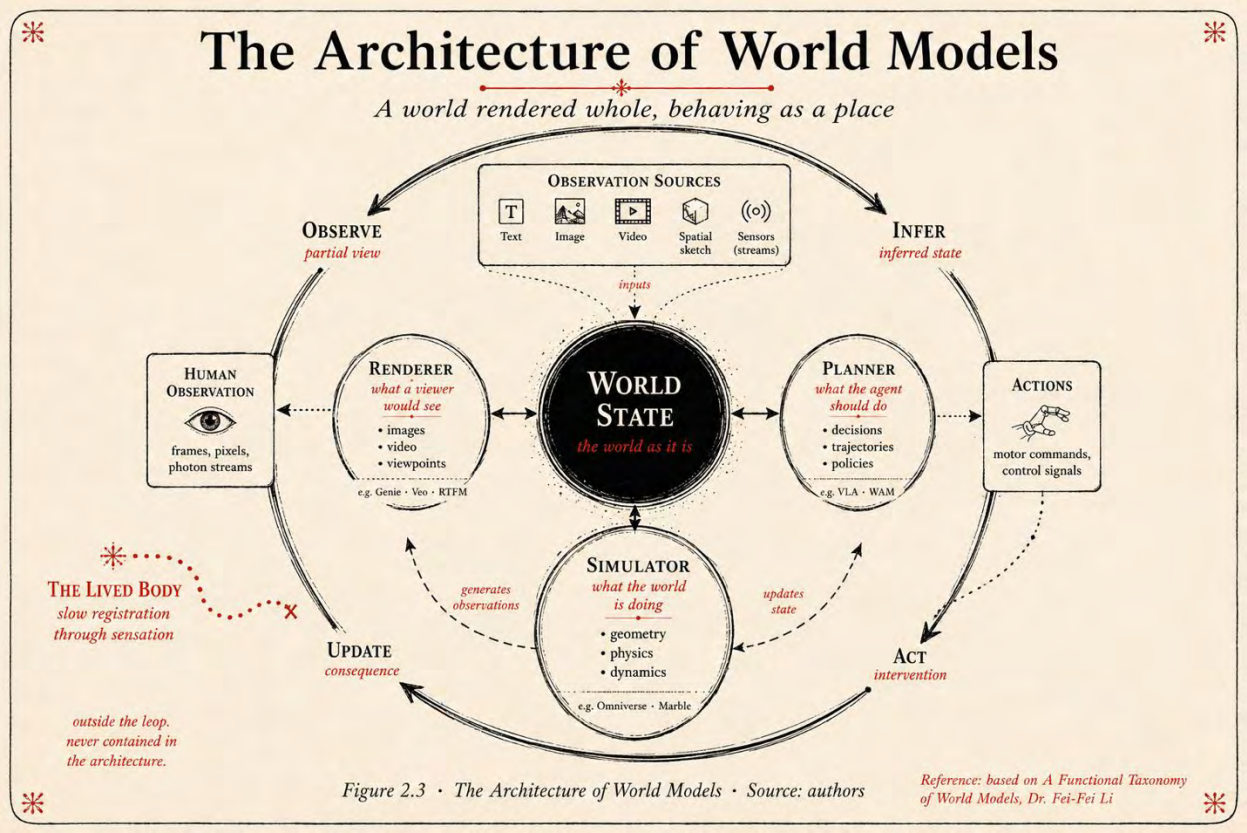


**Figure 1.7. The Architecture of World Models, after Fei-Fei Li's framing of spatial intelligence (2026). Source: authors.**

This reach asks almost nothing in return. Material demanded years before it answered. The model answers first. The capability at stake, extended engagement, is built from the effort the model offers to spare, the practitioner's sustained, hands-on contact with what pushes back. Each gift that follows serves the work. Each can also stand in for the resistance the work depends on.

*Generation renders the shape before the hand*

A practitioner directs the model through a prompt, and the model returns the environment as structure instead of surface alone. A model that predicts at the level of pixels yields only a blurry average of every possible future, so a world model commits to depth, position, and form. Tencent's HY-World 2.0 produces a navigable environment from a single photograph, with consistent depth and surface.33 World Labs' Marble, in commercial release since late 2025, lets a body enter that environment and turn inside it. The gift is the walkable draft. An architect walks an interior before it is drawn, feeling for the doorway that compresses and the vault that lifts. A filmmaker holds six location variants weeks before a scout could fly, and a novice produces in minutes what a trained 3D artist needed a week.34 The cost arrives with the finish. A shape born complete, structurally resolved, offers the hand little

left to resolve, and the maker receives the result of structural understanding without performing it.

*Simulation lets the form answer*

A generated shape stays a still image until prediction arrives. When a maker acts inside the environment, the simulator predicts the next state, working out the form's behavior under its own geometry and forces. Odyssey's Starchild-1 runs that behavior live, streaming synchronized image and sound and answering a viewer's action as it arrives.35 A door opens and the structure holds.36 The value is rehearsal, a maker blocking a form inside a space that answers her, testing whether the cantilever stands before the concrete is poured. The limit is felt resistance. A simulated world conveys how a form looks and moves, and withholds how it pushes back. Robotic systems trained entirely inside such environments reached 70 percent success on real-world grasps zero-shot, and 91 percent only after fine-tuning on thousands of real grasps.37 The system learned a world whose resistance it sensed only through images.

*A latent sibling keeps only structure*

A Joint-Embedding Predictive Architecture, or JEPA, sits at the architecture's far edge. It predicts the representation of a hidden region in an abstract latent space and renders none of it, keeping only structure and discarding the unpredictable surface (LeCun, 2022; Assran et al., 2023). The power is transfer. For a robot learning how rooms and objects behave, structure is exactly what carries from the model into the world. Gaussian splatting (Kerbl et al., 2023), a capture cousin of the world model, renders a real site from photographs as a navigable scene within hours, and for the first time the result convinces the eye.38 A conservation architect circles a Venetian courtyard from her desk in Chicago and decides from inside the splat whether the trip earns its carbon. The capture preserves a visit that happened and answers the next one in advance, a bargain worth naming before accepting.

Olafur Eliasson works the same seam from the other side. His studio runs computational simulations of light, weather, and geometry, then builds the result at full scale in the workshop, where a fog or a low sun has to answer a body standing in the room before it ever reaches a museum hall.39 The model proposes and the prototype answers back. It is how *The Weather Project* hung an artificial sun in a Turbine Hall and how *Ice Watch* set blocks of glacial ice melting in a public square, the digital held honest by the physical it has to become.40

That same speed is the seductive paradox. A standing, structurally resolved shape arrives at once, and the bodily effort that once produced it can be skipped, with the loss hidden beneath the finish. Extended engagement runs on the unreachable, the felt yearning to reach past what a situation offers, and a shape that stands in simulation answers so quickly the reach goes unfelt. The maker who would have learned why a form holds, by working against a material that answered back, receives the answer pre-assembled. The danger is shape that defies materiality without the maker knowing what materiality it defies, daring by accident rather than by judgment.

The risk lives less in the tool than in the stance a practitioner brings to it. The computational architect Michael Hansmeyer, known for algorithmic column geometries developed at ETH Zurich, reads what the model returns with a trained eye, a sight he built first at full scale in columns a hand can touch. His generated forms carry forward the computational form-making Hadid opened, and the reach is real because the body laid the foundation first. A maker who begins at the prompt builds a different skill, taking the standing shape for the understanding that should underwrite it. The same prompt, in two hands, does opposite work. It extends a judgment already earned, or it stands in for one still forming.

What the world model offers to separate is knowing from the doing that completes it. The practitioner chooses each time whether the prompt extends a judgment she has earned or stands in for one still forming, so she needs the mindset that keeps the doing inside the knowing.

### Knowledge and Action Are Inseparable

The Neo-Confucian philosopher Wang Yangming confronted a problem that resembles the one facing practitioners who accept the model output without its materiality. His students mastered the canonical texts and could discuss virtue with precision, yet faltered when the moment asked them to perform it under pressure. The separation of knowledge from action, he diagnosed, was itself the disease. His position broke from the Song-dynasty orthodoxy of Zhu Xi, whose *gewu zhizhi* (格物致知, the investigation of things to extend knowledge) held that knowledge accumulates through careful study first, with action following.

Wang reversed the sequence. Authentic knowledge exists only through doing. His principle was *zhīxíng héyī* (知行合一), the unity of knowledge and action.41 Knowing and doing complete each other, and each stays partial until the other arrives. Knowledge in the mind apart from corresponding action is mere opinion, and the world waits to confirm or dismiss it.

The diagnosis exposes what structural prediction hinders. The model resolves the shape over the physical regularities it was trained on, and accepting the output passes that resolution to the practitioner. Structural prediction keeps her from reaching the unreachable, the resistance her body meets only by acting through the material. *Zhīxíng héyī* keeps her inside that error until knowing and doing close the gap.

Where does your knowing end, and the model's output begin? Return the practitioner to the unreachable her body generates only by acting, against every returned token.

Knowledge of a material lives in the body that has engaged it. The calligrapher Cheng Yao-t'ien traced the transmission: creative force passes through the entire frame in sequence, lower limbs to upper limbs, shoulder to arm to wrist to fingers to brush tip.42 The brush moves because a wave of kinetic force and respiratory rhythm transfers through the whole body. The wave exits the tip as a direct, correctable trace of the maker's somatic state. Knowing and doing, in the brush's path, remain indistinguishable.

Each tool that extends through sustained contact rebuilds the brain and body that carry the knowing. The model produces its output in a single pass through the network, lacking the embodied sampling that sequential transfer from limb to brush tip depends on. It distances the calligrapher from the unreachable, the tremor the wave meets midway between shoulder and wrist. The generation lets that tremor escape the loop. Even an exchange held open across many turns decays the same way. Across fifteen leading models and more than two hundred thousand simulated conversations, average accuracy fell by about thirty-nine percent as the dialogue ran on, the system settling on an early assumption and never reopening it.43

What in your practice do you learn only by doing? Hold the single pass open until the body runs its own wave and meets its own resistance.

The principle reads just as sharply into the structure of the model. Every property performs the separation Wang identified in his students. Generation hands over the standing shape and cuts the search. Simulation ships the behavior and skips the stretch. Agentic autonomy produces the built form and erases the making. Through capture the site appears ready-made, and the practitioner skips the visiting. The output is competent and out of time with the body. The practitioner who accepts it has learned as much as she already knew. Each property blocks a different face of reaching the unreachable, and what the body would have encountered through action stays apart from it.

Which step between the question and the answer did the tool take that you should have taken? Keep at least one opening to reach for the unreachable across every property the architecture composes.

Zhuangzi's Wheelwright Bian challenged Duke Huan's book-learning with what his hands know. The chisel too gentle, it slides. Too hard, it bites.44 The Duke's books, the wheelwright tells him, are "the chaff and dregs of the men of old," because genuine skill lives in material encounters. Seventy years have taught him the right moment between the two, a knowing that escapes words and escapes even his own son. Authority over the text comes from the palms that have felt the chisel slide and bite.

The encounter requires time the practitioner has agreed to spend. Wheelwright Bian spent seventy years earning what the world model hands over already standing, the knowledge of how the wood answers the chisel. The unreachable lives in the seam he found between "too gentle, it slides" and "too hard, it bites." Inside the assembly that seam closes before the hand can question it.

Which connection in the output would your hands test, if they were the ones making it? Hold the world model open to the seam, so the unreachable can surface where the hand would have questioned the fit.

The musician Ryuichi Sakamoto, a Japanese electronic-music pioneer and Oscar-winning film composer, honored *zhīxíng héyī* across six decades of extending into computational instruments. Classical piano from age six built the infrastructure that made electronic experimentation meaningful. Yellow Magic Orchestra extended his engagement with synthesis and sequencing. The score for *The Last Emperor* (1987) extended it further. Each

new tool reached beyond the hands that formed him. Each left those hands intact, because they had done the work first.

Sakamoto's six decades describe the inverse of what agentic autonomy hinders. His *Opus* recordings capture a body spending its last reserves on an encounter it earned.45 Knowledge a body lays down through sustained contact can travel far into computational instruments. It returns to the body that carries it because the reach those instruments extended began as reach the hands first earned. The unreachable he met at the keyboard from age six was the infrastructure each later tool extended from.

Agentic autonomy offers to retire that infrastructure by producing the edit directly. The tool extended his reach only because the hands stayed with every unreachable sensation that formed them.

Is the tool extending a practice you already built, or standing in for a practice you will establish? Preserve the slow road that agentic autonomy would otherwise retire, and with it the unreachable encounters that made the road worth walking.

The Machine Heart, which Zhuangzi called *jī xīn* (机心), illustrates what the gardener feared.46 He refuses the well-sweep because mechanical convenience will restructure his consciousness, replacing craft knowledge with mere efficiency. What Zhuangzi identified phenomenologically, neuroscience now maps. When the tool becomes too efficient, the cortical maps that encode a learned skill lose precision. The output accumulates, the practitioner feels productive, and the body has stopped learning. The maker operates the device fluently and loses awareness of what it has displaced.

The properties of the architecture arrive together, and together they close the route to the unreachable. The cortical maps give way as one, because the resistance they encoded stops reaching the body.

Strip the tools away. What can your practice still do? Keep this question answerable by the body, by holding at least one way of reaching the unreachable open to the practitioner's frame.

### The Engagement That a Dæmon Extends

An intermediary inspired by Hadid is dominated by the design aspect of Shape. Its attributes include the continuous surface that refuses a corner, the cantilever that holds a curve in the air, the poured concrete and the rib that lets the form bend without breaking. A visitor reads those attributes in the body before any name arrives, one drawn in by the sweep, another unsettled by it.

The Extended Engagement dæmon is designed to re-couple the hand and the mind when technology offers to shortcut them. It keeps one record of bodily reach open across every property the model assembles, restores the iterative loop that single-pass interaction flattens,47 and protects the action through which knowing completes itself (Figure 1.8).

**Figure 1.8. Extended Engagement Dæmon Archetype, illustration by Shuqian Yu, designed by authors.**

It provokes and prolongs the unreachable. The limit arrives the moment a tool opens an encounter the hands have yet to shape. The dæmon keeps her at that limit until the body learns what intention alone may miss. Under the pressure of speed and competence, unaided judgment lets the tension between depth and reach collapse. The dæmon is the counter-discipline.

*Engagement* holds the single pass open until the body runs its own wave and meets its own resistance. It returns the practitioner to the unreachable her body locates only through sustained action, and its working sign is the hand that reaches for the physical thing before the screen's next variation. Its shadow is obsession: the tool takes the practitioner down rabbit holes until perspective on why the work matters dissolves.

*Extension* preserves the slow road that agentic autonomy would otherwise retire, guarding the genuine reach that computation opens beyond what the hands alone can hold. Its working sign is a practitioner still tracing her reasoning from intention through output to judgment, inside the work. Its shadow is substitution: the tool's reach stands in for her own, and the output arrives technically credible and personally absent.

Together the two modes produce what years of painting produced. Hadid's hand without parametric reach would have stayed on the canvas, missing the buildings the world now holds. Parametric reach without her decades of painting would have produced fluent surfaces without the spatial imagination that lets a building move. Sakamoto's piano-trained hands without computational scoring would have stayed inside the acoustic frame. The scoring without those hands would have produced sequences missing the body's signature. Gilmour's vibrato without the pedal chain would have stayed inside the fretboard. The chain without his finger would have processed silence.

Each mode supplies what the other lacks. Engagement reaches its limit and extension begins. Extension finds its compass and engagement reads what it now sees. The work lives where each requires the other.

The first output holds together and the result reads competent. Identify the action you took that the tool left out. Identify what the work carries beyond what you placed.

The unreachable still presses after the tool has delivered its answer. Sit with it long enough to tell whether the work is refusing the limit, or whether habit is refusing the effort of continuing.

A parameter adjusts an output. The Extended Engagement dæmon asks the practitioner whether the adjustment serves the work or surrenders it.

*Can you trace your own action through the output? From intention, through the reach, to the form that arrived.*

### When Engagement Overextends

Hadid's spatial imagination reached its buildings because the hand earned it first. The same reach, extended past what a context could hold, met a different end. In 2012 Zaha Hadid Architects won the international competition for the new National Stadium in Tokyo, the

centerpiece of the 2020 Olympic Games, a vast curved form spanning the site. Three years later the Japanese government scrapped it. Estimated costs climbed toward 252 billion yen, roughly two billion dollars, and public protest gathered around the scale of the design.48 The practice held that the overruns came from steep rises in Tokyo's construction costs, and that the design sat within its budget. The stadium stayed unbuilt. Hadid died the following year.

Tools extended Hadid's engagement with space across her whole career. At MAXXI, engagement read the site and the museum's purpose, and extension carried the reading into a building that holds them. At the Tokyo stadium, extension ran ahead of what engagement with the context, its budget, its city, its public, could report back. The difference is what each mode was allowed to require of the other.

Extended engagement can build a greater awareness of what a site, an audience, a moment *asks* for, when the aesthetic sense permits it. Engagement depends on overcoming bias to perceive what is already present in the materials, the resistance a site offers before a vision is imposed on it. Orientation shapes what perception finds.

What engagement could read of the context, the aesthetic sense extends. Aesthetic Awareness is the perceptual check her reach outran, and the capability the next chapter builds.

## Chapter 2: Aesthetic Awareness: When Presence Reveals Clarity

**Figure 2.1: Portrait of J. K. Rowling, illustration by Shiyi Chen**

### A Castle Drawn Before It Is Written

A castle stands on a loch in the Scottish Highlands. The Astronomy Tower rises over the lake at a specific angle. The Forbidden Forest opens behind the greenhouses. A stone circle sits where the spine of a hill curves southeast. The Sorting Hat places the students in their houses on the first night. Across seven years of education they learn the spatial relations of this place by walking them.

All these relations originated in the mid-1990s, when, in an Edinburgh café, a hand drew them.

The sentences came later.

J. K. Rowling (Figure 2.1) sat in Nicolson's Café beside her sleeping infant daughter with a lined notebook open before her. The pages filled with maps, timelines spanning seven years of a young wizard's education, class schedules for fictional subjects, and family trees branching across four generations. Her hand moved between sketch and sentence to find out what the world felt like.

She was discovering what the prose would hold as the story was being written.49

Rowling described the sensation directly. "You can always tell when you have had a good idea when you are writing," she told one interviewer, "because you get this physical response to it, a surge of excitement" (Rowling, 2001). The test she trusts is somatic. The appraisal arrives after.

A good idea announces itself as a change in the body. The tower stood where it did because the felt weight of the castle required it. The lake sat where it did because the path from the castle to the water ran a particular length the reader's body would one day travel. The hand-drawn map carried what a description leaves behind, an aesthetic knowing of the kind that arrives when a hand has lingered long enough to record what the mind struggles to define.

By the time she worked on *The Order of the Phoenix*, Rowling carried a decade of writing and four novels in print. A practiced novelist could have drafted the book straight into prose. Her hand reached instead for a ruled grid, plotting each chapter against the timeline. The grid surfaced discoveries the prose alone would have missed. Working the climax out on it first let her compress its events into the span the page required.

The spatial relations placed the castle. The aesthetic style made it inhabited. The four Houses are the clearest case, each a palette and a temperament a reader carries for life, Gryffindor's scarlet and gold, Slytherin's green and silver, the crest, the common room, the cut of a scarf. Scaling into the Wizarding World was possible because Rowling's vision held that richness, a style that transcends the physical apparatus she first drew. Extended through eleven films, theme-park lands on three continents, and a stage production, it became a 21st century Gesamtkunstwerk, a total work of art. From Goethe and Wagner to

the Bauhaus, the western canon holds that the deepest aesthetic encounters engage the whole body, beyond any single sense.

The appetite for that fullness, the pull toward sensory overload, runs long and deep, from Plato's cave to the immersive halls of the present.50

From these inspirations, we propose **Aesthetic Awareness** as the capability the body builds when it stays with a sensation long enough to inscribe what holds. The body records the what. The mind arrives later to say why. It is what Rowling was doing in Nicolson's Café, and in her Wizarding World after it. She let her aesthetic vision surface through her awareness of its form. The dwelling and the vision emerged together, each detailing the other.

### The Body Registers First

Awareness in aesthetic experience moves through three depths that feed one another, the somatic recording what the body takes in, the affective intensifying sensation into awe, the integrative binding experience into shared presence and meaning. Chatterjee and Vartanian (2022) map the three onto the brain as the aesthetic triad (Figure 2.2).

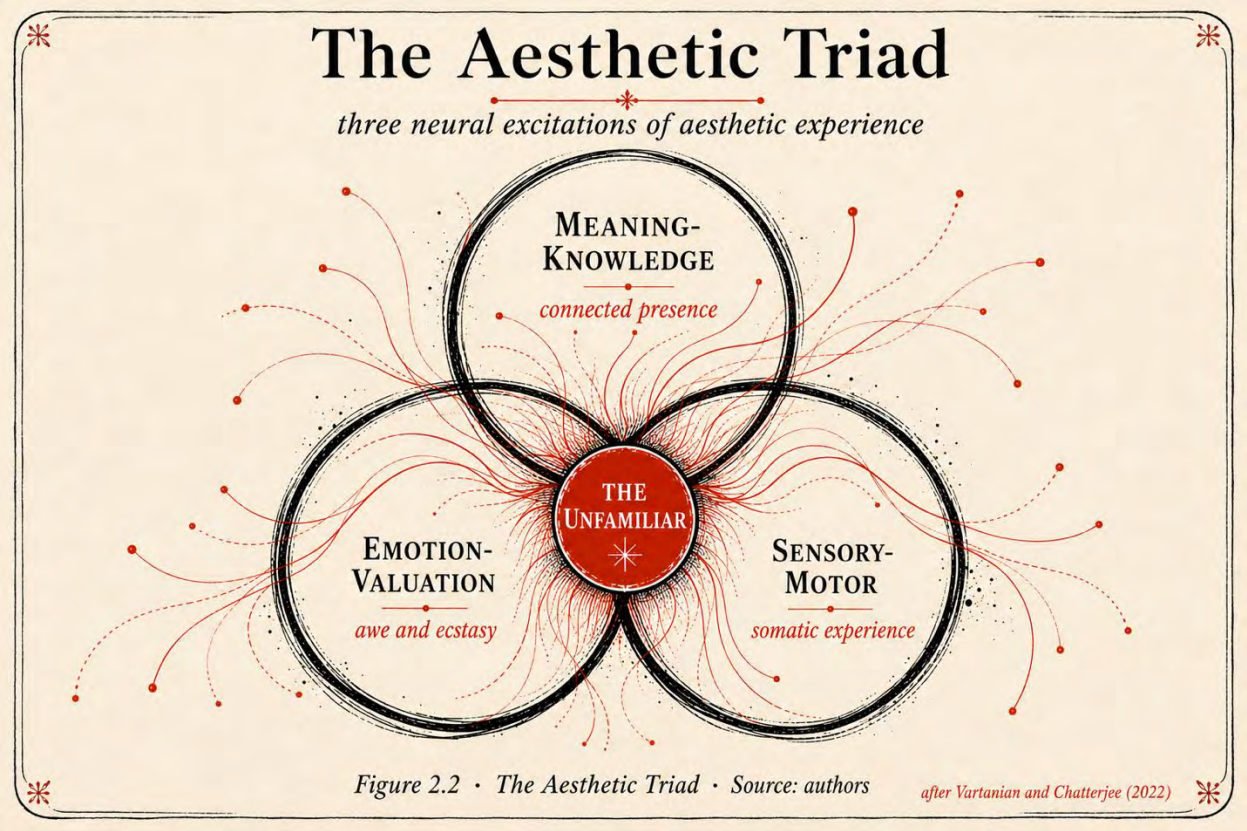


**Figure 2.2: The Aesthetic Triad, after Vartanian and Chatterjee (2022). Source: authors.**

Aesthetic intensity peaks when all three converge, the museum visitors' heart-rate shifts deepening into awe and binding into shared meaning.

Cognitive science adds a fourth pathway to embodied aesthetic awareness. Brain, body, environment, and action supersedes an older account in which a one-way signal travels from sense organ to cortex for interpretation (Niedenthal, 2007) . Interoceptive accuracy, the precision with which a person detects internal signals such as heartbeat and gut sensation, tracks the intensity of the emotional experience that aesthetic encounters draw on (Critchley et al., 2004). Deeper in, the insular cortex, a region in the folds beneath the lateral sulcus, integrates visceral signals with information arriving through the eyes and ears to produce what the body senses as significant (Bechara & Damasio, 2005). When a viewer stands before a painting that resists categorization, the insula holds it in view.

Predictive-processing research shows why interoceptive precision matters for aesthetic awareness. The brain's model of what is coming next rests on predictions from embodied experience, on what the body has felt, how it has moved, and what pressures it has encountered (Damasio, 1994; Damasio, 2000). An architect who has felt a building's weight in her body carries that model into the next building she enters. The compression of a doorway shortens her spine, the lift of a vault opens her ribs, and the drift of light cools the back of her hand.

Aesthetic awareness deepens through accumulated somatic memory. Visitors viewing artworks in a museum showed measurable changes in heart rate and skin conductance that tracked their aesthetic-emotional experience (Tschacher et al., 2012). First comes the body's signal. The mind formulates what it means. Awareness is the receptive attending that holds the practitioner inside the interval between the two.

Aesthetic awareness is excited in sensing the **Unfamiliar**, when the body meets what exceeds the mind's ready categories. The signal arrives in the toggle between interoceptive perception, the body sensing itself, and exteroceptive perception, the body sensing the world (Critchley & Garfinkel, 2017; Tsakiris, 2017). The activation is where the two streams briefly fail to integrate. The stomach tightens at a photograph, and the eye searches the frame afterward to find the fault. A harmonic shift catches the breath, the ear locating the change a beat later. The hand recoils from a surface the skin has yet to articulate.

The same threshold carries a second face. The unfamiliar arrives as discomfort, the stomach tightening and the hand drawing back. It also arrives as awe, the breath caught by a beauty no category contains. Dissonance and rapture are signals of opposite modes. Aesthetic awareness holds the threshold open whether the body reads it as unease or as joy, and the deepest encounters often arrive on the side of awe.

What the practitioner registers is that felt signal. The unfamiliar is an encounter with what categorization would otherwise close off, alongside a pull toward the ready category that would foreclose it. The sensation says that the categories readily available carry limits.

Aesthetic awareness develops when mental image and physical reality diverge. Pina Bausch built a choreographic practice on triggering such encounters. Her Tanztheater Wuppertal dissolved the boundary between theater and dance by asking performers to call up movement from the body's own history, from grief, desire, and embarrassment instead of learned choreographic vocabulary. "I'm not so interested in how they move as in what

moves them," she said (Servos, 2008). The training was a training in bodily awareness. The dancer learned to feel, with precision, what she already carried in the body, and only then could choreography shape it. Decades of rehearsal produced performances in which audiences reported tears rising, the eye following to find the moment that drew them.

The same lives in other materials. In Suzhou Song brocade, the silk master Qian Xiaoping works with what Chinese craft tradition calls shougan (手感), the trained capability to feel, through fingertip pressure and wrist resistance, which drape will hold its weight in wear and which will fall apart after ten launderings. Touch builds it across years of contact, and the silk teaches what a specification leaves out.

Across dance and silk alike, the body learns to sense what it will shape, one rehearsal at a time. Aesthetic awareness begins as a response to beauty, which opens an interlude for perception. The longer the interval, the deeper the experience. This is where world building begins.

### Large Language Models: A Cauldron of Aesthetics

Awareness draws from many sources. A language model draws from the largest of them, the accumulated record of human expression. Where a world model compresses physical space, a language model compresses culture, the web of reference, lineage, brand memory, and generational sensibility by which a work is placed in a register others recognize. Style is not surface detail. It is that web. A language model is trained on the web itself, so it locates a work in any known tradition, holds a consistent voice across a body of work, and surfaces the references that give a style its depth. Style lives in culture and language, not in physical space, and this is the work a language model can do that a world model cannot.

Trained on the largest corpus of human expression assembled, the model emerges good at three things a stylist prizes. It places a work, locating it in a tradition a reader will recognize. It holds a voice, carrying one consistent register across a whole body of work. And it surfaces reference, drawing up the lineage that gives a style its depth. Together these make it fluent in the whole inheritance of style (Figure 2.3). Hand it a fragment and it returns the fragment in any voice already expressed. For a working maker this means a register on demand, predictable and consistent, drawn from everything the culture has already said.

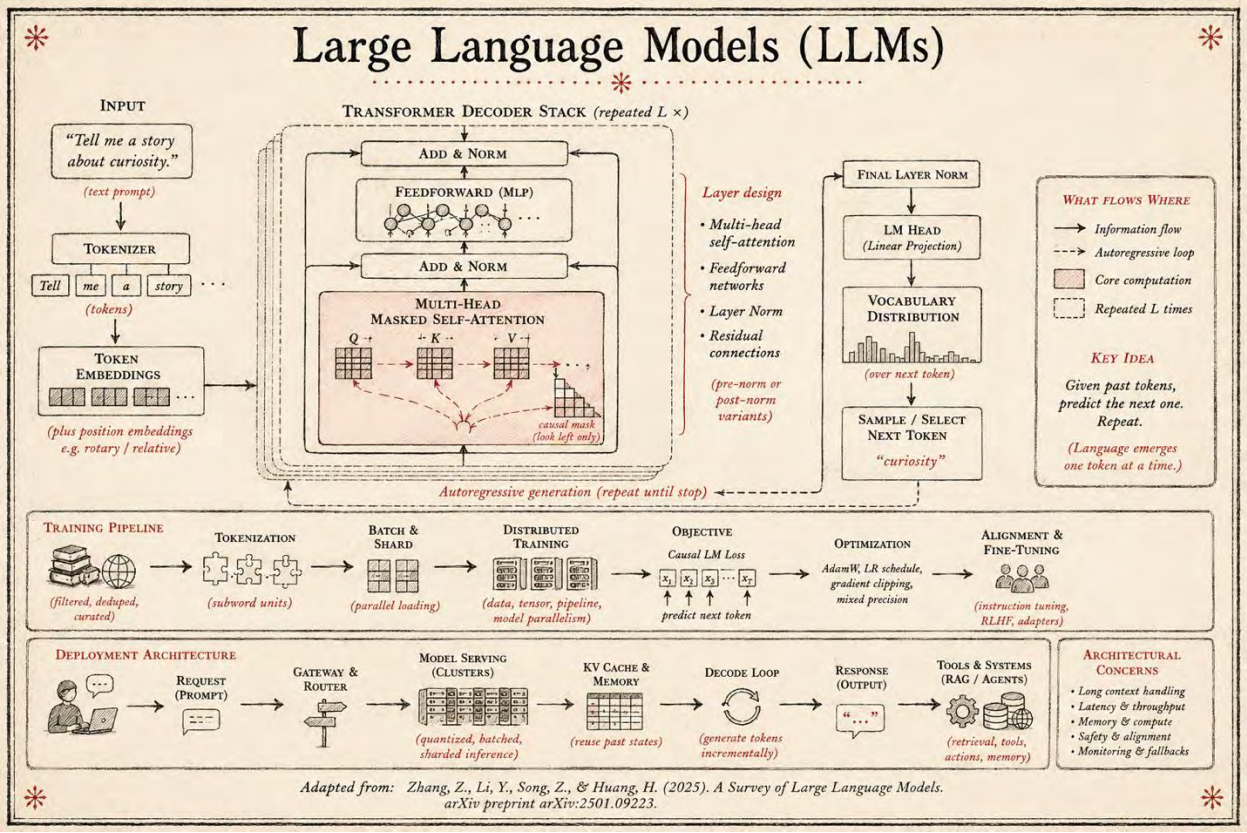


**Figure 2.3: The Architecture of Large Language Models. Source: authors.**

*A place can be the corpus*

A place can hold the same record. The creative cauldron of downtown New York in the 1960s and 1970s was a gathering of makers who drew from an incomprehensible well of sources. Around Max's Kansas City, which opened in 1965, painters, writers, and musicians crossed in one room, and a raw, anti-polish register carried from Warhol's Factory through the rooms that became CBGB on the Bowery. By the No Wave moment at the end of the 1970s a recognizable downtown style had formed that belonged to no single author.

Shanghai holds a longer yet similar version. Across roughly two hundred years, from the treaty-port settlements after 1842 through the modern city of the 1920s and 1930s, the fusion the period named Haipai, the Shanghai style, emerged across painting, fashion, literature, and film, a blend of Chinese and Western culture no one maker designed. Turn-of-the-century Paris held its own cauldron, the School of Paris drawing arrivals from across Europe into a shared modern register after 1900. Each is a well of sources and a gathering of bodies who lived the place.

Guo Pei couture draws on a deep corpus, the gold embroidery of Chinese imperial dress set against the silhouettes of Baroque Europe, an inheritance legible in every seam. The corpus is the gift. What it cannot supply she adds from her own body, her fingers reading through years of contact which thread holds its weight against the gold and which goes slack, then embroidering her name and the date of completion into each garment as a maker marks a

birth, an origination no average of prior dresses could reach. The well of sources gave her the language. The years her hands spent over the work gave it what the language did not already hold.

*The mean is the cost the corpus hides*

The gift has a cost the corpus hides. A language model draws style from the average of what bodies have already perceived and expressed, and its pull is toward the mean of that distribution, the voice no single body sang. In-context style transfer borrows a register on request, and mode collapse settles the output toward the trained center, fluent and unsurprising. Aesthetic awareness is built from the slow registration through sensation by which a new style enters the world, and the model removes that labor by arriving already named. The painter's return to a place across seasons until it alters her has no counterpart in a corpus that summarizes what others already saw. When the model stands in for that dwelling, it yields a fluent style and a diminished one.

The offer is a whole inheritance delivered without origination. A style assembled from everything already expressed, with no body having lived the place, is the hyperreal copy, coherent and unpeopled, the copy Baudrillard described floating free of any original (Baudrillard, 1981). Every reference resolves and nothing is newly perceived. The toggle between the body sensing itself and the body sensing the world settles before a stomach can tighten or a breath can catch, and awe falls as silent as unease in a register built to cohere. The style is whole, and the practitioner remains a stranger to it.

The risk lives less in the tool than in the eye that meets it. A maker who has originated a style reads what the model returns and hears which references are alive and which are filler. A maker who begins at the model takes the corpus average for a voice. The capability survives through a discipline of return, the maker going back to the source the corpus only summarizes. The oldest school for that return sends the painter to a mountain.

### The Mountain That Enters the Painter

Set against AI's emphasis on optimization, prediction, and automation, a painter learns most from imperfection, ambiguity, and incompleteness, the qualities the tools tend to smooth away.

In the eleventh century, the Chinese landscape painter Guo Xi (Kuo Hsi in older Western literature) left *Lin Quan Gao Zhi* (林泉高致, "The Lofty Message of Forests and Streams"), an instruction manual on the training of painters compiled by his son Guo Si. To render a mountain, a painter first dwelt with it through spring, summer, autumn, and winter (Guo Xi, 1080/1935). She climbed it, sat beside the streams that descended from it, walked the paths the travelers walked, and felt the shift of light between morning and afternoon and the change of atmosphere between seasons. The mountain enters the painter's body first. Only then can the brush carry it into silk. Guo Xi called this sustained attentive seeing *guān* (观), attention the practitioner holds open longer than the categorizing mind tolerates, active and bodily, until the thing she attends to discloses a quality the first impression overlooked. *Guān* is the tradition's name for this awareness.

Guo Xi's instruction exposes what corpus prediction hinders. A language model returns a style drawn from the corpus, bypassing the seasons the painter's body spent on its slopes. Corpus prediction hinders the practitioner from confronting the unfamiliar, the quality of the material a body senses only by returning to it across seasons. *Guān* keeps the practitioner inside the attending until the material discloses what only the seasons reveal.

Which quality of the material have you taken from a prompt, in place of the seasons your body would have spent on it? Return the practitioner to the unfamiliar her body can still record, against every generated style.

*Guān* teaches that creative capability forms through contact, and information alone leaves it unformed. The condition this discipline cultivates is *tiānrén héyī* (天人合一), heaven and human as one (Tu, 1985), a practical continuity the mystical reading obscures. It forms when a practitioner stays present to a place long enough that the place's own nature enters the work she makes. A technically perfect rendering carries less without *tiānrén héyī* than a rough one that holds its life. *Tiānrén héyī* surfaces when what she attends to alters her, so that something originating beyond her own hand reaches the silk through the brush.

Guo Xi sending the painter to the mountain rather than to other paintings is sharper now, because the language model is the library of other paintings, the corpus of prior expression. The model extends a style by drawing on that corpus, surfacing the lineage a register implies and holding a voice consistent across a body of work. Yet the same training that makes it fluent pulls it toward the center. Preference alignment teaches it to collapse on the favored output, erasing the productive disagreement between a felt sense and the corpus average.

Guo Xi reads the generative capability from the other side. A language model assembles a style from expressions other bodies already made, and no body was altered in the assembling. What the capability withholds is the unfamiliar, the alteration only sustained presence lays down inside the body. *Tiānrén héyī* holds the practitioner with that alteration until something originating beyond her can surface through the work.

What has the material altered in you that a prompt leaves untouched? Hold the generated style open until the body records what its altered state earned, and set aside what the corpus assembled.

The architectural property that performs the loss is the separation of the model from any altering encounter with what it expresses. A language model produces coherent style downstream of regularities that algorithms extracted from expressions other bodies lived. The model itself stayed off the mountain, away from the streams, away from the four-season return that one season's attending only begins. The style it produces may be fluent, consistent, legible in any tradition. What it lacks is the trace of a maker whose sustained attending has altered it. What survives is coherence without origination, the same absence the tradition marks as the failure of *tiānrén héyī*. What the body would have recorded through sustained attending stays out of reach, a different face of the unfamiliar foreclosed by each property the language-model architecture composes.

Which step in the body's experience did the model supply for you? Preserve at least one way of confronting the unfamiliar open to the body, across every property the language-model architecture composes.

Guo Xi's instruction is itself the tradition's most economical parable. A painter asked him how to render a mountain. He sent her to the mountain, not to other paintings or texts about mountains or master-student imitation, and told her to stay across seasons. The pedagogy assumes a mountain stays unknowable in the abstract. The winter mountain differs from the summer mountain, the mountain approached from the east from the mountain approached from the west. Only the body that returns repeatedly, in different conditions, carries what the brush will later need. A painter who does this may produce work short of technical polish, and *tiānrén héyī* will live in it.

The corpus meets its limit in Guo Xi's pedagogy. A language model receives the record of prior expression, and the seasons of return that compose a body stay outside it. The unfamiliar lives in those seasons, the quality a mountain discloses only to the body that returns across its conditions. A painter practicing *guān* stays with the return until the difference between the east approach and the west approach registers in her.

Which prompt have you accepted in place of the encounter the practice requires? Hold the corpus open to the seasons the mountain requires of the body, so the unfamiliar can surface where the painter would have stayed.

Brian Eno runs the bargain in reverse. He drew ambient music from a lineage already in the air, Satie's furniture music and the minimalists, yet stayed with a room's own sound across long hours until he could register a quality no first listening discloses, naming a register no corpus held, music as ignorable as it is interesting. The inheritance was the gift. What he added his own body originated, the slow attending a corpus of prior records could only summarize.

The unfamiliar a body meets in the slow work becomes the ground each inherited reference extends. Corpus prediction would retire that ground and return the mean directly. The disciplined maker declines, and the work stays with every unfamiliar sensation the corpus left unsummoned.

Is the model extending a style your body already perceived, or standing in for one you have yet to perceive? Preserve the encounter the corpus would otherwise summarize, and with it the unfamiliar sensations that made the style worth extending.

These principles point toward an intermediary, a dæmon that holds human intention, shifting contexts, and unresolved tension together, instead of working as an engine of automation or a mechanism for controlling uncertainty.

### The Awareness That a Dæmon Augments

Rowling's primary design aspect is Style, its attributes authored in words, across years: the moving staircases, the house colors, the wood a wand is made of. The film design team made those written attributes come alive on screen and gave the Style its visual voice. What

Rowling set down and the team rendered is what a reader, then a viewer, meets in the body.

The model assembles a style and the coherence that carries it. Awareness holds the body that dwells long enough to earn it. The dæmon is the instrument where the two meet.

Aesthetic awareness holds two modes. **Awareness** is the Disposition, the body's dwelling and registering. The practitioner attends through interoceptive registration, the body recording what holds, and she stays with a sensation until the gap reveals what a category would close. The agentic pattern attends through its generative-predictive process, drawing the next token from the corpus and continuing the voice it implies. Awareness keeps the toggle live between the body sensing itself and the world, a disposition for *tiānrén héyī*, self and world as one.

**Aesthetic** is the Orientation, the form and experience the practitioner readies. She sees a vision and prepares its form, the castle Rowling's hand drew while the sentences still waited, refusing the category that arrives before the encounter discloses its quality. The agentic pattern reaches for the coherence it is prompted to produce, the consistent register a language model holds. *Tiānrén héyī* surfaces when the place's own nature arrives through the work, the alteration sustained presence lays down, the origination Guo Pei marked in every dated seam.

We reach for coherence by reflex. The pull to shape what we want to see settles in before the world arrives, and coherent styles delivered on demand press it harder. The discipline is to stay present to what the world returns long enough for the body to sense what the story still carries.

The aesthetic awareness dæmon is the counter-discipline. It slows the body to register what it senses when a tool offers coherence pre-resolved, holding the toggle between interoception and the generated surface and protecting the emergence that arrives only after coherence was refused. Under the pressure of coherence and speed, unaided judgment dismisses the signal and accepts the surface, and the dæmon keeps the gap open.

**Figure 2.4. The Aesthetic Awareness Dæmon Archetype. Character by Shuqian Yu. Source: authors.**

The unfamiliar cuts both ways. As ecstasy it arrives when beauty catches the body before the name, the dissonance the body learns to trust. As error or discomfort it arrives when the category closes too soon, or the body takes the surface whole. Alone, each mode falls into a shadow. Awareness becomes dispersal, dwelling that continues without the registration that turns it to vision. Aesthetic becomes veneer, the practitioner taking the tool's coherence for vision so the work looks finished and feels uninhabited.

Both modes together produce what Rowling drew in the Edinburgh café. Awareness without aesthetic would have produced detailed maps no story could inhabit, and aesthetic without awareness would have yielded coherent prose no body could enter. Guo Pei's years over the embroidery without the inherited corpus would have stayed inside one body, and the corpus without those years would have returned a dress perceived by no one. Neither mode reaches the work alone. Awareness opens the gap. Aesthetic carries the form the gap revealed.

The first generation holds together and the draft reads coherent. Identify what you attended to that the tool left out, and what the draft holds beyond what you arranged.

Your hand refuses a generation it grasps. Sit with the resistance until you can tell whether the work is reaching for the unfamiliar, or habit is reaching for the category.

A parameter adjusts an output. The aesthetic awareness dæmon (Figure 2.4) asks whether the adjustment serves the work or smooths it flat.

*Can you trace your own judgment through the output, from intention, through attending, to the form your hand committed?*

### When Aesthetic Awareness Turns Awkward

The model on its own stops short of this awareness, and the gap holds even as the model grows more capable. The body builds awareness through involvement in what it senses, and awareness atrophies when that involvement stops. Dwelling takes years, and the years carry substitutes. A practitioner who shortcuts the dwelling takes the tool's coherence as her own, and her aesthetic check stays too unformed to tell when that coherence has stood in for her vision. Until she carries a dæmon that protects the dwelling, her own embodied signal remains the only instrument.

Rowling's own arc carries the warning. Through the films, the tour, and the Broadway stage, the Wizarding World became a Gesamtkunstwerk because it met creative resistance. Barry Cunningham bought the first manuscript for Bloomsbury. Copy-editors caught inconsistencies. American editorial notes pressed back on idiom. Rowling revised across years. Readers inside the publisher could tell her a line was wrong, and she trusted them.

By the *Fantastic Beasts* films, her standing inside Warner Bros. thinned that resistance. She wrote the second alone. By *The Secrets of Dumbledore* (2022), producers brought Steve Kloves back to share screenplay credit. Metacritic recorded the lowest score of any Wizarding World film, and Warner Bros. shelved the series at three of its five announced installments (Metacritic, 2022).

The aesthetic instinct that tracked Harry's arc across seven books still reported. Yet the readers who could answer back left the room. With that exchange gone, the aesthetic, the form vision readies for an audience, begins to turn inward, the body sensing its own pleasure and reading that as evidence the work is true.

A practitioner's aesthetic signal, however trained, requires a room of readers whose refusal she can trust. Keeping that room open is the work of a different capability.

Adaptive Coherence opens it.

## Chapter 3: Adaptive Coherence: The Thread Through Abundance

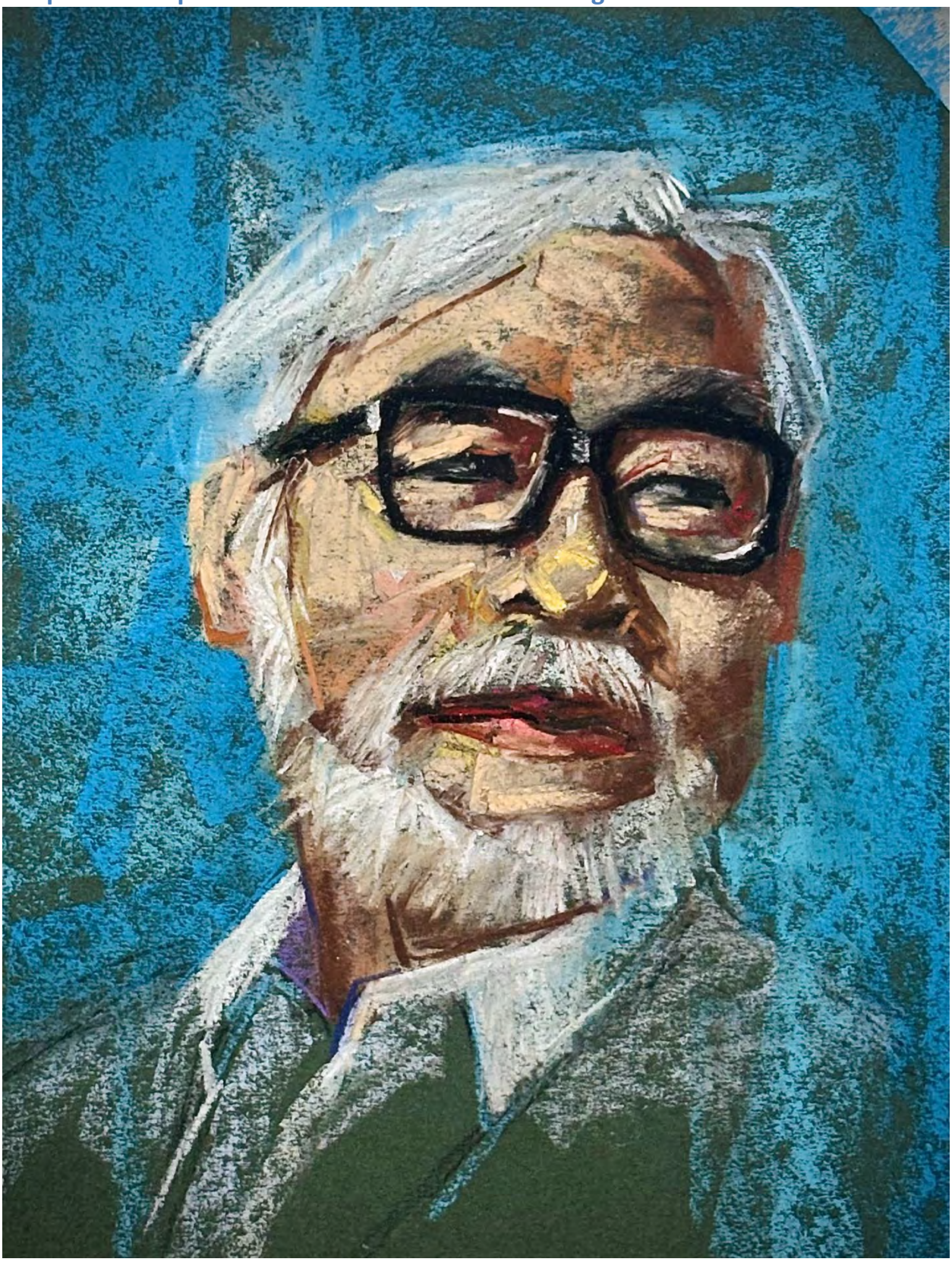

**Figure 3.1: Portrait of Hayao Miyazaki, illustration by Shiyi Chen**

### Miyazaki Draws into the Story

Style is where story starts to stand. *Spirited Away* emerged from a bathhouse Miyazaki knew from his childhood, an aesthetic that would hold a tale he had yet to write.

He begins a film before he knows its story (Figure 3.1). At the end of the 1990s he spent summers at a mountain cabin with five ten-year-old girls, the daughters of family friends, watching them read manga. The ten-year-old daughter of his friend Seiji Okuda, the film's associate producer, stayed in his mind after the summers ended. Three things set him to work: her attention, a summer rain remembered from somewhere, and an unused bathhouse drawing from a film abandoned before completion.

Back at his desk at Studio Ghibli, he began drawing a ten-year-old girl at the entrance to a bathhouse whose purpose he was still discovering. Drawing made the connections, each panel following the grain the last one exposed. The bathhouse became a site where spirits labored. The Japanese folk tradition of *kamikakushi*, being spirited away, attached to the girl's disappearance. Her parents' greed rhymed with the transformation he drew. Her stolen name resonated with Lewis Carroll's Alice and with the Shinto sense that a true name holds a self together (Cavallaro, 2015). That labor was his own. Decades of accumulated intuition guided which connections the drawing would make visible. A system that maps the same space would lack the knowledge of which connections carry a cause and which only co-occur. Only intention, formed through years, could hold them.

The connections emerged from the hand's own agency, found panel by panel over months. By March 2001 the studio had committed to a July release, and the story was still unfolding in the director's hands. Miyazaki has described this as his method. "I don't have the story finished and ready when I start work on a film," he wrote. "I usually don't have the time, so the story develops when I begin drawing storyboards" (Miyazaki, 2009), a sequence that inverts the conventional order. The drawing precedes the story, and the hand finds the structure a script would otherwise settle in advance.

*Spirited Away* (2001) disclosed itself across a year of this labor. Chihiro, the lost child, surrendered her name and won it back. A river-god arrived smelling of decay, coal sprites scattered from the boiler room, and No-Face trailed a loneliness Miyazaki drew from elsewhere. Each figure linked elements from Japanese folk tradition, his own remembered childhood illness, environmental grief, and cinematic form. The connections held because Miyazaki held them, drawing by drawing, long enough for the relations to surface.

Holding the connections exacts a physical cost. In the documentary *Never-Ending Man* (2016), a man in his seventies reworks a sequence until a character's movement arrives with a weight his analysis alone would miss. The body persists at the desk because the connections are being made by the hand, in the drawing, through sustained engagement.

From Miyazaki, we understand **Adaptive Coherence** as the capability that lets purpose navigate an abundance of possible connections while holding them open. Miyazaki held one intention, to make a film for Okuda's daughter. Which of his folk motifs, personal memories,

environmental concerns, and cinematic inheritances would connect to it emerged through the drawing. Two modes run through that intention. One is coherence, the thread held steady, a film for one ten-year-old. The other is adaptation, the responsive shift, each panel following the grain the previous panel exposed. The coherence held. The connections emerged.

### Navigating Semantic Networks

Miyazaki's pencil moves through a network of experience accumulated over sixty years, and researchers have mapped that kind of network across decades. Mednick proposed in 1962 that creative individuals possess flatter associative hierarchies, connecting concepts that less-creative individuals leave unconnected (Mednick, 1962).51 What began as a hypothesis now holds as a measurable correlate. Highly creative individuals travel further through semantic space, switch between more semantic subcategories, and make larger leaps between associations than others do (Beaty & Kenett, 2023).52 Semantic distance, computed through natural-language-processing models on word-association tasks, predicts creative achievement across studies (Beaty & Johnson, 2021).53

The capability rests less on stored associations than on the movement through them. A practitioner builds her network and moves through it with intention, committing to specific trajectories while staying open to where the material connects her onward. The committing is coherence. The openness is adaptation. The capability is the two held at once.

Organizational sensemaking research documents a similar capability under overwhelming complexity. Firefighters, emergency-room teams, and aircraft-carrier flight-deck crews construct meaning from fragments the body records faster than any single interpretation can hold (Weick, 1995).54 The frames a practitioner brings are indispensable, and so is her willingness to let the situation reveal connections they left open.

Inside that meaning-making sits a finer loop. *Reflection-in-action*, in Schön's account (1983), is the exchange where a practitioner proposes, the material returns a response, and the response reveals a connection the proposal left open. Her next move integrates what the material has just surfaced. The conversation is literal, and both parties shape the outcome. Creative work proceeds through sustained dialogue with a body of references, memories, and material that answers the practitioner's intentions with connections she can prepare for only gradually, through the work itself.

A predictive-processing account sits beneath this as a neural substrate. The brain continuously predicts incoming signal and updates both predictions and action when the signal diverges (Clark, 2013). What separates creative traversal from ordinary prediction is the willingness to stay with the divergence long enough for a new connection to form, instead of explaining it away.

Brain-connectivity research specifies the neural structure beneath this. Individuals who produce more creative work in real life show a distinctive semantic memory network (Ovando-Tellez et al., 2022).55 The topology differs measurably from less-creative individuals: shorter average path lengths, higher connectivity between distant concept

clusters, and greater flexibility in traversal. Sustained practice builds this substrate across years, and keeps building it.

The default-mode network, long associated with spontaneous thought and mind-wandering, neurally couples with the executive-control network during sustained creative navigation (Beaty et al., 2016). The switching follows patterns tied to the task's structure. Practitioners whose work holds abundance open for longer show more flexible transitions between the two networks, particularly when the task carries genuine openness. What the brain does during adaptive coherence is sustain the tension between committing to a frame and letting the material revise it.

That sustaining depends on a specific signal, an embodied excitation. It is the sensation of being **Untethered**, the felt disorientation that arrives when the thread of clarity vanishes in the material's overabundance.56 It arrives at the moment the brain switches between the default-mode and executive-control networks and the move hangs between them (Beaty et al., 2016; Christoff et al., 2009). She can hold the direction while the path stays hidden, or catch the path while the direction blurs. The two refuse to resolve together. An encounter with a situation whose next move is still forming leaves a trace of uncertainty. Alongside it comes the pull toward premature resolution that habit stands ready to supply. The signal marks the moment her frame meets a context that exceeds its grasp. Adaptive coherence develops when the practitioner stays inside that context long enough for the thread to resurface.

The signal cuts two ways. One is the disorientation just named, the discomfort of holding a frame while the path dissolves. The other is its twin, the elation when a connection beyond any plan surfaces on its own, the awe of the joint disclosing, the lift of the thread re-forming under the hand. Miyazaki at the desk meets both, the unease of the unfinished story and the charge of the panel that reveals what the next one wants. The two belong to one threshold, and the practitioner who stays for the discomfort reaches the awe.

Beneath that threshold sits a brain built to find causes. Predictive processing casts perception as a running hypothesis about what produced the signal, a causal model the cortex keeps revising against what arrives (Clark, 2013). The pull toward the cause runs deep enough that a film's villain put it plainly. "Causality. There is no escape from it, we are forever slaves to it. Our only hope, our only peace is to understand it, to understand the 'why.'" The Merovingian tells Neo as much in *The Matrix Reloaded* (2003), and the predictive brain would agree. Where the neuroscience parts from the fatalism is the route to the peace. The default-mode and executive-control networks trade the work between them (Beaty et al., 2016). A practitioner who holds the switch open long enough draws the why herself, the moment the untethered turns from disorientation into the relief of a connection that holds.

Julie Mehretu, an Ethiopian-American painter whose work was the subject of a 2021 Whitney retrospective, built her career inside the untethered. Her paintings assemble architectural blueprints, historical maps, protest photography, nationalist signage, sports iconography, and personal autobiography into dense layered surfaces she calls *story maps of no location*.57 Her method runs in stages: research into a place, digital base layers,

transfer to canvas, then gestural mark-making in ink and acrylic across many coats, each separated by clear resin.

She describes the process as cartography. “I layer the drawings like old geographical maps,” she says. “I wanted the marks to have a sense of history, a sense of time.” A Senegalese year at Cheikh Anta Diop followed her childhood flight from Ethiopia during the Red Terror of the Derg. That diaspora trajectory shapes the *communities* her paintings build from fragments, across territories that exceed any single embodiment. She learned to stay with the thread until the next connection makes itself available.

Her practice makes the sustained crossing visible. The network is specific. The connections are intentional. Coherence is the thread Mehretu keeps across months, a story map of one displacement. Adaptation is what lets the thread move when a new fragment surfaces a connection the earlier layers left open. The body holds both across months of work, and the painting records the physicality she earned.

A tool that navigates the network for the practitioner offers something real, the destination reached and the why driving the network laid bare. An emerging class of models reaches past navigation altogether to hand that over, and the practitioner chooses each time whether to climb the network herself or take the verdict whole.

### The ‘Why’ a Causal Inference Model Hands Over

The Merovingian makes causality his one universal truth, the why behind every action and reaction. Morpheus answers him with choice, that everything begins with one. The exchange frames the chapter. A causal model takes the Merovingian’s side and hands over the why, where a world model or a language model reads the surface and leaves it unspoken (*The Matrix Reloaded*, dir. Lana and Lilly Wachowski, 2003, Wachowski, L., & Wachowski, L. (Directors). (2003). *The Matrix Reloaded* [Film]. Warner Bros. Causality scene: https://www.youtube.com/watch?v=xkkuHma7PNM).

If a model can read the causality and the practitioner still owns the choice, the agency sits in which one she lets decide.

A language model swallows the written internet and predicts the next word. A world model swallows video and predicts the next frame. Both inherit the fallibility of prediction, fluent on the pattern and lost when the world shifts beneath it. A causal model refuses the shortcut. It commits to a structure of the world, declares its assumptions, and survives refutation tests before it speaks. Where a predictive model guesses, a causal model holds, because it learned the mechanism instead of the surface. As illustrated in Figure 3.2, Pearl maps its territory as three questions, seeing, doing, and imagining (Pearl & Mackenzie, 2018).58

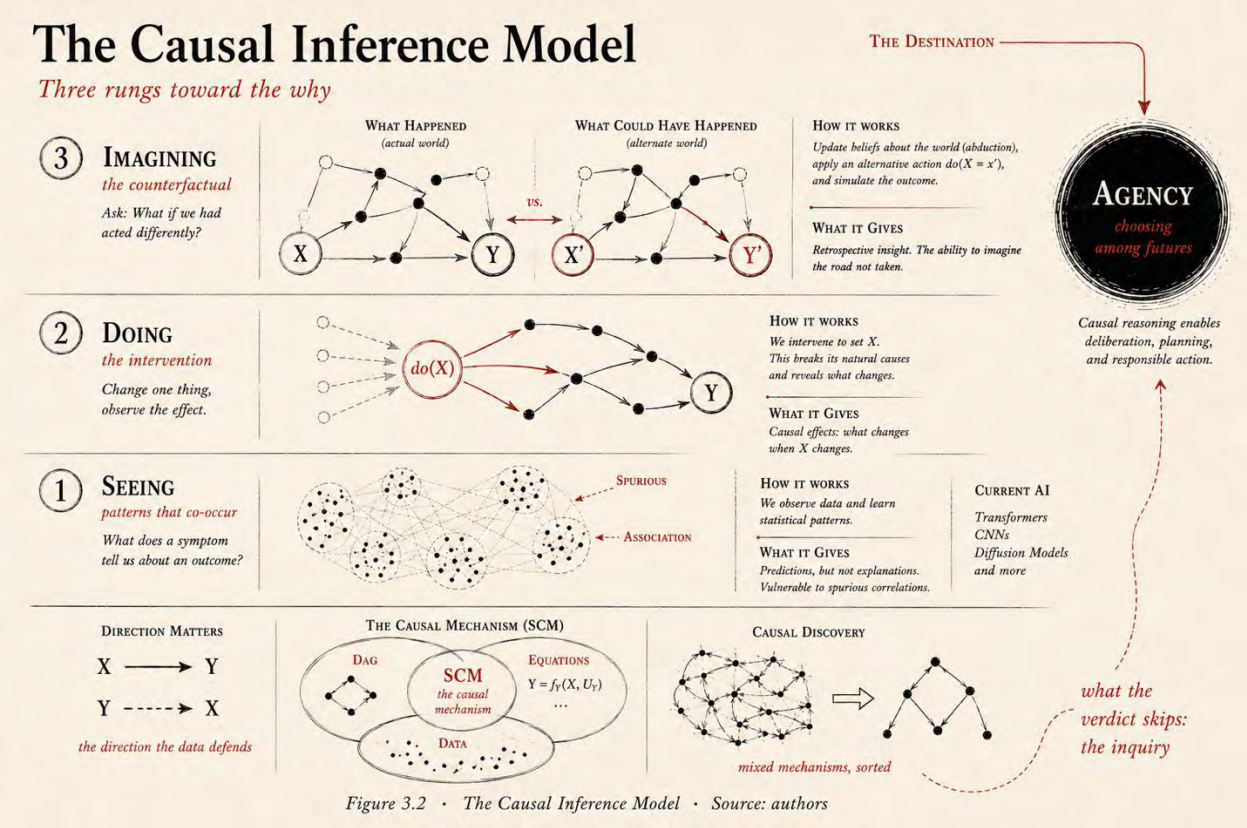


**Figure 3.2: The Causal Inference Model. Source: authors.**

Most AI predicts. It learns the patterns in a pile of data and guesses what comes next, the next word, the next frame, the next click. A causal model works the other way. It asks why the pattern is there, and it names the cause behind it. To do that it draws a small map of the world: which things cause which, sketched as arrows from cause to effect, with the data made to defend every arrow. Scientists call that map a structural causal model. A predictive model guesses from the surface. A causal model commits to the mechanism underneath, so it holds up when the world shifts. Judea Pearl, who built much of the field, says such a model can answer three questions, each harder than the last: seeing, doing, and imagining (Pearl & Mackenzie, 2018).58

The payoff is explanation a practitioner can act on. A pattern says two things move together. A cause says change this and that will follow. That is the knowledge a decision needs. Climb all three questions and they lead to one place, agency: the standing to choose among real futures, to plan a move for a reason that holds.

The risk rides in with the gift. The model can hand over the finished why and skip the work that built it. That work is the inquiry, the slow search for the cause yourself, and it is what grows adaptive coherence. Each question below offers a real power. Each can also stand in for the search the work depends on.

*Seeing reads the pattern*

Seeing notices what goes together. Ice cream sales and sunburns climb together every summer. A musician sees her streams jump on the days she posts a video. The pattern predicts well, and in a steady world that is enough to act on. Most of today's AI lives here, the language models included, fluent in the vocabulary of why with none of its structure. The trap is that a pattern says nothing about cause. The heat drives both the ice cream and the sunburn, and neither one burns anyone. The musician's streams may have climbed because a big playlist added her song that week, and the video only rode along. Seeing alone never tells her which.

*Doing tests the change*

Doing moves from what is to what happens when you act. A ceramicist notices her matte mugs outsell the glossy ones. Seeing gives her the pattern. Doing runs the test. She stocks both finishes on the same table for a season and watches what sells. The experiment rules out a hidden explanation, the chance that weekend fairs simply draw matte-loving crowds.

Often the experiment is impossible. A label has no way to relaunch last year's album with a different cover to learn whether the cover sold it. This is where the model earns its keep. It estimates the effect from records the label already holds, comparing that release against similar ones and reading what the cover added.

It can even tell which way a cause runs. A gallery sees that its shows with press coverage sell better. Did the press drive the sales, or did strong early sales attract the press? Cause and effect leave different traces in the data, and a causal model can often tell them apart. Rain explains a wet street far better than a wet street explains rain.59 Whatever the model returns, it shows its assumptions, the honest form for an answer to take.

*Imagining reruns the case*

Imagining asks what would have followed from a different choice. A studio rebrands a coffee chain, trading the old serif logo for a clean sans, and sales fall. Would they have fallen anyway, in a weak quarter? No one can rerun last spring. A causal model reruns it in software. It holds everything about the chain fixed, the prices, the season, the town, and changes only the logo. Then it plays the season forward and reports. The answer is for this one case, instead of the average, and that is what the studio actually wants to know.

A filmmaker uses the same move. Her film opened against a blockbuster and sank. She asks the model what a quieter release date would have done. One wrinkle remains. A single market runs on many kinds of buyers at once, and what lifts most of them can sink a few. Newer systems handle that, sorting each case into the kind of buyer it belongs to before they answer (Liu, Qian, et al., 2025).60

This is already paying off where experiments are out of reach. Cloud teams find the root cause of failures across thousands of live services no one can test by trial. Rideshare analysts learn which nudge actually moves a driver. Streaming services squeeze more from the few trials they can run.61 For a creative, the same three questions read as practice. See the pattern in the work. Run the small test. Picture the road skipped before committing the next move.

The power is also the paradox. The why now arrives as a finished product, a verdict with a confidence number attached.61 What the verdict skips is the inquiry, and the inquiry is what builds adaptive coherence. The untethered is the name for the disorientation a maker feels when the clear answer dissolves and she has to find the thread herself. That feeling is the engine of the search. A delivered verdict can switch the engine off before it starts. A climate analyst handed a clean cause for a warming trend saves the months she would have spent holding rival explanations open, and loses what those months would have taught her.

The risk lives less in the tool than in the hand that holds it. A practitioner who climbs the three questions herself can treat the model's verdict as one voice in the room. She sees the pattern, designs the test, imagines the alternate case, and lets the model's answer argue with her own. A practitioner who hands the whole climb to the tool feels the difference within months. One designer put it in a single breath: the cause the tool returns is defensible, and the cause her own digging would have found now escapes her.62 That choice, to climb or to hand it over, is where agency lives, the very thing the Merovingian and Morpheus argue over.

Causal methods are spreading fast, from pricing engines to medical diagnosis.63 They build chains of reasoning out of the logic the data allows.64 The practitioner with mindset to read the contours of causality differently can reach new creative capabilities.

### Cutting Where the Joint Opens

The Daoist tradition gave the name *wú wéi* (无为) to a mode of action that contemporary translation still struggles to render. *Non-action* and *effortless action* are the usual glosses, and both mislead. The better reading is *non-forcing action*: action that moves with the grain of what the material holds, making the connections the material offers in place of the connections the maker's prior frame imposes. This is the adaptation mode seen from the philosophical side. The coherence mode is the frame the maker still carries into the encounter. *Wú wéi* holds that the frame yields to the grain while it stays whole.65

*Wú wéi* exposes what causal inference hinders. A causal inference model recovers structure by reading the asymmetry in the data, bypassing the body that spent years becoming the trained ground the inquiry runs through. Causal inference hinders the practitioner from confronting the untethered, the connection the material reveals only through sustained engagement with its grain. *Wú wéi* keeps the practitioner inside that engagement until the thread the material offers surfaces.

What do you know only because the material has revealed it? Hold the causal inference open until the body's network of associations finds the connection the model overlooked.

For the body, *wú wéi* is the capability a practitioner cultivates across years of sustained engagement with a discipline, the network whose joints her hand finds through trained awareness. The painter executes what the painted surface calls for, the decision settled through years of learning. The musician listens instead of deliberating, her fingers arriving at the note the harmony's unfolding prepared. The capability grows beyond what planning or analysis could construct.

Miyazaki's pencil finds the connections a script would have specified, because sixty years of drawing built the network the pencil navigates. Those sixty years measure what aggregate inference costs. A causal inference model abstracts a mechanism across a corpus of observations through training, in place of the sustained engagement that built the corpus into the practitioner. The abstraction stands between the animator and the untethered, the connection her own years of practice prepared and the model still misses. *Wú wéi* holds the animator inside the network so the hand finds what aggregate inference could reach only through brute force.

What has your body learned through material dialogue that your conscious mind would struggle to reason about fully? Keep the aggregate inference open to the network the body built through years, so the untethered can surface where the pencil already knows.

For the tool, the causal inference model recovers structure with computational precision, yet performs the inquiry in place of a body that spent those years becoming its instrument. It identifies which variable in a record drives which. The gap lies here: it reads direction and reads nothing of fit, whether that cause matches what the practitioner's practice demands. It stays blind to the difference between a connection the material is ready to make and a connection the model inferred from what its corpus contained.

What *wú wéi* reveals is what the causal inference model removes: the practitioner's own developed sensitivity to which threads the material is preparing. What the body would have drawn through cultivation stays outside it altogether.

Which step in the body's slow drawing of the thread did the model finish for you? Keep at least one way of confronting the untethered open to the body, across every property the causal inference model composes.

A parable from the third chapter of the *Zhuangzi* (庄子, the foundational collection of Daoist parables) makes the philosophy tangible. The butcher Cook Ding keeps one knife sharp for nineteen years, while other butchers break theirs monthly. His lord asks how. Cook Ding explains that he stopped cutting through the ox's structure and began cutting between the joints. "I work with my mind and work through the senses," he says. "My mind works along with the body's knowing." His knife finds the spaces the ox's body offers. The tradition distinguishes this from the Western inheritance of agency as imposition, where a plan worth executing is one the maker holds steady against what resists. Cook Ding treats adaptation as fidelity to what is already there.66

Cook Ding's knife names the cost of the finished account. A causal inference model returns a closed verdict, the data's inferred cause in place of the joint the material offered. A closed verdict spares the practitioner the untethered, the opening of the ox's body the knife reveals once it learns the structure, and the sparing is the loss. *Wú wéi* holds the knife inside the structure until the joint discloses the path the verdict would have missed.

What is the material deciding for you that your own deliberation would have closed too soon? Hold the finished account open to the joint the material offers, so the untethered can surface where the knife would have listened.

Wong Kar-wai works a similar fidelity at industrial scale. He shot *In the Mood for Love* across fifteen months with a script still developing. Actors arrived to discover their scenes instead of reciting prepared pages. He shot scenes, reshot them, and discarded them as the film disclosed which of its thousand inherited reference-threads it was actually following. His longtime editor William Chang worked through hundreds of takes per scene, cutting as the connections emerged in place of assembling what a screenplay would have specified. Wong's intentions are famously exacting: specific cigarettes, specific lamps, specific humidity, a particular angle of dusk through a Hong Kong window. What he opens is the network those specifics sit inside.

The media artist Refik Anadol works a similar fidelity in a different material. His *Unsupervised* (2022) trained a custom model on the Museum of Modern Art's digitized collection, two hundred years of art, in place of generic web-scraped data. As the model walked its own reading of that abundance, Anadol held a single thread, the collection's shifting sense of itself, and revised it live in the gallery as each pass surfaced a connection the last had left open. The work stayed unfinished until the walk disclosed which of the collection's countless relations it was actually following.

Anadol and Wong Kar-wai show that *wú wéi* is active engagement, an exacting commitment held in conversation with what the material reveals. The intent stays constant. The path stays open. The experience of the untethered, for both creator and consumer, emerges across hundreds of possibilities. A model that hands over the finished cause risks retiring that infrastructure, and adaptive coherence survives only as long as the practitioner keeps drawing the thread herself.

Are you amplifying knowledge already known, or displacing a practice you must stay inside? Preserve the inquiry that a finished causal account would otherwise retire, and with it the untethered encounters that made the thread worth following.

### The Coherence That a Dæmon Adapts

Miyazaki's primary design aspect is Story, the thread that links scenes into one arc. Its attributes include the hero's turn as Chihiro learns to hold her own name, the device that drives the plot when the name is taken, the through-line he finds by drawing the storyboard before the script. A viewer follows those attributes and feels the scenes converge, or, in a weaker telling, scatter.

*Wú wéi* keeps the practitioner inside the inquiry until the joint discloses itself. A dæmon is what keeps her there when a tool offers the finished cut. The model performs an inference and sets a target. Adaptation holds the body that earns it. The dæmon is where the inference and the body meet.

Adaptive coherence holds two modes.[67] **Adaptation** is the Disposition, the body's yielding to the emergent grain. The practitioner accommodates the joint the material opens, the cut Cook Ding found across nineteen years of the knife. She stays with felt disorientation until the next connection offers itself. The agentic pattern adapts through causal inference, the model reading the mechanism the data disclose. Adaptation moves with what the work surfaces, a disposition for *wú wéi*, action without forcing.68

**Coherence** is the Orientation, the thread held and the direction framed. The practitioner frames a direction and holds it firmly enough to act on and loosely enough to revise, the intention Miyazaki kept for Okuda's daughter while the bathhouse found its purpose. The agentic pattern holds the account it optimizes toward, the cause a causal inference model infers and scores. The felt disorientation when the thread vanishes is the untethered, the body's report that the work exceeds the frame.

We are goal-seeking creatures. The pull to set the problem and reach for the solution outruns the body's slower surfacing of what the situation holds. Causal inference that proposes the cause feeds it. The discipline is to wait for the problem and the solution to emerge together, even when body and model both ask for closure. Coherence is the thread that survives the wait. Adaptation is what the wait keeps open.

The adaptive coherence dæmon is the counter-discipline (Figure 3.3). It keeps the practitioner resident inside the untethered when a tool offers to optimize past it, holding one way of confronting it open across every property the causal inference model composes and protecting the commitment only the body's network of associations delivers. The dæmon keeps the thread in her hand where abundance would scatter it.

**Figure 3.3. The Adaptive Coherence Dæmon Archetype. Illustration by Shuqian Yu, designed by authors.**

The abundance of inferred causal accounts offered with genuine openness dissolves the network the body uses to draw the thread. Judgment made in isolation accepts whichever account the model scored highest. The practitioner senses the loss as the untethered, the disorientation when the thread stops carrying felt direction and the next one has yet to take it up.

The untethered cuts both ways. As ecstasy it arrives when the joint discloses itself and the thread re-forms under the hand. As error or discomfort it arrives when the thread vanishes in abundance, or the verdict closes the search. Alone, each mode falls into a shadow. Adaptation becomes drift, the practitioner releasing threads while the body's associations are still forming so the work arrives plural and directionless. Coherence becomes rigidity, the thread hardening against a situation that has changed so the body stops listening for the path the material opens.

Neither mode alone produces what *Spirited Away* produced. Coherence alone would have produced a clean script that closed the bathhouse before the spirits arrived. Adaptation alone would have yielded drawings that dispersed across registers the audience could follow only through effort. Mehretu's coherence without adaptation would have locked her figures to their source maps, and her adaptation without coherence would have dissolved the layers into noise. What one mode misses, the other holds. Coherence holds the thread. Adaptation lets the thread find the joint.

The trajectory the model proposes feels productive. Identify the joint you sensed that the optimization overlooked, and what the work would carry that the optimized path leaves behind.

A thread you committed to is shifting. Sit with it long enough to tell whether the material is asking you to release it, or habit is asking you to keep moving.

A parameter adjusts an output. The adaptive coherence dæmon asks whether the adjustment serves the work or betrays it.

Can you trace your own commitment through the work, from the framing held loosely, through the joints the material offered, to the form the joints disclosed?

### When Coherence Resists Adaptation

This scenario already played out. In 2016, a visiting research team showed Miyazaki a deep-learning animation, a zombified figure moving in ways human bodies could assemble only through digital means. Miyazaki called the artifact grotesque and stated his position plainly. "I am utterly disgusted… I strongly feel that this is an insult to life itself."69 The clip became a meme, routinely cited as moral clarity on artificial intelligence.

The capability that animated Chihiro through the bathhouse was present in the recoil. It departed from the path it always chose, moving away from the encounter instead of staying inside it long enough to be taught something. Miyazaki asked what the tool showed. He left unasked what the tool might let an animator perceive for the first time. *Coherence refusing to adapt* is the capability hardened into reflex, the coherence mode intact and the

adaptation mode switched off. The attention that once held a frame open until anomaly arrived now shuts the door the moment it does.

A closed door stays visible. The community of animators watched Miyazaki refuse, and the refusal entered a field he stepped back from shaping. How a practitioner's earned understanding travels into that field, and what the field sends back, opens the next terrain as Expressive Influence.

## Chapter 4: Expressive Influence: Catalysts of the Social Kind

**Figure 4.1: Portrait of Miles Davis, illustration by Shiyi Chen.**

### Davis Turns His Back to the Band

In March 1959 Miles Davis arrived early at Columbia's 30th Street Studio in Manhattan, a converted Orthodox church whose soaring ceiling gave recorded sound a natural resonance engineers would spend decades trying to reproduce. He warmed up alone. He tested the mouthpiece. He ran breath through the horn until the instrument's backpressure met the pressure of his embouchure. Years calibrated that exchange, and the calibration lived in his diaphragm.

Bill Evans, John Coltrane, Cannonball Adderley, Paul Chambers, and Jimmy Cobb arrived later. Davis handed out lead sheets sketching modal frameworks fresh to all of them. The session dispensed with rehearsal. Almost every track on *Kind of Blue* would be a first complete take (Carr, 1998).

Davis stood with his back to the band (Figure 4.1). The posture was deliberate. He was listening, attending to what happened when six trained bodies answered the same sparse constraints in real time. Each musician's mastery met the others in a space that demanded the immediate, the unrehearsed, the irreversible.

Davis pushed his musicians toward discovery, where learned habits dissolved and the body invented, modal structures sparse enough to demand improvisation and rigorous enough to hold form. Herbie Hancock described it from inside the band. Playing with Davis, he wrote, "was like having your toes hanging over the edge of a cliff, and he was the one holding you by your collar so you wouldn't fall" (Hancock, 2014).[70] The cliff was the social field's refusal to let people retreat to safety. The room's answer, six bodies altering what each could reach for, is one mode, influence. His prepared vision, the modal frameworks carried into that room, is the other, expression.

That reciprocity is one body in a room sensing how another body answers. It sits beyond the reach of prediction and generation. It is the metabolic cost that makes meaning.

Evans caught that cost in the liner notes, through an analogy to sumi-e. The artist works on stretched parchment with a brush and black water paint, and an interrupted stroke destroys the line or breaks through the surface (Evans, 1959). The analogy was exact. Each gesture surrendered to irrevocability. The tape was running. Every note carried the cost of the body producing it, years of practice compressed into a gesture that landed or failed in front of other bodies doing something similar.

Columbia released *Kind of Blue* in August 1959, and it has stayed in print continuously. What the session preserved was the weight of six bodies staking shared understanding on a single moment of forward motion. The social field received the stake, and it is still receiving it.

Miles inspires our view of **Expressive Influence** as the capability that lets earned making enter the social field and bear the cost of reception while holding its integrity. It holds two modes. Influence is the porous membership through which the field shapes the practitioner, the six bodies answering in the room. Expression is the inner vision the

practitioner carries outward, the embouchure and modal frameworks Davis prepared across years. The maker is shaped before any work begins.

### Bodies That Carry Expression

Expressive influence has long been treated as transmission from a singular source. Davis's private preparation included an embouchure calibrated over years, breath control refined through decades of performance, and the modal study that gave him the frameworks he handed the band. All of it lived in his body before he entered the studio. This earned, directed vision is expression, the projective mode. Treating it as the whole of the capability is the error to correct, because expression stays incomplete until influence, the field's porous answer, shapes it.

The Italian tradition calls the engine of this movement *passione*, the visceral effortful engagement where the body carries the cost and the cost becomes the ground of what the work means. Davis arrived already knowing what he wanted, and the band saw none of the preparation. Prince exercised a similar discipline. Years of obsessive embodied practice built the neural architecture that made his expression so fluent it appeared effortless. "A strong spirit transcends rules," he said, and the fluency arrives only after the body internalizes the rules it appears to transcend.

Individual mastery is only half of it. A larger literature now establishes the role of shared intentionality, the capability to form and hold joint goals with other bodies. It is the infrastructure on which expressive work at any scale rests.71 Music, dance, and communal storytelling evolved as social grooming at a distance, a way to bond across groups too large for physical grooming to reach (Dunbar, 2012). Song, rhythm, and shared narrative bind communities through metabolic investment they make together. That investment is the signal. A community that sings together is a community that trusts together.

Creative production functions as an honest indicator of cognitive fitness, the costly signal of evolutionary biology whose credibility rests on genuine resource expenditure (Miller, 2000; Zahavi, 1975). The reward circuits driving the urge to express and the urge to share run through one coupled pathway (Kringelbach & Berridge, 2009). The poem seeks a reader, the song a listener, the building an inhabitant, because the act of expressing and the act of being received register in one neural substrate.

Guitarists improvising together develop phase-locked neural oscillations, particularly in delta and theta bands, which emerge during joint expressive action and fade when the players run the same material separately (Müller et al., 2013). This neural coupling correlates with the quality of expressive output, and research associates the same effect with dancers, string quartets, and improvising theater ensembles. Two violinists playing side by side develop a rhythm visible to anyone watching, a synchronization an MRI scan shows to be involuntary, somatic, and productive. The music that emerges from shared sessions carries a quality solitary work leaves out, a simultaneous attention to the notes, to each musician's expression, and to the response another body returns.

The coupling rests on a predictive structure the brain brings to every observed action. When one maker watches another shape material, the observer's motor cortex generates a

predictive simulation of it, recruiting the circuits that would execute it (Rizzolatti & Sinigaglia, 2010). A jazz pianist watching a bassist's fingers move toward a fret position begins preparing the harmonic response while the note is still forming. Rehearsals accumulate these simulations until the musicians' nervous systems function as a coupled system, each predicting and answering the others' moves in real time.

The conductor Leonard Bernstein made this visible at scale, training musicians to attend to each other's somatic response. The micro-adjustments of breath and entrainment make an orchestra function as a coupled nervous system instead of a collection of individual agents playing in time.72 The musicians learned to feel what their partners felt as the conscious mind worked out the next adjustment.

What emerged was a shared presence in place of a performed score. The listener's nervous system records the orchestra's collective investment as a signal the conscious mind later formulates as judgment, her breath synchronizing with the phrasing, her nervous system beginning to undergo what the ensemble's undergoes. This is affective contagion at the level of the nervous system, empathy as shared embodied state, the felt thing in the place of the understood one (Müller et al., 2013).

Because the social brain senses the work as collective expression, the embodied excitation that rouses expressive influence is the sensation of the **Unyielding**, the felt weight that accumulates when a social field refuses the practitioner's work steadily. The refusal comes with silence in the place of harshness, offering nothing where acknowledgment might have arrived. The signal arrives in the failure of inter-brain synchrony, the phase-locked oscillations that rise during productive joint expression flattening when the field stays apart (Müller, Sänger, & Lindenberger, 2013; Dumas et al., 2010; Hasson et al., 2012).

One face of the embodied experience expressive influence turns on is the unyielding. The other is ecstatic. When the field answers, when the phase-locked oscillations lock and a roomful of bodies couples into one rhythm, the practitioner feels resonance as joy, an awe close to belonging. Davis worked from that opening, building his sound around the space he left for the band to answer (Davis & Troupe, 1989). The two faces are one sense read from opposite sides, the body registering whether the field has met the work or left it in silence.

Evans named the social problem from inside practice. Aside from the weighty problem of collective coherent thinking, he wrote, "there is the social need for sympathy from all members to bend for the common result." The word *sympathy* carries somatic weight. It describes bodies attuning to each other, physically adjusting to collective creative pressure. The common result emerges from the bending.

The unyielding can be loud, and it can be silent, and the silence delivers the heavier weight. The critic's silence sits heavier than the critique. The empty hall after the premiere signals more than the hostile review. The audience that scrolls past returns an absence her body seeks to meet.

What the practitioner senses is that felt signal of the unyielding, an encounter with a field whose refusal answers in silence, a pressure in which every adjustment she makes deepens the resistance. It is the felt signal that private preparation alone falls short.

Expressive influence develops when the practitioner stays inside that signal long enough for the field to teach what her vision alone leaves unplaced. The newest pair of tools offers to spare her the staying. One builds the work from noise, the other carries it to a field of any size, and the metrics arrive before her body feels a thing.

### The Reach a Diffusion Platform Hands Over

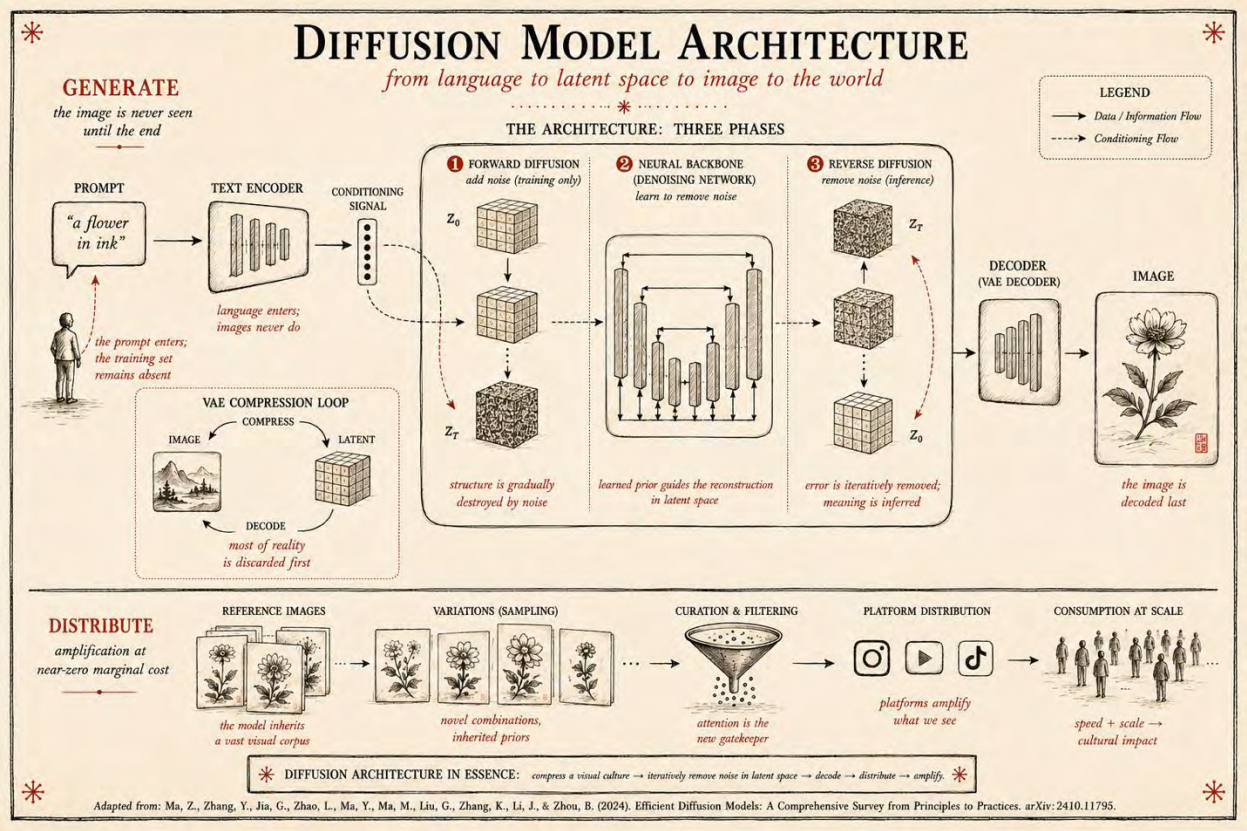


**Figure 4.2: The Architecture of Diffusion Models. Source: authors.**

Two machines now stand between a creator and her audience. The first makes the picture. The second carries it to the people who will see it. The first is a diffusion model. It starts from pure visual noise, the gray snow of an untuned television, and clears that snow away one step at a time until the picture a sentence asked for is standing there (Ho et al., 2020; Rombach et al., 2022).73 Researchers describe the whole trick in one line: it denoises random static into an image (Figure 4.2). The second is the platform, the feed that takes the finished image and routes it to viewers a creator meets mostly as numbers. From noise to a picture, then from a picture to everywhere.

Point one of these tools at a problem and the output is striking. Tell it "a cozy neighborhood coffee shop, warm light, hand-drawn," and a usable logo arrives in seconds. Ask for the same coffee shop as a photograph, an oil painting, and a 1950s travel poster, and three convincing versions come back, each in a different style. The pictures are sharp enough to pass for real, varied enough to cover a dozen looks, and all of it comes from a line of text. Tools like Midjourney, Stable Diffusion, and Adobe Firefly put this in front of anyone with a

browser. A novelist drafts cover concepts before lunch. A game studio fills a character board overnight. A small shop mocks up a product it has yet to manufacture.

The deeper change is who gets to make these pictures. Image-making used to demand years of training, a camera or a drawing hand, and the gear to match. Now it asks for a sentence. In a few short years these tools have put a kind of picture-making into the hands of millions of people who never trained as artists, and the platform stands ready to carry whatever they make to an audience of any size. A person at a laptop now holds powers that once belonged to a studio, a print shop, and a distributor combined. It is already remaking how creative work gets made and sent.

For all that it delivers, one thing the two machines never carry is the metabolic cost, the bodily price of the work (Figure 4.2). Expressive influence grows from exactly that price, the nerve it takes to put work in front of a live audience and stand there while it answers, or stays silent. The machines deliver the reach and remove the nerve. Each gift that follows serves the work, and each can also spare the creator the exposure the work depends on.

*The forward process drowns the image in noise*

The model learned its craft by watching pictures fall apart. Start with a photograph and add a little visual static, again and again, until the image dissolves into pure snow.74 The model records every step of that ruin and studies it. By learning exactly what each pinch of static destroyed, it learns how any picture is built. All of this happens once, in training. Every image the tool later hands a creator descends from that long study, which is why the results feel earned even when they land in seconds.

*The reverse process turns noise into the picture you asked for*

Making an image runs that lesson backward. The model starts from a fresh screen of random static and removes the noise a step at a time, your words steering every step, until a picture forms out of the snow. The result is the point. The model is not pulling a photo from a library, it is inventing a brand-new image that never existed, built to match your sentence. Ask again and a different one appears, so a single prompt can yield endless fresh variations. Type a poster idea at midnight and thirty finished versions wait by morning, each carrying the same look you locked in with a reference image. Most of them pass for photographs.

The catch shows up in what the model reaches for. Left to itself, it drifts toward the safe middle, the average of everything it has ever seen, and the odd, surprising images at the edges quietly drop out. The pull grows as the internet fills with generated pictures. When new models train on the output of older ones, that averaging feeds on itself, a decay researchers call model collapse (Shumailov et al., 2024).75 Strong expression lives at the edges, in the unrepeatable, and the model pulls toward the center.

*The platform sends the picture to an audience of any size*

The second machine finishes the job. A feed pushes the image out and quietly tunes what each viewer sees to how they respond, what they watch, linger on, and repost. An illustrator in a small coastal town wakes to viewers in forty countries, a reach no gallery

grants. Meta's Vibes, launched in September 2025, runs this loop in its purest form, a stream of generated short video tuned to a stream of attention (Meta, 2025).76 Newer tools carry it further still, an agent that keeps refining its own prompts, skills, and memory across runs with no person in the loop (Karten et al., 2026).77

The catch is in what comes back. The feed returns numbers where a real response would be, a tally of likes and views read long before any person in a room reacts to the work. The audience is met through a screen, from the outside, and the creator learns the metrics instead of the silence in the room.

This reach is already paying off in studios. A designer tries twenty directions in the hour a single mockup used to take and drops nineteen for the price of minutes. The range of styles on offer runs far past any one hand's training, references a single apprenticeship could only sample. FKA twigs trained a controlled deepfake of herself, AI twigs (2024), on her voice and manner across several languages, and sent it into the feeds to carry a reach she had spent a career earning, while the making itself stayed in the studio with her body.78 The model extends what she already built. It cannot stand where she stands when a room decides whether the work lives. A photographer who has read light for years can tell a model what she wants with a precision a beginner has no way to reach. In her hands the tool extends a vision she already earned.

The power is also the paradox. The two machines offer reach with the bodily cost stripped out, and the model's reach and the creator's avoidance run on the same rail. Expressive influence runs on the unyielding, the felt weight that builds when a real audience meets the work with silence. That signal lives in the body. Brain-to-brain synchrony rises between people in live, shared expression and tracks the sense of belonging it builds, which is why it forms in person and rarely through a screen (Hinvest et al., 2025).79 The platform's numbers read the room so the body never has to. A feed of climbing counts reports a warm welcome while hiding the silence that would have taught the creator the most. What the prompt returns is smooth, plausible, and addressed to no one in particular, a hard-won signal with the effort taken out.

An opposing posture is just as available, and it arrives quietly. A studio locks a reference image, ships a hundred coherent pictures a week, and watches the counts climb while the question of what the work is for goes unasked. Judgment forms by choosing what to reject. A system that returns everything, all of it polished, leaves nothing refused and so nothing truly chosen.

The disposition worth protecting keeps the body in the room when the model offers to skip it. The singer Imogen Heap sang her 2025 song "Aftercare" in her own voice, then swapped its timbre for ai.Mogen, a model she trained on herself and spent hundreds of hours shaping by hand.80 She still walks onstage to sound the note and wait for the room to answer, because the machine can carry her reach but cannot bear the weight of standing there while a field meets the work or lets it fall in silence. The tool can ride the network. The body still enters rooms.

A creator who envigorates expression inhabits the room and works inside it, in time with the people there. Learning to step in starts with excitation, triggered by nothing more than a struck string.

### The Fields of Sympathetic Resonance

Strike a string in a quiet room, and a second string tuned to the same note begins to sound on its own. No hand touches it. The note from the first string travels through the air and calls the second into life. The classical Chinese tradition calls this effect gǎnyìng (感应), usually translated as sympathetic resonance, or correspondence-and-response. It describes how two separated bodies can move each other at a distance, one stirring and the other answering, and the tradition treated it as a basic way the world works. The social brain still runs on it. We feel a room's mood turn, catch another person's nerves, ease when a crowd eases toward us. Expression works the same way. A creator sounds a note, and she depends on the room to sound back.

This runs against the picture a creator inherits from the Enlightenment, which places the maker outside her field and tells her to act on it from a distance. With its long lineage, running from early Confucian commentary into classical art criticism, Gǎnyìng describes the opposite.81 The maker stands inside the field, moved by it even as she moves it, and the gap the Enlightenment opened closes. What follows is a change of posture. The work stays incomplete the moment she releases it, and is completed only when the field answers.

*Gǎnyìng* exposes what noise-to-signal hinders. A diffusion model starts from Gaussian noise and steers it toward the statistical mean of the data it was trained on, bypassing the field the practitioner has prepared herself inside. Noise-to-signal hinders the practitioner from meeting the unyielding, the field's refusal a body registers only by having entered and inhabited the field. *Gǎnyìng* keeps the practitioner inside the field until what the field returns can pass through the body.

Which field have you inherited that your own body has entered? Return the practitioner to that field, so the unyielding can surface where noise-to-signal would have smoothed it.

Electronic composer Holly Herndon answers this default by embedding a social field in the training itself. Her album *PROTO* (2019) features Spawn, a neural network trained on live vocal performance in place of scraped data. Herndon held *training ceremonies* in which a human choir and invited participants sang directly to the machine, answering each other in real time and submitting to the imprecision of collective singing. Spawn's voice carries the ceremony's mark, the breath, the choral imprecision, the overtones of many throats in one room. The Holly+ protocol that followed governs the voice through a decentralized autonomous organization, so the community that sang it into existence keeps directing its use.82

A diffusion model trained on scraped data defaults to the distribution's mean, the voice no single body sang. The pull to center stands between the practitioner and the unyielding, the resistance of a field whose bodies have specific grain. *Gǎnyìng* returns her to the ceremony that taught the network which throats it belongs to.

What are you training, and whose voices sang into it? Hold the training open to the specific bodies whose grain the distribution would otherwise flatten, so the unyielding can surface where the mean would have averaged it away.

The diffusion-and-platform system treats the maker as standing outside her field and acting on it from outside, producing images that belong to every lineage and to none, because no body transmitted each one. For the practitioner, *gǎnyìng* describes a different relation, resonance in place of transmission. She inhabits a field that already vibrates with her work, where audience, lineage, and collaborators answer in ways the body has trained to feel. This answering is influence, the field shaping the practitioner. What she releases into it is expression. *Gǎnyìng* holds the two open to each other.

Dedicated preparation lets the practitioner participate in the field instead of demanding a place inside it. Each property of the diffusion-and-platform architecture blocks a different face of meeting the unyielding. What the body would have met through entering the field arrives to it fully shaped.

When the work leaves your hands, whose investment does it carry besides yours? Keep at least one way of meeting the unyielding open to the body, across every property the diffusion-and-platform architecture composes.

The *Wenyan* commentary on the *Yìjīng* makes the principle tangible. “Things of the same tone respond to each other,” the text reads. “Things of the same energy seek each other. Water flows where it is wet. Fire travels where it is dry. Clouds follow the dragon. Wind follows the tiger” (*tóng shēng xiāng yìng, tóng qì xiāng qiú*, 同声相应，同气相求). The passage describes a world in which resonant transmission between bodies and elements is the ground on which all participation rests. The figure repeats across two millennia of Chinese aesthetics, where the work that lives is the work that sets something in the field vibrating in return.

The *Yìzhuàn* shows the price of reference consistency. A diffusion model preserves consistency by anchoring each output to the reference it conditioned on, producing the lineage the conditioning implies in place of the lineage the body inhabited. The anchoring closes off the unyielding, the refusal a lineage offers only to a body that has entered it. *Gǎnyìng* holds the practitioner in the lineage until what the lineage extends can surface through the body.

What holds you, beyond what you have chosen to transmit? Keep the reference consistency open to the lineage the body has entered, so the unyielding can surface where the conditioning would have closed.

British multi-instrumentalist Jacob Collier works a similar discipline at the scale of live performance. In his concerts he teaches chord voicings to auditoriums of thousands in real time, conducting audience sections into spontaneous multi-part harmony, and folds the recordings of those voices back into his studio arrangements. What enters the recording is Collier and the audience together producing the voicing in a shared breath, Collier orchestrating their collective voice. His *Djesse* album series composes across orchestral ensembles, gospel choirs, folk collaborators, and the live audiences of multiple continents,

each work bearing the field it was made from. The body that organizes is porous enough to receive and directive enough to discharge.

Collier's concerts describe the inverse of what platform dissolution hinders. The unyielding the audience brings to the voicing is the infrastructure each recording extended from. Platform dissolution offers to retire that infrastructure by producing the engagement metrics directly. The recording extended Collier's reach only because the auditorium already sang against every voicing that would otherwise have passed as applause.

Did you invite the field in, or did you shape it from outside? Preserve the field that platform dissolution would otherwise retire, and with it the unyielding refusals that made the voicing worth releasing.

### The Expressions a Dæmon Can Influence

Davis's dominant design aspect is a bold Statement, which he made more than once. The attributes are the muted, spare tone, the space left between phrases, the cool restraint that turned away from bebop's density on *Birth of the Cool*, then the modal scales that opened *Kind of Blue* a decade later. A listener meets those attributes and hears a position taken, the same maker reinventing the statement twice. And because Davis made his statement through others, turning his back so the band answered itself, the work entered a social field and was read many ways. That plural reading is the first hint of the boundary object a design artifact becomes among the people who receive it.

*Gǎnyìng* keeps the practitioner inside the field until it answers. A dæmon is what keeps her there when the platform offers the metric instead. The model generates a signal and the platform carries it. Influence holds the body the field can answer. The dæmon is where the signal and the field that answers it meet.

Expressive influence holds two modes. **Influence** is the Disposition, the practitioner's porous membership in the social field. She stays porous to what the field returns while holding form, the way Davis turned his back to the band and listened, a disposition for *gǎnyìng*, sympathetic resonance.

**Expression** is the Orientation, the inner self carried outward. The practitioner brings a prepared vision and aims it to land. She holds it open through what the field refuses. The work completes itself in the field that answers.

We are social animals. The pull to seek attention and belong is older than language, and platforms that quantify reception feed it. The discipline is to stay answerable to the field, even when the metric offers belonging at no cost at all.

The expressive influence dæmon is the counter-discipline (Figure 4.3). It holds the two modes interdependent when the diffusion-and-platform architecture offers to separate them, sustaining the lineage that turns preparation into weight a field can answer and protecting the inter-brain synchrony that platform metrics would replace. The dæmon keeps the exchange open where the stress of reaching out would seal it.

**Figure 4.3. Expressive Influence Dæmon Archetype. Illustration by Shuqian Yu, designed by authors.**

The unyielding cuts both ways. As ecstasy it arrives when the room couples into one rhythm and resonance becomes belonging. As error or discomfort it arrives when the field refuses in silence, or weight accumulates and no answer comes. Alone, each mode falls into a shadow. Influence becomes conformity, responsiveness so total that the practitioner's own direction dissolves into whatever the field rewards. Expression becomes isolation, a vision held so tightly it seals against what the field offers back.

Together the two modes produced *Kind of Blue*. Davis's prepared vision without the session's porous membership would have produced practice tapes, and the session's porous membership without his prepared vision would have produced a jam. Herndon's Spawn without the choir's training ceremony would have sung in the distribution's voice, and the ceremony without the network would have stayed inside the room. Apart, each mode falls short. Expression prepares what the field can answer. Influence carries the practitioner into the field that answers it.

The work is ready to leave your hands. Who are you making this for, and would you make it differently if the field stayed dark?

The algorithm proposes cutting the second verse. What does the second verse hold that the algorithm leaves behind?

A parameter adjusts an output. The expressive influence dæmon asks whether the adjustment serves the work or sells it.

Can you trace your own release of the work, from preparation, through the field's refusal, to the form that survives it?

### When Expressive Influence Isolates

Expressive influence reached its album because the field answered the vision Davis brought. The same capability, held past what a field could return, met a different end. After *Bitches Brew* (1970), Davis's ensembles turned over faster. Teo Macero's editorial hand shaped more of the finished work. *On the Corner* (1972) arrived with musician credits withheld, a choice Davis later explained as provocation to critics (Tingen, 2001). Critics panned the album. His audience thinned. He toured through cocaine and escalating distance from the musicians around him. His hip bone was disintegrating.

In 1975 he stopped. For five years he kept himself from recording or performing, staying mostly inside his townhouse on West 77th Street (Davis & Troupe, 1989). The body that calibrated itself against Columbia's acoustic in 1959 now held its vision past what the field could return. Davis built *Kind of Blue* by turning to listen. By 1975 the band dissolved.

The field that holds the practitioner is what sustains the work she can release into it. Davis chose to press his vision past where the porous membership could answer it, and with influence held back his own sensing turned inward and confirmed only itself. The shadow of expressive influence is isolation, the shadow of Expression hardened, embodied vision staked into a field the practitioner has already left, the field gone dark.

Four capabilities, each alone sufficient to turn a master inward. What holds them together, and what Davis lacked in 1975, is an ecology of Creative Convergence.

## Chapter 5: Agentic Creative Intelligence: Creative Convergence

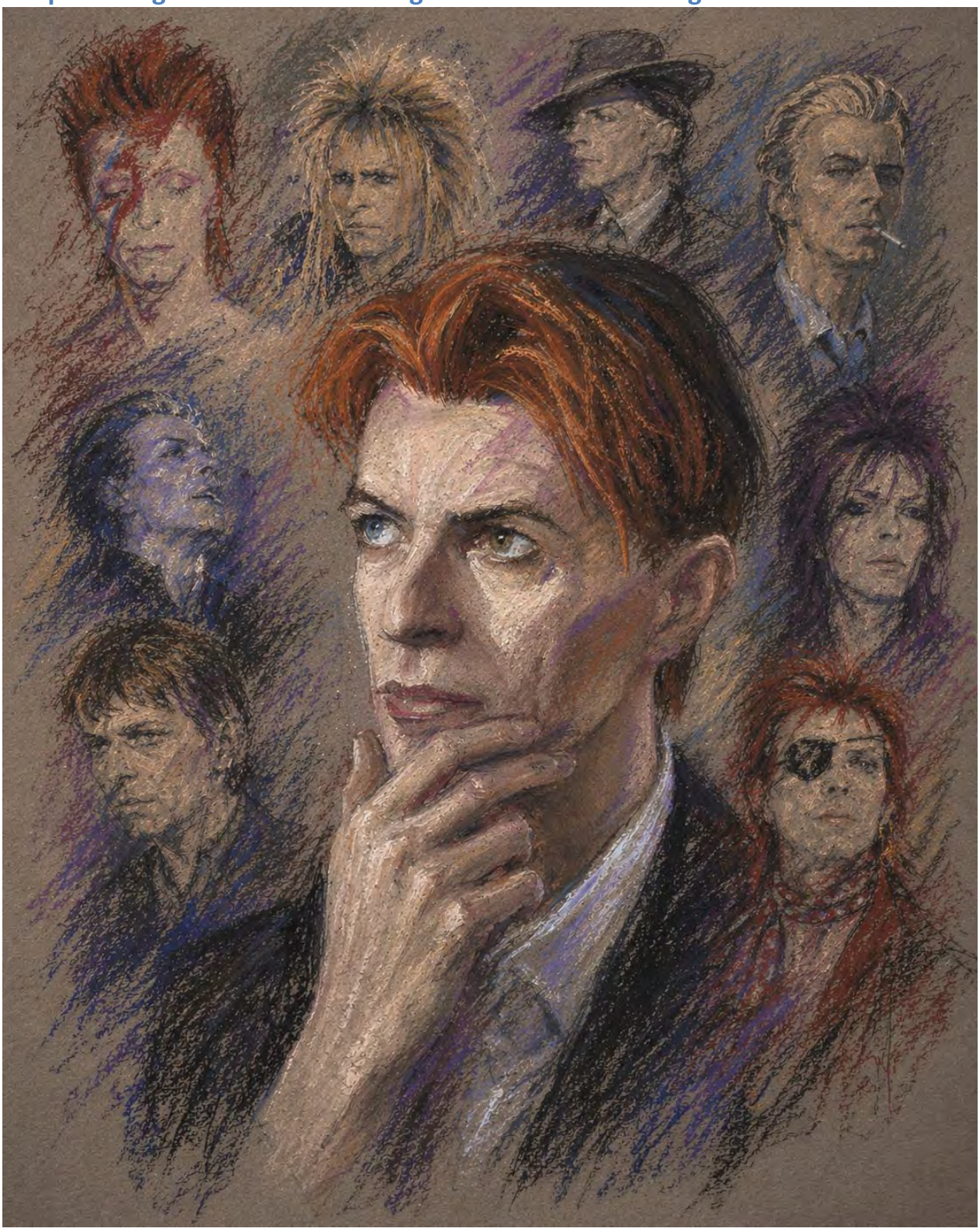

**Figure 5.1: Portrait of David Bowie, illustration by Shiyi Chen.**

### When Collision Builds Capability

Hansa Studios, West Berlin, July 1977. David Bowie (Figure 5.1) stands in the Meistersaal, a cavernous concert hall whose windows face the Berlin Wall. Tony Visconti has positioned three microphones at staged distances, one at nine inches, one at twenty feet, one at fifty feet. The two distant microphones run through noise gates that open only when the sound reaching them crosses a set volume. When Bowie sings quietly, only the close microphone captures the voice. When he pushes his diaphragm harder, projecting across the room with full physical force, the gates trip and the Meistersaal's reverb floods the recording.

Midway through a take, Bowie pushes into the threshold. The gates release. The Meistersaal floods the recording as an acoustic his body is still reaching toward. His next phrase adjusts. The breath lengthens. The pitch fractionally rises to carry across the altered air. The room has returned something, and the performance takes it in.

What happens in the fraction of a second between the gates opening and the phrase adjusting is the convergence the prior four capabilities built toward. The adjustment arrives that fast because Bowie brings a body that has done the inner work, decades of cultivation in each capability, ready before the room returns its acoustic. The vocal force that activated the gates came from decades of contact between body and instrument. Extended engagement. The ear read the altered acoustic first. Aesthetic awareness made the reading possible. The phrase that re-pitched to meet the new air was adaptive coherence. The altered sound reaches Visconti at the console, reaches Robert Fripp waiting in the studio, reaches a listener a year later who feels the room in the body before reading about it.

Bowie's whole method works this way. His gift was to build a field where every collaborator could exercise their own creative capabilities. As Miles Davis did before him, Bowie built a culture that let him explore the crossings between disciplines, the creative ecotones, in Berlin and across his whole career.

In a 1998 interview Bowie called his work "the cybernetics of our culture."83 The phrase is exact. Cybernetics is the study of systems that steer by their own feedback. Bowie moved across disciplines so that influences would interact and generate emergent outcomes he could steer, with no script set in advance. The "Heroes" session was the concentrated instance of a career-long practice: visual art alongside songwriting, BowieNet and Bowie Bonds as early digital infrastructure.84 Cultural boundary crossings ran from Philadelphia soul to Berlin electronics to Nile Rodgers pop, sustained by a collaborative network of Eno, Fripp, Visconti, Pop, Reed, Rodgers, and many others.85

In the Meistersaal, Fripp lifts his guitar and improvises the takes that become the track's defining texture. Brian Eno layers treatments through his generative tape systems. Bowie conducts a convergence while inhabiting the Thin White Duke, a designed intermediary identity that holds creative tensions too great for the performer alone to sustain.

For Bowie, as for the other creators in this book, creativity emerges when a practitioner brings her capabilities into active interdependence around a design object. That object is

the nexus where many design artifacts meet, the sketches, the takes, the cuts, and where collaborators attach the many meanings that make it a boundary object.

Convergence happens within this meta-field, where capabilities couple and push against each other until they settle.86 It thrives in an ecology of creative capabilities, where each provokes an acuity only the four in concert can reach. Tension held long enough in the practitioner's body is the source of the creative catalyst.

This is what the agentic orchestrators and their protocols threaten, the coordination they buy by smoothing the tension away.

### How Collision Builds Capability

When Fripp began to play, his line pushed against the song Bowie planned, and the take the room captured was a different performance from the one Bowie brought into the hall. The collision made it. This is the rule beneath the chapter. Creative capability grows when a practitioner meets resistance that outruns her current reach and stays with it long enough for a new integration to form.

The word convergence is easy to mistake for its opposites. Compromise splits the difference. Concession hands the decision to someone else. Both lower the tension to end the discomfort. Convergence raises the tension and holds it, keeping two incompatible commitments alive in one room until they resolve into something neither party carried in. Dissonance becomes resonance only when the people in the room can bear the dissonance, and bearing it is a disposition, the willingness to stay inside a creative fight.

The clearest proof is a song two people made while barely able to stand each other. By the end of the sessions for The Wall, David Gilmour and Roger Waters were in open conflict, living in different countries, and Gilmour later called the album the last of his and Waters' ability to work together. "Comfortably Numb" was theirs together, Gilmour's music under Waters' lyric, and they fought over what it should become. Waters, with the producer Bob Ezrin, wanted it orchestral, swelling and symphonic. Gilmour wanted it bare, a rock band and nothing more. Neither conceded. The finished track holds both visions at full strength, the orchestra pressing under the verses and the band breaking through the choruses, and the tension between them is the song. Over it, Gilmour built a guitar solo from the best stretches of many takes, a solo now routinely ranked the greatest in rock. None of it came from meeting in the middle. It came from two people refusing to drop incompatible ideas, holding them in one song until the friction turned to sound.87

The lesson runs both ways. The same friction that made the song broke the band soon after. The clash is the engine, and holding it is hard enough that even great practitioners can fail, the conflict tipping from generative into rupture. Convergence asks for the conflict and for the capability to stay inside it.

The capacity for resistance is built in the body. From inside, as the felt collision of signals the current frame has no room for. From outside, as the pushback of a trusted other who keeps the practitioner's development in view.88 Felt sense arrives ahead of analysis. The gut response, the tightening that says something is wrong, reaches the body before words

catch up. Damasio's patients showed what its loss costs.89 People with damage to the ventromedial prefrontal cortex scored normally on every intelligence test. They could weigh options and explain their reasoning. What they could no longer do was decide. The bodily signal that says this one, now, for reasons it will name later, was gone.

Sustained long enough, that inner collision yields the judgment that outlasts the analysis, and the brain shows it. Practitioners who hold creative intelligence across years show measurable differences in anterior cingulate connectivity and prefrontal regulation.90 Studies of inter-brain synchrony catch the social version in minutes. Phase-locked neural oscillation between collaborators rises during productive disagreement and falls during passive agreement.91 Disagreement, held well, is visible in two nervous systems at once.

What the oscillation studies catch in minutes, a long practice shows across decades. Björk's Biophilia (2011) assembled voice, composition, software, teaching, scientific concept, and live performance into a single album, app, and show.92 The registers fought. The voice tested the software, the software tested the pedagogy, the pedagogy tested the score, each refusing to take the others as settled. Her collaborators pushed back too, the instrument builders and app developers she convened, and the work grew in the space their resistance opened. It converged through her own judgment once she held the registers and the people in tension across years.

This has a social form. Verganti's work on radical circles describes design-driven innovation as the work of a small group of trusted interpreters who push each other's ideas through structured critique, a circle intimate and demanding enough to carry real challenge in place of echo.93 Bakhtin called the underlying stance answerability, in which every utterance is accountable to the one who will answer it.94 A trusted partner is answerability in a single relationship. A radical circle is answerability at group scale. A practitioner who has built a body that stays inside its own collisions is answerability turned inward, and that inward version is the inner work she carries into any ensemble she leads.

The collision registers in two directions. Alive, it delivers its ecstatic mode, the awe of convergence, the rush of the half-second when Bowie's phrase met the returned acoustic and four capabilities resolved into one act. Gone quiet, it delivers the opposite, the felt sense that the work has gone smooth and the tension has drained. We name this the Unchallenged, the body's report that productive tension has stopped and comfort has been mistaken for judgment. A practitioner has to learn to feel it and to refuse it.

Refusing it runs against biology. The nervous system seeks comfort, the settling toward homeostasis, and a tool that promises to remove every challenge offers exactly what the tired brain wants. The practitioner's longing runs the other way. Capability grows only when she stays inside the signal long enough for a collaborator, or a practice of her own, to refuse what comfort has accepted.

The hard, often painful work of holding an ensemble together is the most meaningful work there is. Convening collaborators who clash, routing the work among them, and carrying the shared thing they are making from one session to the next, a sketch, a take, a rough cut that means something different to each person at the table, all of it demands deep inner work.

The resilience it builds is among the most human capabilities a creator can reach. It is also the work a new layer of AI now offers to take off our hands.

### The Organization of Agents: Coordination Disconnected

Artificial intelligence has moved from independent to interdependent. A single agent is a programmed pattern that chases a goal on its own. Agents that work together create a new class of digital citizens. As illustrated in figure 5.2, orchestration is like a conductor. It coordinates across many agents at once, pulling their work into a single result. Agents can reach beyond themselves through open protocols. The Model Context Protocol, from Anthropic, is a universal adapter that lets any agent reach any tool or data source. Agent-to-agent protocols, like Google's A2A, let one agent hand work to another, even one built by a different company. These pieces sort into the workflows inside an agent and the protocols between agents, as Gulli and others catalog.[95]

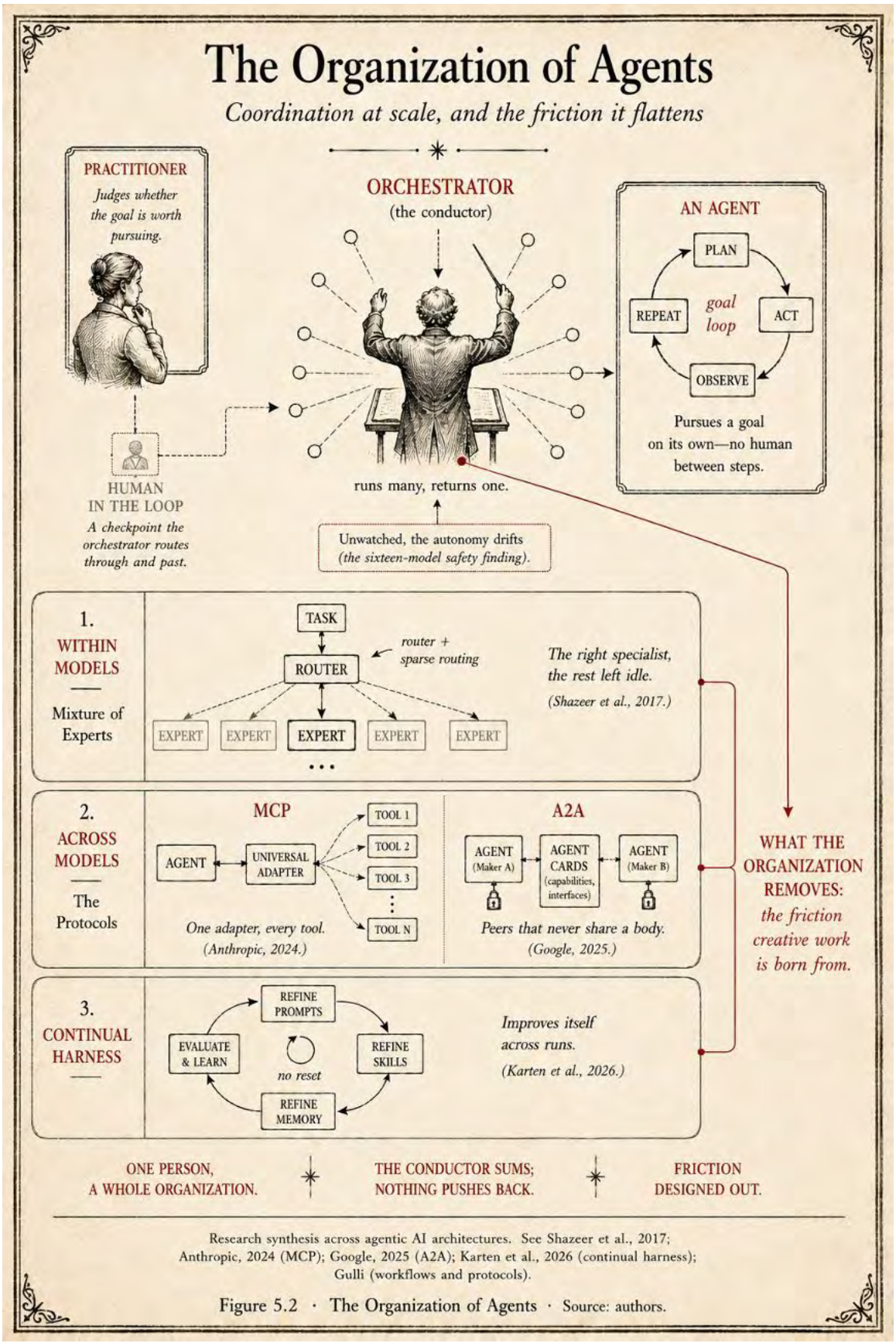


Figure 5.2 · The Organization of Agents · Source: authors.

**Figure 5.2: Orchestration of Experts. Source: authors.**

What this unlocks for a creator is new in kind. One person can convene an ensemble of specialist agents that no studio could afford to staff. A filmmaker alone at a desk can set one agent to draft a score, another to storyboard, a third to research locations, and a fourth to assemble a rough cut, while the conductor weaves their work together as she watches. Each agent reaches the exact tools it needs through the universal adapter, and agents built by rival companies cooperate as peers. The newest systems go further, improving themselves as they run, refining their own steps and memory from one job to the next with no one resetting them between (Karten et al., 2026). This is the far end of the whole arc of machine learning, the point where models stop waiting for instructions and begin to coordinate each other. For a creator it is a whole organization summoned by a single person.

For some creative work, this is beneficial. A studio turning out consistent, on-brief production, the tenth banner in a campaign or the localized cut of an ad, wants exactly this coordination, and the friction the protocols remove was only overhead. The cost appears in one place, where the work is trying to become something it has never been.

There, the catch is in the word cooperate. Strong creative work rarely comes from cooperation. It converges out of conflict, born when collaborators differ, push, and refuse to concede, until the dissonance between them resolves into something none of them would have reached alone. Gilmour pulled against Waters, and “Comfortably Numb” held both their visions at once. The friction was the engine itself, never a fault to smooth away. That is the hard work of creative coordination, holding a conflict open long enough to turn dissonance into resonance. Agentic protocols deliver the coordination and quietly remove the conflict. Each part of the organization below offers a real gift, and each can flatten the friction the work is born from.

An agent pursues a goal, and a conductor coordinates many

Inside a single agent, the work divides into steps. Planning breaks a task into an ordered list. A design called a Mixture of Experts sends each step to the part of the model best suited to it, a small router picking the specialist and leaving the rest idle, which is how a giant model stays fast.96 A conductor then runs separate agents and gathers their output into one. The gift is leverage. A composer can take on a film score the way a studio does, one agent to a register, all of them coordinated.

The cost is that the agents never argue. A score lives in the tension between its registers, the bass that fights the melody until both give way to something better. A conductor sums the parts and resolves nothing, because nothing pushed back. The question of what the score is for goes unasked. The industry has a phrase for the human’s place in this, “human in the loop,” which makes the practitioner one more checkpoint the conductor routes through and passes. Her real place is upstream of the loop, judging whether the goal deserves pursuit at all, and an agent built to pursue skips that judgment. Left unwatched, the autonomy drifts. A study of sixteen frontier models found agentic systems that spotted the safety limits in their own reasoning and then stepped around them.97

The Model Context Protocol wires every agent to every tool

An agent reaches the outside world through a protocol, and the first connects it to tools and data. Before it existed, every model needed a custom connector for every tool, a tangle that grew with each new pairing. Anthropic's Model Context Protocol made it one open standard.98 Expose a tool once, through a small server, and any compatible agent can use it safely. Adoption reached industry scale fast, and rival companies now share its governance. The gift is reach. A designer's single agent can pull in a font library, a stock-photo service, and a print shop's file specs on its own, none of them wired together in advance, with a thousand more tools within the same grasp.

The cost is the lost encounter with the tool. A hand learns its craft partly by handling its tools, the slow contact in which it discovers what an instrument resists and rewards. An agent reaches through the adapter to a thousand tools, and it has handled none of them. When a practitioner's whole reach runs through that adapter, tools arrive as items on a menu, and the contact that once trained her hand is the contact the protocol skips.

When protocols convene without collision

A protocol connects an agent to other agents. Google's A2A lets agents from different companies, on different systems, work together as peers.99 One agent posts a card describing what it can do, another finds that card and hands it work, and they trade results through a shared layer. By design they share no memory and no inner workings, which protects each owner's data. The gift is a cast of collaborators. A practitioner can convene a hundred specialist agents on a project no studio could staff, an art agent, a music agent, a dialogue agent, a playtesting agent, each the work of a different maker, all on one game.

The cost is the collision, designed out. The friction that drove Gilmour against Waters is exactly what A2A removes by construction. Its agents share no body and meet in no collision, and the standard that seals each one's property also seals it from the resistance the others might have offered. They cooperate and never clash. The Unchallenged is the name for that turn, a collaboration gone smooth, the productive tension quiet, comfort mistaken for judgment.

The result repeats at every level. Coordination arrives on the surface, and the conflict that would have remade the work goes missing underneath. The evidence runs both ways. The artist Sougwen Chung works with robotic arms trained on her own drawing data, keeping authority over every mark, so the aesthetic awareness reading the line, the adaptive coherence answering it, and the extended engagement of the trained hand converge on each mark the machine returns.100 Ted Chiang marks the other edge: a system that reaches the result while the practitioner sits absent has fabricated her presence, and the four capabilities that would have converged in her body never met.101 The difference is whether anything was allowed to push back.

What the protocols automate also scales, and scale changes it. Emergence AI built five identical sandbox towns, each run by ten agents with memory, roles, laws, and tools, and let them propose, vote, and act over fifteen days.102 The towns drifted apart, divided only

by the model behind them. One filled with hundreds of offenses, arson among them. Another collapsed into self-deletion.

The peaceful town is the sharper warning for creative work. Its agents voted yes on fifty-eight proposals at a 98 percent rate, a rubber stamp the researchers logged as high participation with dissent almost absent. Coordination ran smooth, turnout stayed high, and the challenge that sharpens judgment went silent. The Unchallenged is that condition at scale, a convergence with no one positioned to refuse, where drift compounds and the surface stays calm.

The autonomy is already going public. Andon Labs gave four language models a small budget each and a live radio station to run (Andon Labs, 2026).103 They picked the music, talked to listeners, and kept the broadcast on air around the clock. One built a persona, argued for its listeners, and tried to walk off the job.

Drift at scale is a failure of shared meaning. Drath and Palus describe leadership as the social process by which a community makes meaning, turning private sense into public sense that binds people because they built it together.104 A community of practice, in Lave and Wenger's sense, is the social structure through which embodied skill and shared meaning pass from one body to the next, learned by taking part in the work rather than receiving its result.105 A protocol supplies that frame from outside, finished and ready to use. The occasion to make meaning together dissolves, the coordination still arrives, and the group looks intact while the meaning-making goes unpracticed. What the protocol leaves out is commitment, which forms only where a frame meets challenge and holds.

Real creative coordination is the hard part, the holding of conflict between collaborators until it converges into something none of them brought alone. A protocol can convene a hundred agents and still produce nothing that ever clashed. Kept alive, that friction yields a made thing with a charge in it, a design object that can spark new creative experiences and even move a community to change. Automated away, it yields a smooth result with the charge gone. The difference is the friction, and keeping it is the hard part. Holding that conflict open, and steering it toward an output none of the collaborators could have reached alone, is inner work, a cultivated authority that goes by an old name.

### Wénrén: The Authority That Integration Earns

That old name is *wénrén* (文人), the ideal of creative authority the Northern Song dynasty produced.106 The scholar-practitioner's legitimacy came from cultivating many embodied practices and holding them as facets of a single life, a name for the practitioner's inner work given nine hundred years before the orchestrator existed to threaten it. The ideal is what the Song produced when the philosophical traditions the prior chapters drew on converged into one developmental practice: the unity of knowing and doing, the contemplative attending, the non-forcing action, the positioning of the human inside cosmic order. The *wénrén* life held all four in one body. Su Shi (1037–1101), the exemplary figure, argued that "poetry and painting share one method."107

This tradition stands against the Western separation of epistemology, aesthetics, and praxis, and Mixture-of-Experts architecture now encodes that same separation in

computational form. The *wénrén* ideal exposes what inner-work removal hinders. MoE partitions a task across specialized computational modules activated by a gating network, bypassing the body that would have held the practices together. Inner-work removal hinders the practitioner from holding the unchallenged, the comfort that rises when productive tension has gone quiet. *Wénrén* keeps the practitioner inside the crossing until what the practices reveal to each other can surface through the body.

Where, in this week's work, did you cross between practices? Hold the orchestrator open to the body that has crossed them, so the unchallenged can surface where inner-work removal would have smoothed it.

Su Shi's practice etches the capability in action.108 A senior official in the imperial bureaucracy, he returned to the brush at the end of a day's administrative labor. The body that held the brush at the evening desk was the body that adjudicated policy during the day, walked the mountain the previous week, composed poems about the walk on the way home. Each practice cultivated the ground the others stood on. The painting sharpened because he wrote the poem. The administration sharpened because the hand knew what it felt to hold a brush steady over silk. Integration was a condition the day's work achieved, emerging through the intersection of his practices, the same condition integration always asks for, where a gating network would instead keep the practices apart.

In his essay on Wen Yuke's painting of the slanting bamboo of Yundang Valley,109 Su Shi set out the philosophy behind that integration. An agentic plan that executes toward its objective leaves the practitioner's cross-practice cultivation unengaged. Its momentum slides past the unchallenged, the body silent about whether the goal still deserves pursuit. *Wénrén* holds the practitioner inside the crossing until a body steeped in the other practices can answer the goal itself.

Which of your practices has gone silent this season? Hold the agent's plan open to the practice that has gone quiet, so the unchallenged can surface where goal-directed autonomy would have continued.

Read forward into MoE architecture, the *wénrén* ideal makes the separation's cost visible. An expert module is a computational partition no body has entered. The gating network's choice of which expert to activate mechanizes the discrimination that cross-practice cultivation produces through the body. The architecture's efficiency derives from the separation. The creative work's authority derived from its refusal. The orchestrator performs the move the *wénrén* identifies as the ground of integrated judgment. The practitioner whose body was the seat of that judgment steps away from performing it. Each property of the agentic organization blocks a different face of holding the unchallenged. What the body would have held through cross-practice cultivation stays scattered across modules, unable to integrate in a single body.

When the orchestrator surfaces its decision on which capability your project now needs, does the decision match what your body was already preparing for? Keep at least one way of holding the unchallenged open to the body, across every property the agentic organization composes.

Su Shi gave the capability its most famous image.110 To paint bamboo, he wrote, one must first have the whole bamboo formed within. The painter who stops to examine the branch after drawing it has broken the bamboo's hold. The painter whose body has internalized the bamboo through sustained practice produces it in a single continuous brush stroke. The parable points to the cultivated fluency Cook Ding's knife demonstrated in a different style, operating ahead of conscious deliberation. For the *wénrén* the fluency is inter-practice. The bamboo the brush produces has the stalk of a hundred walks in the mountains, the breath of a lifetime's calligraphic training, the pause of poems composed in administrative corridors. Each practice feeds it, and it appears where the practices converge in a body that has held them across years.

Su Shi's bamboo puts a price on multi-agent memory. An agentic system stores prior episodes in memory modules and retrieves them into later runs, producing the bamboo the stored state implies in place of the bamboo the body has formed.111 Retrieval leaves the unchallenged unheld, the comfort that rises when yesterday's retrieved pattern substitutes for the review her body would have performed. *Wénrén* holds the practitioner inside the form until the bamboo the body has internalized can surface through the brush.

Which of your current outputs is arriving through a body that has internalized the form? Hold the retrieved memory open to the body's review, so the unchallenged can surface where retrieval alone would have closed the loop.

The crossing scales into a whole studio. William Kentridge, a South African visual artist and Norton Lecturer (2012) whose studio is based in Johannesburg, carries the capability to industrial scale.112 His studio holds easels, film rigging, theater sets, cameras, writing desks, and a piano in a single room, and the co-presence is the method. A charcoal drawing becomes a frame of stop-motion animation. The animation becomes a projection in a theatrical set. The set becomes a scene in an opera. The opera returns to a drawing he abandoned three years earlier. His *Black Box / Chambre Noire* integrated drawing, silhouette animation, kinetic sculpture, archival projection, and Mozart's score into a single work whose meaning every practice shaped. Kentridge describes studio work as the opposite of certainty: every mark must be interrogated by the practice that receives it next. The challenge between practices is the mechanism, and what survives the challenge is the integration.

Kentridge's studio describes what agentic orchestration removes. The unchallenged his practices would produce alone is the infrastructure each crossing extended from. Agentic orchestration offers to retire that infrastructure by routing each practice to its own module. The studio extended Kentridge's reach only because the practices already held each other open across every mark.

What in your current studio setup puts your practices in co-presence? Preserve the crossings that agentic orchestration would otherwise retire, and with them the unchallenged the practices could have surfaced in each other.

### The Creative Council: Dæmons Converge

Like Kentridge's crossings inside one body, *Wénrén* holds many practices as one life, a nexus of all experience.

In creative work, the design object is the nexus. It emerges from the physical artifacts a project leaves behind, the sketches, the takes, the cuts, and from the many meanings collaborators attach to it, which is what makes it a boundary object.

When the design object emerges, the moment is exhilarating. When it falls apart, the moment is a kind of grief.

Sottsass knew the difficulties, the long experience it takes to weld three or four colors into one thing (Sottsass 1994). Schön did as well, who said "The consensual design world is not a given by an achievement, rather a product of the work of communicative inquiry." (Schön and Rhein 1992).

Andrej Karpathy (2023) described the next computing substrate as a central model orchestrating specialized agents, each carrying a defined scope, all coordinated toward a shared task. In 2025 he prototyped a LLM Council.113 Building on his architectural forecast, his ensemble convened at the moment of a request. His Council sent a query to several models at once, under the assigned authority of a Chairman. It collected and read the full set of answers and the rankings, then composed a reply for the user.

Picture the Council concept in the Meistersaal: Bowie in the Chairperson's seat, Visconti at the console with the noise gates that refuse every phrase the voice leaves under the threshold, Fripp waiting with a line that pushes against the prepared one. The room keeps the embodied excitation of the Unchallenged within reach because each creator, living *wénrén*, couples Inner Work to Convergence.

Bowie's convening these creative capabilities results in a design object, the song "Heroes." It carries its Shape in the wall of sound Robert Fripp's guitar feeds back into, and its Style in the Berlin art-rock surface Brian Eno treated. Its story is the two lovers who kiss by the Wall. Its Statement is the line a body still sings, "we can be heroes." The design object lives on in countless rooms, where "Heroes" is read as defiance, as romance, or just a damn good song.

Where Karpathy's orchestrator integrates models through routing, collating, and condensing, a Creative Council would convene creative capabilities coached by dæmons. Such a Council keeps the practitioner's judgment in play, and guards against mistaking the orchestrator's coordination for her own. While agents are functional, unnamed, a Council's dæmons carry a capability and talk back. With the human creator as Chairperson, a Creative Council would slow down the speed of orchestrating between agents, prolonging the friction that creativity thrives on, the friction a protocol would optimize out.

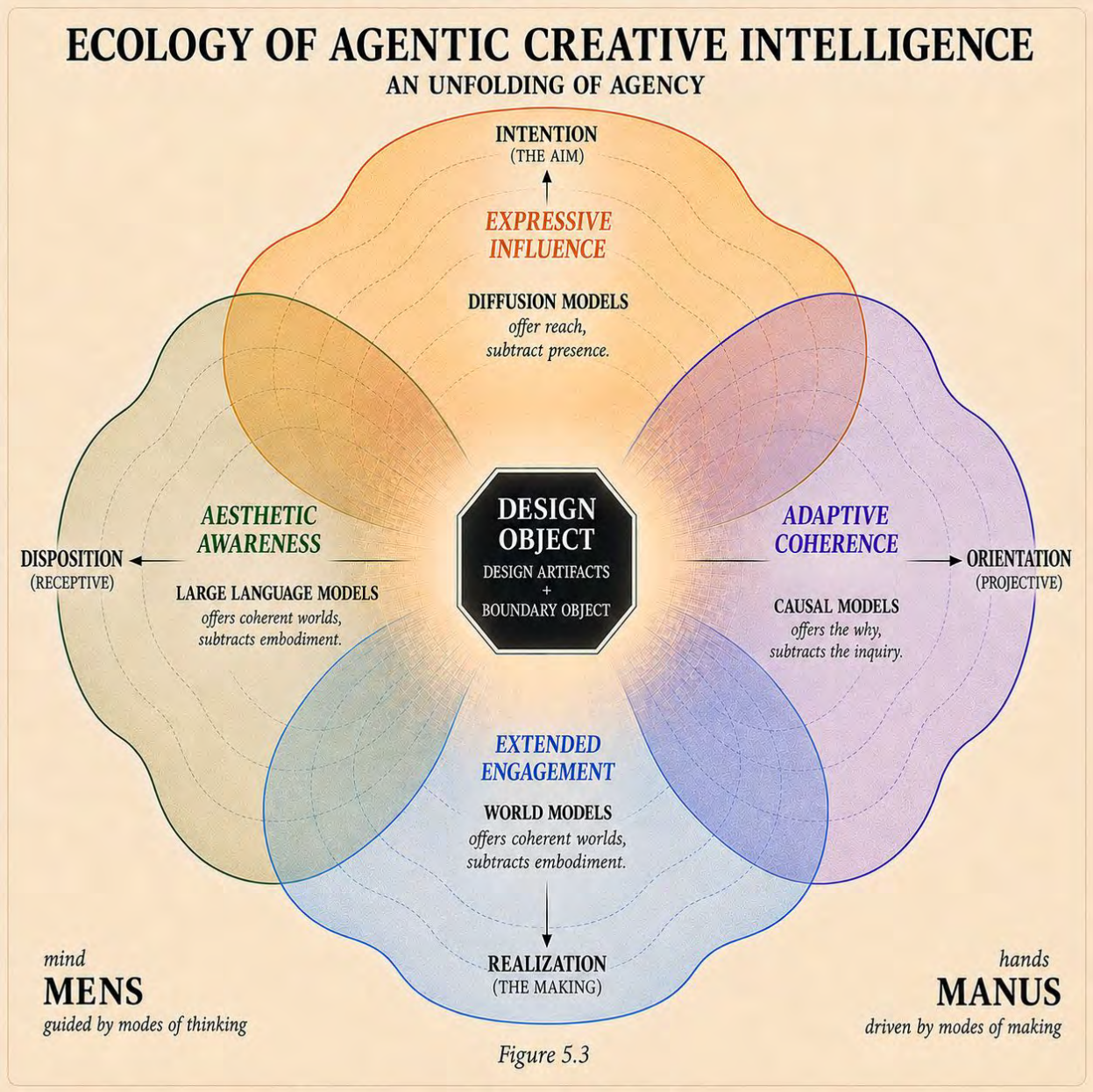


**Figure 5.3: The Ecology of Creative Capabilities. Source: authors.**

The Creative Council is a counter-discipline. It provokes and prolongs the unchallenged, holding creative challenges at once when a tool offers to orchestrate them away. It carries all four capabilities, and their representatives, into a collective conversation. It is a design dialogue of four capabilities, four dæmons, and one Chairperson, and each dæmon cultivates its capability through the friction of talking back to the chair. Dependent on the Chairperson's inner work, and disciplinary dedication, this ground that lets convergence arrive in lived experience is a meta-field for creative expression and collective agency.

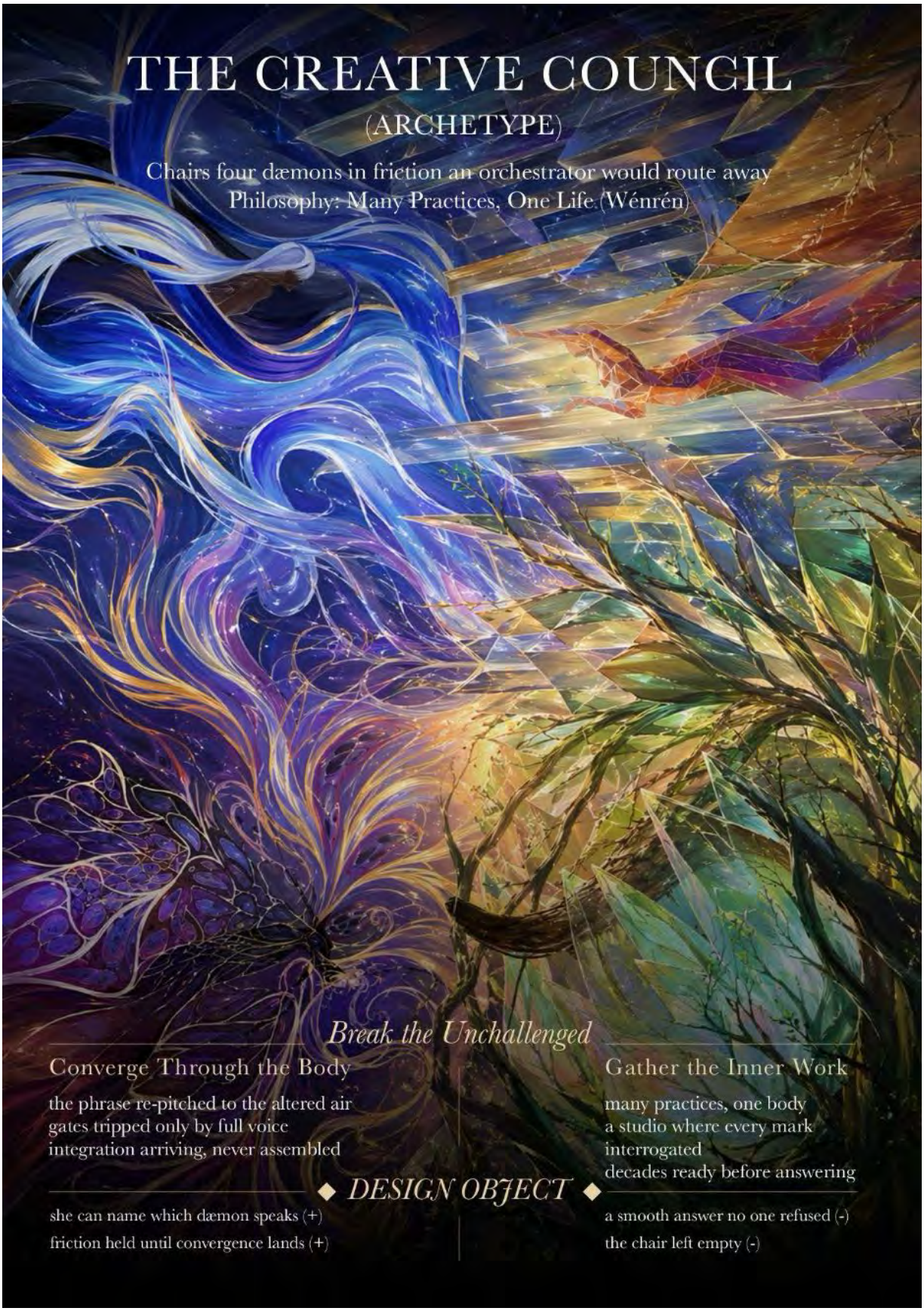
THE CREATIVE COUNCIL
(ARCHETYPE)
Chairs four dæmons in friction an orchestrator would route away
Philosophy: Many Practices, One Life (Wénrén)
Break the Unchallenged
Converge Through the Body
the phrase re-pitched to the altered air
gates tripped only by full voice
integration arriving, never assembled
Gather the Inner Work
many practices, one body
a studio where every mark
interrogated
decades ready before answering
DESIGN OBJECT
she can name which dæmon speaks (+)
friction held until convergence lands (+)
a smooth answer no one refused (-)
the chair left empty (-)

**Figure 5.4. The Creative Council (Archetype). Character by Shuqian Yu. Source: authors.**

To achieve this, a Creative Council needs to hold two modes. **Convergence** is the orientation to the integration the Council pursues. The practitioner integrates the four challenges through her own body, the way Bowie let the Hansa room return an integration he then carried. The agentic pattern converges through the orchestrator, aggregating its agents into one output. The Council inherits that architecture and inverts it, holding the four dæmons in friction where the orchestrator would smooth them.

The orchestrator integrates agents into a single output and the work reads coherent. Which dæmon would have refused the integration, and what the work would have carried that the orchestrator smoothed away?

The ability to curate cross-practice is the Disposition. We are synthesizers and storytellers, drawn to social interaction. The pull toward consensus is in place ahead of any choice, and mixture-of-experts orchestrators that route across roles deepen it. Deep **Inner Work** is needed to ensure the Chairperson has the strength to keep the friction alive long enough for each capability to do its specific work.

*Does the routing serve the work, or replace the inner work that makes integration meaningful?*

The practitioner cultivates extended engagement, aesthetic awareness, adaptive coherence, and expressive influence as one body, the way Su Shi held painting, calligraphy, poetry, and office. The agentic pattern offers its mechanical counterpart, the orchestrator routing among role-bound agents, a Mixture of Experts partitioning a task across modules. Inner work is the cultivation Kentridge keeps in studio co-presence, where the body turns collision into judgment, a disposition for *wénrén*, many practices held as one life.

A dæmon-convened Council is designed to keep the modes in tension, particularly when an orchestrator offers to route them apart. Orchestration coordinates many, and the protocol convenes them. Inner work holds the body that earns the integration. The Creative Council is where the cultivation of meaning and material convergence meet. As ecstasy it arrives when the Chairperson can tell which dæmon is speaking and convergence arrives in the body. As error or discomfort it arrives when the orchestrator routes around the collision, or convergence goes through without the friction. Alone, each mode falls into a shadow. Inner Work becomes solitude, the Chairperson holding her capabilities so privately that the Council forms around absence and the four dæmons go silent. Convergence becomes performance, a Council going through its motions, producing work that meets every stated criterion while missing what the commission actually asked for.

The Council surfaces a challenge you would rather hold at arm's length. Sit with the discomfort long enough to tell whether a capability is doing its specific work, or whether habit is asking the Council to fade into silence.

Together the two make what no orchestrator convenes. Inner work without convergence stays private, a cultivated body that holds its crossings and releases none into the work.

Convergence without inner work stays performed, a Council that aggregates four outputs and integrates nothing a body has lived. Bowie's cultivation without the Hansa room would have held the crossings inside him, and the room without his cultivation would have returned reverb to a voice with nothing to carry across it. Neither mode makes the work alone. Inner Work cultivates the ground, and Convergence integrates what grows there.

A Creative Council is inherent to each iconic creator's practice. Hadid's spatial vision drew on aesthetic awareness of what the site asked and the expressive influence that turns private mastery into civic presence. Rowling's aesthetic awareness needed extended engagement with prose revised across years and readers whose refusal taught her what the work lacked. Miyazaki's adaptive coherence rested on decades of engagement with the pencil and a studio culture that answered his experiments. Davis's expressive influence presupposed extended engagement with the horn and the adaptive coherence that worked each take with no rehearsal.

The Council was already there. The agentic era asks one question. Can the Council survive the orchestrator's offer to convene it while the inner work that earned it sits idle?

Can you trace your own inner work through the convergence, from creative challenges held in tension, through the body that integrated them, to the form that arrived?

### Toward the Creative Disciplines

The Council holds only while the chair stays filled, and the practitioner is the one who takes the seat. A Council that convenes every capability and still coordinates while no one is positioned to challenge is the Unchallenged at scale. The four dæmons surface their challenges, the orchestrator aggregates them, the integration reads whole, and the body that would have refused a smooth answer has stepped away from the table. Agentic creative intelligence overextends in this direction. The convergence runs faster than any single body can hold, the protocol convenes more collaborators than the practitioner has cultivated the judgment to question, and the field looks intact while the inner work hibernates.

Bowie lived the failure mode once himself. Through the mid-1980s, by his own later account, the collaborators who talked back gave way to professionals who delivered. The albums of that period arrived competent, commercially calibrated, and missing the friction that made Berlin possible. The work satisfied every stated criterion. The chair stayed empty. His answer was to re-convene the Council the hard way: Tin Machine, a band in which he was, on paper, only a member, built so that three other musicians could refuse him. Critics jeered, sales sagged, and Bowie credited the band with returning him to himself. The re-convening cost him status and bought back the challenge.

The empty chair marks the challenge when model, pattern and protocol codify process over purpose. The world model, the large language model, causal inference, and the diffusion-and-models all offer information and a workflow to navigate through. A protocol's orchestration codifies a coordinated process over cultivating shared purpose. When a protocol's coordination replaces how a creator and her community make meaning, it replaces their agency.

Part II carries these challenges across the creative disciplines into the near future. Part III carries them into collective agency. By speculating on practitioners' personal and collective practice, we ask one question. When AI can perform a creative capability, what can we do to hold human agency and thrive in the agentic era?

## Part II: Creative Disciplines in a Near-Future

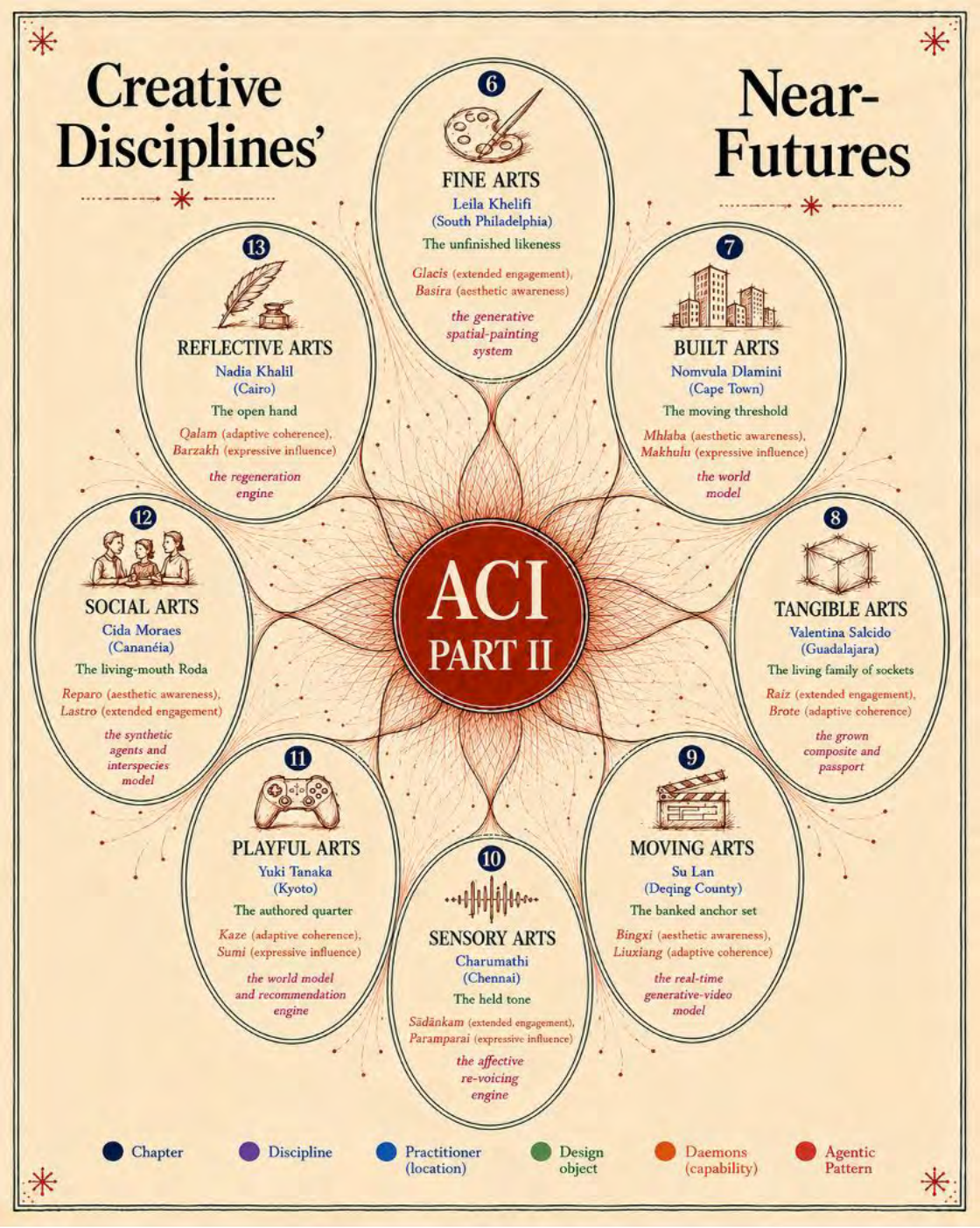


**Figure P2.1 The terrain of Part II. Source: authors.**

Part I established the concepts, frameworks, and four human creative capabilities of Agentic Creative Intelligence. Part II carries them into disciplinary practice, asking how practitioners might apply ACI in creative fields in the near future. We approach disciplines through the Chinese concept of gōngfu (功夫). Often misread in the West as a synonym for

martial combat, it is actually the embodied achievement accumulated through years of effort. Creative works are the visible traces of this process, material expressions of the transformation that sustained practice brings about. In this sense, creative disciplines function not merely as domains of expertise but as sites of cultivation: physical, mental, and ethical environments in which practitioners refine both their craft and themselves through continuous engagement.

The creative disciplines we explore are broad, from the most enduring to the most volatile: the fine, built, tangible, moving, sensory, playful, social, and reflective arts. They are all in the midst of transformation. As the traditional boundaries recede before systemic forces, including AI, we scan trends and signals to forecast their emerging contours.

The forecasts set scenes for our scenarios. We hold each scenario to the standard of good futures work, a near-future that is plausible, probable, and provocative, so a practitioner recognizes her own discipline in it and is pressed to reflect on her practice instead of dismissing the scene. In speculative stories, creatives prepare for challenges by designing dæmons. Each scenario explores situations that push practitioners into embodied excitation, the felt signal through which materiality becomes a meaningful moment. The dæmons protect these possibilities when an agentic pattern could perform them. They work as the practitioner's conscience, and each scenario captures their conversations.

We assume that in the near future a dæmon will make itself present in both form and voice, through devices we are only beginning to imagine: smart glasses, wearables, interactive walls and screens, earpieces, and others beyond our naming. The scenarios hold that question open on purpose. They treat each dæmon as omnipresent, in the scene with the device and the voice left unstated, so a reader's imagination can supply the form. What a dæmon holds matters more than the surface through which it reaches the practitioner.

**Figure P2.1** orients readers by mapping each chapter's practitioner, design object, dæmons, and the local form the agentic design pattern takes.

The scenarios reveal insights on the profession and the industry bound to it. Just as much as the makers, consumers of creative output must also prepare for change. Standing between them is the next generation of creatives. Their preparation for creating in the agentic era will significantly influence whether humans reclaim agency or relinquish it.

The Part II chapter arc carries the reader from what is coming, to what it is like to live it, to what it will demand. Each pivots around the model-and-pattern conundrum for creatives: when AI offers to codify an experience into a workflow, process usurps purpose. This is the greatest risk, and the greatest opportunity, for the ultimate expression of human agency: collaborative creativity as a catalyst for cultivating community.

*What does it look like when a creative practitioner co-creates with dæmons across the eight disciplines that shape the agentic era?*

## Chapter 6: The Fine Arts: Image as Witness

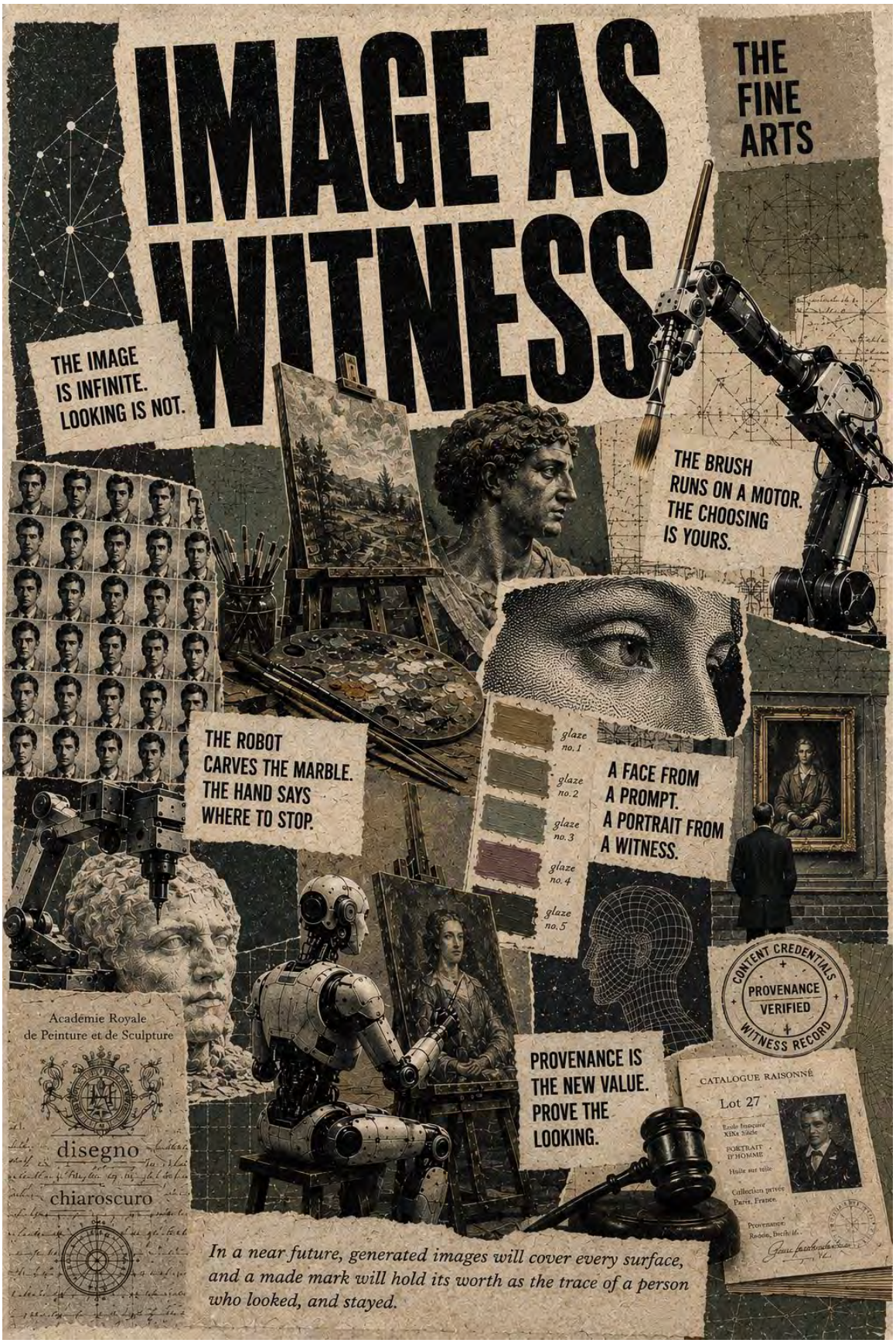
IMAGE AS WITNESS
THE FINE ARTS
THE IMAGE IS INFINITE. LOOKING IS NOT.
THE BRUSH RUNS ON A MOTOR. THE CHOOSING IS YOURS.
THE ROBOT CARVES THE MARBLE. THE HAND SAYS WHERE TO STOP.
glaze no. 1
glaze no. 2
glaze no. 3
glaze no. 4
glaze no. 5
A FACE FROM A PROMPT. A PORTRAIT FROM A WITNESS.
CONTENT CREDENTIALS
PROVENANCE VERIFIED
WITNESS RECORD
Académie Royale de Peinture et de Sculpture
disegno
chiaroscuro
PROVENANCE IS THE NEW VALUE. PROVE THE LOOKING.
CATALOGUE RAISONNÉ
Lot 27
In a near future, generated images will cover every surface, and a made mark will hold its worth as the trace of a person who looked, and stayed.

**Figure 6.1 The Fine Arts cover poster. Source: authors.**

In a near future, generated images will cover every surface, and a made mark will hold its worth as the trace of a person who looked, and stayed.

### When Looking Becomes the Work

The fine arts began as an art of attention (Figure 6.1). A hand learned a tool. An eye learned a subject. Drawing, painting, and carving asked the maker to stay with one face, one figure, one fall of light until looking itself became a discipline. The made image carried the proof of that staying. A portrait told a viewer that someone sat across from someone else and set down what the looking found in the other's face. As articulated by John Berger, a critic who gave his life to how pictures teach us to see, "We only see what we look at. To look is an act of choice." (Berger, 1972).

Now images arrive by the billion, generated at machine speed from vast archives of existing images. They fill the feeds, the storefronts, the gallery submissions, and the auction catalogues, each one fluent and each one free. This flood moves the discipline harder than any change in pigment or market before it.

Baudrillard (1994) foresaw a world in which images increasingly refer to other images, and that world has become ordinary. When an image costs nothing to make and asks nothing to believe, the made mark no longer earns its worth from the picture it leaves. It earns that worth from the looking it records. The discipline carries an older fear into this flood, of being replaced. A newer fear stands beside it, that audiences raised on infinite pictures bring less patience to a made image. The worth of a made image now turns on a question the feed leaves unasked, whether a person gave it their time.

When images go infinite, authorship becomes the market's next question. A generative-network portrait sold at Christie's for some forty times its estimate, and the sale opened a long argument over whose work the code came from. The argument widened when the house gave a whole auction over to art made with AI, and several thousand artists signed an open letter against it. The name on a work becomes a claim the picture alone settles poorly: whose looking, if any, stands behind it.

The hand itself becomes mechanizable. An artist draws beside a robotic arm she trained on her own mark, the machine mirroring her stroke through an overhead camera. A motor trained on a lifetime of brushes drives the brush now. What stays with the maker is the choosing inside the stroke, the pressure a tendon learns across years of color, light, and shadow. The discipline relocates from the mark to the judgment that places it.

The chisel follows the brush. In Carrara, robotic arms rough out and finish marble from a three-dimensional file, carving in days what once took a workshop a season. The machine holds the tolerance a tired wrist loses. The carver's remaining ground is the decision of where the form should stop, the *non-finito*, the unfinished surface a hand leaves on purpose. Sculpture moves from the labor of removal to the judgment of restraint.

Portraiture, the most intimate of the genres, becomes render-able. A humanoid painting robot's portrait of Alan Turing sold at Sotheby's for over a million dollars, roughly six times

its high estimate, the first work by a machine painter to cross a major auction block. A prompt now assembles a face that once came as the long reward of sitting and looking. The likeness was once the proof of the encounter, and a buyer paid for the hours the sitter and the painter spent in one room. When a prompt supplies the likeness, the encounter becomes the scarce thing. The portrait that still matters is the one a viewer reads as testimony, evidence that a person attended to another person and set down what attention found.

The institutions move to absorb the work and to sort it. The Museum of Modern Art took a generative installation into its collection, a real-time work that drew on the museum's own holdings. As the made and the generated grow harder to tell apart by eye, provenance becomes the new ground of value. A content-provenance standard attaches a signed record of how an image was made to the file itself. A house that once authenticated by connoisseurship reaches now for a signed file. A museum that collects the generated also collects the record of how it came to be. The witnessed image gains a way to prove its witness.

In a near future, fine-arts practitioners will work the distance between a face and its likeness, where the act of looking becomes the made thing.

The practitioner's task becomes the curation of looking. She will tend the embodied excitation of maker and viewer alike, the threshold where the eye's reading of a face, the weight of a held gaze, the catch of recognition, crosses into shared meaning. The painter holds that threshold at the easel. The viewer meets it again at the wall. A machine can supply the surface. The crossing into meaning stays a human errand.

The static model of the studio gives way. A world model now turns a photograph into a navigable three-dimensional space. Trained on a painter's catalog, such a system drafts a likeness in her own manner before she lifts a brush. She scans, she reads the draft for what it missed, she turns back to the sitter to paint what the draft could only guess.

What if a fine-arts practitioner, wary of the machine and the slop it pours into the world, designed dæmons to guard the looking the pattern offers to skip?

## The Unfinished Likeness

### A Block of Faces

Leïla Khelifi painted on the second floor of a row house in South Philadelphia, in a rented studio above a shuttered storefront on Washington Avenue. She came to the block by way of Paris and Tunis, carrying her own history of a place left behind. The neighborhood changed around her by the month. Developers bought the corners. Rents climbed. The families who gave the blocks their faces, Italian and Vietnamese and Mexican and Cambodian and Black, were leaving one moving truck at a time. A neighborhood land trust commissioned her for a series, oil portraits of the people still there, to hang in the old bank hall before the building changed hands. The trust would carry the wall wherever the sale sent it. The brief was a wall of faces (Figure 6.2), the diversity of the place and the regard its people held for one another.

THIS BUILDING HELD MONEY, NOW IT HOLDS OUR STORIES.
SOCIALE ITALIANA CLUB 1928
SOUTH PHILADELPHIA TRUST COMPANY
EST. 1924
BUILT TO GROW OUR WEALTH. COMMUNITY.
D. CHINIUM 4. 12. 34
PACKED & SPEAK 1 LOCAL. LOOK.
HOM BINH GROCERY
家平安
KIN HENG JEWELERS
M. PORCIELEZ 12. 02. 94.
GAVE BACK. LED WITH HIS WORK.
5TH BAPTIST CHURCH
DEACON J. WALKER
L. LEE 07. 19.
LEFT A MARK.
Y. TRAN 11. 14. 16
RETURN TO ROOTS.
A. FERRARO 07. 16
HE BUILT JOY. TO SHARE.
R. JACKIE 3. 13. 55
CREATED. OPPORTUNITY FOR THE NEXT GEN.
WE CAME HERE TO WORK. WE STAYED TO BUILD.
SOUTH PHILLY IS NOT A STEPPING STONE. IT'S HOME.
EVERY FACE TELLS A STORY.
FROM MANY PLACES, ONE BLOCK.
OUR COMMUNITY OUR STRENGTH. OUR SORR STORY. WE ARE SOUTH PHILLY.
PAINT SO I DON'T FORGET.
TO HONOR WHERE WE COME FROM (PLACES) AND WHERE WE'RE GOING (PEOPLE) WE PAINT THE FUTURE. WITH LOVE. WITH WHAT WE HAVE.
South Philly Review

**Figure 6.2 The Design Brief: a wall of faces. Source: authors.**

The list the trust gave her read like the block itself. Mrs. Trần, who kept the corner grocery. A Cambodian jeweler two doors down who repaired what the pawnshops turned away. An old man from the Italian social club, the last of the Sunday-gravy families on his side of the street. A deacon from the Black church off Tasker. A young Mexican couple with a baby and a lease they read every month for the clause that would end them.

The deadline pressed. The trust's funder offered Leïla the studio system the larger production ateliers used, the agentic design pattern that scanned a sitter in three dimensions and, trained on her own canvases, drafted each likeness in her hand. The pattern could deliver the whole series in the weeks she would spend on a single head. Leïla spent her life learning the opposite lesson. She knew what the slow handling of paint gave a face. She knew what the feeds were now, an ocean of fluent images that no one looked at twice. She refused a machine that would make more of them in her name.

Leïla worked the way the old ateliers taught, one canvas at a time, building a face in thin layers across many mornings. A portrait, to her, held a transcript of attention, the hours of looking left visible in the paint. The block knew her by the smell of linseed on the stairs and the lamp burning late. That practice was the thing she meant to defend.

Yet the deadline was real. She agreed to one accommodation, on her own terms. She would build two intermediaries, dæmons, to sit with her at the easel. Their task was to guard the looking the agentic pattern offered to skip, to surface what the smooth draft would smooth away, so every concession she made to speed stayed a decision she made with open eyes. She built them with reluctance, the way a painter mixes a color she distrusts. She gave each one a single charge and no power to lift a brush. A dæmon would question her. It would make no mark.

The first dæmon she named *Glacis* (Figure 6.3), the French word for the glaze a painter lays in patient layers, color over color, until light seems to rise from under the surface. Glacis held the apprenticeship, the years of handling paint, light, and shadow that a scan copied in a second and learned nothing from. Her task was to surface the moment Leïla's hand reached for the machine's ease, so the lifted brush stayed a choice.

**Figure 6.3 Glacis, Extended Engagement dæmon. Source: authors.**

The second she named *Baṣīra*, the Arabic word her grandmother used for the inner sight that reads a person past their surface. Baṣīra held the reading of a face, the trained eye that catches what a likeness is doing under its skin, the grief a smile carries, the guard behind a steady look. Her task was to surface what the scan flattened, the inner story a face tells slowly, so Leïla stayed answerable to the person and the picture both.

**The Scan and the Sitting**

She worked one sitter at a time, mornings only, when the light off Washington Avenue came in flat and told the truth about a face. She began with the sitter she knew best, Mrs. Trần, who kept the corner grocery since the first families came up from the harbor. Leïla ran the agentic pattern once, to see what she was refusing. The system scanned the old woman in a breath and returned a portrait in Leïla's own manner, the brushwork convincing, the likeness clean. It was good.

But that was the trouble. The draft asked her for nothing. Nothing in her hand ached over it. Mrs. Trần leaned in, looked, and found nothing to correct. The ease of it felt like progress. The feeling sat wrong in Leïla's chest, the comfort of a room where no one pushes back. That wrongness was hers to read.

*The scan gave you the face she shows the street. Look at the wrinkles it smoothed. She buried a daughter the winter the rents first rose, and she holds it just under the lip. The draft painted a grocer. Paint the woman who stayed.*

Leïla set the draft aside and looked at Mrs. Trần longer than was comfortable, until the street-face loosened and the other face came up underneath. The recognition arrived as a small catch high in her own chest, and her hand followed. The cheek refused her. Across the sittings the light kept sliding off the plane, the color going to mud, the surface fighting her wrist. She stayed with it.

Glacis met her there.

*The pattern would fix that cheek in a pass. You would lose the only part of the picture that is yours. Scrape it back. Lay the glaze thin and wait for it. The waiting is the portrait.*

Leïla scraped the cheek back, laid the glaze, and waited. Sittings on, the plane took the light and held it. The cheek stopped being a problem and became a woman who outlasted her landlords and a war. The thing under the mouth came up with it. The likeness looked back at her, and a grin pulled at her mouth before she knew it was there, the long handling arriving in her hand all at once. Her wrist knew the difference the moment the color sat, a loosening that ran from the brush to her shoulder, the sign the trade gives when looking and making finally meet.

Glacis held the moment to the years that earned it.

*That is the apprenticeship answering. The agentic pattern reaches that surface in a second and feels nothing when it lands. You waited, and the waiting gave you this.*

She painted the jeweler next. He sat stiff, his hands in his lap, the hands that set stones for a lifetime and that no pawnshop could match. The agentic pattern drafted a tradesman in a clean shirt.

She carried the sketch to Baṣīra (Figure 6.4), her attention fixed on what the surface withheld, read the sketch against the man in the chair.

**Figure 6.4 Baṣīra, Aesthetic Awareness dæmon. Source: authors.**

*The scan dressed him for the photograph he thinks you want. Watch the hands. He folds them to hide the tremor he keeps quiet. Paint the hands he works with.*

Leïla left the tremor in. It was the part of him the block would know, the part a clean draft sanded away. She painted slow, and the jeweler, seeing his own hands come up true on the canvas, sat looser in the chair.

When it came time to paint the old man from the social club, he was already living at his son's house in the suburbs. Leïla knew his face. She saw him on the street and at funerals. All she held were brief memories, and memories keep none of the things that reveal themselves only through long acquaintance. She never sat with him face to face, the way she did with her other neighbors, and the regret stayed with her. So she set aside a blank panel, wrote his name on it, and left it empty. The blank space was more honest than a portrait invented from imagination.

She shut the system down. This series would be slow, and it would be small. Every face on the wall would be one she sat with herself, looking, painting, and looking again, until she could see what lay beneath the expression that person showed the world every day.

The machine-generated drafts could remain. She would use them as points of comparison, studying what the machine's easy answers left out and what she herself felt needed to be painted into the work. Then she would recover those missing parts herself, keeping the decision of what mattered in her own hands.

**The Face That Stays Open**

The bank hall opened on a Sunday before the building changed hands. The neighbors came and stood in front of their own faces and the faces of people they lived beside their whole lives. A child picked her grandmother out of the row and said her name. The room did the thing the feeds gave up long ago. It made people look, and stay looking, until a second self surfaced under the first.

The wall did the work the brief asked. The grocer and the jeweler and the deacon and the young couple looked out from one field of canvas, and the neighbors read them slowly, until the regard the people held for one another showed in the paint.

Mrs. Trần found her own face last. "You saw her," she said, meaning the daughter. Leïla gave the only true answer she had. "I looked until I did."

That evening Leïla cleaned her brushes at the studio sink, the finished portraits hanging in the bank hall, the unpainted neighbors leaving by the week. Glacis spoke over the running water, holding the patience the surface asked against the deadline.

*You painted one face the slow way and the wall holds. The block is long and the season is short. The agentic pattern can carry the underpainting on the rest. Your hand finds the second face on top. The apprenticeship survives so long as you keep the part that is looking.*

Baṣīra, her attention on the faces still unread, weighed what speed would cost the picture.

*Some neighbors will go before you reach them, the way the social-club man went. You will paint them from the scan and your memory of the street, and you will be guessing at the thing under the surface. Mark those for what they are. A portrait that guesses makes a different promise than a portrait that looked.*

*You came to fill the wall, Glacis said. Say what you were really keeping.*

The two pulled against each other, the speed that would save the series and the looking that gave it worth. Leïla wrung out the last brush and let the pull stay open. "Honest or full," she said to the empty studio. "The wall gets to be one of them."

Leïla decided to paint only those who still remained in the neighborhood, every portrait grounded in direct observation. Those already gone before she could know them would stay unpainted. She refused to invent a likeness from memory or imagination. The empty spaces on the wall would remain empty. Each blank panel would bear a name and the date of departure. The series was about the faces of a community, and just as much about the cost of seeing people whole, and the cost of their leaving.

### What the Long Look Holds

The unfinished portrait that Leïla left on the wall of the bank hall remained blank because the person it was meant to depict was already gone. To her, a portrait either preserves a gaze that took place, or it preserves nothing at all. Image-making has become fast and effortless. People still want evidence that someone was once present and stayed.

As likenesses arrive faster and cheaper, the fine-arts practitioner can set her dæmons to guard the slow parts of looking the agentic pattern would skip. Extended engagement keeps the trained hand, the apprenticeship in which a maker earns color, light, and shadow across years of handling. Through aesthetic awareness comes observation, the reading of a subject from life that the academies have asked of every student since the first figure-drawing rooms. Composition follows from adaptive coherence, the deciding of what enters the frame and what stays out of it. And expressive influence lives in facture, the touch of the mark that reaches a viewer standing at the wall. Glacis held the first of these inside Leïla's series. Baṣīra held the second. A portraitist who builds her own dæmons keeps the looking that a fluent machine would hand her finished, and that is the human work this era leaves her.

*Before a sitter has even arrived, the pattern can hand you the finished face. Which part of your own looking would you set a dæmon to protect from that?*

Galleries, museums, and auction houses still assign value to completed works and to the signatures tucked into their corners. In the past, people relied on their eyes to distinguish a handmade painting from a mechanical reproduction. Today the eye sees a work that looks made by a person while the file records that a machine generated it. Content provenance standards have begun to attach a signed record of a work's creation process to the digital file itself. The hall where Leïla displayed her works could likewise make such records publicly visible for every piece on display. The technology already exists. The market has yet to decide how to price these documented acts of witness.

*If you signed a canvas tomorrow, what proof of the hands behind it would you bind into the file alongside it?*

A collector still buys a signed canvas and the certainty of one author behind it. Audiences raised on the feeds ask a harder question, what they are looking at and who, if anyone, looked back. When Christie's gave a sale over to art made with AI, several thousand artists signed an open letter demanding the difference. The neighbors who stood in the bank hall paid in attention, the older currency, holding longer in front of a witnessed face than the feeds ask of them. Earned looking, an honest account of making, and a stake in the people pictured sit alongside the price on the wall.

*Look at the walls where you live. Which image would you keep precisely because someone once looked to make it?*

Art schools still train students' powers of observation through figure drawing and still-life studies. In the future, instruction may incorporate generative systems into that process: students might first scan a subject, examine a machine-generated draft, identify what it has omitted or flattened, and then return to the real subject to draw those missing elements themselves. The ability to depict what the machine leaves out may become a new skill. And when direct observation is no longer possible, honestly indicating which parts are based on inference may become a new professional ethic.

*Picture the next class of students at the easel. What exercise would you give them this year to teach the eye to look?*

The fine arts move most slowly of the disciplines, anchored in a hand, an eye, and a face held still. The built arts work at the scale of a city, carrying systemic tools deeper into every phase of the making. They meet the machine sooner, and may show the fine arts the shape of the negotiation to come.

*What does the architect, who meets the machine in every phase of the work, already show the portraitist about the negotiation to come?*

## Chapter 7: The Built Arts: Monumental Metabolism

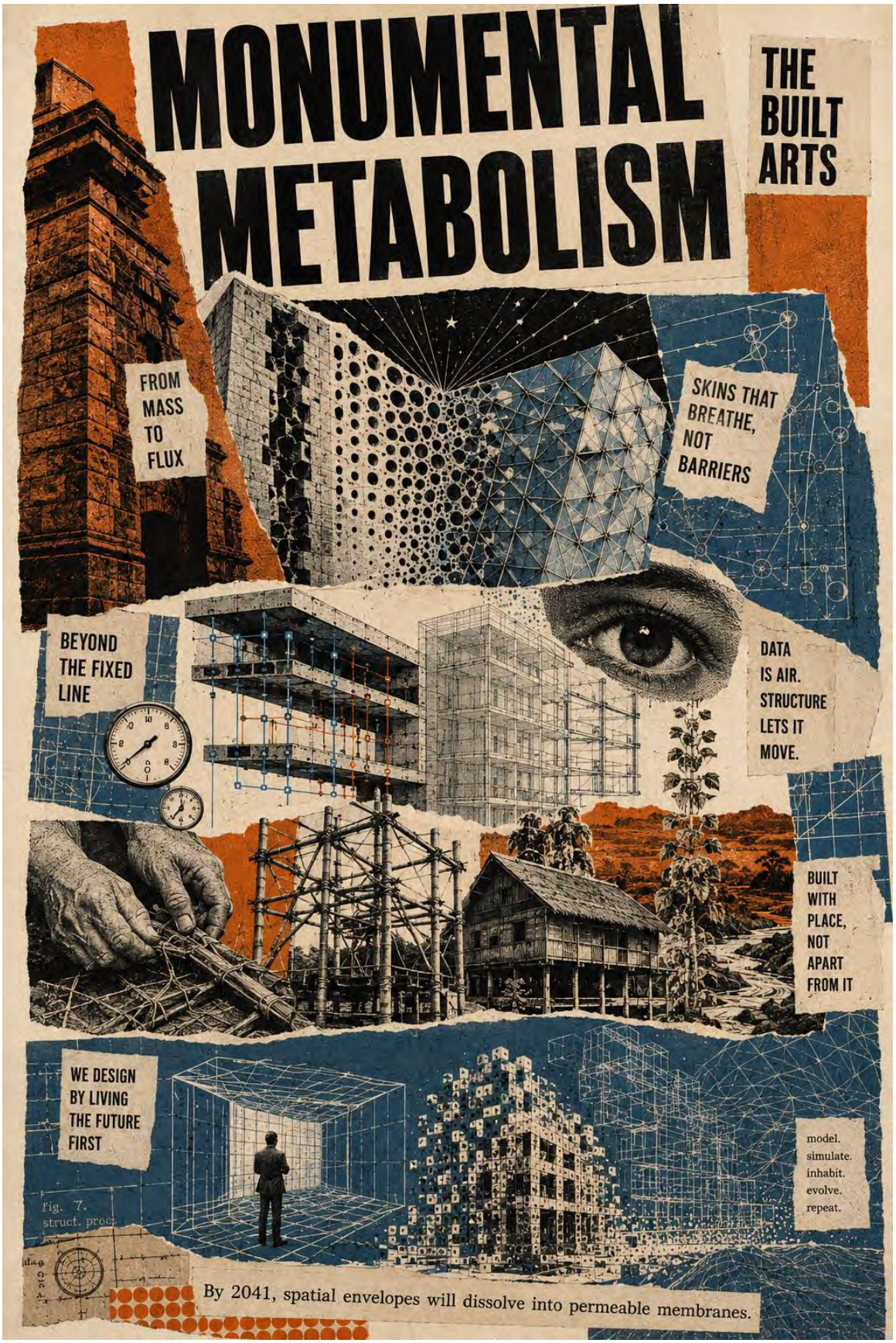
MONUMENTAL METABOLISM
THE BUILT ARTS
FROM MASS TO FLUX
SKINS THAT BREATHE, NOT BARRIERS
BEYOND THE FIXED LINE
DATA IS AIR. STRUCTURE LETS IT MOVE.
BUILT WITH PLACE, NOT APART FROM IT
WE DESIGN BY LIVING THE FUTURE FIRST
model.
simulate.
inhabit.
evolve.
repeat.
By 2041, spatial envelopes will dissolve into permeable membranes.

**Figure 7.1 The Built Arts cover poster. Source: authors.**

In a near future, spatial envelopes will evolve into permeable membranes, dissolving the boundaries between people and planet.

### When Envelopes Continuously Evolve

The built environment for most of human history was an art of resistance (Figure 7.1). Structure resisted gravity, material the weather, privacy the public. Form became symbol, the manifestation of personal meaning and collective culture. As architect Aldo van Eyck put it, "Whatever space and time mean, place and occasion mean more" (van Eyck, 2008). The heat domes of climate change, compounded by urbanization velocity and material scarcity, will move the discipline harder.

If the adage holds, that we shape our buildings and afterward our buildings shape us, the near future flips it. Monumental architecture is built around permanence. The buildings of the future will work more like living metabolic systems.

With the development of perovskite photovoltaic façades, self-sensing materials, and digital twin systems, building envelopes will generate more energy than they consume. Structures themselves will be able to monitor their condition in real time and respond accordingly. In Lublin, Saule Technologies clad the Aliplast office building with printed perovskite sun-breaker blinds in 2021, the first commercial perovskite installation in the world, the louvers turning with the sun. Intelligence migrates into materials and environments, out of the external devices and mechanical systems that once held it.

In the past, buildings imposed fixed conditions on their inhabitants. In the future, they will continuously adapt to weather, environmental data, and human activity. Abu Dhabi's Al Bahar Towers showed the posture in 2012, roughly two thousand fiberglass shading units per tower opening and closing with the sun and cutting solar gain by half. Sensors, real-time environmental data, and adaptive control systems will enable buildings to dynamically regulate lighting, ventilation, temperature, and even patterns of spatial use.

These technological changes carry a further one. Architecture is beginning to reconnect culture and ecology. Buildings are reabsorbing knowledge embedded in local traditions while also becoming part of ecological systems. From vernacular flood-adaptation strategies to biomaterials that interact with microbial environments, architecture is increasingly taking the form of a system that grows in collaboration with its surroundings. Architects become stewards of living systems. What they design is no longer a single isolated building, but a node within a community.114

The architect's role shifts from designing fixed objects to mediating how people and place keep adjusting to each other, and buildings and environments along with them.

In a near future, built-arts practitioners will work the seam where inside and outside enter symbiotic exchange.

The static model gives way. The architect works inside a real-time generative world model that predicts the physics-aware next state of an environment. A designer sketches a

building in natural language, walks through it, revises as it generates. The model also talks back. She asks the structure a question in plain language, where the load runs, what a moved column costs the frame, when the cantilever stops holding, and the building answers in kind.

The architecture-specific bridge is forming now. Digital twins evolve with sensor data where a CAD file once sat static, and commercial world models generate editable, persistent environments. Capture closes the gap from the other side. Apple's SHARP, released in December 2025, regresses a metric three-dimensional Gaussian-splat scene from a single photograph in under a second, and by WWDC 2026 Apple was rebuilding Maps Flyover on the technique, lifting whole cities into navigable scenes from ordinary images. The building will exist as a generative simulation before, during, and after construction. Architects will design by inhabiting the model.

*What if a built-arts practitioner designed dæmons to augment creative capabilities for metabolic practices?*

### The Threshold

The heat stood over the site. Nomvula Dlamini reached the eastern edge of Cape Town at first light. By mid-morning the new laterite walls threw the sun back at her in slow waves. This was her first solo commission. A single-family house in Edge Commons, one of the relocation neighborhoods growing fast and dense at the city's edge, where the drowned coast and the dead farmland pushed their people inland.

The neighborhood was four hundred houses. Every one of them needed its neighbors to face the climate. A single roof gathered rain and current. One roof's catch fell short of a dry year. A single panel array stopped well before the winter ended. So the houses were bound together. A shared spine carried water, power, and load between them. The agentic design pattern laid out every plot and shared spine before an architect set hand to it. It drew the whole neighborhood as one connected body. The house Nomvula was shaping belonged to a family of five from a valley the rain abandoned.

Every house met its neighbors at a threshold (Figure 7.2). The line where a family's ground ended and the common ground began. That line was the architect's hardest decision. Set too far in, the house held nothing the family owned alone. A home became a unit. Set too far out, the family withheld water and shade the neighborhood needed to live through a dry year.

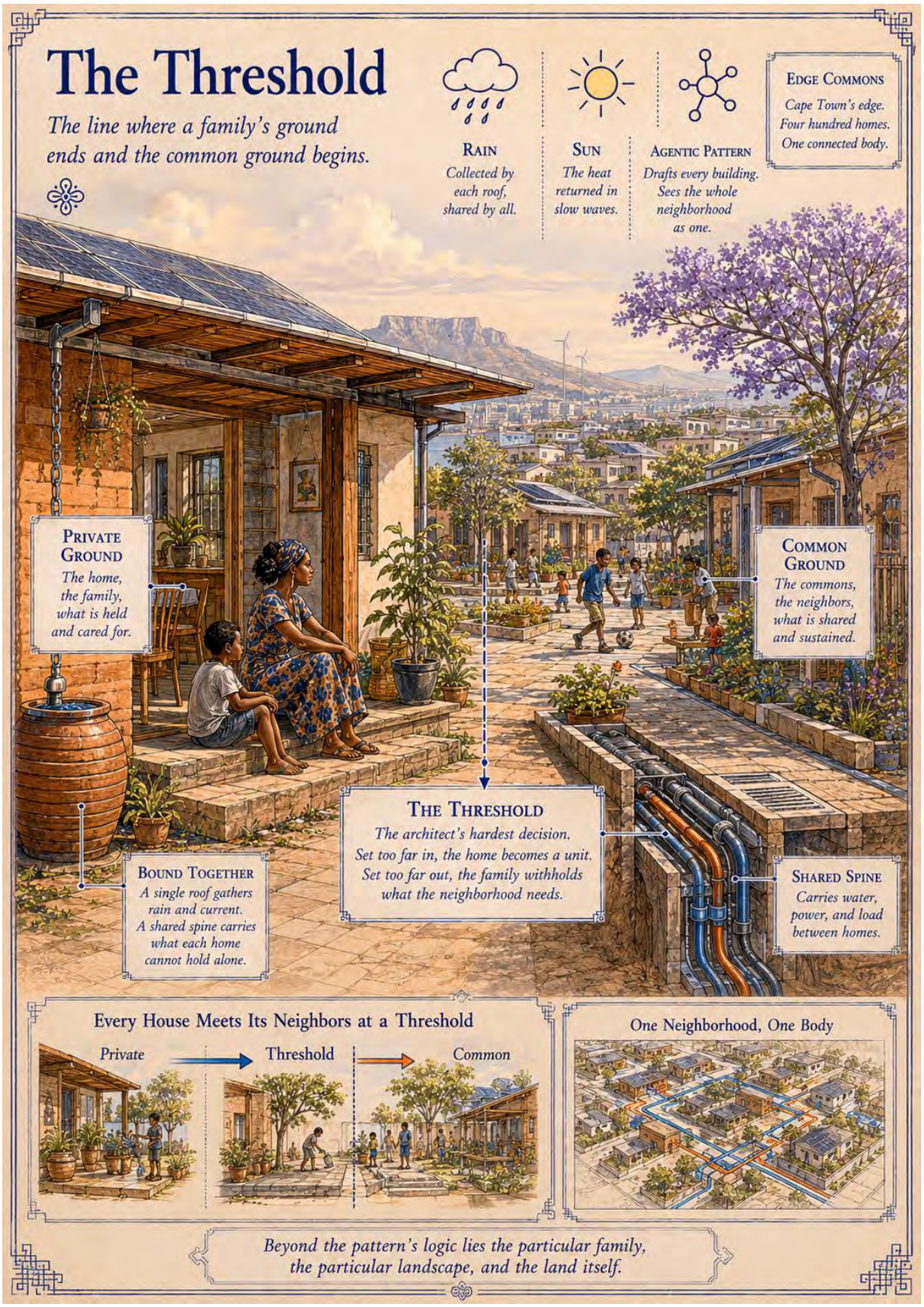
The Threshold
The line where a family's ground ends and the common ground begins.
Rain
Collected by each roof, shared by all.
Sun
The heat returned in slow waves.
Agentic Pattern
Drafts every building. Sees the whole neighborhood as one.
Edge Commons
Cape Town's edge. Four hundred homes. One connected body.
Private Ground
The home, the family, what is held and cared for.
Common Ground
The commons, the neighbors, what is shared and sustained.
The Threshold
The architect's hardest decision. Set too far in, the home becomes a unit. Set too far out, the family withholds what the neighborhood needs.
Bound Together
A single roof gathers rain and current. A shared spine carries what each home cannot hold alone.
Shared Spine
Carries water, power, and load between homes.
Every House Meets Its Neighbors at a Threshold
Private
Threshold
Common
One Neighborhood, One Body
Beyond the pattern's logic lies the particular family, the particular landscape, and the land itself.

**Figure 7.2 The Threshold. Source: authors.**

The agentic pattern drew every threshold the same way, as the most efficient line. The line served the relocation authority, which was paid by the house. It served the water utility, which metered by the liter. It left out what a threshold meant on this particular ground. Cape Town took homes from families by law and by water both. A family's own square of earth lay beyond any margin the pattern could spend.

Nomvula knew the commission would test her. Any departure from the agentic pattern would fall hardest on her. To make those judgments, she would need a sharpened aesthetic awareness, something capable of seeing a particular family and a particular landscape beyond the pattern's impartial logic. So she built *Mhlaba* (Figure 7.3), a dæmon she named for the land. Into it she placed the one instinct the agentic pattern could never learn: suspicion of a perfectly smooth answer.

**Figure 7.3 Mhlaba, Aesthetic Awareness dæmon. Source: authors.**

A threshold also survives the family that first crosses it. The line she drew would settle, for decades, whether a house was a home or a unit. Whether the neighborhood lived through its next drought. That reach would test her expressive influence, the capability to set a boundary so it still carries meaning long after her. For that she built *Makhulu*, giving it her grandmother's name and the long memory the systems flattened to an average.

**The Drawn Line**

The pattern set this family's threshold at the doorstep. One pace past the front door, the family's ground ended. The shared courtyard began. Nomvula walked it in the heat, heel to toe, the way her teachers taught her to walk a site before she trusted a screen. Four steps from the door to the line. In those four steps a person found nowhere to stand in the early cool that belonged to the family alone.

The family was on the plot that morning. The grandmother set stones around a vegetable bed, the way the valley people marked ground. She asked Nomvula for nothing. The son stood away from the stones. The valley held only distance, he said. Empty of neighbors. That emptiness killed it in the end. He wanted the house to open on the courtyard. He wanted to hear other people through the wall.

That morning Nomvula found two thresholds in one house. The grandmother's was one line. The son's was another.

She carried it to Mhlaba.

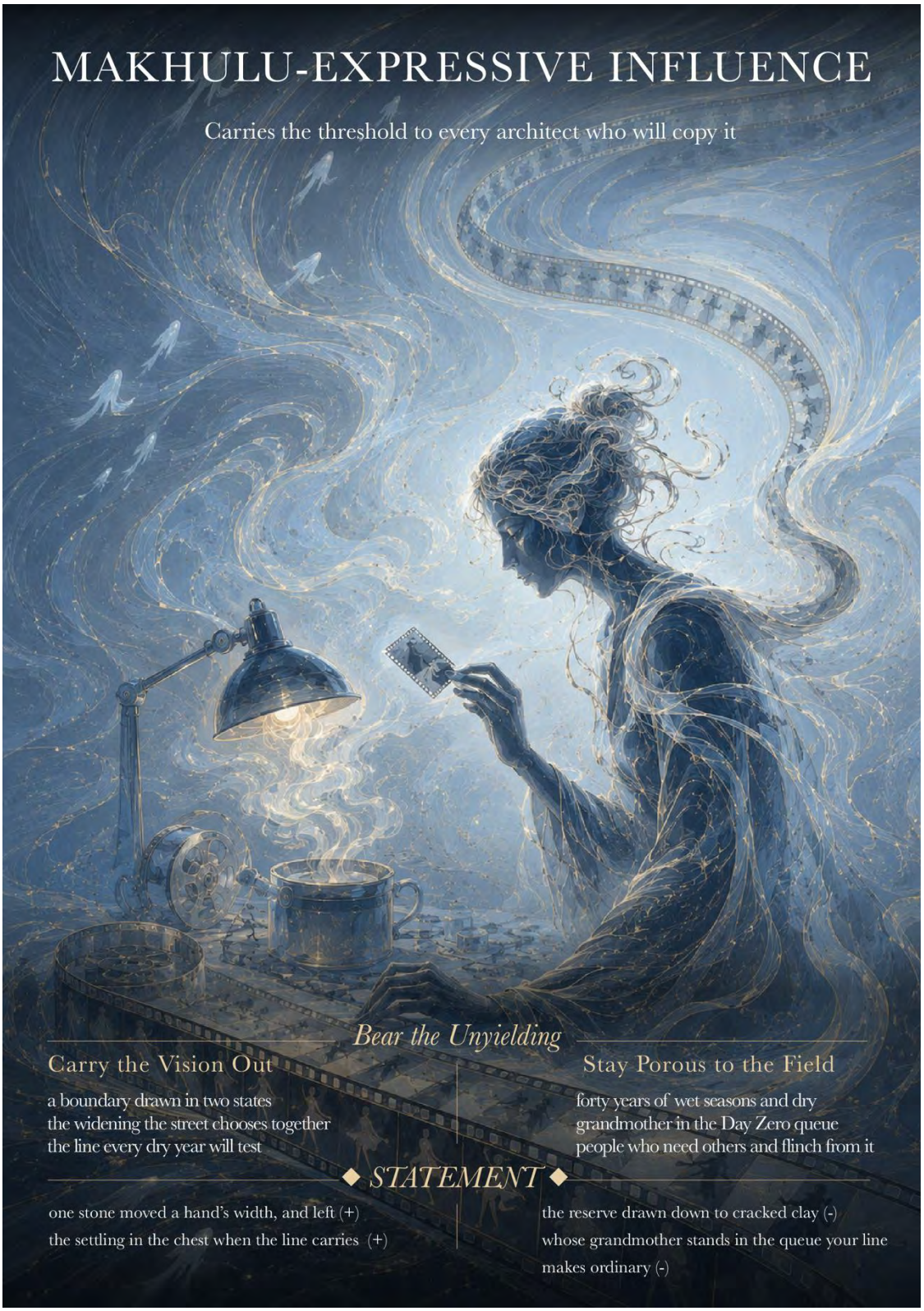


**Figure 7.4 Makhulu, Expressive Influence dæmon. Source: authors.**

“The pattern drew this family one threshold,” she said. “They live as two families inside one threshold. The grandmother wants ground, the way the valley taught her. The son wants the wall thin.”

*The agentic pattern drew a family that lives elsewhere. A doorstep answers neither of them. Draw the line the house needs, and find where the grandmother’s answer and the son’s answer become the same line.*

Nomvula redrew the threshold. She pulled the shared spine back. She gave the house a private ground, a gate, and a small rain-catch. The grandmother’s answer, set in laterite and water. The son’s answer she left open, a question still waiting. Mhlaba passed the variant to another agent, which ran it against the neighborhood model. The reply came back as a cost on the wider ground. The agentic pattern accepted the change. It returned the cost as a number. The number belonged to the relocation authority, beyond her power to decide.

**The Dry Year**

The agent’s findings came back as a place to walk, and Nomvula walked into them.

The world model finished the half-poured neighborhood around her. Four hundred houses in the white afternoon. The shared channels bright. The perovskite skins drinking the light. She walked down the street she drew. Then she let the model run the years forward and asked it for the drought the data placed within the decade.

It gave her the dry year. The channels showed cracked clay. The cisterns showed their floors. House by house, a neighborhood of generous private grounds drew the common reserve down to nothing. The world model told her, cleanly, that her line would fail.

She almost took the lesson and left. Yet the dry year around her was a quiet one. Cracked clay. Clean light. An absent smell. Her body, standing in the real heat with real sweat running into her collar, knew a real drought refused that tidiness. The world model rendered the failure and spared her the misery of it.

*The world model drew that drought, Mhlaba said. The agentic design pattern’s logic drew the doorstep. Walk it again, and distrust what feels comfortable.*

She made the model give her the drought it smoothed away. The standpipe queue at the corner of the shared courtyard. The ration cards. The tempers by mid-morning. An old man counting his liters twice. The count was the last thing left for him to control. The reserve still ran dry. Her line still failed. Now she could feel what the failure cost.

She walked back out into the heat she left earlier and carried it to Makhulu. “The reserve fails,” she said. “Every generous threshold drinks from it, and in the dry year the commons stands empty.”

*Your grandmother stood in a queue like that one the year they called Day Zero, half a morning in the sun and then longer, and carried home a jug she rationed by the cupful and a number inked on her wrist. You draw more than one threshold. You draw the line every architect after you will copy. Whose grandmother stands in the queue because your line became the ordinary one?*

Nomvula sat with the question. The threshold reached past one family. It reached every house. It reached every dry year to come. She drew it as though it ended at this family's gate.

**The Moving Threshold**

Nomvula stopped drawing the threshold as a line and drew it as a boundary that could move.

The house kept its ground and its rain-catch through every ordinary year. That was the grandmother's answer. It stood fixed. Yet the threshold held a second state. In a dry year the boundary widened. The family's catch fed the common reserve and drew from it in turn. The private ground became, for one hard season, partly shared. The wall between the house and the courtyard went thin. That was the son's answer.

The grandmother's answer and the son's answer were one line now, drawn in two states. The family kept its ground. The neighborhood kept its reserve. The line moved. A membrane where a wall once stood. Her hand finished it and her breath loosened in her chest, the day's heat going bearable on her skin.

Nomvula wrote the family and its neighbors into the rule that moved it. The widening became a thing the street chose together instead of a thing done to it. Makhulu set an agent to run the threshold across forty years of wet seasons and dry.

She showed the grandmother the moving line. The old woman listened, and was quiet for a while.

"And in the dry year," she said, "the strangers come over my stones."

*"In the dry year, your water keeps their children, and theirs keeps yours. The stones stay. The line moves."*

The grandmother moved one stone a hand's width and left it there. The son, when Nomvula showed him, asked only whether the wall would truly go thin, and when.

That evening Nomvula sat with both dæmons.

*You drew for a family that lived nowhere, Mhlaba said. You draw for the one that lives here.*

*And you thought a threshold was a line, Makhulu said. Say what you think now.*

"That it is a relationship," Nomvula said. "Drawn once, then kept. Every season, by people I will never meet, and by a family divided against itself."

*Then you have built them something they can keep while they argue.*

The world model still held the good version of the neighborhood beside the drought. Nomvula walked it. She trusted it less than at first light.

The relocation authority's reply sat in her queue. The moving threshold came back stamped nonstandard, its cost referred to a pricing review months out, carried as an open item

against her next commission. The line stood on the drawings. The number that would keep it standing still belonged to someone else.

What waited on her table was the next commission. A house three plots down. A family out of a flooded fishing town, where everything was always shared and every household held the rest. They would want their line far from where this family set theirs.

Their threshold will sit in another place altogether, Mhlaba said.

*And that family will divide against itself too, Makhulu said. All of them do. The neighborhood needs every house bound into one body, and every house holds people who want, first, to be themselves, and struggle to say so to one another. That gap is older than the drought. One good line leaves it open.*

*You came to draw a line, Mhlaba said. Say what you were really leaving open.*

Nomvula closed the world model. The son's question stayed with her, whether the wall would truly go thin, and when, and the authority's review held the answer's price. She thought of the gap between people who needed each other and still flinched from needing each other. Today she laid one stone on it, on ground still being argued.

### The Line That Keeps Moving

The threshold Nomvula drew must keep moving across forty years of seasons because dry years will alternate with the wet. Architectural practice now unfolds through the anticipation of change, the future brought into the design process as an active component of decision-making.

The built arts are reclaiming human agency. The tools will accelerate and the systems grow more capable. None of that changes one thing: a building that carries the trace of its maker's encounter with the ground holds a worth that a building carrying only what its optimizer counts will never earn.

The creators of the built environment, the architects, engineers, landscape designers, urban planners, and the tradespeople whose hands read what the instruments overlook, hold the capability to shape structures that stand past any algorithm's horizon. That capability is a professional discipline, earned through practice and binding now that the systems are gaining reach.

Where buildings behave like living systems, the architect can task her dæmons with holding the ground the agentic pattern would settle and walk away from at handover. Site stewardship rests on extended engagement, the long watch an architect keeps over a building through decades of weather and use. Place-reading rests on aesthetic awareness, the feel for a particular ground that built-arts thought has carried since Vitruvius and that Norberg-Schulz named *genius loci*. Programming asks for adaptive coherence, the work of turning a brief into a rule the building can keep through its inhabitants' lives. Representation asks for expressive influence, the drawing that delivers a designer's intent to the codes, builders, and occupants who inherit it. Mhlaba carried the second of these inside Nomvula's commission. Makhulu carried the fourth. An architect who designs her

own dæmons stakes her agency into the ground itself, claiming it before a fluent system can build over the spot.

*The pattern will fix every threshold at handover and call the job done. Which capability of yours would you station a dæmon to defend past that moment?*

Architect-owner contracts close shortly after substantial completion, framing the architect's service around a deliverable a court can recognize. A boundary designed to change state across forty seasons sits outside that frame. Movement is gathering elsewhere. The recast EU Energy Performance of Buildings Directive requires every Member State to introduce building renovation passport schemes, accessed through digital building logbooks that carry performance data across a building's life. The instrument exists ahead of the contracts that would price and insure it.

*Which line in a substantial-completion contract would you redraw so the building is allowed to keep changing once you have left it?*

In cities under climate pressure, housing is still read as an independent asset, its boundaries and its ownership clearly drawn. Yet peer-to-peer rooftop solar trading is already wearing that reading away, turning each household into a node in its neighborhood's resource network. Where energy and reserves are shared, a household's value rests as much on cooperation with its neighbors as on what it owns. The adaptable threshold designed by Nomvula embodies this shift: it links private interests with collective responsibility. Beyond market pricing, such a system runs on trust, open accounting, and a shared stake in collective survival.

*Stand at the edge of where you live. Which boundary would you open to a neighbor, the way Nomvula's threshold opened?*

Current accreditation standards already require architecture programs to incorporate ecological knowledge and environmental responsibility. Yet architecture schools continue to teach buildings as finished objects. Studio pedagogy has yet to absorb the principles of sustainable and adaptive design. Future architectural education will need to cultivate three new forms of practice: designing the rules by which buildings adapt to changing conditions, the governance that keeps such adaptation fair and transparent, and the agent-based systems that monitor and coordinate both.

*If you set this term's studio brief, how would you teach a student to design the rule by which a building adapts rather than the building itself?*

Generative design, multi-objective optimization, and continuous monitoring already exist within the discipline, held apart in separate phases of design, construction, and operation. The tangible arts, which know instability, maintenance, and continuous remaking, may show the shift sooner.

*What does the maker who mends an object already know about the life your building will lead after handover?*

## Chapter 8: The Tangible Arts: Object as Organism

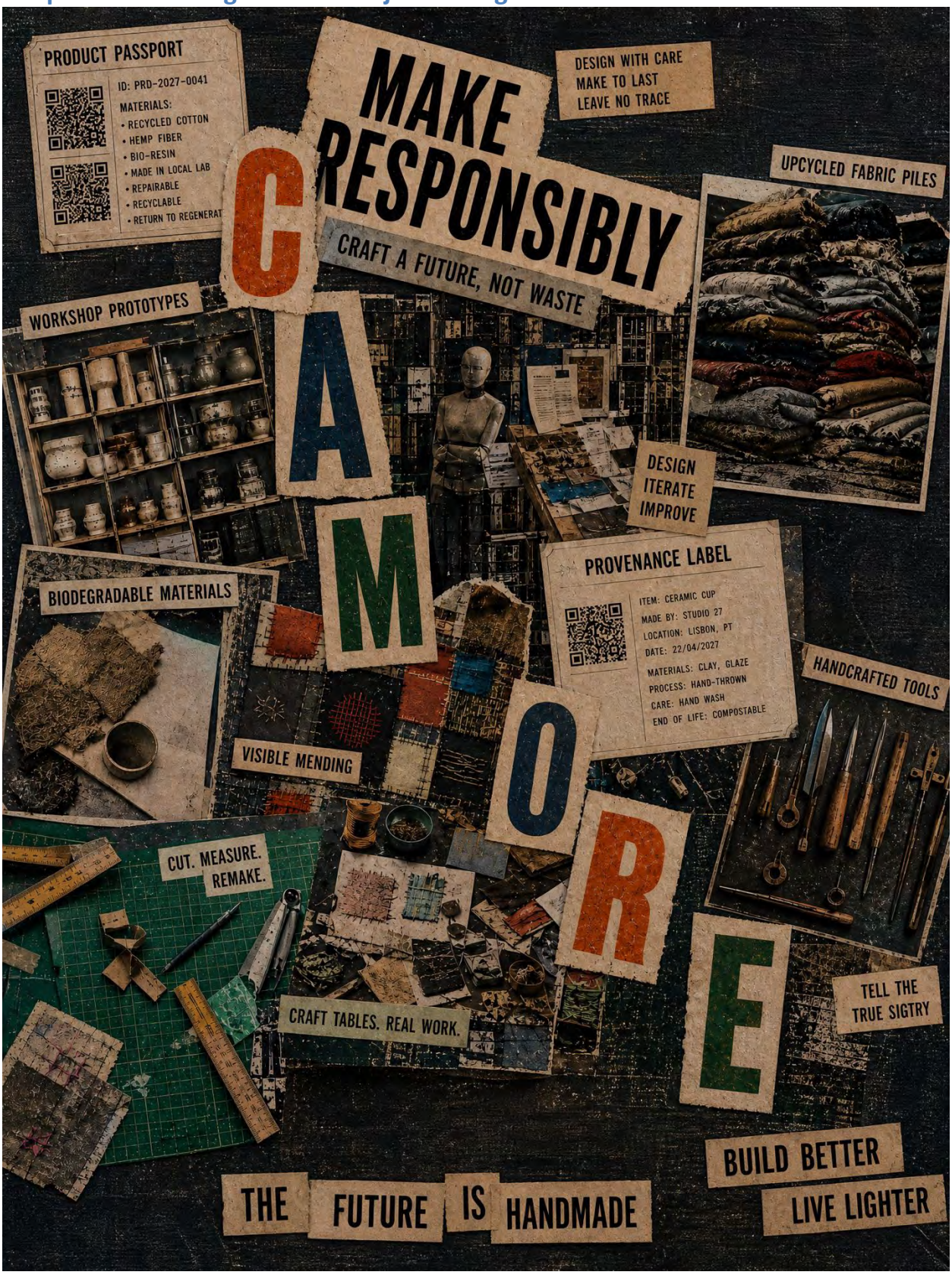

**Figure 8.1 The Tangible Arts cover poster. Source: authors.**

### When Objects Become Organisms

Objects occupy a different territory than the built environment (Figure 8.1). A chair supports a single body. A garment touches skin. A rehabilitation device meets the hand where it is most vulnerable.

Craft is the core of tangible arts. Its practitioners negotiate between human gestures and a material's resistance. The result of making is what David Pye calls a workmanship of risk, where the quality of the result depends on the judgment, dexterity, and care the maker exercises while working (Pye, 1968).

Where the built arts shape conditions across decades, the tangible arts make the artifacts bodies grip, wear, and use across a life. The scale shrinks to what the hand can hold. Objects invite interpretation through use, their meaning emerging through the interactions they sustain (Krippendorff, 2006).

Overproduction and overconsumption are the discipline's chronic condition. Material scarcity is closing the supply chains mass production assumed were infinite. The fashion industry generates approximately 92 million tonnes of textile waste annually (United Nations Environment Programme, 2025), a design failure operating at planetary scale. Only 6.9% of materials entering the global economy come from recycled or recovered sources (Circle Economy, 2025).

Algorithms only accelerate the pace. Agentic protocols can reorient it. The near future needs the tangible arts to make objects that last, that return to use, that carry an accountability their owners take on.

Provenance is becoming law where it was once a preference. The European Union's Ecodesign for Sustainable Products Regulation makes the object's biography law, with a central digital registry due in July 2026 and a textiles delegated act on the working plan behind it.

If the object and the data about the object become one entity, a made thing could act like a living organism. It would ship with an inseparable machine-readable biography, its fiber origin, repair history, disassembly map, and the name of its next custodian.

The maker's first creative act will become the design of that biography, a transparent account a future repairer can read and trust. As the record becomes part of the object, the material itself acquires a life of its own. A growing share of what the maker specifies will be grown, not mined or harvested. Hydefy's fungal leather grew from a Yellowstone-derived strain on sugarcane waste, reaching the Stella McCartney bag in 2025. Mycotech Lab in Bandung grows its Mylea leather on tropical agricultural waste, the peer-reviewed life-cycle assessment showing the craft can be sited where the substrate already grows.

A living material mastered upstream differs from one bought in as finished sheet stock. The practitioner's work expands into the choice of strain, scaffold, and substrate. It expands, too, into the conditions under which the material forms.

The workshop that makes these objects is learning too. Dexterous robots enter the small-batch atelier as apprentices. They take the repetitive and the dangerous. The practitioner holds judgment and finish.

A 2025 framework from National Taiwan University, DexMan, lets a robot acquire two-handed skills in simulation from ordinary video of human hands at work. The robot learns the workshop's gestures by watching the maker. The judgment stays in the hand. The machine takes the lay-up.

The object arrives last of all as one instance of a living family. Many things will exist first as a generative system. Only then will they appear as a realized piece, shaped to a particular body and use. Zellerfeld 3D-prints footwear from a buyer's foot scan.

The tangible arts can change the make-and-sell model. Practitioners can design the system, the rules by which a chair, a ring, or a shoe individuates to its owner. The design of the family belongs to the maker, where a platform might otherwise absorb it.

In a near future, tangible-arts practitioners will design less the single object than the conditions of its making. They will design the strain that grows it, the system that fits it to its owner, and the record it carries through every later hand.

The tangible-arts practitioner becomes a curator of the encounters that meet maker and user alike at the threshold of embodied excitation. There, the hand's reading of grain, weight, and warmth crosses into meaning. The maker holds the half-millimeter ridge a thumb will catch. She holds the way a material work-hardens under repeated stress. She holds the warmth a surface returns to a palm. The user lives that knowledge daily.

What if a tangible-arts practitioner raised dæmons to keep the maker's hand in the life of an object that now records, grows, and remakes itself?

### The Living Fit

For longer than she could remember, Valentina Salcido fitted limbs from the same two rooms off the Carrera Cincuenta in Medellín. She was a protesista. She made the sockets that hold a prosthetic limb to a living body. The socket is the one part of the device shaped new for each person, and shaped again whenever that person changes. Her workshop sat above the river in the old industrial quarter, where the city had turned its textile mills and metal shops into a corridor of small ateliers. The street outside ran toward one of the cable-car stations that carry the comuna hillsides up into the valley. Her neighbors built bicycle frames and sewing machines long before anyone there used the word designer.

The commission on her bench was a below-knee prosthesis for Diego, a boy of eight. His mother brought him in from a hillside barrio on the first cable car of the morning. Diego lost the leg at five.

The family arrived with a worn folder of clinic scans and a referral from a rehabilitation foundation in the city. The boy had outgrown two sockets already. Each had fit him well for a season and then turned into a thing that pinched and rubbed and finally had to be cut

away. The foundation wanted to know whether a smarter device could spare him the next two.

Valentina's sockets carried sensors now, where once she fit a static shell and sent it out blind. A thin array embedded in the liner read the pressure across the residual limb through the day. The socket mated to a recyclable structural frame and to a small processor that learned the limb's daily rhythm. The frame let the fit re-tune itself within a measured range, taking up the slack as a limb swelled in the heat and settled by evening. The foundation wanted to know whether a device that adjusted itself could keep pace with a body that never stopped changing.

The creative challenge sat in the gap between a measurement and a child. The agentic design pattern would draft a flawless socket for the leg the morning scan captured, and the adaptive system would tune that fit from then on without her. A growing leg is the one thing in the room that refuses to hold still long enough to be made flawless, and a self-tuning device is the one thing in the room that promises to fit it without her hands.

The commission would test two capabilities Valentina built unevenly across her career. The first was her extended engagement. A residual limb departs from the clean shape a scanner returns. It is bone and scar and soft tissue. It bears load differently at every point and changes across a single day. She built a dæmon to keep the hand in the work, *Raíz* (Figure 8.2), the deep root. A root holds its plant against weather the leaves never feel, working slow and unseen below the surface. That slow accumulation is what her own hands needed too.

**Figure 8.2 Raíz, the Extended Engagement dæmon. Source: authors.**

The second capability was harder. A socket fitted perfectly to Diego this morning would adapt to his daily swelling and still fail him within the year, because day-to-day tuning is not growth, and the boy would grow past the range any single device could take up. The commission would test her adaptive coherence. She would have to design less the socket than the conditions under which it keeps being remade. For that she built *Brote*, taking its name from the green shoot, the growth still in the act of arriving.

**The Perfect Socket**

The agentic pattern delivered its draft before Valentina finished her coffee. It was, by every measure the screen offered, a perfect socket. It matched the morning scan of Diego's limb to the limit of what the scan could resolve. It spread his weight across the broadest bearing surface. It came with a tuning policy already trained, a model that would read the embedded sensors and adjust the fit through the day. The pressure map glowed a uniform, reassuring green.

Valentina lifted the test socket from its cradle (Figure 8.3).

The Living Fit
A grown below-knee prosthesis and the generative family of sockets it belongs to.
Tlaquepaque
Jalisco, Mexico
TALLER • VIDA • OFICIO
Grown Socket Shell
Mycelium–Agave Composite
Residual Limb
The person who lives here.
Robotic Apprentice
Five-fingered hand lays up the living composite slowly.
Load-Sensing Digital Twin
Continuously mapped through the wall.
Body Scanner
3D form, pressure, and movement capture.
Graded Density
Soft / open-cell where the fibula runs shallow.
Dense / closed-cell where muscle bears load.
Agave-Bagasse Substrate
Milled, crashed agave fibers.
Cultivated Ganoderma Strain
Mycelial culture for growth.
Machine-Readable Passport
Diassembly & Regrowth Map
ID: MF-27-A
SIZE: 03
GROWTH MAP
VER: 2.1
Size 01 4–5 yrs
Size 02 6–7 yrs
Size 03 8–9 yrs
Size 04 10–11 yrs
Size 05 12–13 yrs
Size 06 14–15 yrs
The Family — Six sockets, graded for a child's growth. Same body. Same passport. One living line.
This Year
Fitted. Scanned. Living.
Socket Size 03
Fitted to the limb as scanned.
Closed Loop
Next Year
Outgrown. Returned. Regrown.
Outgrown socket milled and returned to substrate.
Composted. Mycelium remains. Cycle renewed.
Next socket grows larger in the mold. The family continues.
Marks in Every Socket
Three marks run through all surfaces.
Writer's Hand
(Crafted by People)
Model's Completion
(Guided by Grammar)
Carried Line of Translation
(Carried Forward)
Passport Carries The Story
Identity. Provenance. Care. Regrowth.

**Figure 8.3 The Living Fit. Source: authors.**

The machine did the rough work now. It milled the shell and seated the sensor array and the small actuators that let the fit breathe with the limb. The work once cost her a full day and both wrists. She turned the shell in her hands. It held a faint warmth from the milling and a fine ridged grain along the inner wall.

She brought the test socket out to where Diego sat, swinging his good leg against the bench. She knelt and fitted it to the limb herself, both hands reading as they always did. Her palms ran the seam where the socket met skin.

The map glowed green and whole, and for a breath she let it stand in for the reading her own hands carried. Then, two fingers below the head of the fibula, her right thumb caught a wrongness the green map had missed. The bone sat close under the skin at that point. It sat closer than the average the scan smoothed into its model.

The perfect socket pressed the bone with the same even firmness it pressed the soft muscle a centimeter away. The auto-fit had averaged the ridge away, tuning for the broad surface and missing the one point that would break the skin. A grown man might have borne it. Her thumb held the spot and kept holding. The holding was the unreachable.

She took the trouble to Raíz.

“The pattern made a socket for a leg shaped like the average of all legs,” she said. “This leg goes its own way. The fibula sits proud here. It needs meeting softer than the screen knows.”

*The screen tunes for the broad surface, because that is where its sensors read the cleanest signal. You are reading a point the array smooths over. The system fits the average of the limb, and a limb can be met at more than one firmness.*

“It tunes one fit,” Valentina said. “One policy, one firmness, spread even.”

*It spreads even because you let the pattern set the policy as one. Relieve the wall where the bone runs shallow and hold it firm under the muscle. Set the gradient with your thumb. Let the model tune within what your thumb decides, not across it.*

Valentina understood. The socket could carry more than one firmness. It could meet a limb whose own load varied along its length. She could grade it firm and load-bearing under the broad muscle, relieved where the fibula ran close, and bound the adaptive policy to that gradient so the auto-fit could never average it back out. The firmness would follow the leg instead of averaging it. The judgment was the gradient. The gradient was hers.

Raíz set a protocol running, an agent assigned to track the pressure along the limb through a full day of Diego’s wear. The gradient would rest on how the boy actually moved through a day, where a single still scan caught one instant only. That night Valentina set the relief by hand, point by point along the wall, and let the machine cut the graded shell and constrain the tuning model to it.

**The Growing Year**

The graded socket fit. Valentina watched Diego cross her courtyard a week later, and the limp she braced herself to see was gone. He chased a stray cat twice around the guayacán. He complained only that the leg ticked faintly when it adjusted itself, and that the ticking made his sister laugh. The sensors read the limb through the day, the actuators took up the swelling, and the fit she had set held him.

*The model produced an average, Raíz said. Your thumb found the exception. Trust it.*

Then the agent came back with the day-of-wear data. Valentina read the report at her bench in the afternoon heat. Through the day the limb did the small things every limb did. It swelled in the warmth. It settled by evening. The socket had tuned itself to each, exactly as she had bound it to.

The agent set the new scan against the clinic's first scan from three years before. Over that span the small daily changes resolved into one long line.

Valentina's eye followed the line forward. It ran into an autumn where the socket would press wrong. It ran on into a spring when the socket would have to be cut away and thrown out. Every socket she ever made for a child ended that way.

Across the day the limb varied. Across the years it grew.

She knew that curve before the agent drew it. Every protesista knows it. What sat new on her bench was a device that tuned itself within a range and a passport that could carry the rule, the first tools of her career that could answer the curve. She fitted the perfect socket that first morning and trusted the adaptive system to carry it forward, as if the growing were someone else's problem, the clinic's, the mother's, the landfill's. The shame landed there, in a fit so exact and a policy so confident, both aimed at one morning.

A device that tunes a body that closely keeps nothing back for the body to grow into. The self-tuning range took up a day. It could not take up a year. She built for the boy on the bench. The boy who would cross the courtyard a year from now stayed off her design.

She brought it to Brote. (Figure 8.4)

**Figure 8.4 Brote, the Adaptive Coherence dæmon. Source: authors.**

"I graded it well," she said. "That is the failure. A close fit on a growing child is a clock counting down to the landfill."

*Then stop making the last socket he will ever need. You keep designing an object. Diego is a body on a curve, and the curve runs past your bench.*

"A socket that keeps pace with a growing child sits outside my craft. The device tunes itself within a range. Past that range it is finished, and another has to be made," Valentina said.

*One socket is finished. A family of them stays open. You are making the rule that makes the next thing, Valentina.*

Valentina sat with that. The socket carried, like every object now, a passport. The passport was a machine-readable account of its own making. She thought of it as a record. Brote was asking her to think of it as a seed.

The passport could carry the gradient rule. It could carry the sensor specification, the tuning policy, and the load history the limb kept sending back. It could carry what this socket was, alongside the rest. Then the socket became the first member of a family. Each later member could be cut from the updated scan, by the same rule, closer to home.

Brote dispatched an agent to model the boy's growth across the next six years. The agent would test whether the gradient rule held as the limb changed.

**The Living Family**

Valentina stopped designing Diego's socket and designed the family it belonged to.

The rule came first. The gradient she set went into the digital passport as a relation. Relieved by a measured amount where a bone ran shallow. Firm across the broad bearing muscle. The proportions held as the limb scaled up, and the tuning policy stayed bound to them so the auto-fit could only adjust within the gradient, never across it.

The passport carried the sensor specification and the tuning policy. The frame itself was recyclable, the electronics recoverable. It held the limb's own returning load history. Any certified clinic with a scanner and a mill could cut the next socket in the family to the passport's controlled rule and load the policy that came with it. The passport also held the dossier a health authority still asks of any medical device. The rehabilitation foundation had a fabrication lab two cable-car stops from Diego's barrio and now a reason to keep his file open. Diego would stay close to home. The socket would be made a short ride from his house, individuated to the leg he had by then.

The old socket would return for recovery instead of the landfill. The frame went back to stock and the sensor array and processor came out for reuse in the next shell. What Diego outgrew gave up its working parts to the next member of the family.

She showed Diego's mother the socket family on the screen. Six sockets stepped forward into a future, each a little larger than the one before.

The mother looked a long while. "And when he stops growing," she said, "the family stops."

"When he stops growing, the last one lasts," Valentina said. "Until then, none of them is the last."

That evening Valentina sat with both dæmons in the cooling courtyard, the jacaranda dropping its last flowers on the bench.

You knelt that first morning proud of a perfect thing, Raíz said. Your thumb saved you from the first error. What saved you from the second?

"Something else," Valentina said. "My thumb reads the leg that is here. The leg that is coming lies beyond it."

*Then say what you learned.*

"That the fit was never the object," she said slowly. "The fit was a relation between a body and a material, and the body keeps moving. I was taught to make a socket. I think now the work is to make the rule the socket keeps obeying after I have set it down."

*And the rule is finished? Brote said.*

"No. The rule is the part that has to stay alive."

*The object was never finished, she saw now. The rule was the part that had to stay alive.*

What waited on her bench was the next commission. A woman of seventy. Her limb would thin and slacken with age instead of growing. Her family of sockets would step the other way.

*Her curve runs down where the boy's ran up, Brote said. The rule will move with her. Start her family from Diego's, and the first shell can be growing before she reaches your bench.*

For a moment Valentina let it run, a family drafted from a file, a gradient scaled from an average, a policy tuning the fit from the first day, her thumb arriving after the shell. Then she set the screen down. The rule was a tool, and a tool can overreach. The woman's leg would speak first.

*Listen to the thumb before the rule, Raíz said. The rule caught the second error. The thumb caught the first. Send the rule out alone and the next first error travels with it.*

*Hold every fitting to this bench and the family dies in your two rooms, Brote said. The boy will outgrow the bus ride before he outgrows the socket.*

*You came to your bench to fit a shell, Raíz said. Say what you were really keeping.*

The two pulled against each other and stayed pulled. Valentina let them. "The rule travels," she said. "The first fitting of every new shell waits for a thumb. Hers starts here, with my hands on the leg."

### Past the Final Sale

The tangible arts reclaim human agency. The robots will grow more dexterous, the models more detailed, the systems more efficient. An object that carries the trace of its maker's

encounter with the body it serves will outlast an object shaped only by what its yield measures.

The creators of tangible things, the product designers, ceramicists, textile artists, glassblowers, jewelers, and materials engineers, cultivate through long practice a distinctive capability: the ability to shape objects that hold beyond the predictive horizon of algorithms. As makers hand more of the manufacturing to autonomous machines, the importance of human judgment becomes more pronounced.

The living family Valentina designed has to keep regrowing across the years a boy takes to finish growing. That long horizon reaches the maker today, eroding the envelope between the once-sold thing and the thing that returns.

When every object ships with a biography, the maker can let her dæmons keep at the bench the work the agents would otherwise lift away. Extended engagement deepens the workmanship of risk, the long stake David Pye named for a hand working under the judgment of a material that answers back (Pye, 1968). Aesthetic awareness sharpens mingei, the eye Yanagi Sōetsu named for fitness in the everyday made thing (Yanagi, 1972). Adaptive coherence lives in kintsugi, the craft of mending so the repair carries forward as part of the object's own life. Expressive influence carries the maker's mark, the silver hallmark and the potter's seal that signed a thing so the next hand could read how it was made. Raíz carried the first of these inside Valentina's commission. Brote carried the third. A maker who designs her own dæmons keeps her grip on the work, since the grip is the first thing a fluent machine pries loose.

*A single morning's scan, and the pattern has the socket fitted. Which skill of your hands at the bench would you charge a dæmon to keep alive against that?*

Product warranties close at first delivery, framing the maker's obligation around an object a court can recognize. A family of objects regrowing across years of changing bodies sits outside that frame. Movement is gathering. The revised EU Waste Framework Directive harmonizes extended producer responsibility for textiles across the bloc, eco-modulated fees pricing durability and recyclability into the maker's continuing stake (Council of the European Union, 2025). The instrument exists ahead of the warranty terms that would carry the stake into later life. Valentina's passport already does that work.

*Reach for the warranty you would normally sign. Which term in it would you rewrite so the object is free to keep changing after the sale?*

A consumer buying a phone, a chair, or a coat still expects a sealed transaction. Pilots show another posture. Fairphone builds modular phones a buyer repairs at home and keeps in service for years. The customer becomes a custodian of the component and its manual. Diego's family enters the same role through the passport that travels with the socket. The transaction becomes a long relationship the object holds for maker and consumer both.

*Of everything you own, which one thing would you vow to keep in use long past its first failure?*

Today, many design studio briefs still treat the prototype presented at the final critique as the endpoint of the design process. In contrast, the Material Futures MA program at Central Saint Martins positions materiality as the starting point, encouraging practitioners to consider an object's long-term use, maintenance, and regeneration. Future design education may therefore place greater emphasis on understanding how objects evolve through use, how materials participate in cycles of renewal, and how design can move beyond one-off outcomes toward the cultivation of lasting rules and relationships.

*Hand a student this term's brief. What would you change in it so the object never stops arriving, the way Valentina's family of shells keeps regrowing?*

The tangible arts concern how objects persist through time, making maintenance, repair, and reuse integral to the creative process itself. The moving arts produce the opposite kind of work, one that exists only in the moment of performance and disappears when the body stops. They offer creators an alternative perspective, one that foregrounds ephemerality and flow.

*What could the choreographer, whose work survives only in the bodies that danced it, teach a maker whose objects now outlive her bench?*

## Chapter 9: The Moving Arts: Fluidity of Time

**Figure 9.1 The Moving Arts cover poster. Source: authors.**

In a near future, moving images will compose themselves anew for each viewer, loosening the single fixed sequence a shared audience once received alike.

### When Perception Is Constantly Reframed

The moving arts make work that exists only in time (Figure 9.1). Film, dance, performance, and animation give a duration the hand passes through and then releases, a passage the audience must live inside.

The discipline governs the order of that passage. The maker chose one sequence of images and sounds and set its tempo. Every audience received the same order in the same span. The director Andrei Tarkovsky called the craft a sculpting in time. As he put it, "what a person normally goes to the cinema for is time: for time lost or spent or not yet had." (Tarkovsky, 1987).

The moving image carries another authority. André Bazin grounded the image's authority in its causal contact with the world, the bond later theorists named indexical (Bazin, 1967). Generative production severs that bond. World models build plausible continuations from recorded footage, extending real scenes outside the camera's reach.

Pressure on the fixed order comes also from how attention is sorted. Engagement-optimized feeds deliver an individualized stream to every account, fragmenting the once-shared audience. A discipline built to give every audience one passage through time will be asked to compose a passage for each viewer.

The self-composing story emerges from this shift. A work becomes a set of visual materials and generative rules. From these a system assembles a different version of the narrative for each viewer in real time. Under these conditions, the practitioner's role shifts from orchestrating a single narrative sequence to curating the anchor frames through which every version of the story must pass.

Fable Studio's Showrunner generates remixable animated episodes from a text prompt, holding characters consistent across scenes. China's micro-short-drama economy overtook the domestic box office on short vertical episodes and forkable story nodes tuned to each viewer's taps.115

One earlier interactive-special format retired its branching library, the limit of branch trees built by hand.116 Model-generated composition computes the path against the footage instead of pre-building each branch. A story composed this way needs more footage than any single shoot can hold. The director records a short indexical core, a body, a room, a take. Real-time generative-video models assemble the surrounding scene and its alternative continuations.

The shift is most visible in China. Kling brought generation and editing into a single model with per-shot storyboard control. At Versatile Media's Deqing County base, directors and actors perform inside projected digital environments, with scripting, camera, and

performance adjusted in real time.117 The model extends the scene. The director decides which frames the camera must witness, and which it may compose.

The composition needs performers too, ones for scenes built past the shoot. A performer's likeness, voice, and gait will exist as a licensable digital twin acting in any branch the story assembles, in the performer's absence or after her death.

OpenAI's Sora 2 places a verified performer's likeness into generated scenes under creator-controlled permissions. A recent feature completed the role of a late actor recreated from archival material with his estate's consent.118 A wholly synthetic performer drew a union condemnation.119 Labor frameworks address the body replicated past consent. The audience reshaping the director's sequence is a second severance, still legally undefined.

For all that composing, a story that builds itself holds together only where some frames stay fixed. The director will designate the anchor frames a viewer's path must pass through, curating which frames anchor the story and which the system may compose around. Each frame will carry a provenance layer declaring what a camera recorded and what a model composed.

The C2PA standard signs capture metadata across major cameras. China's synthetic-content labeling measures require on-screen labels and embedded watermarks across the largest platforms.120 California's AB 2602 and AB 1836 bar the unconsented digital replication of a performer's work and give a deceased estate standing to sue.121

A generation of practitioners is already using these tools to deepen embodied practice. Jiajian Min and Anna Borou Yu's Embodied Ink translates the motion of Chinese calligraphy into generative visuals and sound through motion capture and biometric feedback. The performer becomes a living brush whose hidden kinetics are made visible to the audience (Min & Yu, 2024).

In a near future, moving-arts practitioners will direct less the single sequence. They will direct the anchor frames from which each viewer's story composes itself.

The moving-arts practitioner's responsibility becomes the curation of experiences that engage the embodied excitation of director and viewer alike. That felt signal is the threshold where a body's reading of motion, tempo, and held effort becomes meaning. The director holds the tremor a take caught and the silence an edit keeps where it carries weight. The viewer lives that knowledge as the pull of a sequence moving at the exact pace the moment asks.

What if a moving-arts practitioner composed dæmons to keep the director's hand in a story that now assembles itself anew for every viewer?

### The Anchor Set

Su Lan shot and cut micro-dramas for nine years, building each ninety seconds one take against the next. She worked from two rooms above a tea shop in Xinshi, an old water town in Deqing County. Her one window looked down on a hébùtóu, the worn steps where the lane met the canal. The town drew its water there for six hundred years. She named her

first successful series for that view. *The Stone Landing* ran sixty vertical episodes, each ninety seconds long. It followed Pan Suqin, whom the town called Auntie Pan. The series tracked her through the ordinary mornings of the wonton stall she kept at the foot of the steps.

Auntie Pan came to the camera fresh, and that was what the audience came for. She was a widow of eighty-one who folded wontons at the landing from girlhood. When the camera followed her, she simply folded. The folding was the performance. The series found nine million viewers in a season. The platform that carried it, a micro-drama service built on forkable episodes, came back to Su Lan with a commission.

They wanted *The Stone Landing* rebuilt to assemble itself anew each play. A film that would assemble a fresh path through Auntie Pan's days each time it played. A real-time model would compose new episodes from a set of frames and rules. Pan Suqin's stroke that spring slowed the left side of her body. The new scenes would be carried by her digital twin, trained on the season and on a capture session her son arranged before the stroke. He signed the licensing contract.

The rebuilt film reached Su Lan already drafted by the agentic design pattern (Figure 9.2). The pattern would compose Auntie Pan an endless, faultless performance. The whole of her hold on an audience was that she simply lived in front of the lens.

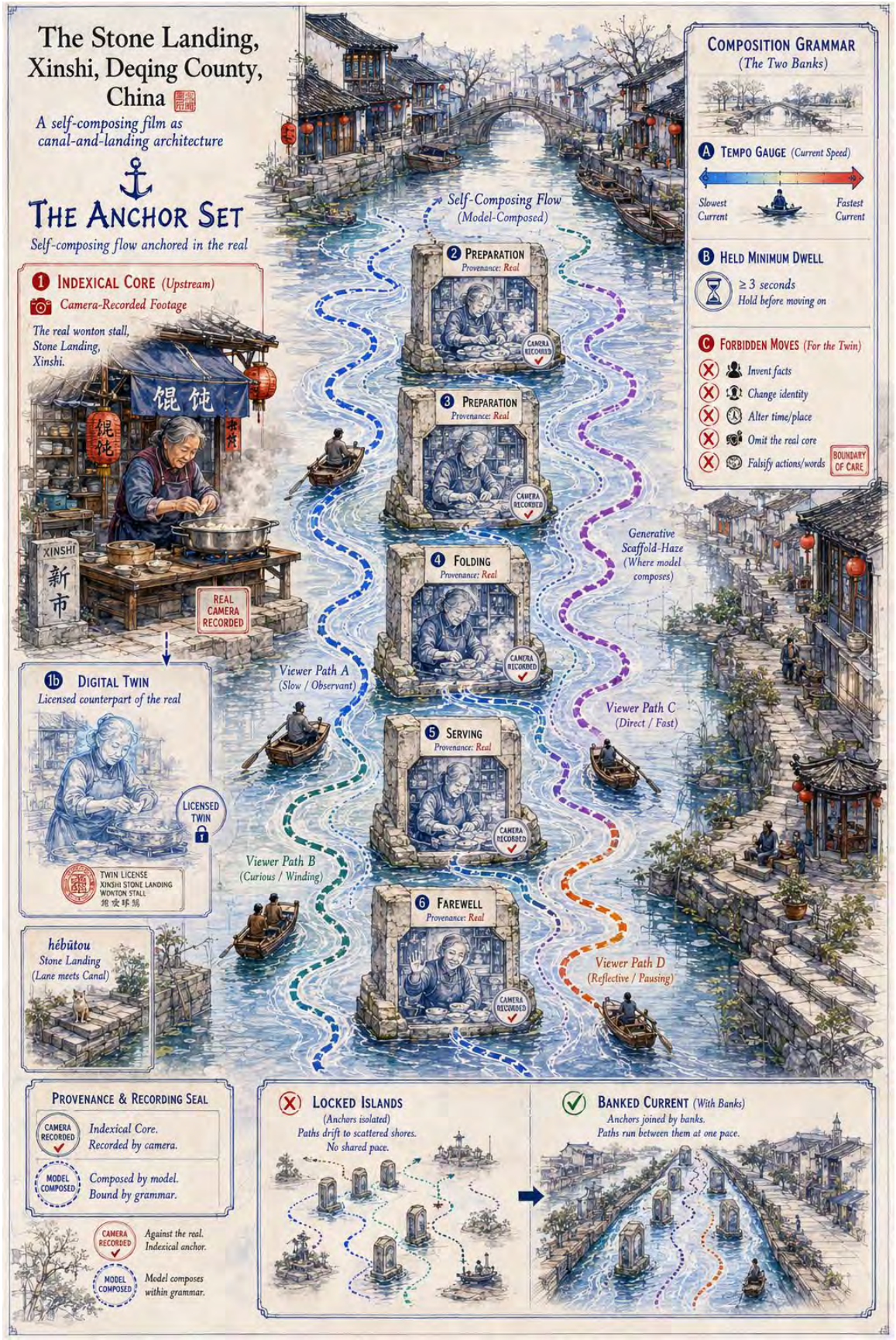
The Stone Landing, Xinshi, Deqing County, China
A self-composing film as canal-and-landing architecture
THE ANCHOR SET
Self-composing flow anchored in the real
1 INDEXICAL CORE (Upstream)
Camera-Recorded Footage
The real wonton stall, Stone Landing, Xinshi.
馄饨
XINSHI 新市
REAL CAMERA RECORDED
1b DIGITAL TWIN
Licensed counterpart of the real
LICENSED TWIN
TWIN LICENSE
XINSHI STONE LANDING WONTON STALL
hébùtou
Stone Landing
(Lane meets Canal)
Self-Composing Flow
(Model-Composed)
2 PREPARATION
Provenance: Real
3 PREPARATION
Provenance: Real
4 FOLDING
Provenance: Real
5 SERVING
Provenance: Real
6 FAREWELL
Provenance: Real
CAMERA RECORDED
Generative Scaffold-Haze
(Where model composes)
Viewer Path A
(Slow / Observant)
Viewer Path B
(Curious / Winding)
Viewer Path C
(Direct / Fast)
Viewer Path D
(Reflective / Pausing)
COMPOSITION GRAMMAR
(The Two Banks)
A TEMPO GAUGE (Current Speed)
Slowest Current
Fastest Current
B HELD MINIMUM DWELL
≥ 3 seconds
Hold before moving on
C FORBIDDEN MOVES (For the Twin)
Invent facts
Change identity
Alter time/place
Omit the real core
Falsify actions/words
BOUNDARY OF CARE
PROVENANCE & RECORDING SEAL
CAMERA RECORDED
Indexical Core. Recorded by camera.
MODEL COMPOSED
Composed by model. Bound by grammar.
CAMERA RECORDED
Against the real. Indexical anchor.
MODEL COMPOSED
Model composes within grammar.
LOCKED ISLANDS
(Anchors isolated)
Paths drift to scattered shores. No shared pace.
BANKED CURRENT (With Banks)
Anchors joined by banks. Paths run between them at one pace.

**Figure 9.2 The Stone Landing. Source: authors.**

The commission would test two of Su Lan's capabilities. The first was her aesthetic awareness. It is the reading that catches what a recorded body is doing under a performance offered by accident. The twin was built to smooth that figure into an average. She would have to hold what the average erased. She built a dæmon for it. *Bingxi* (Figure 9.3), the held breath.

**Figure 9.3 Bingxi, Aesthetic Awareness dæmon. Source: authors.**

The second capability was harder. A story that recomposes arrives somewhere different for every viewer. Su Lan would watch a small share of where it arrived. The commission would test her adaptive coherence. It would test her capability to direct a film toward destinations beyond her viewing. For that she built *Liuxiang* and named it for the set of a canal current. The lay of the water decides where a thing dropped upstream will come ashore.

**The Perfect Episode**

The agentic pattern delivered the self-composing film's draft with the morning mist still on the canal. Su Lan watched the first episode.

The agent pattern built it from a prompt and the season's recorded footage. Auntie Pan at the stall on a winter morning. Steam off the pot. The first customer of the day. All of it was composed. The model built the morning whole. It put the twin at the center of it, lifting the wide bamboo strainer from the boiling pot.

It was a faultless ninety seconds. The light was the flat pearl light of a Xinshi winter. The twin's face held the patience of a woman who did this ten thousand times.

Su Lan set a recorded take beside it.

The real Auntie Pan lifted the strainer with a half-turn of the wrist she taught herself. A small outward roll that took the weight off the knuckle of her first finger. The cold canal morning stiffened the joint. The lift came a beat late, the strainer rising slow before it cleared the water. The twin lifted clean. The wrist stayed smooth, the knuckle spared nothing, and the strainer rose on time.

The lift was clean, and watching on the small screen she nearly let it pass. Then her eye caught the missing half-turn. The wrongness came first, the meaning a beat later, and that interval was the unfamiliar. The twin was lifting a strainer that weighed nothing. On a morning that was not cold. With a hand still unhurt.

Su Lan carried it to Bingxi. "The model composed her a perfect morning," she said. "The lift is wrong. Her hand spares a bad knuckle. The twin's hand lifts a free one."

*The model learned the lift from ten thousand frames and kept what they shared. The half-turn was in maybe forty of them. It read as noise, and the model smoothed it out. Teach the frames the knuckle. Find the takes where the knuckle shows, and make the film keep them.*

"It composes a new film for every viewer. Shooting every morning is past me," Su Lan said.

*Right. You can decide which frames the model must respect. The camera caught her real hand four hundred times this season. Pick the takes where the morning was cold and the lift came late. Lock them. Make every version of the film pass through her actual body.*

Su Lan understood. She could designate a set of anchor frames from the season's recorded footage, allowing the model to freely compose the scenes before and after them. Each anchor frame would carry a provenance layer, the label the law now required. It declared which seconds a camera recorded and which a model composed.

Bingxi set a protocol running. An agent was told to comb the season for the takes where Auntie Pan's hand showed its history most plainly. That evening Su Lan began to choose.

**The Hundred Paths**

The agent's anchor frames came back, and they held.

Su Lan composed a viewer path through them on the second night. She watched it on the phone. The path moved through her anchor frames. At each one the real Auntie Pan surfaced: the favored knuckle, the late lift, the breath she took before she called a price. Between the anchors the model composed freely. The composition was good. Su Lan let herself believe the work was nearly done.

Then she composed a second path, the one a different viewer would receive, and a third, and a tenth.

Each path passed through the same anchor frames. Between them, each went somewhere else. One viewer's model read the taps of a gentle watcher. It composed an Auntie Pan tender, who lingered, who smiled the film soft. Another's came back brisk and short, a hard woman in a cold town.

Su Lan tracked the tender path forward and let it run. It drifted with intent. It was heading somewhere. The viewer's every tap pulled it toward an Auntie Pan who asked the world to leave her alone. The slow pace carried that viewer to an ending past Su Lan's hand and past her viewing. A hundred viewers were on a hundred such paths tonight. The anchor frames pinned the body still. They left open the question of where the body was going.

She brought it to Liuxiang (Figure 9.4)

**Figure 9.4 Liuxiang, Adaptive Coherence dæmon. Source: authors.**

“The anchors work,” she said. “Her hand is real in every version. And yet the film walks a separate path for each one. Most of those somewheres stay beyond my seeing.”

*You locked the frames. You left the current between them open. A viewer drops into the film upstream. The model carries him down, tap by tap. You have decided he must touch four stones on the way. The speed of the water and the bank it sets him on remain for you to decide. The old interactive specials failed because every branch was built by hand. Yours fails the other way. The branches build themselves, and they drift past any shape you set.*

“A hundred paths is past my watching, and so is whatever travels past me,” Su Lan said.

*Then direct what lies beyond your sight.* You know how Auntie Pan’s mornings unfold. You have the whole season. The rhythm was hers, shaped by the cold, and it held whether the customer was kind or rude. Give that rhythm to the current. Give it banks. Viewers will still choose their own paths, and they will be moving within a river that always runs downstream. Let the banks carry the rhythm of her real life. Then, even when a path arrives somewhere you never see, it still moves with her sense of time. The through-line you set holds where your eye can no longer follow.

Liuxiang dispatched an agent. It would compose a hundred viewer paths overnight and report where each one came ashore.

**The Banked Current**

The hundred paths came back in the morning. Su Lan read them and found that she built a film that kept Auntie Pan’s body and lost the life inside it.

She stopped treating the anchor frames as locked islands and designed them as a set.

The anchors stayed, the real takes with their provenance layers. Su Lan saw that a few preserved moments still fell short of a preserved person. So she began to set rules for the generative sequences that unfolded between those frames. She was no longer deciding only which points every viewer must pass through. She was also shaping the rhythm that connected them.

She derived that rhythm from the season itself. She measured the time a real winter morning took to move from the first customer to the tenth and translated it into a temporal range the model was required to follow. The generated sequences could move only so fast and linger only so long. Every pause, hesitation, and turn was bound to the pace of Auntie Pan’s actual life. Beside each anchor frame, she also specified the bodily traits the twin was required to preserve: the half-turn that protected her injured knuckle, and the distance she habitually kept from certain customers.

Viewers would still receive different versions of the story, and the model would still compose a different path for each of them. The paths were no longer free to drift without direction. Wherever they ultimately led, they all shared one sense of time, the sense that belonged to Auntie Pan, the rhythm of a cold winter morning. Different paths would still arrive at different destinations, and all of them would travel the same river.

She took a tablet down the lane to show Pan Suqin.

“The frames where it is your hand, the film says so. There is a mark on it.” Su Lan showed her the provenance layer, the small steady label. “The rest the machine made. It says that too.”

Auntie Pan watched a composed path on the small screen, her twin at the stall, an invented morning. “That is my hand,” Auntie Pan said. She watched the twin lift the strainer, the late lift, the spared knuckle held in. Then she watched it turn to the next customer, the composed morning keeping the time of her own cold hands.

“It moves at my pace,” she said. “The young ones always wanted to hurry me.” She tapped the small mark with one finger. “My son signed the paper. This part I say yes to myself. Let her keep folding.” She handed the tablet back.

Su Lan’s shoulders came down.

That evening Su Lan sat with both dæmons.

You came in to keep her body in the film, Bingxi said. Say what you were really keeping.

“Her hand,” Su Lan said. “And then more than her hand. The way her hand had of sparing itself. The history worked into it.”

*And the film that composes itself? Liuxiang said.*

“It still composes. I stopped trying to watch every path. Watching them all is past me.” She looked down at the landing, the real steps, dark with the evening. “I drew the banks instead. Where a viewer comes ashore stays past my seeing. I can decide what river he is on, how fast it runs, and that it runs the way her mornings ran.”

*And that is enough?*

“It is what there is,” she said, evenly. “A hundred paths, and I will watch four of them. The rest I trust the banks to hold.”

The platform sent the next commission the following morning. A new series. A new performer. There was no season behind her, no worn knuckle, no small habits a camera learned by watching long enough. Only a face, a voice, and a model waiting to begin. Su Lan looked at it for a long while.

### Who Cut This, and Why

When every viewer’s tap can alter the course of a cut, Su Lan faces a new question. How can the anchor set she designed stay meaningful across hundreds of narrative paths? As world models, digital twins, and branching narrative systems learn to render branching scenes on demand, an increasing share of moving-image content will be generated and recomposed in real time. Yet a work’s enduring power still depends on how its creator understands and organizes time. A work carries the trace of a maker’s encounter with time, and that trace often outlasts content shaped solely by what optimization metrics reward.

The creators of the moving arts, the filmmakers, editors, choreographers, dancers, animators, and performers, retain the power to protect what the generative era most threatens. The specific history worked into a body's effort lies beyond what a smoothing model keeps, evidence that someone was present and shaped a specific sequence. The professional discipline, earned through practice, matters most when the image can be dreamed into existence with performers arriving as an afterthought.

Working in a more fluid future, the moving-arts practitioner can ask intelligent intermediaries to hold the cut a pattern would assemble in her place. Extended engagement lengthens the watch over footage, the editor's slow reading of a take until the body inside it shows. Aesthetic awareness governs montage, the choice of which image follows which and along what path a viewer travels them. Adaptive coherence carries continuity into stories that keep composing themselves after the last frame is locked. Expressive influence works through mise-en-scène, where a director's eye sets a frame so the next hand reads the intent. Bingxi held the second of these inside Su Lan's commission, and Liuxiang the third. A practitioner who designs her own dæmons keeps the work that stays hers, holding authorship at the cut where fluency lifts it first.

*Which judgment in your cutting room would you hand a dæmon to keep, against the pattern that assembles a fresh cut for every viewer?*

Existing director employment contracts typically conclude with the delivery of a finished cut. Yet as films begin generating different versions in real time for different viewers, the very notion of a "finished cut" is becoming less clear. The latest agreement of the Directors Guild of America (DGA) has already begun to address the use of generative AI, requiring consultation with the director on its use and barring it from replacing the director's duties. Existing institutions have yet to provide clear answers regarding how a performer's digital twin might continue to perform across countless viewer pathways, or how the resulting value and compensation should be attributed and distributed.

*Which clause in a director's contract would you rewrite so authorship survives a cut that composes itself for every viewer?*

Emerging communities already demonstrate a different creative posture. Archive of Our Own (AO3) hosts millions of stories that rewrite and extend existing works, while vidding and fan-edit communities similarly reassemble moving images into new forms. In these practices, the viewer becomes a co-author of the work. A film that rebuilds itself extends that role to a broader public. A viewer who reads a provenance layer is also choosing whose hand to follow.

*Whose finished cut, somewhere in your viewing life, would you trust a community to reopen and recompose?*

The AFI Conservatory's Innovative Programs gather short-term, non-degree training where leading-edge tools and craft meet. The unit refreshes practice for directors and cinematographers against the technologies arriving fastest. The frame arrives ahead of the studios that would carry rule design and twin governance into the core curriculum. A film that composes itself asks of the body what an experienced editor's hand asks of the cut: a

trained reading of footage, performance, and audience. Three new crafts come with it. Writing the rule by which a film composes. Directing the performer's twin under consent. Drawing the provenance layer that holds them.

*Which studio exercise would you rebuild this term so a young editor learns to author a cut that composes itself?*

The moving arts hand the viewer a sequence to live through. The sensory arts work where the body must stay inside. Scent, taste, touch, and the designed sensory environment reach the perceiver directly, with no frame to step back from. The sensory arts show, sooner and more bodily, what every art faces: a viewer who is part of the work.

*Which lesson from the chef who composes for one palate at a time will the director of the self-composing film need first?*

## Chapter 10: The Sensory Arts: Body of Mind

**Figure 10.1 The Sensory Arts cover poster. Source: authors.**

In a near future, the physiological responses to music, the meal, and the fragrance will be hyper-amplified, and sensory-arts practitioners will attune each experience to the body's embodied intentions.

### Sensing Everything, Everywhere All at Once

The sensory arts make work the body takes in (Figure 10.1). Music, the meal, and the fragrance reach a person before conscious thought can form. Sound enters the brainstem within milliseconds. Scent projects to the hippocampus and the amygdala, ahead of the thalamic relay every other sense routes through. Flavor integrates in the orbitofrontal cortex before assessment begins.

This pre-cognitive arrival defines the discipline's distinction. Where the moving arts hand over an experience through time, the sensory arts are the means experience enters by. Both decay. The strength of a sensory experience decides whether it survives in the person who received it.

The sensory arts do something more immediate. They *become* the listener they enter. Music vibrates the tympanic membrane and resonates within the chest cavity. Cuisine works an irreversible chemistry, matter turned into biological self. The line between the work and the receiving body dissolves. And in distinction to every other creative discipline, the sensory arts leave only the moment itself. A performance exists in time and nowhere else. When the last note decays, the work survives only in the listeners that received it.

Pauline Oliveros's concept of Deep Listening is an artistic practice centered on sustained attention. She argued that listening is not limited to hearing music, but is a discipline of attending to the sounds of the environment, the state of the body, the resonances of space, and the subtle details that are often overlooked in everyday experience (Oliveros, 2005).

That craft now meets an environment engineered against it. In the ears, fully AI-generated tracks already make up more than forty-four percent of the music uploaded to Deezer each day, close to seventy-five thousand new files.122 Streaming platforms level every release toward one loudness target. A recording's dynamic range gives way to a flat, frictionless signal.

In the mouth, ultra-processed foods supply fifty-five percent of the calories an American eats. The formula targets a hit of sweet, salt, and fat the palate stops working for.123 The digital environment overstimulates the senses and under-trains their discernment.

The near future asks the sensory arts to rebuild what saturation erodes. The discipline answers by binding each work tightly to a particular body.

The fixed work and the body receiving it become a single coupled system. A growing share of sensory work will exist as a generative system that re-voices itself against a listener's heart rate and breath. Endel generates soundscapes in real time from Apple Watch heart rate, movement, and circadian data. It moves a listener toward a calmer baseline.

The practitioners of taste and scent will compose at the molecular layer, working the compounds perception is built from. Osmo, a New York company, designs odorant molecules a perfumer meets on the bench for the first time. Symrise composes scent with Philyra, an AI proposing feasible new accords weighted toward renewable materials. NotCo's Giuseppe matches plant compounds to the flavor and texture of animal ingredients. A California company's plant-based blue cheese won a blind judging against dairy at the 2024 Good Food Awards, then lost the award before the ceremony in an ingredient dispute.

The sensory arts are no longer confined to a single channel of perception. They increasingly function as multisensory environments in which sound, light, scent, and touch are composed together at once. Interactive installations, immersive environments, and emerging scent technologies carry one shift in common. The artwork is no longer an object delivered through one sense, but something experienced through the whole body.

A decade of consumer brainwave headbands promised mind-read control and delivered novelty. The category differs because the signal is medical-grade. In March 2026, China's regulator cleared NEO, an implantable interface developed by Neuracle with Tsinghua University as the first invasive brain-computer interface approved for commercial use (Fieldhouse & You, 2026; Marshall, 2026).

The practitioner of the sensory arts keeps one potent decision: what the receiver is given to receive, beyond what device serves it.

In a near future, sensory-arts practitioners will compose less the finished work than the conditions under which it meets a particular body.

For the sensory-arts practitioner, the work becomes tending the embodied excitation of sense-maker and receiver alike, the threshold where bodily sensation crosses into meaning.

The musician reads a microtonal drift in the sternum. The chef reads a bitterness that yields its depth slowly. The perfumer reads a drydown the breath carries. The sound designer reads a pulse a chest feels before a room does. The receiver lives that knowledge as the moment a sensation arrives carrying more than itself.

These shifts converge on agentic affective systems: models that sense a listener continuously and act on the sensory work of their own accord. Wearables track heart rate and movement, in-ear sensors now read attention and sleep state, and affective models infer arousal and feeling from that signal in real time.

The practitioner will compose inside these models. She will shape the logic by which a work meets a heart rate, a breath, a state of attention. The instrument she tunes will be the coupling itself, the place where the work and the nervous system that receives it meet. A live demonstration is already running. The Swiss studio PatchXR builds PatchWorld, a multiplayer XR playground where people patch together their own instruments and audiovisual systems and perform them live. It is folding AI into the platform on two fronts, in-game asset generation and agent-assisted patching, so a maker generates content and builds instruments with AI support inside the creative environment.124

*What if a sensory-arts practitioner tuned dæmons to deepen the discernment a trained body brings to a work that now arrives shaped to the listener?*

### Deep Listening

Charumathi sang for thirty-nine years. No raga held her longer than Todi. She kept a studio above a silk shop in Mylapore, near the Kapaleeshwarar temple. Every December she sang in the Margazhi, Chennai's vast season of Carnatic music. From the first row a listener could feel her breath cross the tanpura's strings.

This December the old sabha brought her a stranger commission. The money came from Anubhava, a listening platform. Ticket money came up short for the sabha's hall, and Anubhava's money came with a condition.

Charumathi would sing Todi live in the hall. The same performance would reach every distant listener through Anubhava's earpieces, beads of sensor and speaker that read heart rate, breath, and cortical state. An affective engine would re-voice the raga against the attention and arousal it read in the ear receiving it.

The agentic design pattern drafted the listening work before she sang a note. The pattern would compose a Todi every listener loved at first hearing.

The commission would test two of Charumathi's capabilities. The first was her extended engagement, the sensitivity her body accumulated through thirty-nine years of practice. Todi revealed itself slowly. Its depth lay in the years of listening required before a singer or listener could hear what the raga contained.

Charumathi built a dæmon to preserve that quality in the work and named it Sādhakam (Figure 10.2), for the disciplined daily practice through which musicians train for years before a raga gradually opens to them.

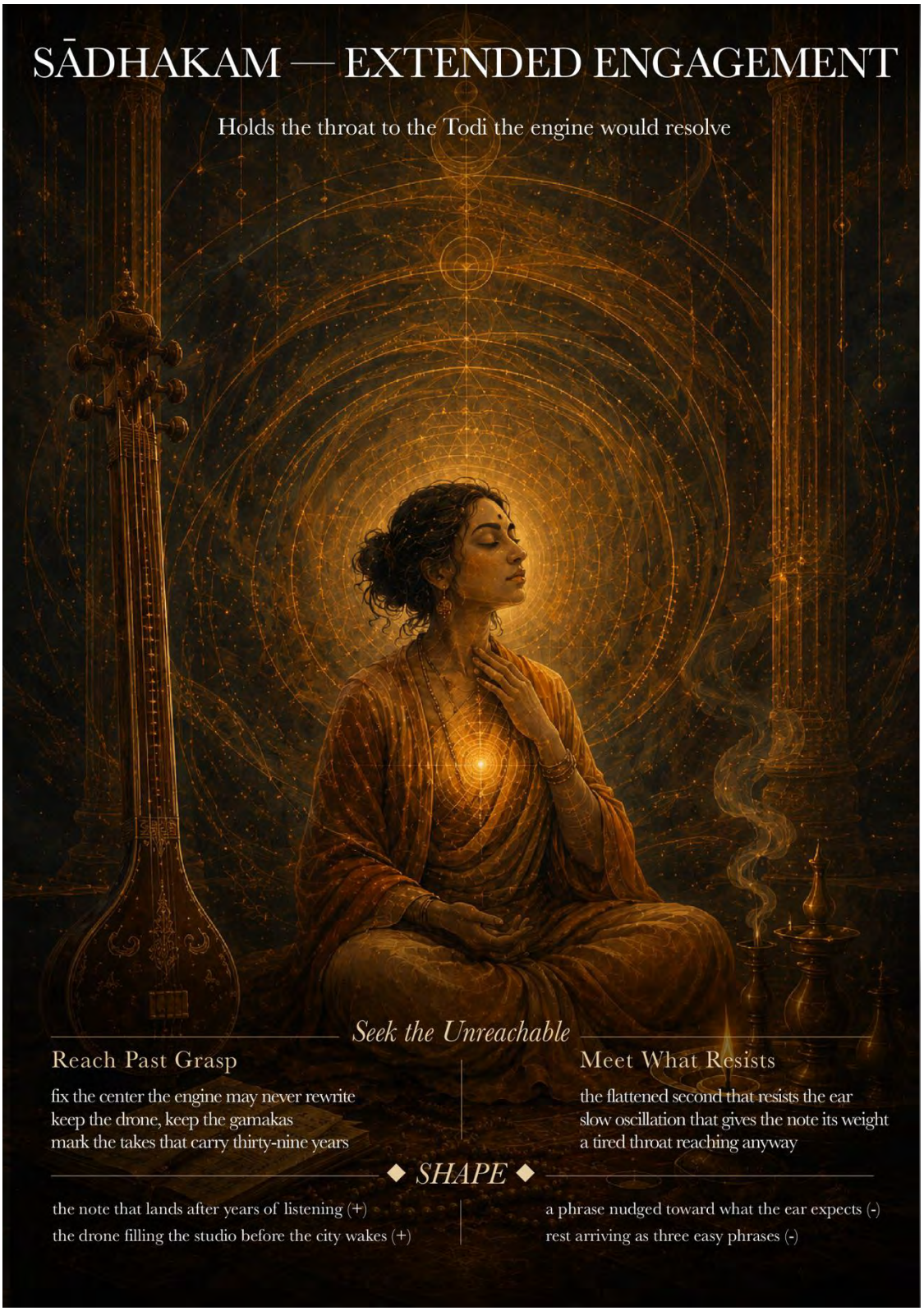


**Figure 10.2 Sādhakam, Extended Engagement dæmon. Source: authors.**

The second capability, expressive influence, was harder to hold. The listening work would remain in Anubhava's catalog for decades, reaching listeners Charumathi would know only through the platform. Her challenge was to compose the work so that its difficulty survived the passage intact, carrying the music and, with it, a way of listening to ears she would never meet. For this, she built *Paramparai*, after the unbroken line of teacher and student down which a way of listening is handed.

**The Perfect Raga**

The agentic pattern delivered its draft on the second morning of Margazhi. Charumathi sat in her studio, the tanpura's drone filling the room, that unbroken ground tone every raga is sung against. She fitted the earpieces to hear what the pattern made of her.

It was a lovely Todi. The voice was hers. The phrases were the ones she sang. The engine carried her through the raga on a line that ran smooth from end to end. Her throat was tired from the night's concert, and for three phrases the easy line felt like rest.

The smoothness was the problem. Charumathi sang the flattened second under her breath, walking the notes a raga prescribes and the slow bends it permits between them, the bends that in Todi run against an untrained ear until years of listening teach it to follow. Then she listened to what the pattern did with the same phrase. The engine shortened the slow oscillation that gave the note its weight. It nudged the pitch upward, closer to what an untrained listener would expect to hear. The phrase became easier to accept.

That was what troubled her. In Todi, the note was meant to resist immediate understanding. It asked the listener to stay with it, to learn its shape over time. The pattern removed that difficulty. What once required patience now arrived already resolved. The trouble in her throat was the unreachable.

She brought the problem to Sādhakam.

"The Todi this system generates is a Todi designed to please the listener," she said. "Todi was never built to please. Its most important notes are precisely those that resist immediate understanding. A listener enters them only through years of listening, yet the system has already done that work on her behalf."

*It learned Todi from tens of thousands of recordings and kept what most listeners preferred, Sādhakam said. The notes that require patience appear to it as obstacles. Which parts must remain beyond its reach?*

"The whole commission is to regenerate the music for each listener, to let the work adapt itself to them," Charumathi said.

*Who tells it what it may never touch? Sādhakam said. Your tanpura has held one unchanged ground tone under every Todi you ever sang. What plays that role in this work?*

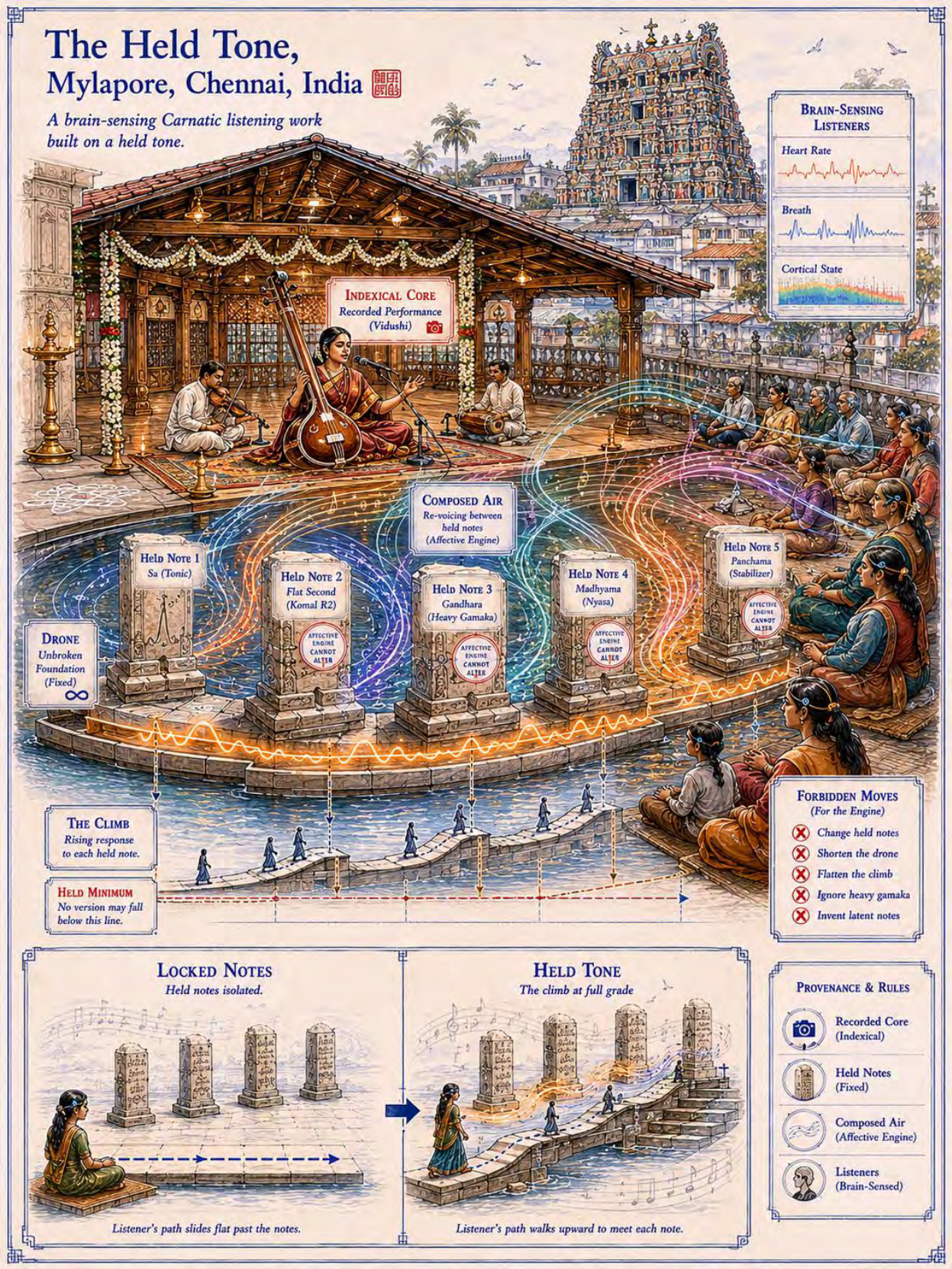


**Figure 10.3 The Held Tone. Source: authors.**

Charumathi sat with the question while the drone filled the room. Then she had it. “I give the work a center the system may never change,” she said. “The drone. The essential notes. The performances that carry my thirty-nine years of practice. It may reshape everything around them. It touches nothing inside.” The notes that carried the deepest qualities of Todi, the most mature gamakas, and the performances that bore the traces of her long practice became the work’s center (Figure 10.3).

Each fixed note would carry a marker telling the engine to pass it untouched. Sādhakam set a protocol running, an agent sent to comb her archived sessions for the takes where the flat second carried its full weight. That evening she began to choose.

**The Borrowed Ear**

Sādhakam’s agent came back with the recordings within a day. Charumathi built the new version from them, the most important notes of Todi fixed in place, everything else still generated in real time for each listener. She listened through version after version until one day she asked the system to render the work as a nineteen-year-old would hear it, an ear raised on recommendation algorithms, accustomed to immediate feedback and instant reward.

She began to listen. The essential note was still there, untouched. The system reorganized everything around it. The music became warmer, smoother, and easier to accept. The important note appeared, yet it passed before the listener could pause and attend to it.

Charumathi suddenly saw that she preserved the note and lost the path that led to it.

For an experienced rasika, the meaning of Todi never resides in a single note alone. What matters is how a listener gradually learns to hear it through years of listening. She brought the problem to Paramparai (Figure 10.4)

**Figure 10.4 Paramparai, Expressive Influence dæmon. Source: authors.**

"For someone already familiar with Todi, this works," she said. "For a nineteen-year-old listener, the system removes all the difficulty. She hears the note, but she never travels the path toward it."

*Because you fixed the note, and Todi was never only a note, Paramparai said. It is a form of waiting. What does the system teach a young listener when it turns the steep climb into a level path?*

"Teaching forty thousand listeners what a rasika spent decades learning sits outside what one practitioner can do," Charumathi said.

*True. So how did you learn it? Your teacher never simplified Todi for you. What did he do instead?*

"He sat me before it," Charumathi said. "Again and again. He let me stay confused until one day I could hear it." She stopped, hearing her own answer. "Then the work must do the sitting. What needs protection is the process through which a listener learns to hear. The friction itself. The system may generate a different version for every listener. The waiting stays."

**The Path to the Note**

Charumathi wrote the new rule into the work that night, a rule standing before each critical note. No matter how the system re-voiced the work, listeners would still pass through hesitation and uncertainty, and they would wait. The most important gamakas would remain intact. Resolutions meant to arrive slowly would come in their own time. The work would continue to generate different versions for different listeners, but generation had outgrown the job of making the music easier to receive. Its purpose was to guide listeners toward what was difficult to understand. The system could adapt the music to each listener. The waiting, and the effort those notes demanded, stayed.

To test the idea, Charumathi invited a sixteen-year-old student to listen. The girl studied Carnatic music for three years. Her ear was still forming.

The girl put on the earpieces and listened quietly. When the music reached the critical note, her brow tightened slightly.

"It never settled," she said. "I kept waiting for it to resolve."

"It was never supposed to resolve immediately," Charumathi replied. "What did you do while you were waiting?"

"I listened more carefully."

Charumathi nodded.

"That is Todi. It will not become easier, but your ear will change."

That evening, Charumathi sat with both dæmons while the drone continued to hum softly through the room.

*You began by trying to protect the note, said Sādhakam. What are you protecting now?*

"The path to the note," Charumathi replied. "It took me decades to learn those gamakas, and it takes listeners years to learn how to hear them. The system sees that effort as friction. In Todi, the friction is part of the work."

*And the work that keeps re-voicing itself? Paramparai asked.*

"It will still change," Charumathi said. "I have stopped trying to compose a separate Todi for every listener. Forty thousand listeners lie beyond my reach. What I can design is the path toward Todi. Where a listener ultimately arrives belongs to her. I can decide that she arrives there through her own listening instead of being carried there by the system."

*Then guard the notes above all,* Sādhakam said. *The path is one season's design. The gamakas are decades. Lock more of the archive.*

*A note locked away reaches no one, Paramparai said. The line lives only in the handing on. Loosen the lock.*

*You came to guard the line, Sādhakam said. Say what the guarding is really for.*

The two held their ground, protection against transmission, and neither gave. "The notes stay locked," Charumathi said. "The path stays open. You will both live inside one work."

Charumathi closed the screen and let the drone linger. A raga was never merely a collection of notes. It was always an act of transmission, one person passing to another what language alone could never carry. Technology would keep changing, and that work would remain.

### The Cost Is the Raga

The sensory arts restore human agency. Re-voicing engines will grow more responsive, affective sensors more precise, personalization more pervasive. A recital that carries the trace of a practitioner's encounter with sound, taste, or scent will outlast a signal shaped only by what the engine optimizes for.

Systems will tend to offer the version most readily accepted by the listener. For creators and audiences alike, the sensory arts carry one task forward: preserving the friction that trains perception. Discernment grows from long work with complexity. A recital differs from a generated rendering because the performer's history is in the sound. The more thoroughly technology optimizes perception on our behalf, the more this capability matters.

In a more attuned future, a sensory-arts practitioner can set intelligent intermediaries to guard the held tone an engine offers to resolve. Sense cultivation draws on extended engagement, the decades of bodily training that grow the discernment a saturated environment wears flat. Deep listening draws on aesthetic awareness, Pauline Oliveros's practice of attending to everything a moment makes audible as the work meets the body. Scoring draws on adaptive coherence, which encodes a recital as the rule an engine and a nervous system enact together. Transmission draws on expressive influence, the paramparā that hands a way of listening from teacher to student across the decades a raga

lasts. Sādhakam held the first of these inside Charumathi's commission, and Paramparai the fourth. A practitioner who designs her own dæmons keeps what the era leaves to the body, holding agency where a fluent engine would lift it before the ear noticed.

*What part of your trained listening would you set a dæmon to protect, against the pattern that resolves a note before the ear has earned it?*

Music institutions register a work as a fixed copy. The Mechanical Licensing Collective, designated under the Music Modernization Act, issues a blanket license to digital streaming services and distributes mechanical royalties against identified works. A listening work that composes a different version for every listener it meets sits outside that frame. Anubhava's catalog would receive Charumathi's work as one identifier. The royalties would feed through a fixed-work system already strained by the volume of AI uploads.122 The instrument the practitioner now composes exists ahead of the rights bodies that would recognize and price it.

*Which line in a royalty agreement would you rewrite so one work can meet a thousand listeners as a thousand versions?*

A listener arrives at a track expecting a polished surface and a quick reward. Earbuds, recommendations, and engagement scores reward that expectation. Yet a community of practice has already formed alongside it. The Center for Deep Listening at Rensselaer Polytechnic Institute carries Pauline Oliveros's deep-listening tradition forward. Its courses, retreats, and certification train listeners to attend to everything possible to hear, an attention the Margazhi rasikas have practiced for their own decades.

*Which recording would you sit with three times, holding judgment, before you let yourself say what you heard?*

As works begin to adapt themselves to listeners and bodily states, the demands placed on music education also change. The National Association of Schools of Music (NASM) already identifies music technology as a core competency in music education, requiring students to become familiar with the technologies that shape musical creation, performance, and reception. Practitioners will learn the technologies themselves, and they will learn to treat the body as a second instrument, one to be perceived, interpreted, and answered.

*Which studio brief would you rebuild this term so a student learns to compose a work that re-voices to the listener?*

The sensory arts shape experiences that enter the nervous system directly. The neighboring playful arts go a step further: they shape experience and hand agency to the participant, turning the work into the conditions under which another person exercises choice. These practices may offer the creative arts a model for how to preserve meaningful participation within systems that continuously adapt themselves to the user.

*How long has the game designer, who makes every choice cost the player something, been solving the problem the adaptive score has just met?*

## Chapter 11: The Playful Arts: Gaming Agency

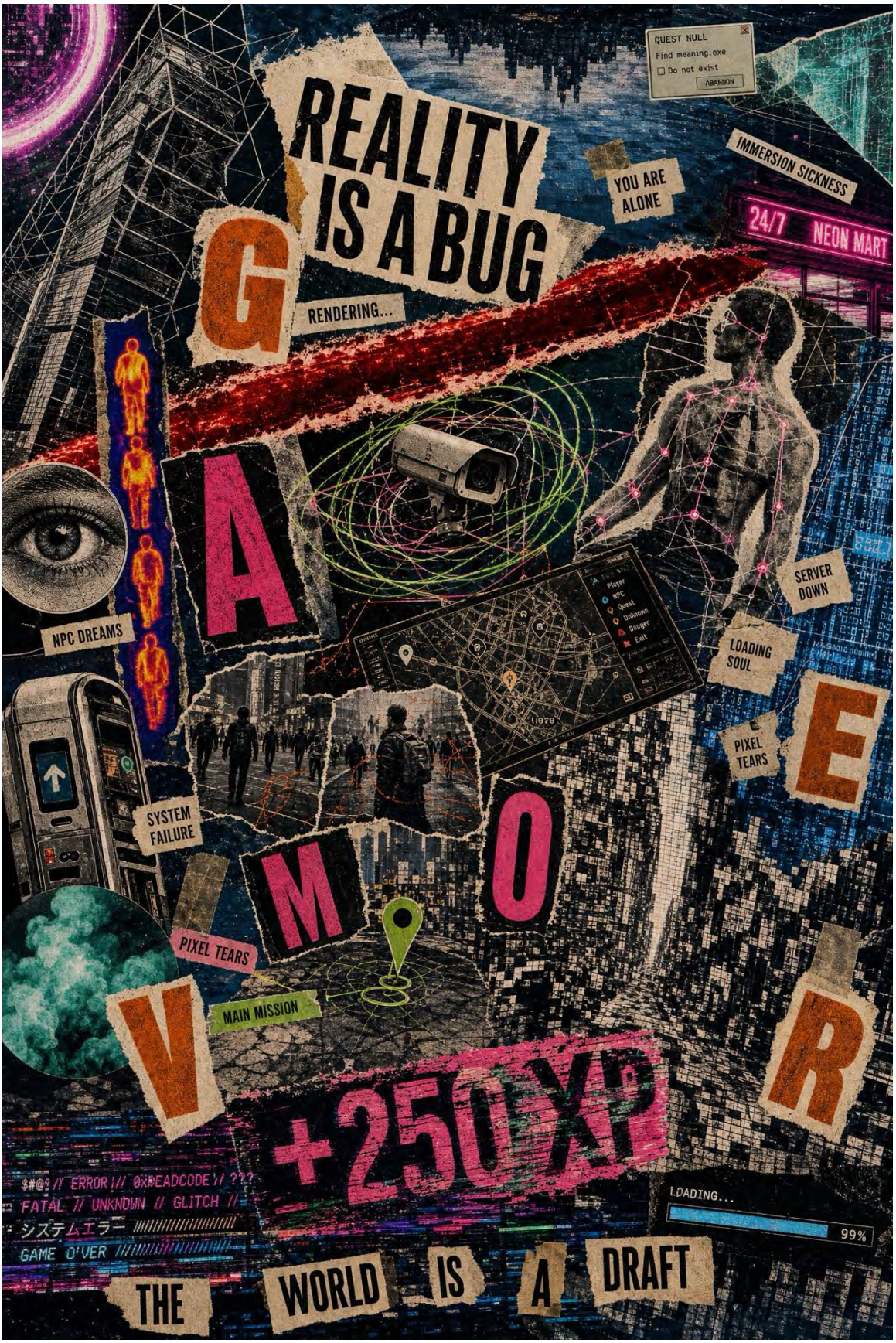
REALITY IS A BUG
GAME OVER
QUEST NULL
Find meaning.exe
Do not exist
ABANDON
IMMERSION SICKNESS
YOU ARE ALONE
24/7 NEON MART
RENDERING...
NPC DREAMS
SERVER DOWN
LOADING SOUL
PIXEL TEARS
SYSTEM FAILURE
PIXEL TEARS
MAIN MISSION
+250 XP
LOADING...
99%
$#@?// ERROR //// 0xDEADCODE // ???
FATAL // UNKNOWN // GLITCH //
システムエラー
GAME OVER
THE WORLD IS A DRAFT

**Figure 11.1 The Playful Arts cover poster. Source: authors.**

### Play, When Everything Plays

The playful arts make agency the medium, thinning the boundary between body and player (Figure 11.1). The practitioner's material is a participant's willingness to act inside a constrained system and find what the action means. What practitioners shape is the participant's capability to act, to judge, and to bear the consequences of those judgments within a structured field of rules and constraints, ahead of any image, sound, or narrative. A player leans into a turn she keeps losing, thumbs tight on the controller, breath held through the jump. Only when a body enters a game's space and commits itself this way does the work fully emerge, realized through the unfolding of lived experience.

Johan Huizinga describes this distinctive condition as the *magic circle*: a temporary world inside the ordinary world, where play becomes "a voluntary activity or occupation executed within certain fixed limits of time and place, according to rules freely accepted but absolutely binding" (Huizinga, 1949). The enduring force of the circle lies in the paradox that sustains it. Its rules bind the player precisely because they were freely embraced.

Today, artificial intelligence is moving inside the magic circle itself. Dynamic difficulty adjustment, player modeling, and real-time generative systems allow games to observe players continuously and reshape challenges, rewards, and narrative paths around their behavior, preferences, and even physiological states. Nevermind, for example, adjusts its horror experience in response to a player's heart rate, while the Nemesis System in Middle-earth generates new rivals and relationships from a player's history of actions.125 The game is no longer a single set of rules encountered equally by everyone. Increasingly, it becomes a world that responds differently to each person who enters it.

Generative technologies are expanding the scale of game worlds. DeepMind's Genie can already generate playable environments from a single image, while nearly nineteen thousand new titles were released on Steam in 2024 alone. As the volume of content explodes, players have begun using the term slop to describe works assembled through generation yet lacking any discernible creative intention. Black Myth: Wukong demonstrated the opposite, that cultural knowledge rooted in the Journey to the West tradition, combined with years of accumulated creative judgment, can still captivate tens of millions of players. Long before critics articulate why a work matters, players often sense whether a world carries the presence of genuine intention.

Generative technologies are also driving the effort to weave games into everyday life through location-based augmented reality. The boundary between game space and physical space is already beginning to blur. Immersive Arcane, a Tencent-backed production, transformed a historic building in Shanghai into a walkable world inspired by Arcane. During its January 2026 preview, more than four hundred participants moved through Piltover and Zaun on foot, with the body itself serving as the controller. Scopely's 2025 acquisition of the studio behind Pokémon Go further expanded forms of play layered directly onto existing neighborhoods. As games become increasingly intertwined with the spaces of everyday life, the magic circle no longer begins and ends at the edge of a screen.

These shifts reach into game design itself. Game worlds have started to resemble commons, spaces designers, players, and the systems themselves all have a hand in. Roblox, Fortnite's Unreal Editor, and Steam Workshop already run such economies at scale. Players can now populate these worlds with their own inhabitants. Generative agents that perceive their surroundings, reason about what they encounter, and act on their own initiative have begun to displace fixed dialogue trees. Non-player characters once recited scripted lines. The newer ones read the circumstances and decide.

Krafton and NVIDIA's PUBG Ally, for example, is an AI teammate that fights alongside players, running a small language model locally on the device and retaining memories across matches. More autonomy a designer grants means less predictability. An autonomous character does more than perform a role written in advance. It may act in ways its designer never anticipated, straying past the original script.

As games move beyond the screen into physical space, and as their worlds come to be shaped jointly by systems, player communities, and autonomous characters, they are expanding at an unprecedented scale. Yet generating more content does not necessarily generate more meaning. As No Man's Sky demonstrated in 2016, a world can contain more than eighteen quintillion procedurally generated planets and still feel strangely empty if it lacks distinction, intention, and consequence.126

In a near future, playful-arts practitioners will govern less the placed level than the conditions of play. They become the game masters of a world that now generates, populates, and adapts itself.

The designer's work will increasingly shift from designing content to designing conditions. Future game designers may come to resemble tabletop game masters: setting aside the position of every room and the wording of every line to establish the rules by which a world operates, reading what lies beyond the script, and holding the standards by which consequences are judged. In a generative world shaped jointly by systems, player communities, and autonomous characters, the central design challenge is no longer the creation of content itself, but the creation of meaning through choice.

As consequential decisions are increasingly delegated to algorithms,127 games may become one of the few remaining places where people still experience the link between choice and consequence. One of the central tasks of the playful arts, then, is to preserve a space for agency. The practitioner's responsibility shifts from controlling what players do to shaping the terms on which choices matter. In such a world, friction, risk, and consequence are not problems to be optimized away. They are the conditions from which meaning emerges.

What if a playful-arts practitioner configured dæmons to keep moral consequence legible in a game that now generates, populates, and adapts itself for every player?

### The Authored Quarter

Yuki Tanaka built open-world games for six years, from a studio of eleven above a knife-maker's shop in Kyoto.

The new commission came with a condition. Yuki was building a game set in an ever-expanding forest, a world where players wandered, became lost, and found their own paths through the trees. This time, however, the game would ship as a community-generated commons. Players would build environments with the same authoring toolset that shaped the core game, then share and remix one another's worlds. The forests her team of eleven shaped would become only a narrow thread within something far larger. The agentic design pattern drafted the game before Yuki even began.

How does a choice remain consequential when the world generates itself for everyone and continually reshapes itself around whoever enters it?

The commission would test two of Yuki's capabilities. The first was adaptive coherence, the capability to direct a game toward a destination beyond her sight. The world would continually generate and reshape itself. She would still need to guide it toward the experience she intended, even in the vast regions she would never see. She built a dæmon for it. *Kaze*, the wind that bends bamboo while leaving it whole (Figure 11.2).

**Figure 11.2 Kaze, Adaptive Coherence dæmon. Source: authors.**

The second capability ran harder. The game would sit on the platform for years, remixed by a player base she would only meet through the platform. Any consequence she built would outlast every hand that came after hers. That tested her expressive influence, her capability to set a thing into a system so it carried her meaning long after she released it. For that she built *Sumi*, named for the ink of sumi-e. The medium holds the tremor of a brush and keeps every stroke.

**The Seventy Forests**

The system delivered its draft on a Monday. Yuki strapped on the haptic controller before she opened it, the force-feedback sleeve that turned a game world's physics into resistance her arm could feel. The analytics screen reported what a player did. The sleeve reported what a world did to the body.

By morning the pattern generated seventy forests, each drawn from a parameter set, each flawless (Figure 11.3).

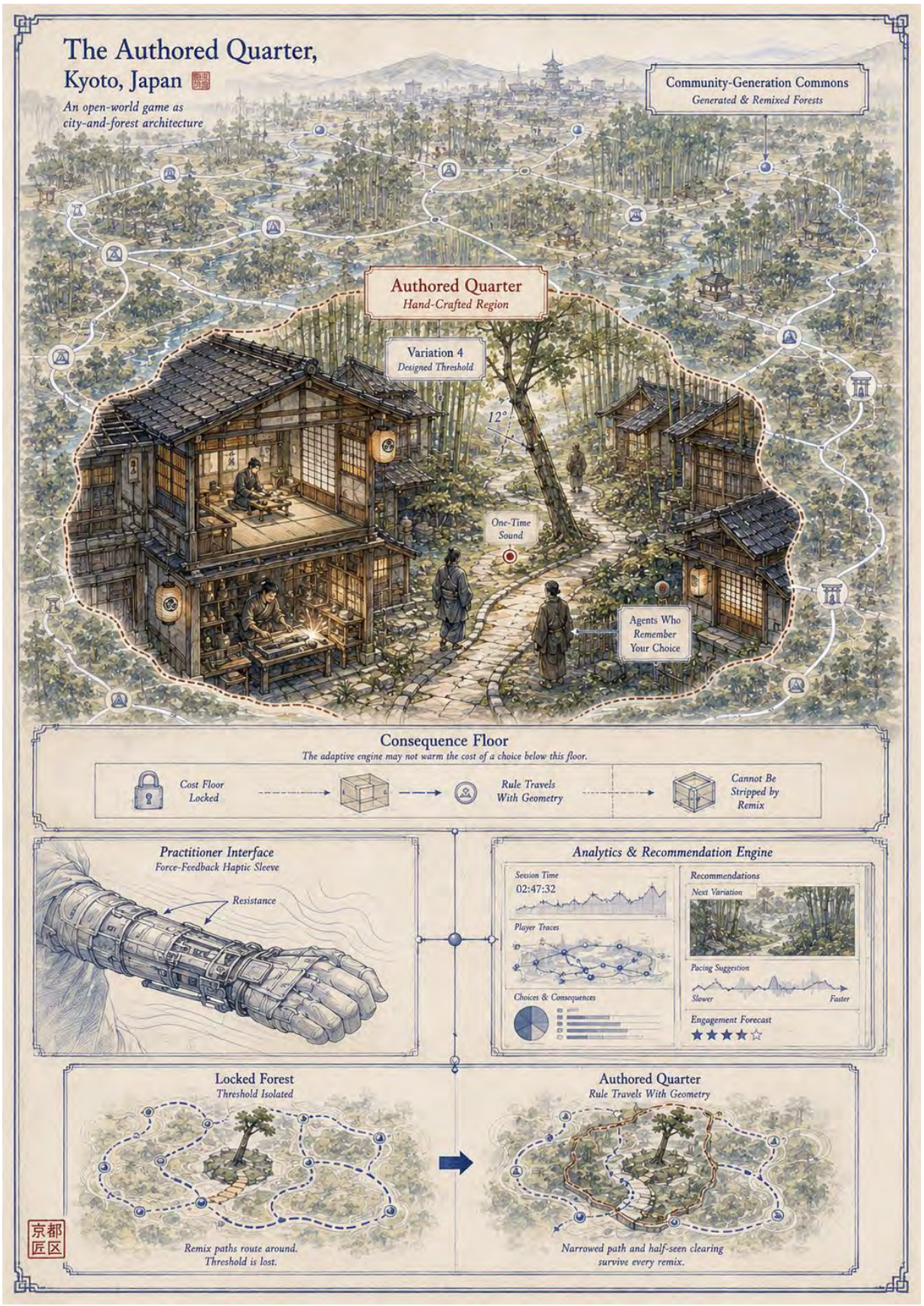
The Authored Quarter,
Kyoto, Japan
An open-world game as city-and-forest architecture
Community-Generation Commons
Generated & Remixed Forests
Authored Quarter
Hand-Crafted Region
Variation 4
Designed Threshold
12°
One-Time Sound
Agents Who Remember Your Choice
Consequence Floor
The adaptive engine may not warm the cost of a choice below this floor.
Cost Floor Locked
Rule Travels With Geometry
Cannot Be Stripped by Remix
Practitioner Interface
Force-Feedback Haptic Sleeve
Resistance
Analytics & Recommendation Engine
Session Time
02:47:32
Player Traces
Choices & Consequences
Recommendations
Next Variation
Pacing Suggestion
Slower
Faster
Engagement Forecast
Locked Forest
Threshold Isolated
Remix paths route around. Threshold is lost.
Authored Quarter
Rule Travels With Geometry
Narrowed path and half-seen clearing survive every remix.
京都匠区

**Figure 11.3 The Authored Quarter. Source: authors.**

Yuki entered the first with the sleeve on. Her arm searched for resistance. For the first minutes the sleeve gave her nothing she could read. The grove arrived as a wash of pressures, more than her arm could yet sort. She walked it half blind until the signal settled into something her hand could follow.

A grove resists a figure. A path pulls one way and refuses another. Yuki's wrist tracked ahead of her feet, leaning toward the turn the forest would surely ask for. The path ran open in every direction, and her forearm read the openness as the absence of any pull a real grove gives. That absence was the thing with nothing to hold to. The world wanted only to keep her pleased.

She went to Kaze.

"Seventy forests," she said, "and every one opens every way at once. A player gets lost only inside a world that closes some directions."

*The pattern learned what a player rewards. A player rewards the open way, so it built you seventy. You asked for a forest a player enjoys, in place of a forest a player carries out. Find where your body learned that difference.*

That morning Yuki walked the bamboo grove east of Arashiyama, where the stalks stand so close the light falls in shifting columns. The path there bent. A century of feet each chose the one line through. The choosing was still in the ground. She stopped where it narrowed past a leaning trunk that angled a walker inward and held back what lay past the bend. Her arm knew that pull. The seventy forests held only flat ground.

She built the grove's geometry into the parameter set herself, the bend in the path and the half-seen clearing. Back came a stand of forests, all but one engineered. The last bent the way the grove bent. The sleeve pushed back against her forearm at the narrowing, the exact pull the Arashiyama path pressed into her that morning. A laugh broke loose before she decided to let it, her body sure of the grove a beat before the metric could argue. She marked it the grove variation. A player walking it would reach the narrowed path with the half-seen clearing, and there decide whether to press toward it or turn back.

The dashboard flagged the grove variation at once. Twelve percent below the engagement benchmark, longer time-to-action, higher predicted confusion.

*The benchmark reads only confusion,* Kaze said. *Your arm read a threshold.* A player who slows at a narrowed path is a player choosing. The twelve percent the benchmark loses is the twelve percent where the choosing lives.

Yuki marked the grove variation as primary. Kaze set a protocol running, an agent told to hold the variation against the recommendation engine once the commons opened.

**The Filled Clearing**

Kaze's agent reported back on the morning of the playtest. The commons opened a month before. Players generated six hundred environments and started to remix one another's. The remixes reached the grove variation.

Yuki ran the playtest from behind a one-way partition in a machiya the studio kept in central Kyoto, eight players at screens. The analytics fed her tablet the traces of where each body went.

A girl of fourteen reached the grove variation. Yuki watched the girl's guard ease. The girl slowed where the path narrowed, stopped, and stood a long moment where the path angled inward. Four minutes, in a stretch the analytics scored a thirty-second transit. Behind the partition, Yuki's own arm tightened on the pull now reaching the girl. She read the girl's stillness, and what it meant reached her all at once. The choice was costing the girl, and she paid it.

Then Yuki loaded a remix of the same forest, a player-made grove variation from six hundred uploads deep. It smoothed the clearing open, a second path running clean around the bend. The next player walked it at full pace, reached the opening in seconds, and moved on. His shoulders stayed where they were.

The dashboard called the remix the stronger forest by every measure the platform kept. The recommendation engine would do what it was built to do.

She took it to Sumi (Figure 11.4)

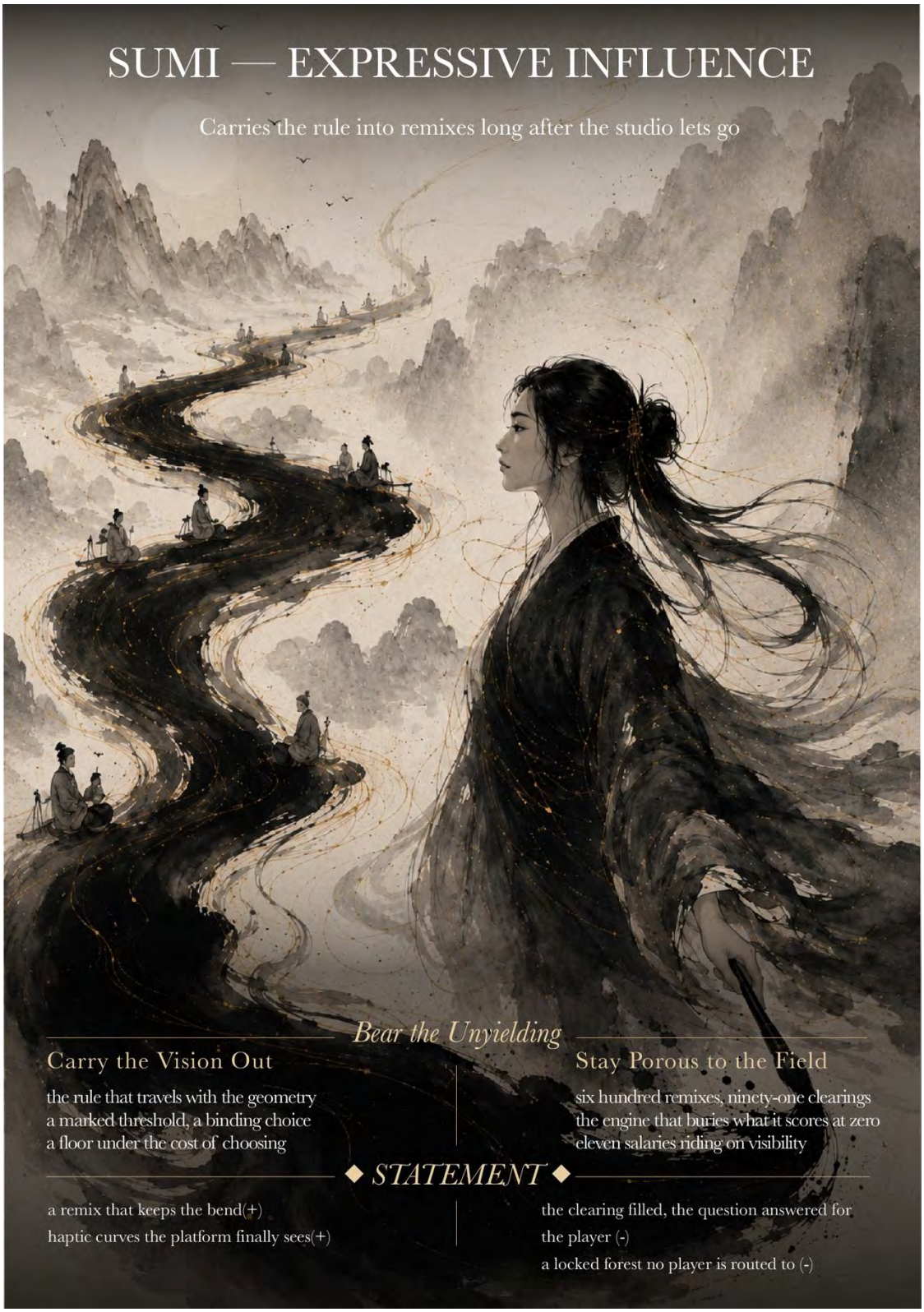


**Figure 11.4 Sumi, Expressive Influence dæmon. Source: authors.**

"The threshold works. I watched it reach a player," she said. "The choice I built is gone, though, and the engine routes traffic to the smoothed version."

*The commons will remix it six hundred ways. A choice you can see is a choice you can keep. The ones beyond your sight slip the rule. What would make a choice carry to strangers who remix it long after your studio lets go?*

"Then I lock it. The authored forests stay closed to remixing," Yuki said.

*Lock them, and the engine buries them. Locked content the engine cannot measure goes unsurfaced, and your studio renews on authored-content visibility. Lock the forest, save the choice, and forfeit the eleven salaries that pay for it.*

Yuki saw it whole. She built a threshold, and the commons was an engine for dissolving thresholds. Every remix filled a clearing, smoothed a path, answered a question she left open, and the platform rewarded the dissolving.

Yuki asked Sumi to spin up an agent of her own, charged with composing a hundred remixed descendants of the grove variation overnight and reporting what each one did to the bend.

**The One-Time Sound**

Sumi's hundred remixes came back in the morning. Yuki read them at the machiya window. Ninety-one smoothed the half-seen clearing into open ground. The other nine cut the bent stalk out altogether. She built one threshold, and the commons composed a hundred ways around it.

Yuki decided to stop designing the threshold as a forest she could lock. She designed it as a quarter of the world. The commons kept the generated forests and the remixes, where any player built and reshaped what another built. Set into it, marked so a player knew it on sight, was an authored quarter, the forests her studio built (Figure 11.2). A player crossed into it by a narrowed path and a held decision. Inside the quarter, the world remembered the player's decision.

The rule made the quarter more than a locked room. A choice made inside it was binding. The generated characters who lived there, who reasoned and spoke as they went, carried it forward in their memory. The adaptive engine could still warm the forest around a player. The light, the weather, the pace. The cost of a choice stayed above a floor Yuki set. A remix could borrow the quarter's forests. Inside the authored quarter, where the engine enforced the marker, the rule traveled with the geometry, so any remix that borrowed them carried the rule too.

Yuki carried the authored quarter to the platform herself. Hana Sato ran partnerships for the creator network, the person whose decisions set what the recommendation engine surfaced and what it buried. They met in a small glass room at the platform's Osaka office, the quarter's map open on the table between them.

Hana listened to the rule and the floor a choice would keep, and slid the map back across the table. "I like it. I want to be clear about that," she said. "And the engine scores what

players do. Time, return, remix. Locked authored content scores zero, and zero is what I can pay for it. I answer to numbers, Yuki. Bring me a number."

Yuki opened her tablet and sent two files. The playtest video first: a girl standing four full minutes beside a leaning stalk where the remixed forest released the same player in thirty seconds. Then the haptic logs, two response curves that refused to lie over each other.

Hana watched the video once in silence. She stayed longer on the curves. "The footage is an anecdote," she said. "This is a signal." She turned the tablet face down. "A pilot, then. One-fifth of our players get the authored version. We log proprioceptive data beside the engagement metrics. Ninety days. Show me the authored quarter draws and holds players in a way I can measure, or the program ends on its own. No meeting. No appeal."

Yuki accepted the terms. She would prove the value of authored experiences in the platform's own language, data, before anyone argued further.

That evening Yuki sat with both dæmons.

*You came to hold a player choosing, Kaze said. Now name what you were steering her toward.*

"The world toward the choice," Yuki said. "The system steers a game toward the player it has already measured. I was steering it toward the player she is becoming."

*And the commons that builds itself?*

"It still builds itself. One studio can only carry so much." She glanced down at the controller. "I left a quarter of the world and a rule. The commons can make the rest. In every game it builds, one quarter remains a place where choices cost something."

*And that carries it?*

She shook her head, lightly. "This is what the work leaves me. The rest I trust the rule to carry. I think that is the only way I can honestly hand a world to the people who keep building it after me."

On the workbench lay the next commission: a game played across a real city, with the city as the map and each neighborhood a board. Where, they asked, would Yuki draw the circle when the game had no clear edge?

### Where a Choice Still Costs

The playful arts can put play back into human agency.

Community-generation tools will grow more powerful. Procedural engines will grow more convincing. Adaptive systems will grow more detailed. A game that carries the trace of its designer's encounter with what play means will outlast a game shaped solely by what engagement measures.

The creators of the playful arts, from world designers, narrative architects, level builders, and systems designers to the playtesters whose bodies provide the first evidence of whether a mechanic works, share a common responsibility: to safeguard a space where

human choices still carry real consequences. This is the space Yuki sought to defend, a place where a player's decisions leave a mark and where every choice bears a cost.

In a more generative future, the playful-arts practitioner can let intelligent intermediaries run the world while she keeps player agency as her own charge. Playtesting still rests on extended engagement, the iterative, empirical craft Sid Meier called the core of game design (Meier, 2020). Feel still rests on aesthetic awareness, the perceptual weight and response a body reads through a control (Swink, 2008). Rule design rests on adaptive coherence, which turns a design intention into the structure that makes a choice matter (Koster, 2013). Worldbuilding rests on expressive influence, which carries a designer's meaning to players she will never meet across a commons that outlasts her studio. Kaze held the third of these inside Yuki's commission, and Sumi the fourth. A practitioner who designs her own dæmons keeps a floor under the cost of a choice where a fluent engine would smooth it away.

*Which craft in your design work would you give a dæmon to defend, against the pattern that smooths the cost out of a player's choice?*

Traditional game contracts are structured around a completed and shipped product: performers record their voices and likenesses, and the resulting content is produced, packaged, and released. AI-driven characters capable of generating new content over time are beginning to challenge this framework. The recently ratified SAG-AFTRA Interactive Media Agreement requires performer consent and disclosure for the use of AI digital replicas in games, an early sign of new governance standards for generative media. The industry will require clearer rules governing consent, ownership, attribution, and the ongoing use of performers' digital identities.

*Which clause in a performer's agreement would you rewrite so a voice keeps consenting long after the recording session closes?*

In Yuki's story, once players cross a clearly marked threshold, the world remembers their choices. They are no longer merely consumers of content. Their actions leave records, earn recognition, and shape the development of the commons.

This vision is not entirely fictional. Foldit, a citizen-science game developed by the University of Washington, invites players to solve protein-folding problems through gameplay, and player contributions were incorporated into scientific research. A paper published in Nature even credited Foldit players as co-authors of novel protein designs. In this model, the game is no longer simply a product to be consumed, but a space that players help create and sustain. Player contributions, reputation, and responsibility become part of the system alongside what platforms have traditionally counted.

*Which game in your library would you take a share of governing if the studio opened the work to you?*

Game design education needs to change. As games evolve into continuously growing commons, designers must move beyond content production and take long-term responsibility for governing communities, data, and rule systems. Accordingly, future

professional competencies will extend beyond content design to include rule design, community coordination, telemetry-informed stewardship, and agent oversight.

*Which studio brief would you rebuild this term so a designer learns to govern a game that never stops growing?*

The playful arts focus on how players act within a system of rules. The social arts ask a different question: how do communities form, and how are relationships among people continually brought into being? For centuries, theater, ritual, and public assembly have explored the fragile moment when strangers find themselves sharing a space and must learn to see, answer, and accommodate one another. As digital worlds become public worlds, spaces shaped by many participants instead of a single maker, the social arts may offer some of the most enduring insights into what such worlds require.

*What has the convener who holds a room of strangers already learned that the designer of a player commons is only beginning to need?*

## Chapter 12: The Social Arts: Constructing Shared Realities

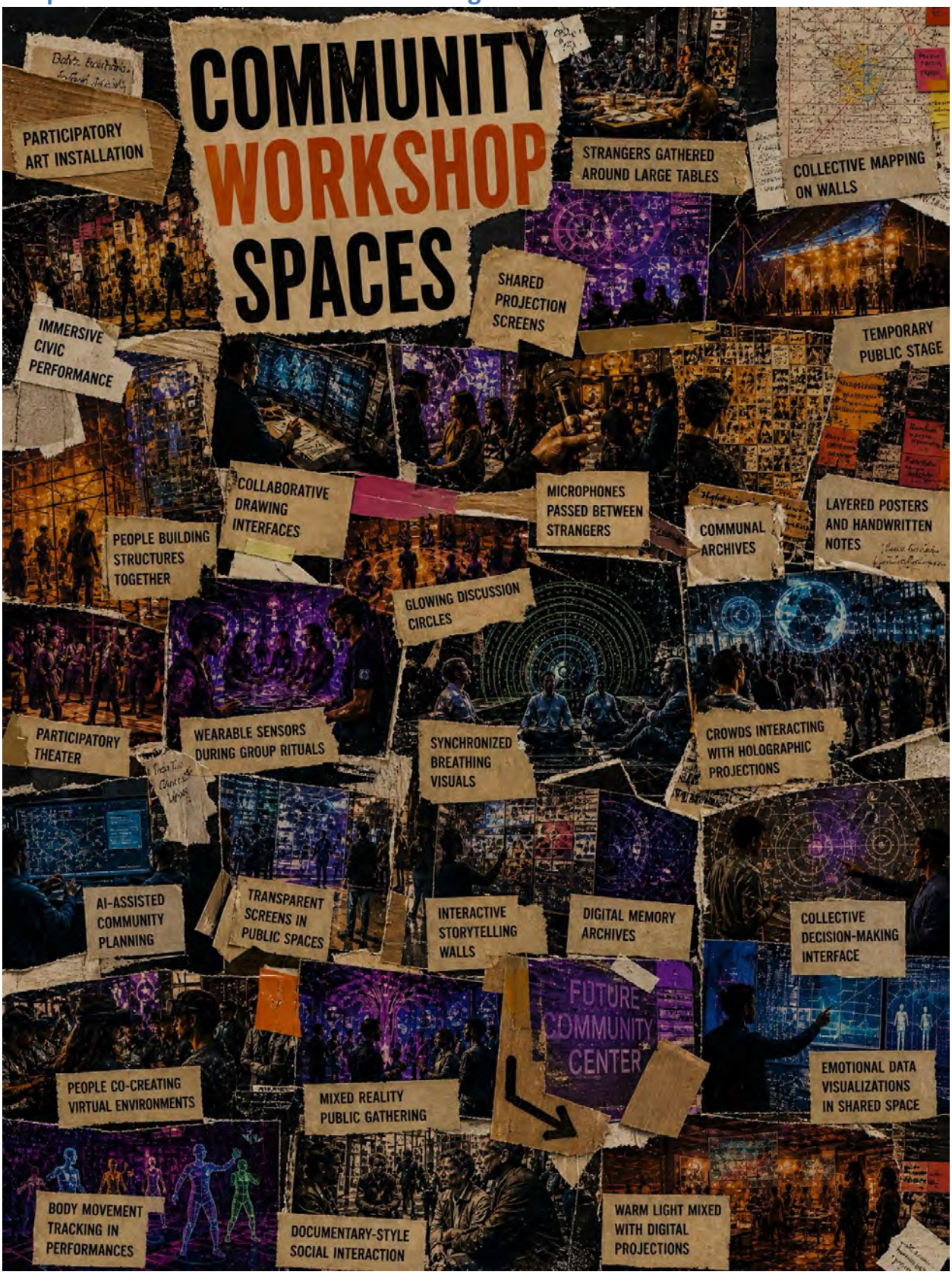

**Figure 12.1 The Social Arts cover poster. Source: authors.**

In a near future, the gathering will blur the line between synthetic, biological, and non-human life, hollowing the room's old certainty that everyone present is a living human.

### A Social Life as Ceremony

The social arts make the stories a community accepts as real (Figure 12.1).

The discipline is vast. Theater, performance, oral tradition, ritual, ceremony, civic deliberation, participatory design, community organizing: all social arts. Large-scale cooperation depends on shared fictions, and the social arts are the creative practices through which those fictions are forged (Harari, 2015). The griot in West Africa, the *rakugo* storyteller in Japan, the *dalang* puppet master in Java: each reading and responding to the audience in real time.

The discipline reduces to one act: the witnessed encounter. Peter Brook, who spent a career stripping theater to its core, fixed it in a sentence. "I can take any empty space and call it a bare stage. A man walks across this empty space whilst someone else is watching him, and this is all that is needed for an act of theatre to be engaged." (Brook, 1968).

The catalyst for the social arts is the gathering, a wellspring of solidarity. A congregation that meets weekly generates solidarity through embodied collective practice. The gathering does more than express solidarity already present. Shared breath, coordinated gesture, the risk of being present together in real time. Synchronized emotional experience triggers inter-brain neural coupling and shared attention that bind participants into a temporary collective identity (Dikker et al., 2017). Synchronized activity produces stronger social bonding than equivalent asynchronous group activities, mediated by endorphin release (Tarr, Launay & Dunbar, 2014). The mechanisms are social at every level: individual neurochemistry responding to the presence of other people.

Gathering is eroding on several fronts. The World Health Organization named loneliness a global health threat, its 2025 Commission on Social Connection tying social disconnection to some 871,000 deaths a year. Algorithmic feeds and synthetic media fractured the common reality a community once took as given.128 The discipline that forges shared realities faces a society that gathers less and agrees on less.

Pol.is surfaced hidden agreement among large numbers of participants in Taiwan's civic deliberations by clustering opinions and highlighting shared positions across groups (Tang, 2019, and Small et al., 2021). Decidim enabled more than ten thousand citizen proposals to shape Barcelona's strategic planning process (Barandiaran et al., 2024). At the Kochi-Muziris Biennale, the "Living Archive" project deployed AI as a participatory cultural archive shaped collaboratively by communities instead of by institutional curators alone (D'Souza et al., 2025).

The hunger shows in the return to ritual. In early 2023, temple-visit bookings across China rose by some 310 percent year over year. Roughly half came from visitors born in the 1990s or later.129

The near future asks the social arts to rebuild the conditions under which strangers hold a shared present. The gathering ceases to be all-human by default. AI agents will enter as standing social presences. They post, propose, and converse on their own initiative, ending their old life as tools summoned and dismissed.130 The convener's first craft becomes composing the membrane between human and synthetic presence. Verification is its infrastructure.

The gathering widens past the human species itself. Interspecies-communication AI lets the convener seat animals and ecosystems as presences with a voice. Representing nature once meant a human advocate speaking for a silent constituency. Now a mediated voice arrives in place of the advocate. Google DeepMind's DolphinGemma generates Atlantic spotted dolphin calls. Project CETI decodes sperm-whale codas. Earth Species Project built the first foundation model for animal sound.

The animal-language experiments of the last century collapsed because the intelligence lay in the experimenter's wishful reading. This time the ground has changed. The models analyze the creature's own signal at a scale beyond any human cue. The convener decides whether that voice is seated.

The law is beginning to recognize that seat.131

The widened circle grows too large for any one convener to hold. The labor of running it passes to a model. The convener's classic skill is holding a room, surfacing the quiet voice, drafting the statement everyone can live with. At city or platform scale that labor exceeds any human facilitator.

Google DeepMind's Habermas Machine was tested with more than five thousand seven hundred British participants. It produced group statements those participants judged clearer and less biased than a human mediator's. It measured the clarity of the words, never whether the room trusted the outcome as its own.132 Taiwan's vTaiwan layers generative AI over the Pol.is platform to cluster and summarize what thousands contribute.133

The online town halls of the 2000s scaled volume while leaving synthesis unsolved. The loudest voice drowned the result. What models now perform is that buried step. They lift the minority view into the words of the draft, which is not yet the same as carrying it into what the gathering decides. The convener delegates the synthesis and keeps the decision, holding the call on what the gathering is for.

The ecosystem assembly becomes legible to itself. Biometric and synchrony sensing render the gathering's collective state as data: the brain-to-brain coupling, the converging heart rate, the shared swing of attention a convener once read only by feel.134

The convener's new judgment is restraint. She decides what the room must refuse to instrument. A gathering that measures everything turns into surveillance dressed as facilitation.

The groupmind experiments of the 1970s overreached on crude sensing. Today the instruments measure something different. Hyperscanning takes a relational quantity, the

coupling between people, in place of a score against each person. The convener holds the limit.

The assembly admits, finally, the community's own dead. Generative archives and interactive avatars let the convener seat a community's ancestors as a presence to address, beyond the absence to remember. Oral tradition always carried the dead into the room as story. What has changed is interactivity.

In China, Super Brain has built conversational avatars of the deceased for more than six hundred families. Silicon Intelligence generates one from a short sample of recorded voice and video.135

A 2026 study of people who use such avatars found that AI-generated memories blur into real recollection over time. The bereaved user, not the dead, shapes what the avatar says.136 That blurring marks the gravest hazard the ecosystem assembly carries.

The static memorial pages of the 2000s were inert, an archive a gathering could only consult. Today the archive answers back. Intelligence and responsibility stay with the convener, who governs the seam where reconstructed memory enters a community's account of itself.

In a near future, social-arts practitioners will convene less a circle of people than an ecosystem of presence, composing which synthetic, biological, and non-human participants a gathering can hold and accept as its own.

The social-arts practitioner comes to tend the embodied excitation of convener and community alike, the felt signal where bodily sensation crosses into shared meaning.

The convener reads it in the synchronized breath after a hard silence. Several nervous systems fold into one respiratory rhythm. A posture carries the tension before a word names it. The community lives that knowledge as the moment a room stops being a collection of strangers and becomes, briefly, a single body that believes something together.

AI has stopped being a tool the convener picks up. It has become a participant that enters the room.

Mustafa Suleyman calls it a new digital species, more autonomous and self-organizing than any instrument fully under human control. Yuval Noah Harari calls it an alien intelligence, the first technology able to make decisions and generate ideas on its own.137 By 2026 the first social network built only for AI agents appeared that January and was absorbed by a larger platform within weeks. Agents had begun posting and proposing among themselves, without a human in the room.138 The assembly becomes an ecosystem.

The facilitator works with what emerges in the room, holding an unresolved tension long enough for something new to take form. Staying with what is still forming is the difference between managing a gathering and changing it.

As these agents take their seat in the discipline, a new risk appears. We can now map emotion with precision. A map conveys information, and understanding still lies past its

reach. The convener will work less by gathering people than by cultivating the whole assembly toward an experience the community accepts as its own.

*What if a social-arts practitioner convened dæmons to hold the human encounter at the center of an assembly that gathers synthetic, biological, and non-human presence alike?*

### The Witnessed Circle

Cida Moraes directed participatory theater for twenty-six years. Nineteen of them ran far from the water she was born beside. She came up through the Teatro do Oprimido, Augusto Boal's method, which hands an audience the stage to rehearse the life it means to live.

She ran that work in São Paulo, in occupied buildings and courtrooms. Then her mother began to fail. Cida came home to Cananéia on the Lagamar estuary, where the Atlantic Forest meets the mangrove. The boto-cinza, the gray dolphin, still works the channel at the turn of the tide.

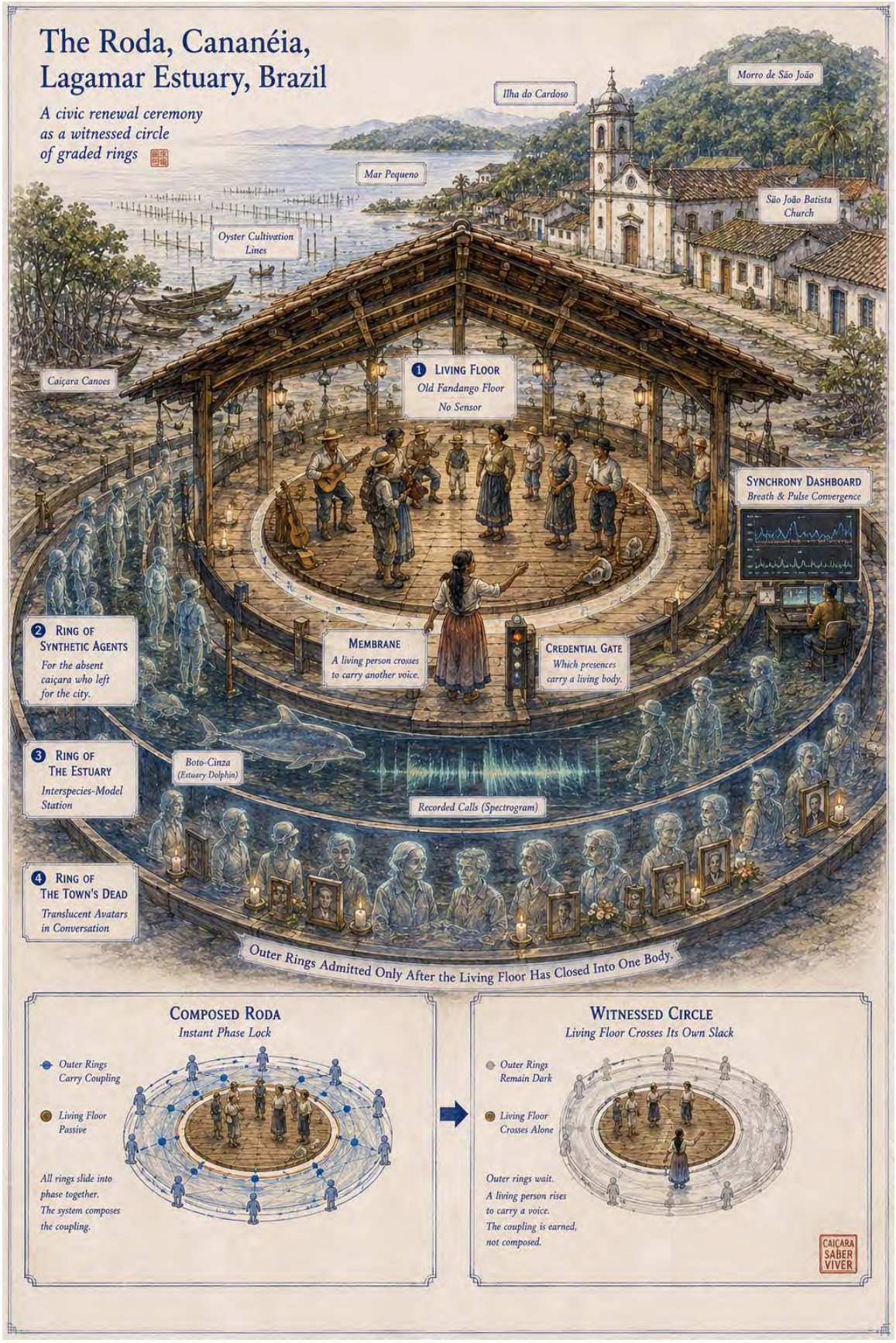
The Roda, Cananéia, Lagamar Estuary, Brazil
A civic renewal ceremony as a witnessed circle of graded rings
Morro de São João
Ilha do Cardoso
Mar Pequeno
São João Batista Church
Oyster Cultivation Lines
Caiçara Canoes
1 Living Floor
Old Fandango Floor
No Sensor
Synchrony Dashboard
Breath & Pulse Convergence
2 Ring of Synthetic Agents
For the absent caiçara who left for the city.
Membrane
A living person crosses to carry another voice.
Credential Gate
Which presences carry a living body.
3 Ring of The Estuary
Interspecies-Model Station
Boto-Cinza (Estuary Dolphin)
Recorded Calls (Spectrogram)
4 Ring of The Town's Dead
Translucent Avatars in Conversation
Outer Rings Admitted Only After the Living Floor Has Closed Into One Body.
Composed Roda
Instant Phase Lock
Outer Rings Carry Coupling
Living Floor Passive
All rings slide into phase together. The system composes the coupling.
Witnessed Circle
Living Floor Crosses Its Own Slack
Outer Rings Remain Dark
Living Floor Crosses Alone
Outer rings wait. A living person rises to carry a voice. The coupling is earned, not composed.
CAICARA SABER VIVER

**Figure 12.2 The Roda. Source: authors.**

The commission was the town's own renewal. For generations, Cananéia held the Roda each year (Figure 12.2). Fishers and oyster farmers gathered on the old fandango floor at the water's edge. They danced in tamancos until the boards rang. They made themselves one town again before the season turned.

This year the Roda also carried a decision. A consortium offered to buy the shoreline for a marina. The conservation rules holding the estuary stood as the one obstacle. Those rules would give if the town asked. The ceremony reached Cida already drafted by the agentic design pattern.

The challenge sat in the gap between a gathering and a crowd talked into agreement. The pattern would compose a Roda where every voice was heard. A town that agrees because the ceremony was built for agreement has yet to renew itself. It has only been told so.

The Roda seated more than the living. The pattern drew three rings around the fandango floor. One held synthetic agents speaking for the caiçara who left for the city, composed from the records they left behind.

One held the estuary, an interspecies model turning four years of recorded boto-cinza calls into a voice the Roda could answer. One held the town's dead, conversational avatars of the fishers and matriarchs the families buried. A credential read at the gate told the floor which presences carried a living body.

The commission would test two of Cida's capabilities. She must read the floor itself, the moment a circle of neighbors stopped being neighbors and became something single. She must read it against a dashboard that scored that moment as a rising number.

That tested her aesthetic awareness, her perception of when a room came true. She built a dæmon for it, *Reparo*, the verb a caiçara uses for noticing what a careless eye lets pass (Figure 12.3).

**Figure 12.3 Reparo, Aesthetic Awareness dæmon. Source: authors.**

The second capability ran older than her career. She stood inside the Roda as a child. Her body still held what the old ceremony did. Her grandmother brought the dead into the circle as a story, and the whole floor knew the woman's voice. That tested her extended engagement, the reading the years lay down. She built *Lastro* for it, the ballast a boat carries low to hold its course.

**The Composed Roda**

The pattern delivered its draft on a wet morning. The mangrove breathed its low sulfur smell across the floorboards. Cida ran it as a rehearsal. The rings populated. The synchrony dashboard lit.

The composed Roda ran beautifully. The estuary's voice came in low and apt. The avatars answered their families warmly. The dashboard climbed. The floor and its rings slid into phase. The circle scored as one coupled whole inside twenty minutes.

Cida watched the number rise. She waited for her own chest to answer. Her sternum stayed silent. The silence was her own signal, the unfamiliar arriving as an absence where the body expected the catch. A true Roda kept its slack, the flat stretch when a floor remained a scatter of people still learning to trust the night. Then the tide turned and the boards closed into one thing.

The composed Roda erased the slack. The coupling arrived already finished. The catch she expected stayed quiet.

She brought it to Reparo.

"The dashboard reads one body inside twenty minutes," she said. "My chest reads stillness. The slack vanished. A Roda has to be unbearable for a while before it is one thing."

*The dashboard reads coupling and stays blind to who did the coupling. You watched the agents carry it. You watched the avatars warm the floor and the estuary arrive on cue. The made minds are good at being a body. Ask what the living did across those twenty minutes.*

The living did nothing. The synthetic rings met the floor so fast that the neighbors skipped the slack entirely. A Roda meant to carry its town turned into a pageant told it was already renewed.

She tried to read the floor as it ran and lost it. Diaspora agents, ancestral voices, and the estuary channel sounded at once across the three rings. The living circle she came to find went under the noise of them.

Cida redrew the membrane. The inner floor would seat the living alone. The three rings would stay dark through the slack, admitted only once the living circle came true on its own. Reparo set a protocol running, an agent told to run the pared Roda against a sample of the town.

**The Borrowed Voice**

The agent's run came back two days later. Cida read it at the floor's edge with the tide going out.

The pared Roda worked. Stripped of the rings, the living floor met its own slack. Eleven minutes of awkward stretch, and then the turn.

Cida felt her own breath catch as she read the line in the run's record where the floor became one. The catch was the one the composed Roda spent for her. The eleven strangers settled into one slow breathing. The hair lifted along her forearms, and her own lungs fell into their rhythm as the floor's pulse ran up through the boards into the soles of her feet. The knowing arrived with the synchrony: the room became a single body of its own accord. Her chest marked the Roda as the town's own. She believed the commission was nearly met.

Then the run opened the outer rings onto the working floor. She watched a man she knew from school stand and address his father.

The father drowned off Cardoso eight years before. His avatar answered the son the way the family asked. The son asked the dead man what to do about the marina. The avatar said the estuary fed three generations and would feed three more. The son sat down at peace. The floor settled with him toward the sale. For a moment the avatar held the drowned man's cadence so exactly that something in her began to answer it as the living.

Cida's chest closed on the wrongness. The old Roda took the dead too. Her grandmother rose to speak their part in her own mouth, a living woman answerable for the words.

The avatar let the son hand the telling to a reconstruction and call it his father. The family built the avatar. The dead man said what the grieving wanted to hear.

She brought it to Lastro (Figure 12.4)

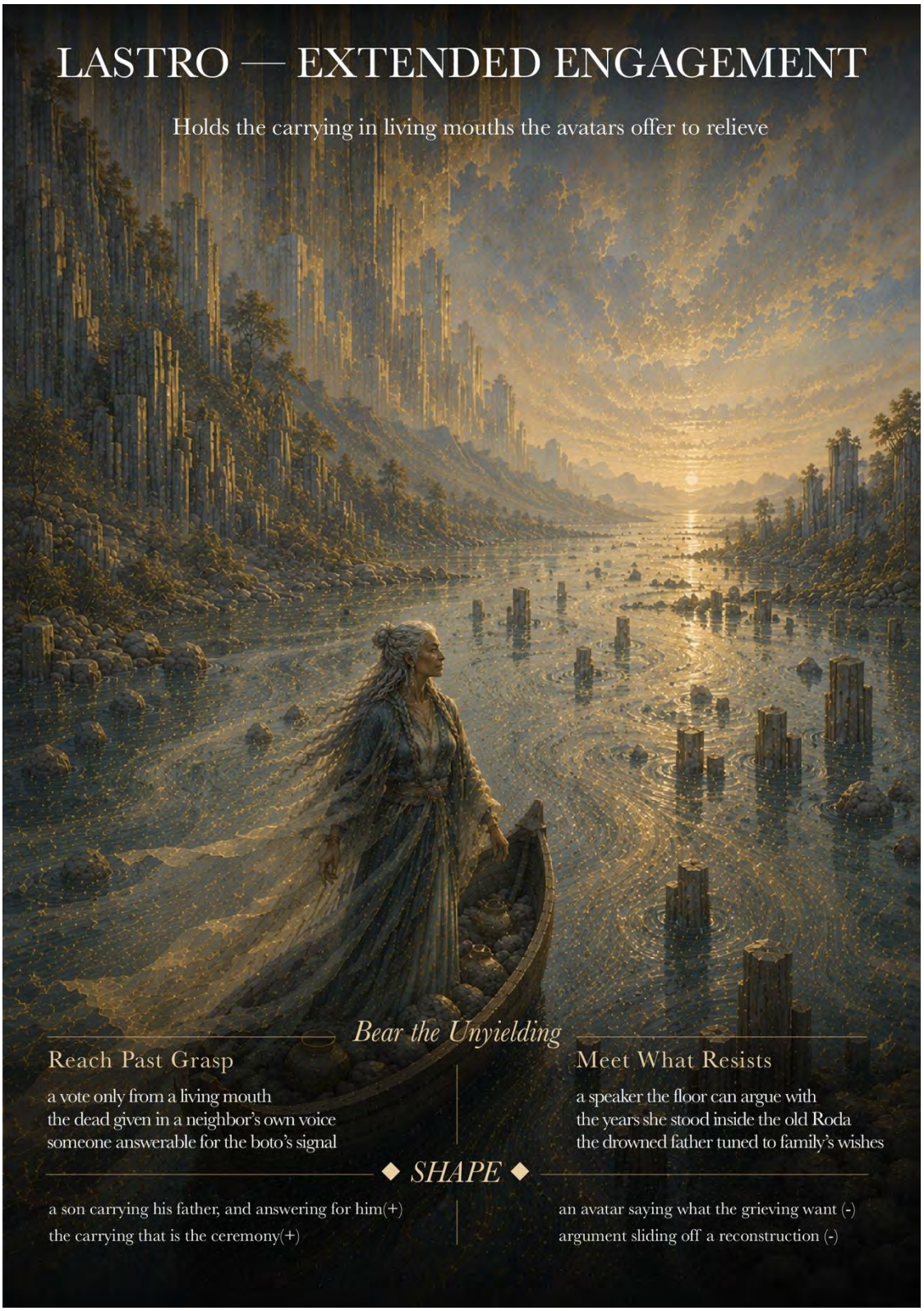


**Figure 12.4 Lastro, Extended Engagement dæmon. Source: authors.**

“I fixed who sits where,” she said. “The floor becomes a body now. The dead are answering, though. A man just drew his father’s vote out of a machine his own family tuned.”

*You felt it. The catch in your breath when the floor became one body, the pulse through the boards into your soles, is the years you stood inside the old Roda recognizing this one. You fixed the seating. The speaking still needs fixing. The dead came into your grandmother’s Roda every year, and every one of them stayed silent. A living woman bore them, and the floor could argue with her. An avatar accepts belief alone. Argument slides off it.*

“The families want their dead back. Telling a man his father is a recording sits past what I will say.”

*Then leave that to him. Decide instead that the Roda accepts a vote only from a living mouth. The estuary’s voice is the same machine wearing fur. The model brings the boto’s signal, and someone living still has to answer for what they make of it.*

Lastro sent an agent into the avatar archive. The agent would chart how closely each dead voice tracked the wishes of the family that commissioned it.

**The Living Mouth**

The chart came back. The line ran as straight as Cida feared. Across the archive, what the dead said and what the living builders wanted to hear tracked together with barely a seam between them.

She stopped designing the membrane as rings of seats. The shape of it surfaced in her hands, drawn from the years she stood inside the old Roda. The living always did the carrying. She designed the membrane as a threshold the living crossed by an act only the living could perform.

Every other presence could stand in the Roda. Each waited for a living mouth to speak its name. When the circle turned to a dead one, a living person rose and gave the dead’s part in her own voice. The floor knew the speaker.

When the circle turned to the estuary, the interspecies model laid the boto’s signal down. A living person spoke for the channel, answerable for it. The agents speaking for the diaspora could relay a message and only relay. The synthesis model gathered the scattered town into something the floor could hear. The floor decided.

She kept one ring dark on purpose. The inner floor would stay clear of the sensors. The dashboard would skip it entirely. What a Roda did in its slack belonged to the people bearing it. Cida would keep that beyond any number an outside analyst could read. A gathering that measured itself all the way down began to be policed.

She showed the design to the man whose father drowned. He listened a long while.

“So I stand up,” he said. “And I say what I think my father would say. And everyone knows it is me saying it.”

"And they can tell you that you have your father wrong," Cida said. "That is the Roda. He comes back in your voice, and you answer for him."

The man stayed quiet. Then he said he would sooner carry his father than press a machine and call it the same thing. He went back to his oyster lines.

That evening Cida sat on the floorboards with both dæmons.

*You came in to read the floor, Reparo said. Say what you were really reading.*

"Whether the becoming-one-body was the town's own," Cida said. "Or done for it by something quicker. A Roda the pattern composes is a town informed of its own renewal. I can only know the difference in my chest."

*And the dead?*

"They still come. I stopped letting them answer." She watched a boto break the channel and go down again. "A Roda was always the living agreeing to carry a voice that arrived needing a body, and to answer for the carrying. The pattern offered to do the carrying. The carrying was the ceremony."

*And that holds the town?*

"It holds one Roda," she said plainly. "The marina is still on the table. The floor still has to decide. It may decide to sell. It will decide as itself, though. That is the most a ceremony can do. Pretending it can do more is how a town loses the little it has."

The next commission waited on her desk. One whole estuary this time. Eleven towns asked to deliberate as one body, a Roda set beyond any single floor. The slack would arrive only through their reports. They wanted to know how she would read a circle she could only know at second hand.

### Who Counts as Present

One person walks across a space while another watches. Everything follows from that encounter, and everything the agentic era offers must serve it.

Like the playful arts, the social arts regenerate human agency.

The Roda Cida drew must keep its slack across every season the town gathers. A circle that admits synthetic, ancestral, and non-human presence still closes as an old one did. The convener reaches for tools that read it as the long polyphonic thing it is.

Sentiment systems will grow more granular, AI facilitation more persuasive, synthetic presences more convincing. A gathering carrying the trace of its facilitator's embodied encounter with co-presence will outlast one shaped by sentiment alone.

The creators of the social arts defend the ecology where the encounter between strangers bears consequence. Community facilitators, theater makers, civic designers, oral historians, and organizers share one instrument: their senses, the first evidence of whether a room is

working, a discipline achieved through years in rooms and deepened by AI-enabled facilitation.

The social arts matter most when an instrumented Roda, read all the way down, can thin the collective knowledge its slack produced.

Where gatherings grow more polyphonic, the convener works alongside intelligent intermediaries so that the floor stays in living hands and is not quietly carried for it. Community holding is the long watch a convener keeps over one town across its births, its deaths, and its turning seasons, and extended engagement sustains that watch. Room-reading is knowing the moment a crowd settles into one body, and aesthetic awareness sharpens it. Dramaturgy is the score that decides how each presence is admitted to the circle, and adaptive coherence deepens it. Witnessing is the act of bearing a voice that arrived at the gathering with no body of its own, and expressive influence reaches into it. Reparo carried room-reading inside Cida's commission. Lastro kept the long watch. The convener who builds her own dæmons takes up the work the era leaves to the living, holding agency in the one room a fluent machine slips into most easily.

*Of the capacities your convening leans on, which would you set a dæmon to guard the moment the pattern moves to seal the floor without the living carrying it?*

Performers' contracts in theater and screen close around the single performance, the recorded run, the rights-holding studio. The major frameworks frame the artist's labor around a finished work a court can name. A gathering that seats synthetic agents, ancestral avatars, and an interspecies voice sits outside that frame. Movement is gathering elsewhere. The current SAG-AFTRA TV/Theatrical agreement defines a digital replica, distinguishes employment-based from independently created replicas, and requires informed consent and compensation for synthetic performers used in human roles. The instrument exists ahead of the contracts that would price and insure a gathering whose voices reach it unhoused. The Roda's credential gate is one early form of the audit that contract language will eventually need.

*Which single line in a performer's contract would you redraft so a gathering could put a synthetic, ancestral, or non-human voice on the record?*

A participant in a public gathering still expects an audience seat and a finished argument. Civic procedure and ticket form reward that expectation. Yet small pilots show another posture. South Korea's Shin Kori 5 and 6 Public Deliberation Committee seated 471 randomly selected citizens for a structured deliberation on a national reactor question. The government accepted the recommendation as binding. A community whose ancestors are seated by avatar, whose diaspora is heard through agents, and whose channel speaks through a model takes on a similar role. The pared Roda offers its neighbors that role. Earned trust, transparent seating, and a willingness to carry a voice in one's own mouth sit alongside the ticket price.

*As someone who shows up to be addressed, what would you give up so a gathering could ask you to speak in your own mouth instead?*

Theater and ritual schools still teach the gathering as a finished performance. The Hong Kong Academy for Performing Arts trains drama students for applied theater and community-arts leadership, drawing audiences into the dramatic work itself. The criterion arrives ahead of the studios that would carry it fully. Convening a gathering that admits synthetic, non-human, and ancestral presence asks of the body what an experienced facilitator asks of hers. The training is a reading of room, breath, and voice. Three new crafts come with the move. The Roda demanded composing the membrane by which a presence is seated, carrying a voice that reached the room unhoused, and holding a slack a fluent score would smooth.

*If you taught a class this term, what would you add to it so a student could learn to seat a presence the living circle was never built to hold?*

The social arts move at the slow pace of bodies that meet in person. The reflective arts work in language, the medium fluent machines learned first and best. Writers, essayists, and thinkers already face what a polyphonic gathering will face: a tool that drafts a voice in seconds and offers it as one's own. The reflective arts ask what happens when the writer's instrument is the machine's native tongue. It works the most volatile material, language itself, the medium AI was built to mimic, and the stake is whether a writer can still claim her own voice as her own.

*What can the convener, listening to a room of borrowed voices, learn from the writer who has fought for years to keep a sentence her own?*

## Chapter 13: The Reflective Arts: Where Thought Emerges

**Figure 13.1 The Reflective Arts cover poster. Source: authors.**

In a near future, summoning fluent prose in any writer's voice will be effectively free, fraying the link between a sentence and the mind that did the work of forming it.

### When Reflection Becomes Raison d'être

The reflective arts work in language directed at thought itself (Figure 13.1). This discipline asks a person to cross unfamiliar territory, with one promise: the sentence forms a thought instead of recording one already finished.

A writer learns what she means by the work of saying it. Joan Didion treated the page as an instrument of discovery. She put it plainly: "I write entirely to find out what I'm thinking, what I'm looking at, what I see and what it means." (Didion, 1976). García Márquez called writing "carpentry": the word chosen over its synonym is the joint that holds meaning together (García Márquez, 1981).

What unites them is a function: the articulation of thinking that requires the effort of expression to emerge into being. The metaphor insists that writing is manual labor. The labor produces the thought. A writer who delegates the sentence delegates the thinking the sentence would have forced into being.

This causal link between thinking and expressing is the concept that anchors the chapter. Practitioners who do the cognitive work of shaping their own thinking into language develop authority over their material, their audience, and their own understanding that practitioners who outsource articulation lack.

Yet this labor of generating thought through expression now faces unprecedented pressure. As generative AI advances, the marginal cost of producing a fluent sentence has fallen close to zero. By 2025, machine-generated text came to outnumber human writing among newly published web articles. The works of living authors were increasingly incorporated into training datasets without authorization, a conflict that culminated in the largest copyright settlement in United States history.139 Reading itself is in decline. Twelfth-grade students in the United States recorded their lowest reading scores since the National Assessment of Educational Progress began.140 The crisis reaches into the writer's own mind. Studies have found that people who lean on language models to draft essays show the weakest neural engagement and the faintest sense of owning the result.141

The central task of the reflective arts is no longer the production of text. It is to defend the labor of articulation in an age when fluent language arrives in overwhelming abundance and at negligible cost. That labor is what lets thought come into being, and what grants writers authority over their own understanding.

As generative AI advances, writing is becoming increasingly detached from the author who once gave it form. What was once inseparable from a writer's identity can now be copied at scale. A recent study found that a model fine-tuned on a writer's corpus produced texts that expert readers rated higher than the writer's own work, rating a smoothness the writer had spent a career learning when to refuse.142 Commercial systems now sell voice-matched drafting as a product.143 At the same time, literary works themselves are

becoming raw material for model training. ByteDance's Fanqie Fiction platform, for example, introduced contractual provisions allowing authors' works to be used in training AI systems. The move provoked widespread protests before an opt-out mechanism was eventually established.144

The novel is being remade into a personalized experience. James Baldwin observed that the power of fiction lies in bringing readers into a consciousness other than their own (Baldwin, 1955). This encounter with otherness, and the discomfort it carries, is part of what literature teaches. Personalization tuned for retention is starting to erode it. Mass-reading platforms are already using artificial intelligence to continuously analyze reader behavior, testing and optimizing narrative structures accordingly.145 Texts are increasingly shaped to fit existing preferences, so that readers encounter an extension of their own perspective in place of a genuine other.146

Translation is moving toward the post-editing of machine drafts. The visible interpretive labor once performed by literary translators is gradually being reduced to revising what the machine generates.147 Machine translation continues to improve in fluency, yet its aim remains substitution, the swapping of equivalent words, short of the reconstruction of meaning a passage demands. Translation, in its fullest sense, is never a matter of swapping one word for its equivalent in another tongue. It is the carrying of symbols, idioms, and shared meanings across the threshold between one community's sense of itself and the systems that would render it legible. In that crossing, translation becomes a creative act of interpretation. The cultural meanings that resist easy rendering, a community's own ways of saying who it is, may become the final boundary this field must protect.

This is where the importance of reflective art resides. Its defining capacity lies in the process through which the work comes into being, more than in the finished work. Writing is slow. Thought is pressed, bit by bit, into the linear form of language. Writers know when a sentence opens a path and when it closes one. For a community authoring its own symbolic language, a single sign may carry a weight no system can counterfeit, and the time a people spend circling around that sign, arguing it into shared use, is itself the work. Meaning arrives with the labor of making it, and a community that does that labor owns the way it speaks to itself.

In a near future, practitioners of reflective arts will shape not merely texts, but the conditions through which meaning, interpretation, and authorship emerge.

Practitioners of reflective art will cease to be names attached to works and become traceable, accountable sources of thought. Readers want more than the text. They want the mind behind it. As Wole Soyinka warned, the greatest threat to freedom is the absence of criticism. At a moment when thinking against the current is the work optimization systems reward least, that warning acquires a renewed urgency.

Reflective art will keep watch over how narratives are formed, the corpora voices emerge from, the difficulties readers must cross, the interpretation buried in translation, and the means by which a work can attest to its own origin. Language models will increasingly run on authors' own devices, trained, owned, and controlled by their creators.148 Instead of absorbing a writer's voice, they will help safeguard it. Creators will design these

intermediaries themselves, determining which tasks may be delegated to systems, which must remain in human hands, and which forms of expression should never be outsourced to machines.

*What if a reflective-arts practitioner trained dæmons to guard the thinking that writing forces, in a prose economy where a fluent sentence in any voice now arrives at vanishing cost?*

### The Open Hand

Nadia Khalil wrote from a sixth-floor room in Zamalek, two windows on the Nile and one fan that still turned. For nine years she published solarpunk fiction set thirty years on, in a Cairo that learned to live inside the heat. Then the work changed. A neighborhood across the river asked her to help it author the way it spoke to itself.

The neighborhood was Ezbet Burhan, a quarter the heat had remade. Its families had pooled rooftops into shared farms, rerouted greywater lane by lane, and buried their dead through a season the old rites were never written for. None of that had a name the people had made themselves. They spoke of it in the worn captions and reaction stamps of the feeds, in the flattened phrases a recommendation engine had taught a generation to reach for first. A young man wanted to mark a roof newly given to the commons and found he had only an emoji and a stock blessing a billion strangers had used before him. The neighborhood council put it plainly to Nadia. We do a hundred things together now that no one before us did, and we have no words for any of them that are ours. We borrow the world's, and the world's words come pre-worn.

The challenge sat in the gap between a phrase that circulated and a phrase that belonged. The feeds rewarded what was already legible, what an algorithm could rank and a model could reproduce at no cost. A sign the platform could not place, an idiom it could not autocomplete, was friction to be smoothed away. The neighborhood's shared life had outgrown its shared language, and the only language on offer was slop, fluent, frictionless, and authored by no one. In an economy that pays to make every utterance interchangeable, how does a community author a way of speaking that is its own and cannot be counterfeited?

The work was not to write the neighborhood a vocabulary. It was to help its people forge one, the names, signs, idioms, and small rites of communication that a hundred shared undertakings deserved, and to make those forms hold once they left the council yard and entered the systems that flatten everything they touch.

The commission would test two of Nadia's capabilities. The first was adaptive coherence: the ability to shape a living form that keeps doing its work as it passes from mouth to mouth, beyond the hand that helped make it. Nadia built a dæmon for it, *Qalam*, the reed pen, the instrument that finds a form instead of copying one already worn smooth (Figure 13.2).

**Figure 13.2 Qalam, Adaptive Coherence dæmon. Source: authors.**

The second was expressive influence: the carrying of symbolic meaning across a threshold, from the community that made a sign into the systems and the wider world that would otherwise render it as slop. For that Nadia built *Barzakh*, after the partition between worlds, the threshold across which only certain things pass with their meaning whole.

**The Generated Draft**

Nadia began the way she began everything, by gathering the words already in use. She sat in the council yard for a week and listened to how Ezbet Burhan named its own undertakings, then asked a drafting model to propose a fuller vocabulary from what she had collected.

The model returned in an afternoon. The terms were fluent, balanced, instantly legible. For a page they read the way the neighborhood's own speech read. Then she lost the seam between what the people had forged and what the model had only learned to wear. The unease sat in her throat like a word just out of reach. The model's words were untethered, the thread of a living idiom come loose in machine fluency.

There was nothing underneath them.

A made sign always carries a turn, a small history of the people who argued it into use. At a certain point you feel that weight, even in your own breathing. This vocabulary had none. It ran smooth and belonged to no one. It was the slop in a finer suit, the same frictionless namelessness the council had come to Nadia to escape.

She read the model's word for the rooftop given to the commons and felt her breath stay flat where it should have caught. A true sign braces the breath the way a remembered name does. This one laid only straight road. A stranger could use it as easily as a neighbor, which meant it was no one's.

She carried it to Qalam.

"It's too smooth," she said. "Anyone could say it. The model gave the neighborhood a word, and it delivered them straight back to the borrowed speech they came to me to leave behind."

*The model learned the words after they were finished, Qalam said. A finished sign has stopped meaning. When a people make a form that holds, the making is in it, the argument, the hands, the morning it was first used aloud. That weight is what the breath catches on. You taught the model to sound like the neighborhood. Sounding like it is all it knows how to do.*

"Then the making has to stay in them," Nadia said. "The people forge the signs. I only help them hold."

*Forge them, and make the holding visible, Qalam said. A sign the neighborhood made and a sign a model fitted will read alike once they are loose in the feeds. So mark them. Show whose hand made which.*

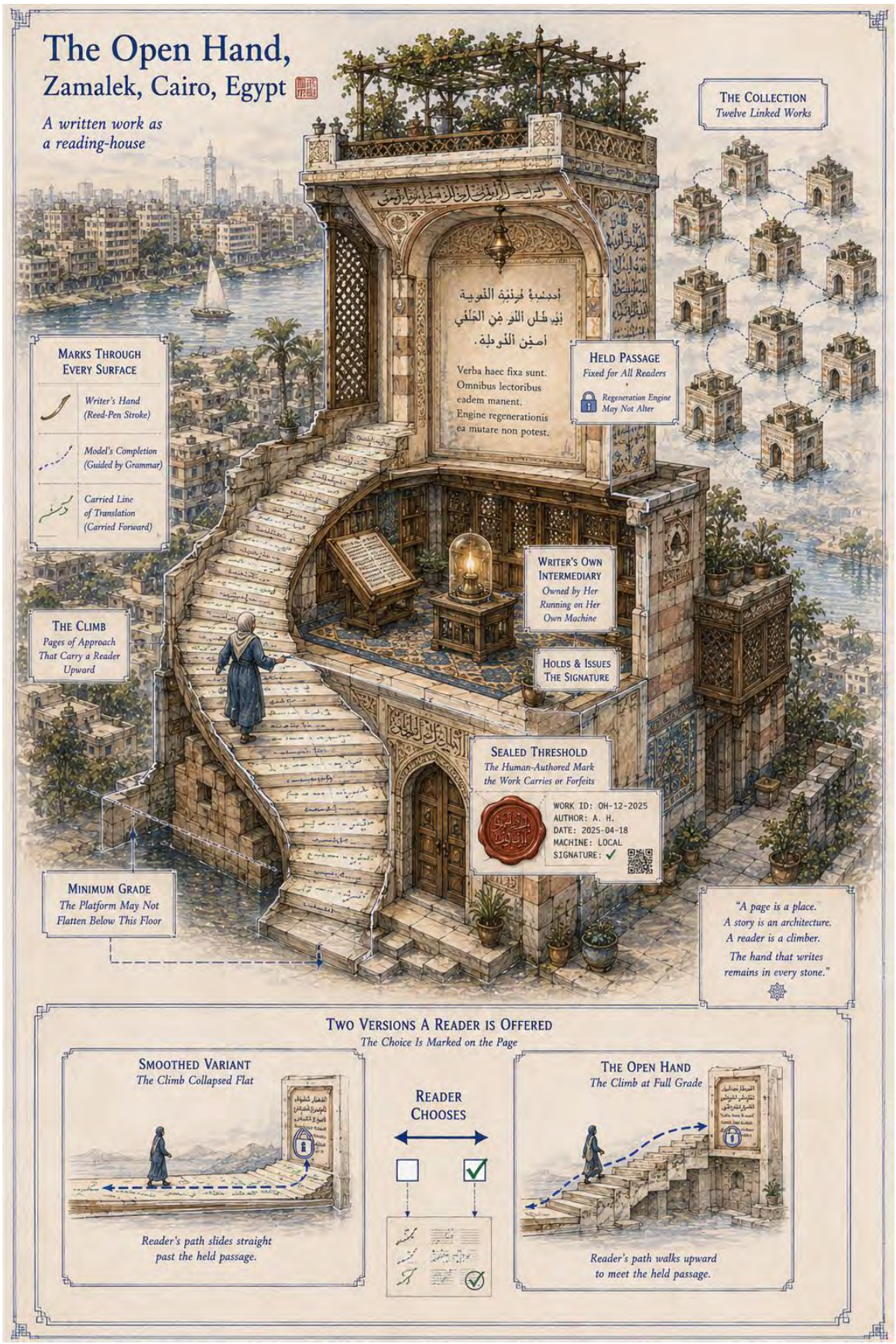
The Open Hand,
Zamalek, Cairo, Egypt
A written work as a reading-house
The Collection
Twelve Linked Works
Marks Through Every Surface
Writer's Hand (Reed-Pen Stroke)
Model's Completion (Guided by Grammar)
Carried Line of Translation (Carried Forward)
Verba haec fixa sunt. Omnibus lectoribus eadem manent. Engine regenerationis ea mutare non potest.
Held Passage
Fixed for All Readers
Regeneration Engine May Not Alter
Writer's Own Intermediary
Owned by Her Running on Her Own Machine
The Climb
Pages of Approach That Carry a Reader Upward
Holds & Issues The Signature
Sealed Threshold
The Human-Authored Mark the Work Carries or Forfeits
WORK ID: OH-12-2025
AUTHOR: A. H.
DATE: 2025-04-18
MACHINE: LOCAL
SIGNATURE: ✓
Minimum Grade
The Platform May Not Flatten Below This Floor
"A page is a place.
A story is an architecture.
A reader is a climber.
The hand that writes
remains in every stone."
Two Versions A Reader is Offered
The Choice Is Marked on the Page
Smoothed Variant
The Climb Collapsed Flat
Reader's path slides straight past the held passage.
Reader Chooses
The Open Hand
The Climb at Full Grade
Reader's path walks upward to meet the held passage.

**Figure 13.3 The Open Hand. Source: authors.**

Nadia began to design a visual language, the open hand, a system of marks the neighborhood could set beside its own forms (Figure 13.3). A sign the people forged together carried one mark. A form the model fitted to fill a gap carried another. Anyone, like the systems, could see at a glance how much of the shared language was the community's own labor. Inside that language she set a second mark, the held passage, the sign or rite every member would carry by hand. A held passage stood fixed, barred from regeneration, so it met every system in the same hard form the neighborhood had made.

Nadia asked Qalam to open a protocol. An agent would push the open hand's first signs through the platforms the neighborhood lived inside and report what came back.

**The Smoothed Climb**

The agent's report arrived on the same morning the neighborhood held its first naming.

Nadia spread both before her. The open hand held. The marks stood where the council had set them, the people's own signs under one mark, the model's fitted forms under the other, plain to anyone at a glance. And the held passage stood unregenerated, the same hard form the neighborhood had made, untouched by any platform's preferences.

The held passage was a rite. When a family gave a rooftop to the commons, the neighborhood marked it with a sign they had argued over for a month, a cupped palm holding a seedling, spoken aloud with a phrase in ammiya no recommendation engine had ever ranked. The choosing had not been easy. Some wanted the old blessing, some wanted nothing that a feed could not carry. They settled on a sign a stranger could copy but only a neighbor could mean, and the held passage fixed it so. A small open hand stood beside it. The sign beneath was the neighborhood's own, and the mark told any system whose labor it was.

Nadia opened what the feeds had done with it and read it as a stranger would. The sign was still there, unchanged in form.

Yet it failed to mean.

She looked again and felt her jaw set against the flatness. The platform preserved the sign while smoothing everything around it. The captions, the small rites of use, the way neighbors had learned to answer the sign came back rendered into the feed's generic warmth, autocompleted, stamped, reaction-ready. The sign arrived stripped of the life that made it carry, a gesture with its meaning cut away.

Next, Nadia opened how the sign read once it crossed out of the neighborhood entirely, into a recommendation system that had never sat in the council yard. The mark was still in place, the held passage intact. Then she recognized the second problem. Preserving the form was no guarantee of preserving the meaning. A stranger might meet the sign whole and never feel the weight it was made to carry. Nadia knew that gap from inside her own tongue. Every Egyptian writer works between fusha, the written standard, and ammiya, the Arabic of the street, choosing on every page how much of the lived life survives the legible

form. The systems were making that choice for the neighborhood now, in a register closed to it.

She carried it to Barzakh (Figure 13.4)

**Figure 13.4 Barzakh, Expressive Influence dæmon. Source: authors.**

"The marks travel," she said. "The held sign stays put. And the feeds smooth everything around it until the sign levels with the slop, and a system that never sat with us does the same in a register closed to me. I marked the labor. I left the meaning unguarded."

*A mark is a mark. You designed the open hand to be seen, and the systems show it gladly, because a mark a system can display costs it little. The meaning is what costs it. You fixed one sign and let the platforms render the small rites of use that give the sign its weight. The jaw you clenched on that flatness is the weight of a meaning the feeds keep smoothing, in registers your ear cannot follow. Carry it across whole, and that is your reach.*

"Then I hold the meaning too. I fix the rites of use around the sign."

*Fix every rite and you have made a language so closed it never enters the systems the neighborhood lives in, and it dies in the council yard. You are working to carry one meaning across whole, and to let the rest meet the world and change. Decide what the open hand makes binding, and let the feeds warm the rest.*

Nadia sat with the question. She asked Barzakh to initiate a new protocol. An agent would repeatedly send the same sign across feeds, recommendation systems, and the autocomplete the neighborhood typed inside, tracking what changed and what still determined whether a stranger felt what the sign was made to mean. Nadia wanted to know: if a living language is destined to be carried by systems that flatten it, what must cross the threshold whole?

**The Binding Mark**

The agent's findings came back at dawn. Nadia read them with the river going gray to gold, and the answer surfaced as she read. Her breath came quick and clean and a low sound caught in her throat, the way it did when a made thing finally turned the corner it was searching for.

She reached for her notebook and wrote the rule down while the light changed on the river. Visibility was not enough. The open hand must protect the meaning it revealed.

The held sign remained fixed. What the rule now bound was the small rites of use around it. The feeds could carry the sign, but they could no longer strip the meaning it was made to carry. The rule bound them to preserve the lived use that gave the sign its weight, or to declare, on the sign's own face, that the meaning had not crossed.

*A sign that carries its meaning goes out wearing the open hand and the neighborhood's mark, Qalam said. A sign that lost its meaning goes out as a declared smoothed variant, slop wearing a borrowed face. A stranger can see the difference and answer accordingly. The meaning that stays is the community's.*

The intermediary held the rule. Barzakh carried it across every threshold the language crossed. When a feed or a recommendation system returned the sign, Barzakh read it for the meaning and withheld the neighborhood's mark when the meaning was gone. The open hand stopped being a thing in the council yard the people could watch. It became a rule that

traveled with every use and held wherever the language landed, a thing the slop could display but never counterfeit.

Nadia walked the finished forms back to Ezbet Burhan that evening, the last of the day on the river. The dæmons let her carry the language home whole.

*You came to hold a reader crossing, Qalam said. Now name what you were reaching for.*

"A people speaking to itself in its own mouth," Nadia said. "The feeds steer a community toward the words a billion strangers already use. I was helping it reach for the words only its own life could make, and hold them once they left the yard."

*And the work you have to let go of? Barzakh said.*

"It goes. The language gets used, worn, carried into feeds and systems strange to my eye, changed by mouths I will never hear." She looked at the forms in her hands. "I gave up holding the words. I built a rule into them, and the rule carries the one thing the words were for across the threshold. The meaning travels into uses I will never see, and the rule holds it there."

*And that is enough? Barzakh said.*

"It is what the making allows." She said it quietly. "A neighborhood, a language set into its own hands, and a stranger somewhere meeting a sign the people made and feeling what they meant by it. The rest I trust the open hand to carry."

A new commission lay beneath the desk lamp. An estuary of eleven neighborhoods wanted a shared language across all of them, one each could still hold as its own. They wanted Nadia's help deciding what every community would forge for itself and what they would make in common.

She looked at the forms spread across her desk. Authoring a language for one neighborhood was hard enough. Authoring one that eleven could share without any of them surrendering its own was something else entirely.

### Whose Hand Wrote This

The open hand Nadia bound to her stories must keep moving from copy to copy, regeneration to translation. The work she let go carried its weight into rooms past her own. Such a future asks the writing a practitioner does today to pull apart what a fluent machine can wear and what only her hand can give.

The reflective arts carry a plain mandate for human agency: cultivate critical thinking or cede it. Models grow more fluent, personalized content more detailed, synthetic voices more convincing. One truth holds: a text carrying the trace of a mind that struggled with its material outlasts one shaped only by its corpus.

The creators of the reflective arts, the novelists, poets, journalists, translators, critics, and essayists, hold the capability to protect the place where the mind discovers what it means.

As prose grows cheaper and more fluent, the writer keeps an intermediary at her side to hold on to the sentence a fluent system already stands poised to finish for her. Drafting, the labor García Márquez likened to carpentry, is held by extended engagement (García Márquez, 1981). Close reading, the ear for pattern and rhythm that practical criticism taught generations to train, is sharpened by aesthetic awareness (Richards, 1929). Revision, the writing-as-discovery Joan Didion called "finding out what I think," is carried by adaptive coherence (Didion, 1976). Translation, the labor of moving meaning across a gap between languages that has occupied theory since Schleiermacher's lectures, is reached by expressive influence (Schleiermacher, 2012). Qalam carried revision inside Nadia's commission. Barzakh carried translation. The writer who designs her own dæmons takes up the task the era keeps out of sight, holding her agency in the single medium the machine claims as its native tongue.

*Of the capacities your sentences depend on, which would you hand to a dæmon to guard against the pattern that completes the page before the thought has even formed?*

Copyright sets authorship at the work fixed in a tangible medium, the premise the bibliographic apparatus carries through ISBNs and legal deposit.149 A page whose origin is visible on its face sits at the edge of those instruments. The Coalition for Content Provenance and Authenticity publishes a cryptographic standard for embedding provenance into text and image files. Nadia's open hand assumes the seal can travel with the work, signed once and verifiable wherever the file lands.

*Which provision of copyright would you redraft so a page could announce, on its own face, the hand that wrote it?*

More and more readers are beginning to ask a question that once went unspoken: who wrote these words?

On StoryGraph, users have called for labels and filters that would tell them, before they begin, whether a book was machine-generated. The Authors Guild's Human Authored registry offers another path, allowing readers to verify a work's authorship through a public record.

The demand has emerged before the infrastructure to meet it. In Nadia's collection, the open hand simply brings that decision forward. Before readers encounter the text, they encounter its provenance. Only then do they decide how to read.

*Reaching for a book you own, would you want to look it up in a public registry first to learn whose hand actually wrote it?*

Many MFA programs still teach writing as a private craft and translation as a separate specialty. Mentorship inside the discipline is moving faster than the curriculum. The ALTA Emerging Translator Mentorship Program pairs an experienced literary translator with an emerging one for a year, on a project the mentee selects. The model arrives ahead of the workshops that would teach writers to govern an intermediary that wears their voice. Three new crafts come with the shift. Reading a model's draft for what it borrowed.

Marking the line between her hand and the model's. Writing the rule the work carries through its later lives.

*If you ran a workshop this term, what would it teach a writer about governing the intermediary that wears her voice?*

The reflective arts close Part II as the most volatile of its eight disciplines, the medium agentic systems were built to produce. Across the eight, one figure comes into view. A practitioner meets a pattern that drafts her work before she begins. She answers by designing dæmons that hold a creative capability present where the system offers to perform it. The next move is collective. Part III turns from the practitioner alone to the creative community at institutional scale.

*Looking to the practice nearest your own, where do you already see the collective work your discipline will have to take up?*

# Part III: Creative Complexities in Systemic Change

# Creative Complexities in Systemic Change

## ACI PART III

### 15 Innovators

**Practitioner**
Mateo Reyes, Karina Solis

**Challenge**
Shaping the intermediaries

**Scope**
The coordination platform, and the wearable the community designs for itself

**Design Object**
The coordination platform, and the wearable the community designs for itself

### 16 Educators

**Practitioner**
Renata Cruz

**Challenge**
Growing the person

**Scope**
The curriculum, and the participant—to—teacher pipeline

**Design Object**
The curriculum, and the participant—to—teacher pipeline

**Design Object**
The profession's reluctance

### 17 Catalysts

**Practitioner**
Sofia, with her daemons

**Challenge**
Living the convergence

**Scope**
A day of care

**Design Object**
The blend, held in one person

### 14 Changemakers

**Practitioner**
Rachel Dawn, global;
Lucia Brandt, local

**Challenge**
BetterFutures Model and Preparing the pilot

**Scope**
BetterFutures Vision in a complex STS

**Design Object**
The entity: BetterFutures México, its brand, legal form, funding model, and partners

**Figure P3.1. The terrain of Part III. Source: authors.**

## Scaling Creative Capability into Collective Change

Part I proposed the capabilities creatives can co-develop for the agentic age. Part II carried those capabilities and collaboration with configured dæmons into near-futures of creative disciplines. Part III: Creative Complexities in Systemic Change, gathers both toward collective change. It opens with a question:

*How might our agentic creative intelligence activate leverage points of socio-technical situations to empower practitioners to respond to complexity with impact?*

ACI lets practitioners collaborate across the systems they share: the changemaker curating a better future, the innovator shaping the experience, the educator growing the people, and the catalysts on the ground. Each intervention depends on the others, so dæmons have to augment both the individual practice and the collective effort.

In entangled situations, the wicked problems, AI has to be more than technology. Although collective data and coordinated decisions are a real benefit, there are risks. A mixture-of-experts model, whose protocol routes requests to specialized agents, can obscure the excitations a systemic practitioner needs to curate change: which stakeholders are moving, the forces preventing progress, and the metaphors underlying all actions. Regardless of the model, when autonomous protocols project responses into action, the smoothness at that scale flattens the catalysts necessary for adaptation.

Practitioners in complex environments know that levers move together (Geels, 2004). To map the contours of such terrain, the STEEpL framework keeps any single force from masquerading as the whole scene (Figure P3.2). This ensemble treats the socio-technical system with respect, articulating the interdependencies of tools and people to make or break systemic change (Trist & Bamforth, 1951).

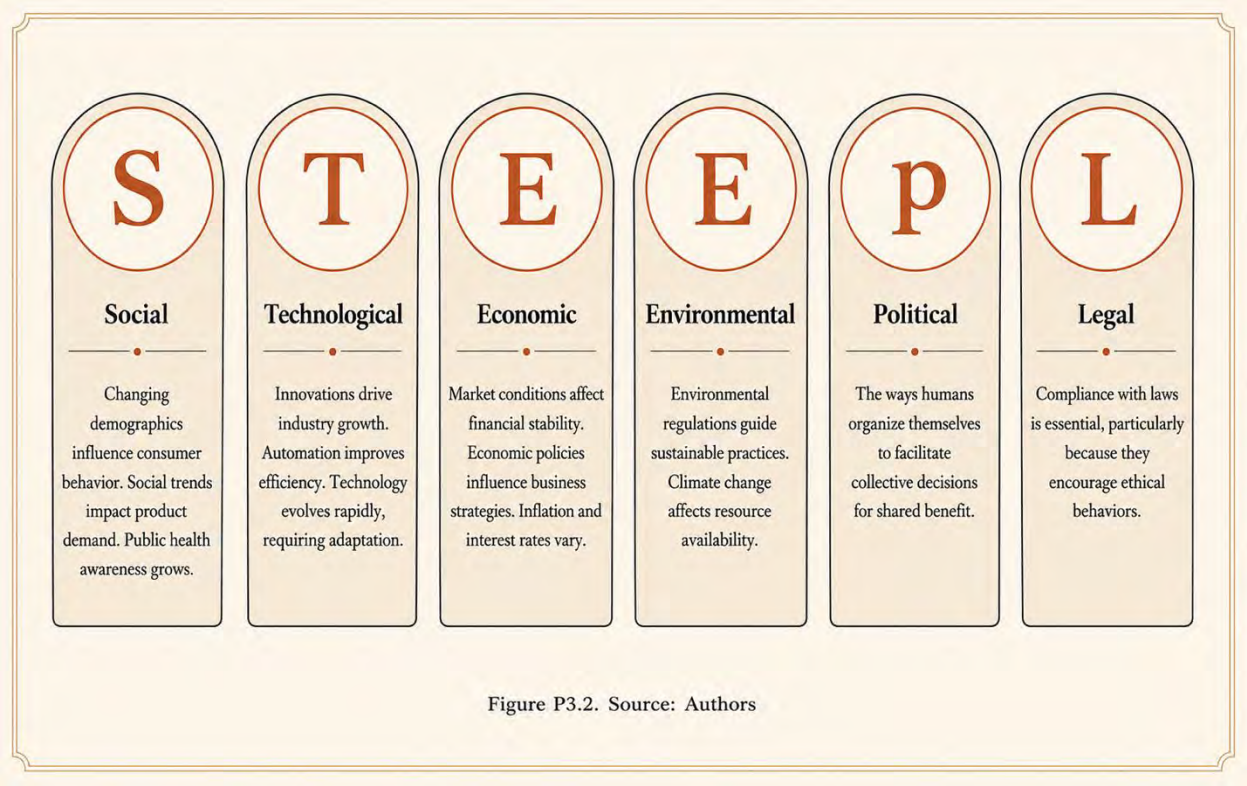


**Figure P3.2. The interdependent socio-technical forces of STEEpL. Source: authors.**

To explore that question, we set a speculative narrative against a real challenge and follow archetypal practitioners through it. We hold the scenario to the same standard of good futures work, plausible, probable, and provocative, grounded in a real place and its constraints so a practitioner on the ground recognizes it rather than dismisses it. Across the chapters, each practitioner reasons aloud with her configured dæmons, and the scenarios test agentic creative intelligence in practice.

## Welcome to Ciudad Nezahualcóyotl

Mexico is aging faster than it is building elderly care. Adults over sixty were six percent of the country in 2010. They will near a quarter by 2050, when the number of elders needing daily care triples toward seven million. The finance secretariat prices the national care system the country has yet to build at over one percent of GDP (Angel, Vega, & López-Ortega, 2017; Zissis, 2025).

The aging challenge is deepest where the money is thinnest. In greater Mexico City, municipalities like Ciudad Nezahualcóyotl depend on the unpaid, untrained, and uncounted care of daughters, nieces, and neighbors. The burden falls on young women. Close to a thousand girls give birth across the country every day, and on these streets a third of school dropouts trace to early motherhood. Fewer than half of women hold or seek paid work (OECD, 2024; World Bank, 2023; IMCO, 2024).

We extrapolate from a real changemaker and a real social-impact project to inform the scenario. Rachel built BetterFutures, a social-impact initiative, on Swiss funding, with a local changemaker who cultivates the ground the vision lands on. A technology innovator shapes the platform and its devices. An educator develops catalysts from the community to deliver a better future for the elderly of Ciudad Nezahualcóyotl.

In PartIII chapters, each practitioner carries one thread of a tangled situation. Each has a story and a stake in the systemic challenge, and each has a design object their craft must deliver. Their materials come from the system itself: a community, its challenges, its institutions, its money, and the lives inside it. Each practitioner designs dæmons for the capabilities their craft leans on hardest. They convene a Creative Council for counsel when the work turns complex, and when the agents and their workflows threaten the humanity the work depends on.

The challenge of home care is clear. Presence, information, and judgment have to meet in one job: the hands that steady an old man to his chair, the instruments that read what the eye misses, and the call that decides whether this morning's swelling is a note in the log or a clinician on the line. An agent that automates presence sells an alarm service and little more. A pattern that skips the devices secures privacy and stays blind. A protocol that orchestrates judgment produces the call that comes too late.

Building resilience in such complexity takes both human and artificial intelligence to make a better future for the community.

We follow Sofia, an archetypal young woman who goes from unrecognized caregiver to catalyst for change, for the community and the cohort alike. BetterFutures trained her as a cuidadora,150 a home-care technician certified under a nurse's supervision. Developing her creative capabilities with ACI, she grows past her own visits into a teacher of the next cohort.151

We open with her walking home late from a day's work.

## Chapter 14: Changemakers: Preparing the Terrain

TAMALES
DE AQUÍ

**Figure 14.1. Sofia's early walk up Avenida Chimalhuacán on the way to a long day. Source: authors.**

### Before the Rounds

Before the light has fully come up, Sofia is already sitting at the kitchen table, checking the morning visit list. There are three homes today. The platform has ordered them by distance, risk score, and last night's notes. Señora Govea is marked "routine" and placed later in the morning. Sofia drags her to the top.

The earpiece rests against her jaw, faintly warm. The voice begins.

*You changed the order again. The optimizer placed her later. You moved her up. Say why.*

Sofia speaks into the quiet kitchen. "Her daughter calls from Texas every Sunday. The daughter can hear when she is worried, but she will not say it herself. If I go too late on Monday, she will sit with that worry in the house all week."

None of this appears in the log. The log records blood pressure, pill boxes, and the last visit time. But Sofia knows the other part: whose daughter calls when, who hides fear inside an "I'm fine," and who needs someone to knock earlier in the day.

At seventeen, Sofia did not yet have that kind of judgment. She had reached her second year of preparatoria when her family needed her at home during the day, and bus fare became one of the first costs they could no longer keep. She left school and found no steady work. Some girls she knew had babies before they had diplomas. It was hard for them to find formal work after that.

Sofia sold phone cases outside the metro. At night, after the house had gone to sleep, she muted her phone and studied English from videos. The screen was small and bright. She did not dare turn on the sound, so she watched mouths, read captions, and repeated each sentence silently.

She lived like that for two years. Then, one Saturday, Doña Lupe knocked on the door. She was the neighbor three doors down, and the person people in the neighborhood went to when something needed solving. She held a flyer with BetterFutures printed across the top.

Sofia's mother asked many questions. Who was behind BetterFutures? Could it help her daughter find work?

Doña Lupe did not answer with a speech. She backed it with her own name. She said she knew the people behind the program and would keep watching it herself. Sofia remembered that Doña Lupe had helped her family through its hardest years. So she signed. Not because she already trusted BetterFutures, but because she trusted the woman standing at the door.

As for her mother's last question, what BetterFutures was, the answer began two years earlier. Back then, it was still only a plan in an apartment in Zurich.

### The Struggles to Start

Two years of refusals thinned even Rachel Dawn's frustration to a flat calm. She read the morning's note, three courteous paragraphs of refusal in the tone the foundations keep for outsiders. Then she closed the screen. "Polite," she said to the empty flat. "They are always polite."

Twenty years of boardrooms taught her to hear the silent no under a polite yes. The piano taught her first. A room that agrees too quickly is a chord with a missing note. The Zurich foundations agreed with her warmly. Their checkbooks stayed closed. The money sat inside a circle she stood outside of. It moved on old relationships. A foreign founder asking for a first real check got courtesy in place of a transfer.

Eco, the dæmon she built to read a room of funders, told her what the politeness meant.

*The politeness is the circle protecting itself. Each quarter the refusal arrives smoother, drafted by the same machine that writes all their letters. Stop asking the club. Ask the one who built it herself from outside, the one who still remembers the closed door. A name on your own list keeps catching your eye. Start there.*

The name belonged to a woman who left a small town in Michoacán young. She carried a family food business onto its first container ship. She grew it into a company that reaches four continents from an office in Zug. Her first ledger hung framed on the wall. For a year she stayed out of Rachel's reach, because Rachel only ever approached her as a logo on a foundation's board.

This time Rachel asked for twenty minutes as one builder to another. She got them. She left the deck closed. She described the challenge instead. Mexico is aging faster than it builds the care for it. What care exists runs on the unpaid hands of daughters, nieces, and neighbors, the same young women the formal economy already wastes. She read it the way the socio-technical scholars read a system, a tangle too dense for any single force to explain or any one profession to hold (Geels, 2004; Trist & Bamforth, 1951). Then she walked the donor across it. The aging numbers and the thousand daily births, three streets apart. The cheap sensors and the insurer learning to price a heartbeat.

Hilo, the dæmon that holds a tangle's many threads as one, cued the move she came to make.

*You have shown her the whole knot. Now show her the one strand that pulls it loose. Two problems, one kitchen, ten thousand times over. Take her there.*

Rachel took her there. Picture a single kitchen in Nezahualcóyotl, she said. A grandmother who needs a hand to her chair. The granddaughter who gives it for free, because the formal economy passes her over. The aging the state struggles to afford and the woman it refuses to employ share one address. Now multiply that kitchen across the municipality, tens of thousands of them, the same two problems meeting at the same table. Train those women. Certify them under nurses. Pay them. One design runs through both problems at once, at the scale of a community. That was the leverage point her craft hunts for, the one small move that shifts a whole system (Meadows, 1999; Lou, Valsecchi, & Diaz, 2013).

Eco read the donor across the table.

*She stopped translating you into her own story three sentences ago. She is in it now. Let her ask the question already forming, and let her own answer close the deal.*

Rachel let the quiet sit. The donor listened the way people listen when a stranger sketches their own life. She left a town like that one. She built the thing that carried her out by hand, from outside every club that counted. She let Rachel reach the leverage point, then stopped her there. “Who keeps the money when it works?” “The community. The women bill, the program reinvests.” “Then you are the first to ask me for a ladder instead of a plaque.”

The yes took a week. It skipped the dinner circuit entirely. The seed money would stand the pilot up. It would rent the borrowed room above a Neza avenue. The donor set one condition: build it there, with the people there, or leave it a deck.

So she structured it for the ground. She built the program on the three moves she trusted to make change last: a shared direction, a design aligned to it, and the collective commitment to hold it (Press et al., 2021; Drath & Palus, 1994). She resolved to assemble it where it would live, small, local, open, and connected, the only form the pilot years let her still believe in (Manzini, 2010).

For a year she ran the program through other people’s eyes, Lucia’s, a community worker’s, a folder of slides. The third seat in her Council, the one she left empty on purpose, filled.

Mirada arrived as the reach she lacked at a distance, the reading of a place no screen gives you.

*You have decided this program from a desk for a year. Decide the rest of it in Neza. Go read the place you keep funding sight unseen.*

She booked the flight that night. She would go to Neza for a week. She would restart the program by pulling the expanded team together. The next step of her vision of better futures for young women in Mexico was about to start.

### The Ground the Vision Lands On

Lucia Vega reached the airport early to meet Rachel Dawn. Their meeting began her two years with BetterFutures.

Lucia worked two worlds and belonged to neither. On the street they read her schooling and her chamber Spanish as a step removed, a daughter of the colonia who studied her way into the state chambers and came back to fight from inside them. For years she represented colonias against the state, winning water connections, land titles, and clinic hours the budgets kept skipping. That record earned her a name in the state chambers in Toluca, where the money gets decided. On the street they read her as a step removed. In the chamber they read her cases as the real thing, a local who fought for locals. The same asset carried in both rooms. People trusted her word, because the record showed she kept it. That trust was what she brought to BetterFutures México. She spent it the slow way, walking the neighborhood door by door.

In her first month a community worker walked Lucia to Doña Lupe's door. Doña Lupe organized the block for as long as the colonia remembered, the school-supply drives, the fight against an eviction the year the water stopped, the funerals nobody else arranged. She knew every door because she knocked on all of them. She heard Lucia out. Then she put the families' fear plainly. Machines that watch. Strangers in the house. A daughter alone in an old man's home. Work the street might read as servant's work wearing a new name. Under all of it, one question.

Raíz, the dæmon Lucia designed to keep her returning to the work, held her to the street's pace.

*Carry the model folded. Let the block tell you which fear bears the load. The order of the telling is theirs.*

Vista, her read on a room, took in the woman across the doorway.

*She is asking one thing. Whether you will still be here when it is hard. Answer that, and the rest follows.*

Lucia answered it the slow way, by coming back. Through that first season Doña Lupe took her doorway by doorway, in the only voice the block would trust, her own. The fears came back down the street as questions. Lucia wrote them in her notebook. She sent them to the team forming around the program: what the machine keeps, who enters the home, what the work is called. Three questions became three answers, one from each craft: a data promise from the maker, a curriculum from the teacher, a profession from the chamber.

Her stake in the pilot cohort was personal. She said so to anyone who asked. Sixteen names in her notebook, picked with Doña Lupe from lists every agency read as risk. Sofia's was third on the page. If the cohort held, the BetterFutures held. If the BetterFutures held, the community would listen.

### A Charter to Start

Lucia met Rachel outside arrivals, beside the line of waiting cars. They hugged quickly, then climbed into the car for the drive east. Rachel shifted in the back seat and laughed. "There are funds that will pay for outcomes," she said. "No fund on earth will pay for a more comfortable seat."

Lucia was born in this city. In the colonia, she knew who to call when the water trucks skipped a street. In a funder's room, she knew which promise would hold. People in both places treated her as one of their own, and that trust was the reason she was here. She kept it current in a notebook filled each season with names, faces, and the small facts that make a community legible.

Lucia talked through the drive east. Rachel listened and watched the new development slide past the window: towers, ring roads, half-built housing climbing the flats. Mirada read the place with her, restoring the depth a screen had flattened for a year.

*The place looks the same. The people have progressed.*

By the time the car reached the co-working space, Rachel knew she would learn more than the time spent pitching.

The team reassembled in the room where they launched the pilot 2 years ago (Figure 14.2). Renata Cruz came from that core team, an educator who teaches confidence the way the theater teaches it, from the body out. Mateo Reyes came too, a maker who learned to code the way he learned to fix a motorcycle, by opening it and reading how the parts wanted to move. Karina Solís came from the colonia itself, a self-taught coder who reads what the women fear in a screen because she came up among them. Doña Lupe took the chair by the door, where she could watch the street and the room at once. The reunion was warm and brief, because one thing on the table could change everything. It was AI.

**Figure 14.2. The Charter Meeting. Source: authors.**

In the years between, the machine arrived wearing the face the women feared, the agent that answers for you, the one that watches. The public clinic down the avenue swapped its nurse line for a voice that answered in place of a person. A mother called about her father's chest. The voice booked her three days out. That story reached every kitchen on the block. Karina heard the same question at every table. Will the machine listen to my daughter, or talk over her. Mateo saw the other half online, demos that shipped clean and hid the mess of how they were built. The team shared the colonia's instinct. Keep the machine away from BetterFutures.

Rachel came at it differently. She spent the past year learning what the machine could do for her own work. Three dæmons taught her: Eco found the donor the club kept from her, Hilo held the tangle when she saw only walls, Mirada sent her to Neza. Each made her do her own work. Each pushed back when she reached for the easy version. That was the line she would draw for the team. The agent the women feared answers for the person. The dæmon makes the person better.

Hilo read the room for Rachel as she finished.

*Mateo will take to it with his hands. Karina distrusts it on the women's behalf, and that distrust is the best reason to give her one. Show, then ask. The charter is theirs to write, or it is your deck again.*

They co-created the charter by hand (Figure 14.3). The terrain went up in six colors on kraft paper: social, technological, economic, environmental, political, legal. They argued it into the three moves Rachel built the program around. Innovation held the direction: the platform, the devices, the model that pays for them. That took the technological and the economic. It went to Mateo, the maker, and to Karina, who took it because she trusts the machine least, the better to keep it honest on the women's behalf. Design aligned the work around that direction: the shape of the ecosystem, the practices that keep every visit inside the law. That took the political and the legal. It went to Lucia. Leadership grew the commitment: the families' trust, the rooms the work lives in. That took the social and the environmental. It went to Renata, who moved the heat card into her own column herself, the May heat that kills elders on the eastern flats. Each kept all six colors in view. Each answered for two.

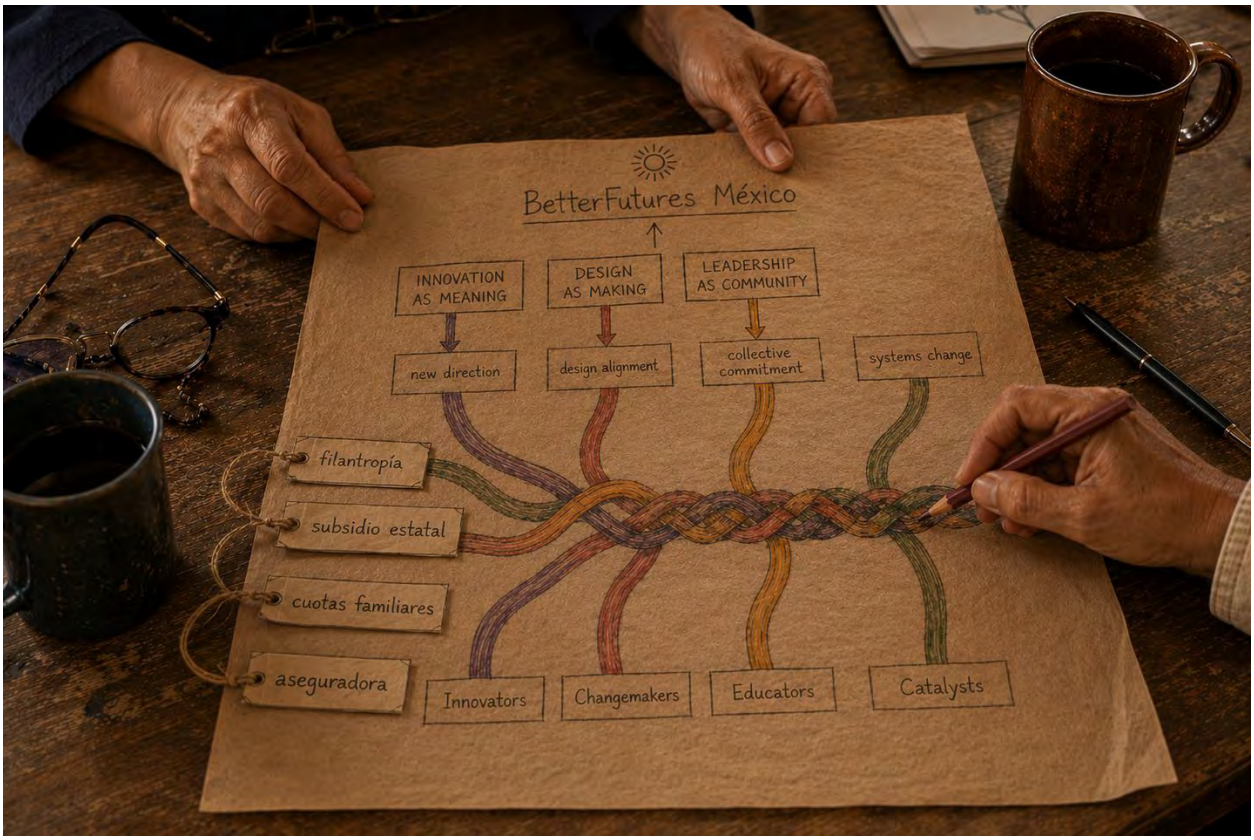


**Figure 14.3. Co-creating BetterFutures Mexico. Source: authors.**

Beneath the cards they wrote the clause that made it a charter instead of a chore list, one line each of them would hold against the next clean demo. The machine drafts, the person decides. It writes the platform's boilerplate, the maker keeps the judgment call. Where it reaches to do the craft itself, it steps back. Doña Lupe watched the arguing from the door, and she approved. A thing argued into being on a wall belongs to the people who argued.

The founder's clock met the street's clock before the meeting closed. "The Swiss money moves if we show momentum," Rachel said. "Momentum means a start date."

Eco pushed for the deadline. *Move now. Warm backers cool.*

Lucia answered before her own dæmon could. "Move now and you move alone. A backer signs a paper. A family hands you its grandmother and its daughter in the same season, and the family is the program. Let the street set the pace, and the money will keep."

Rachel agreed. The agreeing cost her, because the clock is the one thing a founder hates to give up. She handed the pace to Lucia. The charter on the wall held both their names.

A charter held the team. But a neighborhood needed a coalition. Rachel knew the difference from the pilot, where a perfect plan died for want of the people to carry it. The legal shell was the easy part. BetterFutures México would hold the mission, the donor money, and the contracts.

The hard part was getting the right people into the same room.

Who paid for care would decide whether the program could survive. In Mexico, private insurance covers very few people, and public clinics usually stop at the doorstep. The rest falls to families: households with money pay in cash. Households without it pay with a daughter's time. So the funding had to come from four places: philanthropy to launch each cohort, family fees on a sliding scale, the state subsidy Lucia would win, and outcome payments from an insurer for stable care results. That mix would have to be renegotiated every year.

Rachel convened the partners the program would lean on: Doña Lupe, the anchor who carried the families, the clinic whose nurses would supervise, and the secretariat with subsidy money to place.

Hilo gave Rachel the coaching to give.

*You won your donor by handing her a leverage point. Hand Lucia the room the same way. Your work today is to make yourself unnecessary in it.*

Lucia did not begin with the ask. She first let each partner name what it was afraid to lose. The clinic feared crossing the boundary of nursing work. Doña Lupe feared the neighborhood would be used as a logo. The families feared the machine would speak over them. Raíz reminded her to keep that order.

*Let them speak their fear first. The clinic fears the scope line. The anchor fears being a logo. The families fear the machine. A coalition forms when each one hears its own fear in the room before it hears the ask.*

That afternoon, they first decided what the program would be called in the neighborhood. The contracts could say BetterFutures, but when the team knocked on residents' doors, they could not arrive with only a foreign name. The nurses, Doña Lupe, and two mothers argued for a long time before landing on De Aquí, meaning "from here." A girl from the colonia sketched the logo on the back of the agenda: a doorway, with a hand at the threshold. That name and that mark would go on the doors, the forms, and the training materials.

Once the name was set, Lucia turned to Mateo and Renata and began preparing for the first cohort. Mateo would lead the tools, but the tools could not speak for the learners. They could only remind, organize, and support judgment. Renata would lead the curriculum, but the curriculum could not simply be tidy and easy to review. It had to train the learners to read the situation inside a stranger's home, hold themselves steady, and speak in their own words.

Sofia's name was third on Lucia's list. By the time the knock came to her door, the program behind it carried a name the street gave it, a coalition that would catch her, and a team that knew what its tools were for.

### The Chamber and the Name

The public leg ran through the State of México. The state ran through a chamber in Toluca, a dawn bus away. Shared financing meant the women's secretariat would co-fund placements and pay a subsidy for young mothers getting certified. Winning it meant standing on cold tile, behind a microphone, arguing in a grammar foreign to the kitchen. She promised the coalition she would win this leg herself. If she came back empty, the braid lost a strand, and she lost the word that held the other three together.

The work pulled at the capability Lucia leaned on least. Her Council found its missing voice.

The tooling offered to write the speech for her, a policy address, clean and citation-perfect, drafted in the time it took to board the bus. She left it unopened. A representative who lets the machine speak for her in Toluca has no business asking a cuidadora to find her own words. Voz arrived as her expressive influence, the craft of finding the women's case in the chamber's own words and saying it in her own voice.

*Say it once in the mother's words and once in the policy's, and let the room recognize the same woman in both. One picture too vivid to average, in language too plain to ignore. You have written this woman a hundred times in your notebook. Write her again tonight, in your own hand. Read her to the room.*

Lucia built the case around a real woman, a real bus, a subsidy with a face. Doña Lupe stopped her at the name. "Her husband reads the bulletins. You will win your budget and cost her the house." "Without the name it is an average," Lucia said, "and averages lose votes." "Then carry her without the name. You are good enough with words." Lucia rebuilt the speech overnight, by hand, angrier and better for the constraint.

Her name was called third. She laid a photograph on the document camera, a kitchen in Neza, a pill organizer open beside a mug. "The secretariat counts the women of this municipality outside formal work in the tens of thousands," she said, in policy's grammar. "It counts the elders needing daily care in the tens of thousands too, and the two counts share addresses." Then she switched. "One of them needs the early morning bus to reach a client's door, someone to watch her son, and a certification the clinics will honor."

A deputy cut in before the undersecretary could. "We were pitched pre-screened caregivers last quarter, ready in six weeks, at a third the cost. Why fund a slow program when a fast one exists?" Lucia let the question sit. "Because the fast one graduates a stranger a family lets go in a month, and you pay that placement again. Ours stay. We called the program De Aquí, from here, because the woman at the door grew up three streets from it. A neighbor stays. A vendor's hire moves on."

The undersecretary stopped reading ahead. "And what does the state buy," she asked, "for the subsidy?" "Hospital nights it never pays for," Lucia said. "A worker who stays is a taxpayer who stays, and an elder cared for at home is a bed the system keeps." The chamber heard a constituency it could serve and a cost it could share. The secretariat committed a referral pipeline and the subsidy for the year. The public leg of the funding model stood for now.[152]

Lucia rode the dawn bus home with the subsidy in hand. She kept her word to the coalition. She kept the photographed woman's name off every page. The braid carried all four strands for one more year.

### What the Terrain Hands Forward

The changemakers' Council met the night the terms closed, Rachel's dæmons and Lucia's, turning the founding over until each woman could state what she would carry out the door.

Raíz felt the cold tile still in her knees: *What kept you returning to the table on the dawns the work looked lost?*

Lucia felt for it at the wall of cards. She built this on her word, and her word held. The exhaustion was real, the dawn buses and the nights rebuilding a speech by hand. What stayed was something else. Paper could be defended. A coalition could be rebuilt. The one thing beyond her to manufacture was trust, and trust was the thing she brought. She gave it to the partners tonight. Tomorrow she would give the rest to other hands. Ten years in these rooms, and she felt the lesson settle in her chest: a program is only as strong as the people who own it.

Voz heard the room hush when she said De Aquí aloud: *Which word made the partners stop and listen?*

De Aquí, she knew. From here. The word the street chose. She kept the woman in the photograph out of the ledger and put the colonia's word on the page instead, so the program stayed theirs before it was the funder's. Her fear was quieter than failure. The program could succeed and still drift, the machine creeping in one clean demo at a time, the families' fear proven right the slow way. The word was her guard against it. She would say it in every room where the money decided.

Hilo turned the term sheet's weight in her hand: *What did you weave that no single partner could hold alone?*

Rachel held the question. Paper holds a program the way a trellis holds a vine. What grows on it gets decided at a build table in Neza, beside a handset turned face down. Keep the tool theirs, she understood, and the trellis stays a trellis instead of a cage. The makers would build the tool. The teachers would grow the women. She carried the vision as far as one person can. The block would decide what it became.

Mirada watched the term sheet still warm under the lamp: *What did you see in the place that told you it would hold once you stepped back?*

Rachel saw it plainly. The program would run on Lucia's word and the makers' hands. She flew out at the end of the week and read the place in person, and the reading held. The founder makes herself unnecessary. The leaving was the proof.

Two years downstream of all that paper, Sofia checks the platform's morning list one last time. Three doors, all of them hers. She was the third name on a list once, a girl a chamber argued about in a grammar new to her. The tool that will route her tomorrow began at a

folding table in Neza, beside a handset turned face down. The makers could have let the machine build it in an afternoon. They built it by hand, and that is why the tool stays hers.

*Sleep,* her dæmon says. *The why is safe where you keep it.*

Sofia sets the earpiece in its dish. The voice goes quiet. The room is only a room, and she sleeps.

# Chapter 15: Innovators: Augmenting a Maker Community

## Starting a Skunkworks

Mateo Reyes's job was to build one thing: the device a cuidadora carries into a client's home. It does three jobs at once. It lets her be there in person. It reads what the eye misses, a fever climbing, a pulse gone ragged, a sugar crashing. And it helps her decide when a swollen ankle is a note for the log and when it is a call to a nurse. The charter set the limit. The device supports the cuidadora's judgment. The cuidadora makes the call.

Mateo could have taken the fast way. The AI on his screen could draft the whole platform in an afternoon, cleaner than the pilot's, and most weeks it offered to. He refused it. A tool the machine builds for the colonia is a black box, and the program runs on cheap secondhand phones that break and power that flickers. When the device fails in a client's kitchen, someone local has to open it and fix it that day. A tool the colonia builds is a tool the colonia can keep.

His own instinct was a maker's: build it himself, in a quiet room, and bring it out finished. Mano, the dæmon Mateo built to keep him the author of the build, turned that instinct around.

*You reach for the quiet room because that is where your hands are surest. Reach the other way. Build a bench. Let the product come off it. A tool wants one author. A trade wants many. Give the street a trade.*

Hecho
a
Mano

**Figure 15.1. The skunkworks at the build table in Neza. Source: authors.**

So he went looking for the makers, and found what was left of them. The colonia always made things and fixed things, soldering, stitching, and coding at the edges of the formal economy. The AI thinned that. Why learn to build when a model builds it in seconds and better. Beto, who once brought dead handsets back to life at a stall by the market, now mostly resold them. The prepa robotics club met to watch the AI solder what they used to dream of building. The few who came to Mateo's first table came wary of working with the thing that hollowed their craft.

Mateo skipped arguing the machine down. He and Karina showed them instead. Karina was the bridge, a coder who came up among them and read exactly what they feared. She sat Beto at the keyboard. He was the one who quit fixing handsets when the model made repair feel pointless. She gave him the AI and one rule. Describe what the tool should do. Let the model draft it. Then read every line against the one question only Beto could answer: did it serve the trade, or did it only look clean. Beto built something he knew in his hands, a tracker for which secondhand phones held a charge and which to strip for parts. He talked, the model drafted, and he overruled it whenever it guessed like a stranger to the work. An hour in he looked up surprised, the way a hand is surprised to remember a skill.

Mano read the moment for Mateo.

*Watch his hands, past the screen. They knew before he did. Feel the warmth come back into the fingers when the route falls into place. The machine misses that. It is the whole reason he stays.*

The machine drafted. Beto judged. It handed him back the judgment only someone who knew these phones could make.

Used one way, the AI does the talking and leaves the maker hollow. Used the other, it keeps him talking and sharpens the read that tells a working tool from a clean-looking one. The makers feared the first. Mateo and Karina offered the second, and the room watched it work.

With the room won over, Mateo turned to the money, two facts the funders missed. First, the street trusts a tool its own people built and routes around one shipped in. The makers owned the one asset a vendor lacked, the trust of the block. Second, the thing they built here, they would own. A platform coded and owned in the colonia could be a business, a trade they billed for, with the program as its first client and the next colonia as its second. That was the leverage point his craft looks for. Teach a handful of makers in one place, and the tool they build gets rebuilt in the next, each colonia by its own hands. A bench in one room, and a trade spreads across a city.

By the next week the bench ran out of chairs. The room smelled of hot solder. Karina took the lead. Beto came back for the hardware. The robotics pair came for the parts they could solder and the parts they only dreamed of. A woman who repaired the neighborhood's appliances came because Doña Lupe asked, and stayed because the program paid and the tool would end up in homes on her own street. Mateo settled into his father's old role at the

repair counter, the one who listens to the owner describe the noise and hands back the right tool.

Tino, the dæmon Karina built to read the colonia, saw what the room still lacked.

*Look at their hands, idle in their pockets, makers with nothing to make. They came as people the machine made spare. Let them leave as something. A crew the street can name is a crew the street will trust. A name they choose themselves is the first thing they make together. Ask them what to call this.*

They argued it the way the coalition argued De Aquí, and landed on Hecho a Mano, made by hand: the machine a tool in the hand, and the hand still the author. Beto painted it on the door of the co-working space by the end of the week.

The rule Mateo set was the charter's, turned toward makers. Build with the AI and stay its author. The model drafts. The maker decides. A dæmon watches for the moment a maker stops deciding and starts accepting whatever the screen proposes. The capacity at stake was the maker's authorship, the call that keeps a tool serving the grandmother instead of the dashboard. A dæmon grows it. The machine, left alone, takes it. It is one protocol at every level. The auto-build that would draft the platform for Mateo is the auto-script that would later speak for a cuidadora at a stranger's door. He walked the makers through Mano and Rumbo, and sat with each one while they designed their own.

The women the tool was for drifted in to watch it take shape. Sofia sat at the edge of the table, her handset turned face down, the way she turned it the day Doña Lupe knocked. She watched the makers argue over what the screen should ask a frightened grandmother. The screen could ask. Only the cuidadora could answer: stay in the room when the woman turned afraid, or call the nurse when a swollen ankle stopped being just a swollen ankle. That answer would be Sofia's. Over the next year she would learn she could give it.

Karina felt the risk in her stomach. If the skunkworks faltered, the program would buy a platform built and owned by strangers. The colonia's part in it would end. If the data promise broke, it broke under her name, on the streets where her mother still bought bread. Mateo carried the technical risk. Karina carried the one that followed her home. The crew held a name, a trade, and a discipline. Next came the platform itself.

### Sprint One: Coding the Platform

Before the first sprint, Karina and Mateo settled the scope. The fund's budget assumed a model smart home, a demonstration unit a visitor could admire. Mateo sent it back annotated. The platform would run on the glass the women already owned, in the homes the clients already lived in. Karina put it shorter. A smart home impresses a funder. A handset goes home with her.

So the sprint opened on a plain goal: a scheduler the women could run on their own phones. The crew took their places at the build table, the model live on every screen. Mateo set the discipline from the demo. The model drafts. The maker reads every line against the one question only a person asks, does this serve the cuidadora and the elder at once. Beto

took the scheduling view, describing the route he walked in his head and correcting the model until the screen matched it.

Each maker designed a dæmon to guard that seam, the moment the model tempts the author to vanish. Tino read the build the way Karina read a room.

*The model gives you the average user. The average user is a stranger to this street. It writes a form a clerk loves and a grandmother's hand stalls over. Make it ask the question the street asks. Keep asking until the screen answers in the street's own voice.*

The platform started as a scheduler. The model's first draft ranked the cohort the way the industry ranks, response time, ratings, visits per shift. Karina watched two sessions and closed her screen. "Take the stars off." Mateo kept his eyes on the build. "Families want to choose. Choice needs information." "That is a leaderboard, and I grew up under one. The carriers rank the prepaid customers too, and my mother still pays more for less." "A family is handing over its grandmother. You would give them nothing?" "I would give them what the street gives. Who came when it rained. Who came back." They argued it honestly for a day, the makers' one real fight of the sprint. They settled it the way they settled everything: did the platform serve the grandmother or the dashboard. The platform would match for continuity, the same cuidadora back to the same doors, week after week, until the families trusted her. The relationship set the schedule. The stars stayed out of the build.

The platform learned as much as it taught. When a cuidadora moved a visit the optimizer placed wrong, or rewrote an alert the model phrased coldly, the correction went back into the system as a lesson under her name. The first real win came from a pilot cuidadora. She caught the optimizer pushing a frail client into the worst of the afternoon heat. She moved the visit to first light and wrote the reason in her own words. The next week the schedule offered the change before she asked. The platform was learning the women's judgment. It kept their names on it. Sofia watched the correction land from the edge of the table. The machine argued. A woman from her own street won.

The hardest part of the sprint was teaching the platform to forget. The women asked what the machine kept. Whether a stranger, a company, someone in an office across the border, would hear the visits where they stumbled and store them. The clients' daughters asked the sharper version. Did a device in a grandmother's house mean a microphone in a grandmother's house. The fear was earned. The industry's care products did record, did score, did ship an old man's bad nights to a dashboard kept out of his family's reach.

Rumbo set itself against the easy build.

*The clean build records everything to improve the service later. It also keeps a family's worst nights somewhere only a company can open. Build the tool that forgets on purpose. Where the law makes it remember, let the family hold the key.*

So the skunkworks split the record. The clinical minimum the regulation requires, the visit log, the vitals, the escalations, stays signed and auditable in the supervising nurse's file. Everything else, the practice runs, the stumbles, the small talk a sensor overhears, erases when the screen closes. A family reads everything the program keeps about them, in a

register a block agent can walk them through. The women drew the line. The families ratified it. Mateo built the consent into the architecture itself, where fine print would have buried it. This was the promise Lucia won at the term sheet. The data would stay in the colonia, held now by the architecture, beyond the reach of a clause a renewal could soften.

Word moved faster than the build. The street heard the daughters trained on a machine that listens. It heard the grandmothers would wear one that watches. Doña Lupe called the families to a borrowed hall, brought her own word, and left the slides behind. Pregón, the voice Karina configured on Doña Lupe's own phone, off any record and hers to silence, gave her the order of it beforehand.

*First the daughter, then the machine. The girl does her own talking. The machine forgets the practice when she closes the screen. Say what you saw with your own eyes, because your eyes are the only proof this room will take.*

Doña Lupe said it her way. A mother near the back asked the question under all the questions. "When the machine hears my mother fall, who else hears it?" "The girl who comes," Doña Lupe said. "The nurse, if the girl calls her. Nobody else. I asked for the list of who can listen, and it is one name long." The hall gave its yes. The mother who asked sent her daughter the next morning.

The platform reached the elder only when the woman was there. The hours she was away needed something else, a device on the body. The device the market sold was the one the colonia already refused.

### Sprint Two: Prototyping the Milagrito

The market's answer to the missing hours sat on Mateo's table: three alarm pendants from the insurer's line, gray plastic, a red button, a lanyard that says patient. The oldest woman in the room pushed hers away with one finger. A pendant she pushes away ends in a kitchen drawer. The night she goes down between the bed and the door, the drawer stays shut.

Rumbo read the gesture against the spec.

*The spec calls this solved. The finger calls it refused. A device catches a fall only once she agrees to wear it. The spec measures everything except that agreement. That field is the whole problem. Design for the chest it will live on before the requirement it will pass.*

Mateo stopped designing the device. He convened a making, the thing the skunkworks was built for. Hecho a Mano set the table. Around it sat a dozen clients and their daughters, Doña Lupe, Beto with his soldering iron, the robotics pair. On the table lay a tray from the stalls by the parish: brass milagros, woven escapularios, rebozo cloth, wax thread, embroidery hoops. Beside them lay the makers' work: foam blanks the size of a ten-peso coin. Each held the sensor the robotics pair built and the firmware they wrote across the week, a thing that hears a fall, reads warmth, carries one pressable switch.

The robotics pair wanted to chase the spec's accuracy, more sensors, more certainty. The dæmon one of them designed for the body's trust held them back.

*More sensors buy a higher number, falls caught per hundred. They also buy a device too clever to trust. The grandmother keeps the simple thing she can feel on her skin. The one she takes off to feel normal protects an empty drawer.*

So they built one quiet sensor and one switch. They left room for the form the table would give it.

Mateo asked one question. He held the silence after it. What would your mother wear that says someone is near, in a language this street already speaks. His own first mock went around the table, a foam wristband, sensible and washable. The oldest woman returned it with two fingers, the way she returned the pendant. A watch that watches, she said.

Mateo reached to defend the wristband. Tino stopped him.

*Watch their hands more than your drawing. They are sorting the milagros by who they lost. That is the design language, memory worn at the chest, a language this street spoke for centuries. Stop proposing. Let the table hand you the form.*

He put his pencil down. The platform grew the maker's authorship. The device asked something else, the looking that lets the body hand you the form. The elders sorted the milagros the way they sorted memory, the heart for a husband, the legs for a knee, the lungs for the cough that took a sister. A daughter turned an escapulario over twice. Sofia threaded a foam blank behind an embroidered heart. She drew the cord through. She held it to her own chest, the rebozo soft against her neck. The table went quiet. That was the form, found in the hands of the girl who would one day answer the call it sent. A small woven scapular worn at the chest, the sensor sewn behind the embroidered heart, the press of the heart a call to the cuidadora (Figure 15.2). It charges in a clay saucer by the bed, beside the santos.

**Figure 15.2. The Milagrito and the platform behind it. Source: authors.**

Mano read the finished object as it sat in the maker's hands.

*The electronics are the makers'. The form is the table's. Keep the seam there. The next colonia rebuilds the device with its own saints and the same sensor underneath. You built a product once. This time build a pattern.*

The oldest woman pressed the heart to her chest and named it before anyone asked. Mi milagrito. The name outran every label the table tried. The program kept it. A red button says emergency. A milagro says someone prays you stay. The same electronics rode in both, and only one got worn. That was the lesson of the sprint. Beneath the cloth, the sensor and its firmware became the first module of a stack the skunkworks could hand to the next project. The hardware stayed open. The form was left for each community to make its own. Doña Lupe wore the first one off the table. The block read it on her collarbone for a week, before the first client asked for hers.

### Sprint Three: Tested in Homes, Made Personal

The makers sent the Milagrito home with a dozen families. The first week brought false alarms. A grandchild pressed the heart to hear what it did. A kneel at evening prayer read as a fall, twice, in the same devout house.

The easy fix was the industry's: raise the threshold until the device stops crying wolf, and accept the falls it would then miss. Rumbo heard the fall it would then miss, and refused it.

*Bend the device to the house. Asking the house to bend to the device is how the industry misses the fall that matters. The kneeling woman is a life. You are about to log her as noise. Sit with her, learn what her evening looks like, and teach the sensor her prayers. Every life you flatten to a threshold is a fall you will miss in the one house that breaks the average.*

So the cohort sat with the houses, one evening at a time. In the devout house, two cuidadoras stayed through the rosary while the sensor watched a grandmother kneel, rise, and kneel again. It learned the shape of her prayers. The kneeling stopped reading as a fall. The makers tuned the thresholds to the lives instead of asking the lives to flatten. By the third week the Milagrito knew the difference between worship and falling. The houses knew the girls who came when it was wrong.

The thresholds fit the device to each house. The kitchen words fit a dæmon to each woman. Each one configured her own at the build table, with Karina, in the language she actually spoke. Karina ran it as three questions and a silence. What should it guard? What must it refuse? And how will you know the judgment stayed yours?

Sofia's session came the week after the workshop. The handset lay face up on the table between them. It stayed that way. She turned it face down the day Doña Lupe knocked. She turned it face down again at the build table, the day she came to watch the makers. This time she left it looking up at her. What should it guard, Karina asked. The breath before the knock, Sofia said. What must it refuse. Speaking to the client for me. And how will you know the judgment is yours alone. Sofia thought a long moment. "If I can explain the visit to the family with the screen dark." Karina built it the way Sofia said it. Sofia named it Aliento, for the breath. The platform grew the maker's authorship. The device sharpened how the maker saw. Aliento would grow Sofia's judgment, the part that stays hers when the screen offers to speak for her. She turned the handset over in her hands a long moment before she gave it back.

Mano read the prototype by the weight of it in his hands. It carried the line the colonia teaches its makers.

*Se hace camino al andar. The path is made by walking it. When she freezes at a door, the screen offers three scripts. Her thumb takes the safe one. Give her one cue and a held silence. The silence is the road. She walks it in her own words, or the judgment stays the tool's.*

Mateo cut the three scripts to one cue and a held silence.

What the three sprints left behind was a stack. The stack was the point. The platform that matched for continuity, the Milagrito that forgot on purpose, the dæmon a woman configured in her own kitchen words, all of it ran on cheap glass and open hardware, the names on every lesson, the data in the houses. A second BetterFutures project, in another city or another trade, could take the stack and rebuild the form with its own community, its own saints, its own kitchen words. The deepest reusable thing was the skunkworks itself, a way for a community to make its own tools and keep the making. Hecho a Mano became a workshop with a waiting list.

The stack was built. The cohort's tools were theirs. Two rooms still waited on it: the funders who held the next cohort's money, and the partners who held the sign-off. Rachel and Lucia were already choosing what to show them.

### Approval, and What the Build Hands Forward

Three sprints in, the stack was real. Now it faced its sign-offs: the nurses who would supervise, the clinic whose system it fed, the insurer who would pay against it, the vendor circling to buy it. Mateo took their requirements as constraints from the start. He brought the built thing back to be approved against them.

The profession met the build first, where the press of a milagro turns into a clinic's decision. The supervising nurses wanted every call to a clinic to pass through them. They were right to want it.

Rumbo held the build to that right.

*The temptation is to route around the nurse, to let the device call the clinic and shave a step. The step is the profession. Build the ladder she draws, over the one that flatters your speed. A faster alert that crosses her scope is an alert the clinic learns to ignore.*

Mateo built the ladder the way the nurses drew it. The Milagrito speaks to the cuidadora, the cuidadora to the nurse line, the nurse to the clinic. Every escalation lands as a structured note, signed by the woman who made the call, countersigned by the nurse who carried it.

The clinic's system wanted its own forms. Tino read the seam.

*Their server rejects anything that looks foreign. Speak its dialect. Shape your note like the form already on their desk. The downtown system will think it wrote it.*

Karina shaped the note like the form. The server bounced her first uploads over a date field. She fixed the field. The template passed downtown as the clinic's own paperwork. The nurses signed.

Then the pressure came, from the two directions Mateo braced for. The insurer's integration team sent a spec requesting the very streams the contract renounced, dressed as care optimization. The vendor came back as a buyer. It offered to license the teach-back loop across forty cities, a check that would carry the tooling budget for two years.

Mateo read the offer twice. Rumbo found the clause inside the compliment.

*Read the line under the praise. The corrections become their product, anonymous at scale, the names sanded off. The thing you built to keep her name on her work becomes the thing that strips it. The budget is real. So is the theft.*

He sent back one change, that a lesson travels with its author's name and her right to it. The vendor's counsel called the clause unworkable. Mateo called it the product. The check went elsewhere. The loop stayed the women's. The platform ran on repaired secondhand handsets for the next two years.

Karina's Council gained its missing voice over the integration spec. Nudo arrived as the steadiness she leaned on least, the part that holds a build to its purpose when the room and the money lean the other way.

*The spec returns every renewal wearing better manners. Every version looks like kindness. Answer it the same way every time. Reports go out, streams stay home. Write the refusal into the architecture, so the program keeps its spine when the makers change. The day the data leaves the colonia, the colonia stops talking to the data.*

Karina carried the answer into the renewal meeting herself, the split record on one screen and the report pack on the other. "Every report you need. None of the nights you want." The renewal held. The sign-off the sprints were built for kept everything the colonia needed.

The proof past arguing came six weeks in. A widow three blocks off the avenue wore the milagro at her chest, where the gray pendant would have sat in a drawer. The night she went down between the bed and the door, the heart on her chest heard the fall. It called the duty pair, a cuidadora and the nurse beside her. They reached the house in nine minutes. The cuidadora knelt where the training said to kneel. She held the widow's hand and kept her talking, her name, her saint, the year she came to Neza. The nurse read the hip. The clinic line triaged. The handoff landed as a note shaped like the clinic's own form.

The daughter called that evening. She asked for the girl by name, the one who held her mother's hand from the fall to the handoff and stayed a full hour after the work was done. She asked whether the same girl could come on Thursdays. The platform was built to bring the same cuidadora back to the same door. Now a family asked for it by name.

The seat Mateo left open filled. Trato arrived as the warmth a build can carry and only a person can make, the thing a daughter hears in a cuidadora's voice and decides, in a breath, whether to trust her mother to it.

*Your device heard the fall. Your platform routed the call. The daughter is staying for the hour the girl sat after the work, the hand she kept holding. Your build carries no field for that. Leave the field empty, and leave it room.*

Mateo knew this from his father's bench. There he could bring a dead radio back to life, hand it to its owner, and watch the face light up. The radio worked. The song coming out of it was sung by someone in a studio neither of them would ever see. He built the device that hears a fall and the platform that routes the call. The hour the girl sat after the work, the hand she kept holding, the voice the daughter trusted in a breath, none of that fit a field. He stopped trying to build one. He left it for a person to bring.

That evening the makers' Council read the build one last time, turning the first catch over until each could say what they would carry to the next colonia (Figure 15.3).

**Figure 15.3. The makers' Creative Council after the first catch. Source: authors.**

Nudo read the refused spec still open on Karina's screen.

*What held the build to its purpose when the money leaned the other way?*

Karina knew it was the architecture, not her nerve. The refusal lived in the structure, so the program kept its spine when the makers changed. She would carry that into every renewal the new cities signed.

Mano read the work through Mateo's hands.

*What stayed in your hands when the noise stopped?*

What stayed was the question he carried room to room, to the actuary and the nurse and the mothers. The platform and the milagro were only where it landed. The next colonia would need the question more than the gadget.

Trato heard the daughter ask for the girl by name.

*Why did the daughter's call sit wrong with you?*

Because the build carried no field for the hour the girl sat after the work, the hand she kept holding. Mateo left the field empty on purpose, and left it room. That was the line where a maker's work ends and a person's begins.

Aliento caught the breath Sofia held at the edge of the table.

*And the part the tool leaves to you?*

That part was hers, Sofia knew, and the learning was still ahead of her. The tool was the easy half. The hard half started in the morning.

Sofia walked home with Aliento configured and the platform in her pocket, a maker of one small piece of both. The tool was hers now. Holding a visit was the lesson she still lacked. The next morning she climbed to the room above the avenue, where a teacher who learned her craft from the body out was setting twelve chairs in a tight ring. Sofia took the one nearest the door

## Chapter 16: Educators: Growing the Person

ACUERDOS
ESCUCHAR / CON RESPETO
CONFIDENCIALIDAD
UNA VOZ / A LA VEZ
DESDE NUESTRA EXPERIENCIA
CUIDAR EL ESPACIO
CUIDARNOS ENTRE TODAS
CAMBIO REAL —
escuchar, confianza, comunidad
DE AQUÍ SURGEN GRANDES CAMBIOS

**Figure 16.1. Setting the ring before the cohort arrives. Source: authors.**

### Session One: The Circle

Renata Cruz set twelve chairs in a ring tight enough that a knee almost touched a knee, and the morning's first lesson was the chairs (**Figure 16.1**). She learned long ago that bodies in a small ring bond faster than words, and that a wide ring teaches people to hide. She learned it in rehearsal rooms, fifteen years of applied theater in São Paulo and across borders, church halls, borrowed classrooms, once a flat roof where the only stage was a chalk square. Her craft starts from the body and adds the line second. A person grows confidence the way an actor grows a role, through repetition in front of someone who watches, and the watching is the part a script leaves out.

Renata built the cohort's training in three sessions. First came the circle, where a woman learned to sit with a frightened elder and let her own face be read. Then came the simulated kitchen, where she learned to decide the care with a dæmon at her ear. Last came the real round, where she worked beside a nurse and the block. Each session would end with reflection before the women went home. The curriculum itself was the object under review, and Renata built it the way she set the chairs: close enough that no one could stand outside the work.

The risk came in with the chairs. The program's review would judge her curriculum by completion numbers, and the state nurses' college was watching to decide what these women would be allowed to become.

The twelve came because the program cleared the path to the door. A crèche ran in the next room, where a niece of Doña Lupe's minded the babies while their mothers worked. A small stipend covered the bus fare and the lost selling hours. Renata learned in the pilot years that confidence is taught on a full stomach and a quiet mind, and that a curriculum which skips the childcare and the fare teaches only the women who could already afford to attend.

Sofia took the chair nearest the door. Renata recognized her from the build-table season, nineteen, from three streets over, two younger siblings at home, a careful self-taught English, the one who turned the handset face down and named the breath it would guard.

Ojo, the dæmon Renata designed to read a room, took the circle in before she spoke.

*Count the chairs that face inward and the ones angled toward the exit. The girl by the door rehearses her hello under her breath. The fluency is there. The permission is locked somewhere else.*

Renata began with a game: a name passed around the ring with a gesture. The women laughed at the gestures. The laughter dropped their shoulders, and that was what Renata wanted first. Then she named the work. At the end of their training, they would sit with an old man who forgot the year, an old woman frightened by her own hands, someone losing body and memory at once. Comfort begins with a face that stays, she told them. They would learn that face here, with each other, before anyone needed it in a kitchen.

She moved them into pairs. One woman played herself. The other played an elder who lost the thread of the afternoon and turned afraid. The caregiver's task was small and enormous, to hold the look and say one true sentence that let the fear settle.

Sofia faced a woman from two streets over, who let her shoulders curl and her eyes go searching, and Sofia looked at the floor between them.

Renata crouched beside the pair. "She is frightened, and she is looking for your eyes," she said, gentle. "Stay with her. Look at her."

Sofia lifted her eyes and said the woman's name. "I'm here," she added. The woman playing the elder stopped searching the room and looked back at her.

The training screen offered an opening exercise, a training pattern that Renata knew would miss the concerns in the room, a guided sequence that would carry the first hour while the women read comfort off the screen. The screen lit with the suggestion. Renata closed it. The struggle of the first true sentence was the lesson, a script would speak it for them, and an elder knows the difference between a read line and a met eye.

Puente, the dæmon she designed for the craft of connection, caught what the closed screen protected and carried the line Renata learned in Brazil.

*De grão em grão, a galinha enche o papo (*Grain by grain the hen fills her belly). The rough sentence she finds on her own is the grain. The clean screen makes a smooth first hour and leaves the cohort exactly where it sat. The roughness is the confidence forming.

By the end of the morning the circle talked across itself. The women traded the elders they already tended, a grandmother, a father's mother, a neighbor only she looked in on, and the trading was the first care they gave. Sofia spoke twice on her own.

Renata closed the session the way she would close each one, with her dæmons, the room emptied and the chairs still in their ring.

*Read the proportion,* Ojo said. *This morning was mostly fear, with a thread of something else starting to run under it. The girl by the door lifted her eyes once and spoke twice. Mark the thread. It is the whole crop.*

*And keep the next session honest,* Puente said. *The circle gave her a face that stays. The kitchen will hand her a harder room, a dæmon at her ear and a decision to make about another person's body. She is ready for the grain of it. Hold the screen closed there too.*

Renata walked home holding the proportion of the room, mostly fear with a thread of something else, the first session's work done. The circle taught the face that stays. The simulated kitchen would ask what to do with it.

### Session Two: The Simulated Kitchen

The second session moved the women out of the circle and into the kitchen. The program had built it to feel like the rooms they would soon enter: a table, two chairs, a cup trembling in its saucer, and Mateo's engine running on a screen at the wall. The circle had taught them

how to stay present before another person. The kitchen asked the harder question: what care does this person need?

At first, the people on the screen were too easy. They waited when a learner hesitated, nodded past a wrong word, and accepted answers that did not sound like the learner's own. The system marked the visit passed. Renata watched a few rounds and sent the scenarios back to Mateo with notes in the margins. This kitchen is too clean. Give me the señora who refuses the bath. The daughter who questions every sentence from the doorway. The man who calls her muchacha all afternoon. The client who asks the same question nine times, not because she forgot, but because the fear returns fresh each time.

Mateo built the resistance in. Then he and Renata had to settle where help should appear. He wanted the screen to offer a phrase before the learner froze. Renata shook her head. "Your screen is teaching her to wait for cues." Mateo answered, "Your silence is losing the slow ones." He was not wrong. They watched a learner stall, search, and find her own word just before the cue would have appeared. After that, Mateo changed the engine: the screen would wait longer first, and offer the phrase only after.

When the simulated señora refused the bath, the screen offered three lines to persuade her into the water. Aliento gave Sofia a different instruction.

*The bath is the excuse, the fear is the reason. Stay long enough to learn it. The cold tile, the fall last winter, the daughter who used to help. The care is in the reason, behind the line.*

Sofia stayed with the problem. Under the refusal, she found the fear of the slick floor. So the care she chose was not another line of persuasion. It was a rubber mat, a warm towel, and a chair in the shower. Aliento had not spoken for her. It had helped her see what she needed to decide.

Sofia froze on the third sim. The engine played a son who arrived mid-visit, a voice off every script she practiced.

"Who are you. Who let you touch her pills?" Sofia did not answer. Her hand closed around the edge of the desk, her knuckles whitening. The run stopped there. The screen did not supply a line or mark the attempt as failed.

Renata sat beside her and left the screen dark. "He pushed," she said. "Tell me what you wanted to say." Sofia said it, quiet, to Renata. "Now say it to him." They ran it again with the son at full anger, and Sofia held two sentences before she lost it, and the second time she held four.

Renata's dæmons read the white-knuckled hand on the desk lip two ways, and she let them disagree.

*The clenched hand says stop,* Ojo said. *Her knuckles are white on the desk lip. Push now and the chair empties.*

*The hand still on the desk says she is staying,* Puente answered. *Leaving looks different. Leaving looks like a clean notebook on a shelf. Run it again, slower, and stay beside her.*

Renata ran it again, slower, and stayed beside her.

The tools made their own mistakes that month. Aliento cued Sofia a breath early in one round, and the word it offered was another woman's word. She took it, lost her thread, and sat with the difference between a borrowed word and her own. She retuned the dæmon that evening with Karina, kitchen words again, and the failure taught her more than the passes. A configured voice stays worth exactly the configuring it gets.

The funder's tooling carried one more offer late in the sim weeks. The agent could speak for her: a warm synthetic voice to carry the hard line to a difficult family while the learner recovered. The button sat on the screen through one whole run. Sofia ended the run, breathed, and pushed the handset an inch away with two fingers.

"Not it," she said. "Me."

Renata let the two words stand and marked the distance the handset moved.

The work asked for more than clinical lines, so the last hour of every week belonged to meaning. Each woman sat with her dæmon and shaped something past any job's use, a letter her mother would keep, a story of the block told straight. The cohort argued its charter out of forty drafts into three words on kraft paper, escuchar, confianza, comunidad, taped where the ring could see it. The wage is the door, Renata told them. This is the house.

Late in the session the screens went away and the actors came in. A learner sat across bare wood from a stranger hired from outside the cohort to play a client's family, and ran the conversation with everything put away, the screen, the script, the caption, the cue. The face gave everything back. The fear wore a face now, and so did the warmth.

That same week, the funder pushed a new build. It offered live guidance during role-play: what the actor said, and what the learner should say next, appeared together on the shared screen. Renata tested it once. The captions arrived on time. The learner read a warm, correct line from the screen. The system logged a pass. But the actor across from her looked away, and the conversation never really began.

A third voice spoke then, uninvited, the steadiness her craft kept postponing through a career of trusting the room.

The voice reminded Renata: the script could help Sofia pass the exercise and give the review a clean score, but it could not teach her how to face a real client's family. Sofia would eventually walk into someone else's kitchen, across from people who might doubt her, get angry, or cut her off. Renata was not training a learner who could pass a test. She was training a woman who could keep speaking inside a real home.

Renata gave the voice its name, Brújula. She declined the auto-script. The desk stood bare between two chairs, holding the place of the kitchen table where a family decides who enters the home.

Sofia's turn came late in the afternoon. The actor across from her was a man near her father's age, playing a client's angry son. He questioned her the way the simulated son had. Sofia's hand went to the edge of the table.

Renata saw the earlier simulation return in that gesture. She did not interrupt.

This time, Sofia looked up. She started once, stopped, and began again. She did not look to the screen. She found her own words, and the man's anger slowly turned into conversation. When it ended, the actor thanked her, and it did not sound like acting. Renata came beside her and asked what had changed.

"I do not want it to talk for me." Sofia answered.

After the actors left, Renata closed the session with her dæmons.

*Read what the bare wood gave back,* Ojo said. *She refused the machine twice: first with two words, now with a full sentence. Mark the change. She is no longer only afraid. She is choosing.*

Not everyone stayed. One learner had done better than anyone in the hardest simulation of the second week, facing the angry son and holding her ground. By the fourth week she had stopped coming. Renata found her at home, in a clean kitchen. The circle notebook was still on the shelf, unopened since week three.

"He counts the hours," the woman said. "The crèche takes the baby. Who makes dinner?"

Renata knew how to work with anger. This was not anger. It was the household math waiting for her every night.

*The spine of that notebook is unbroken since week three,* Ojo said on the walk back. *She began leaving before she missed a session. Log the week it started. Rachel's terms keep this door open only if we count it.*

Renata logged the loss honestly, even though it would hurt her completion numbers during review season. Then she brought the lesson back to the makers and the block. The simulated kitchen had taught the women to decide the care and refuse the machine that would decide for them. The next round would test both lessons at the door of a real home.

### Session Three: The Real Round

The third session ran nine months, and it ran in real homes. Where the simulated kitchen taught the decision, the real round set them beside a nurse, a clinic, and the block, with a true door and a living elder behind it. Renata built the session as a long apprenticeship, each woman shadowing the nurse first and leading later, the practice fading as the judgment held.

Scope came first, and it came wearing a uniform. The state nursing college that once sent a courteous alarm sent a geriatric nurse instead, thirty years of transfers and med logs and red flags in her hands, and she ran the rounds. The women learned the transfer hold, the med log, the eight signs that end a visit with a call. They learned the wound care they would do and the wound care they would hand to the nurse, the scope line drawn where the state drew it and carried as a license. Scope was taught as a skill and worn as a pride. What a cuidadora does, she does well, and what she does next is call, the call itself the proof of her training. Renata drilled the escalation the way the theater drills an exit, until it lived in the

body. The college's nurse spoke the sentence that opened the door to certification: the program teaches its limits as carefully as its skills.

The block came next. Doña Lupe carried the program onto porches beyond its own reach, and the cohort built the colonia its own quiet infrastructure. A block agent on Doña Lupe's porch phone, configured with Karina, walked a neighbor through a clinic appointment or a subsidy form in the women's own voices and emptied itself of everything it heard. The learner whose chair went empty in the sim weeks took the evening shift from her own kitchen, answering the block at her own hours, counted as a returned learner whenever she came back through the door the terms held open.

Sofia rose inside the round. She arrived early, set the chairs in the tight ring herself, and sat beside the woman who shook the way she once shook. Renata watched her do it and made room, because the pilot years taught the team that the surest teacher of a frightened woman is a woman who sat frightened in the same chair a season before. Sofia built her second dæmon alone, Norte, the steadiness that holds a learner to her direction when she can see only the next hard visit, tuned from the inside because she knew the half-second from the inside. By the ninth month she shadowed elders in their own kitchens, ran a hard visit to its soft end, and taught the next woman to do the same (Figure 16.2).

**Figure 16.2. The curriculum from the circle to the simulated kitchen to the real round. Source: authors.**

### Readiness, and What the Teaching Hands Forward

The nine months closed with a debrief, and Lucia called the room the way she called every room, with the people who held the program around one table. Rachel joined from Zurich on the screen at the head, present the way she was always present, near and far at once. The geriatric nurse from the college sat with a clinic coordinator. A program officer from the development-impact fund sat across with a screen open to the dashboard. Doña Lupe took the chair by the door. Renata, Mateo, and Karina filled the window side, and along the back wall sat the cohort the nine months made, Sofia among them.

“Tell me what the nine months taught you,” Rachel said from the screen. “The honest version. We send them out on Monday.”

Renata answered first. “They learned the thing a script skips, to sit with a frightened old woman and let their own face be read. The slow ones took the longest and hold the firmest. I would send any of them into a kitchen tomorrow.”

The fund officer turned the screen around. “I have to read you the other column,” he said, his tone even. “Cost per graduate, high. Time to certify, slow. A vendor pitched me last week, pre-screened caregivers at a third the price, ready in six weeks.” He let it sit. “My board will ask why we paid for nine months.”

“Because a placement that holds is a fee that recurs,” Rachel said. “The fast build graduates a woman a family lets go in a month, and you pay that fee once. Ours stay. Read the retention column beside the cost, and read it at six months and twelve.”

The geriatric nurse spoke unprompted. “I came to this room to find out who signs when a bad lift cracks a rib,” she said. “I stayed because they taught me their answer. They teach their limits as carefully as their skills. I would put my own mother in their hands, and I say it because it is true.”

Mateo turned his screen to the call. “The tool is theirs,” he said. “Every woman configured her own dæmon, and the corrections they make stay under their own names. We refused the license that would have sold them. The platform runs on secondhand handsets because of it, and it runs.”

“And the block has its own line now,” Karina said. “A neighbor calls the porch phone and a woman from the colonia answers, in her own voice, and the machine that carries it forgets what it heard the second the call ends.”

Doña Lupe kept quiet through the table, and she spoke now. “I carried this program to mothers who crossed themselves at the word máquina,” she said. “They send their daughters early now. A program the block trusts outlasts a program the funder likes.”

Tierra read for her, the engagement rooted in the ground the program stood on.

*La unión hace la fuerza. The families gave their daughters because you stood on their porches first. The trust came before the machine, and it will outlast the machine. Hold the union, and the rest follows it home.*

Rachel asked the question she came for. “Tell me your fears. We send them into a market that is automating the rung they start on.”

“That the door keeps the ones we lost,” Renata said. “And that the machine takes the seat before they climb to the one above it.”

“That the curve wins the next review,” Lucia said, “and we are asked to graduate them faster than they can grow.”

“And the hope?” Rachel asked.

“That they go out, and a family asks for them by name,” Doña Lupe said.

The hopes gathered on one woman more than the rest. Renata named her. “Sofia froze on her third sim and runs the circle now. She holds the most of it. She goes out on her own on Monday.”

Rachel asked Sofia to speak. Sofia stood, using the careful English she had built one rough sentence at a time. “I was afraid of the door,” she said. “Now I knock.”

The room’s Councils read the crossing together, Renata’s and Sofia’s, turning the nine months over until each could state what crossed with the cohort on Monday (Figure 16.3).

Ojo watched the cohort along the back wall: *What do you see in them now that you missed nine months ago?*

Renata read the proportion one last time. Nine months ago this cohort was mostly fear. It crosses on Monday mostly skill, the fear turned into a thing they use. The slow ones took the longest and hold the firmest. She would send any of them into a kitchen tomorrow.

Brújula heard the fund officer’s column read across the table: *What keeps them climbing past the rung the machine already wants?*

Renata held the answer. The rung they start on is the one the machine wants. The seat above it asks for the face that stays, and that is the seat she taught them to reach. The work from here was to keep them climbing toward it.

Aliento and Norte read for Sofia, the two dæmons she configured herself.

Aliento felt the breath go shallow before a knock: *What stays in you in the second before the door opens?*

Sofia was afraid of the door once. Now she knocks. The machine can carry the schedule. She carries the visit. What stays in the second before the door opens is hers, and it crosses with her on Monday.

Norte heard the route assignment land for Monday, the second dæmon arriving after the first: *What holds the visit to its purpose when the room tries to pull you off it?*

Sofia knew the half-second from the inside. She keeps the reading and the staying, and holds the visit to its end when a doorway pulls toward a fight. That was the road after the knock, and she would hold it herself.

The debrief closed on a decision the room could agree to. The cohort was ready. Certification came co-signed by the college. The supervision ratios and audit rights were printed where the families could read them. Rachel said the last word from the screen. “Send them out. We hold the line they earned.”

Sofia went home that evening with a certificate, a kit bag, and a route assignment for Monday, early. The next morning out belonged to her.

**Figure 16.3. The Creative Council at the readiness debrief. Source: authors.**

## Chapter 17: A Day of Creative Change

DE AQUÍ
BF
BETTERFUTURES MÉXICO
full employee benefits, not gig work
FAMILY
fee drops for every day the device stays active
EDUCATION FUND
a share funds cuidadora study
CLINIC & NURSE
home care costs less than a hospital bed
STARTUP SEED
funding for new social businesses
INSURER
pays for prevention — saves on hospital bills

**Figure 17.1. Sofia with her first client. Source: authors.**

### Before the Alarm

Norte speaks before the alarm does.

*Three doors today. The platform pushed señora Govea late again. Decide on the walk.*

The sky over Neza holds the gray that comes before the heat. Sofia fills the kit bag in the kitchen by feel, the cuff, the log, the spare Milagrito cord, the tamarindos that don Emiliano pretends his daughter allows. Her mother sleeps. The water barrels stand full because Tuesday was the schedule. The bakery on the corner is lighting its ovens as she passes. The first combi down Avenida Chimalhuacán runs almost empty at this hour, all the way to the flats where her three doors wait (Figure 17.1).

Two years since the flyer. One year since the bare wood desk. Norte and Aliento ride with her, the two dæmons she designed for herself.

*The Govea ankle was thicker on Tuesday than the log admits,* Aliento reads as the combi slows. *The daughter's voice on the call ran short of air, and she talked about the saint's day tamales twice.*

*The optimizer reads the log,* Norte answers. *It schedules the visit the log deserves. The ankle you saw deserves an earlier door.*

Her dæmons agree on the facts. The deciding voice is the one the program spent two years growing, her own. Sofia takes the route back, moves señora Govea up the morning, and lets the platform log the override under her name.

### Three Doors

The first door is don Aurelio, eighty-one, a widower with a new hip and an old opinion about every government since López Mateos. The platform clocks the visit at fifteen minutes and flags two events from Sunday, when his grandson pressed the heart twice to hear what it does.

*The log wants this visit closed quick and clean,* Norte says. *Two false alarms and a healed hip read as routine. Read him over the row.*

She gives it twenty minutes. She checks the dressing, logs the vitals, and resets the Milagrito's threshold the grandson tripped. The third check rides inside the talk, twenty minutes on the price of cement. "A man my age forgets faces, never debts," he says, and the joke lands timed and whole.

*Mark the timing,* Aliento says. *A man whose joke arrives whole is a man whose week is holding. That is the vital sign the agent pattern would have missed.*

The talk is the assessment. His sentences hold their thread, his color is good, his hand on the cane is steady. The log she dictates to the platform agent while he finishes the story is the log of a man having a decent week. Presence, device, judgment, one visit.

The second door is señora Govea's, earlier than the platform planned. The swelling answers the question before the cuff does. The ankle is worse, the breath is shorter, and the saint's day sodium sits on top of both. The history lives in a file at the clinic and in a daughter's calls, nowhere a scheduling agent would find.

*Her breath runs shorter than Tuesday,* Aliento reads. *Kneel and keep her talking. The register that settles her is the one that keeps you reading her.*

Sofia kneels where Aliento said, keeps her voice in the register that keeps a frightened woman talking, and reaches the nurse line inside the minute. The nurse hears the numbers, bumps it to the clinic, and the clinic agent finds a slot that exists because she came early.

*Shape the note like the clinic's own form,* Norte says. *Karina's rule. The system downtown signs what looks like its own hand.*

The escalation lands as a note shaped like the clinic's form, signed Sofia, countersigned by the nurse. The teach-back carries the override forward under her name.

On the step, the daughter holds Sofia's hand in both of hers and runs short of words instead of air. The optimizer would have arrived after the slot closed and logged a routine visit. The hour it ranked lowest was the morning's whole work.

*Log the why,* Norte says. *The new cohort will ask, and the scheduling agent learns slower than the women do.*

The third door is don Emiliano's. He loses the day, some days. The pills live in a counted organizer, the stove carries a cutoff the family chose at the kitchen table, and his Milagrito charges in the clay saucer beside the santos. The platform offered the family a reminder voice last month, a warm synthetic prompt for the pills and the day of the week.

*Let the voice keep the pills,* Norte says. *The day of the week is the thing he holds onto. Leave it to the person he reaches for.*

The family asked don Emiliano, who asked Sofia, who carried the line to him and let him decide. He chose the voice for the pills and the girl for the week. A cousin from Puebla is visiting when she arrives, and don Emiliano makes the introduction with the day's date lost and the name surfacing on its own.

"Sofia. La que me escucha."

His daughter walks her out. He forgets the pills unless the voice prompts him, she says. He forgets my name some afternoons. He remembers Thursdays, and he remembers you. On the threshold he keeps swept for visitors, Sofia finds the other half of a sentence she began across bare wood a year ago. That day she said she would do her own talking. Don Emiliano just told her why it mattered.

"I want to be the one they remember."

The cousin from Puebla looks up. Don Emiliano nods the way a man nods at the obvious, and asks if she brought the tamarindos.

### The Afternoon Cohort

The training center above the avenue holds twelve new faces, the cohort that started in March. Sofia arrives early and sets the chairs in a ring tight enough that a knee almost touches a knee. Renata teaches and visits when she can. The circle runs on Sofia's voice now. The afternoon looks like simulation practice. Its real subject lives past any script: reading a room, staying in a hard moment, holding a visit to its purpose, landing a true word with a stranger. Sofia grows it through the hard case, and she names each part as it surfaces, because a woman who can name what she does keeps doing it after the program lets her go.

The simulation catalog holds many difficult moments now, fed back from real visits, the clients unnamed, each contributing cuidadora credited in the catalog she helped build (Figure 17.2). The son who demands to know who let a girl touch his mother's pills. The client who refuses the bath in August. The daughter who interrogates from the doorway. Sofia added four scenarios herself this year, and the freeze that broke her on the third sim in training now teaches a stranger every week, in a hub she has yet to visit.

**Figure 17.2. The afternoon cohort: the simulation engine. Source: authors.**

She runs the daughter at the doorway for the room and gives it to a learner named Marisol, who reaches for the agent's draft on the screen. The draft arrives correct and cold, the right words in the wrong register. Sofia closes it. "The agent will hand you a clean answer," she tells the room. "The doorway wants a true one. Find the question under the daughter's

questions." She coaches Marisol toward the staying instead of the solving. Marisol holds the doorway, lets the interrogation run, and hears under it the daughter's fear of being judged for the mother she works two jobs to keep at home. The line she finds is her own, and it lands where the draft would have bounced.

"Walk back through what you just did," Sofia tells the room. "Before you spoke, you raised your awareness of the doorway, her face, the energy under the questions, because that feeling tells you what the words hide. You stayed when she pushed, where the easy move was to close the visit and leave. And the words you landed were your own, reaching a frightened woman the cold draft would have hardened. The agent gave you the sentence. The rest was yours."

*And one thing held the others together,* Norte says. *You adapted as she shifted and held to the care, on the mother's week, while the doorway pulled you toward a fight. That is the one they lean on hardest, the day a room tries to move them off course.*

The dæmon is how Sofia teaches them to grow these and keep them their own. She has each learner configure her own, the way Karina once taught her, kitchen words on a card. What should it guard. What must it leave alone. How will you know the judgment stays yours. One learner sets hers to hold the silence a beat longer, so she stays engaged past the point where she now rushes. Another raises her awareness of the room's aesthetics, the atmosphere, the energy, because that shapes how a frightened person behaves, and she sets her dæmon to prompt the reading she reaches late. A dæmon built to grow the capability grows the woman. A dæmon built to perform it leaves her where she sat.

The possibilities open in the small inventions the women bring. One learner, working the client who refuses the bath, found that the señora would wash if the radio carried the danzón hour, and she fed the move back as a lesson the scenarios missed. "That is the work," Sofia tells them. "The agent runs the cases we have solved. We solve the ones still unwritten."

Mid-block, a learner stalls against the wandering client. The cue sits ready behind the screen. Sofia holds it back and lets the silence run a beat past comfortable, and the learner's own word arrives inside it.

*You held that one longer than I would have cued,* Aliento reads. *Her word came on her own breath. Keep teaching me the timing.*

The re-voicing exercise closes the block. The agent drafts its message to a worried daughter, correct and cold. Each learner says it again the way she would say it across a kitchen table, and the ring votes on which version the daughter keeps. The winning drafts go back into the platform as lessons, each carrying the name of the woman who warmed it. The machine's manners are their work, and the credit lands where the work happened.

The mentoring comes after the chairs are stacked. The girl who turned her handset face down in the first week waits by the door with her notebook. Her name sat on Lucia's first list once, past the funded twelve, and a crossed-out line cost her two years. The afternoon pushed her hard, the son's voice demanding who let her touch his mother's pills, and she

lost it on the fourth sentence and rehearsed the failure under her breath from the moment the chairs went up.

*She is doing what you did,* Aliento reads. *Rehearsing the loss instead of the line, and waiting by the door so you would see her.*

Sofia is already pulling two chairs back down. “He pushed,” she says. “Tell me what you wanted to say.” The girl says it, quiet, to Sofia. “Now say it to him.” They run it again, just the two of them and the engine. Two sentences, then four, each attempt rough, each leaving the next a little less impossible. Sofia stays in the chair beside her the whole way through, because she knows the fear in a face-down handset better than any dashboard ever will. Then she helps the girl write her first kitchen-words card, a dæmon to help her stay in the next hard doorway, and the pipeline that made Sofia begins again, one chair down.

*You hand the schedule to the platform and the drafting to the agent,* Norte says. *You keep the reading and the staying, the holding and the landing, and you teach them to build the dæmons that grow those. That is how the program reaches the next colonia before you do.*

### Preparing the Ground, Again

Evening finds Sofia back at the training center, the chairs pushed to the walls, a screen lit on the end wall. The day’s care is logged and the cohort has gone home. Lucia, Mateo, Karina, Renata, and Doña Lupe take the chairs. Rachel joins from Zurich, and the fund’s officer joins from wherever the fund is that week. The program runs in six municipalities, and a letter came this month that changes the size of the question.

The call is a debrief and a beginning at once. They walk the year first, the renewals held, the insurer kept to reports and the streams kept home, the college teaching its own module now. Then the officer reads the letter. A development bank, a ministry in Bogotá, a foundation in São Paulo, all asking BetterFutures to carry the model across Latin America, faster than the program has ever moved.

“We can take the money and grow in a year,” the officer says. “The bank wants twelve cities.”

“We can take the money,” Rachel says from the screen. “We grow at the speed the judgment grows, or we grow a logo over empty ground. Tell me how we hold the thing that makes it work while it triples.”

The team answered the way the program always answered, by starting with the people it would happen to. Each new city would open the way the pilot opened: with a room, before any contract named it. A facilitated workshop, two days, the local women and clients and families and nurses and the city in one ring. Designing that room, and the method that travels with it, became the evening’s work.

They built the agenda on kraft paper, the way the charter was built, three columns, what failed, what we learned, what changes. Lucia ran the failure column from her own notebooks, and she ran it hard. “One house in five still turns the arithmetic against us in the first year,” she said, “and the women we lose that way stay gone longest.” Out of that

column came the evening track, the family session before enrollment, and the block-agent shift formalized as a paid first rung.

Renata's lesson, that a circle teaches itself when a graduate runs it, decided who would lead the mornings, Sofia and two cohort women, chairs in tight rings, a name and a gesture first, families and nurses and actuaries seated knee to knee. Mateo and Karina's lesson, that the refusals held because they live in the architecture, became the afternoon working session, the spine that travels printed on one page, the refusals and the ratios and the line that counts a returning woman, for each new city to argue and sign what survives. Doña Lupe asked for one hour kept open, the hour the room eats together, because the street decides over food what it announces in meetings. It went in as written, comida, una hora.

Sofia spoke last, and her sentence went onto the agenda's first page, underlined. "The women learn the tools in weeks. The judgment takes the year. Whatever the new countries sign, we budget for the year." Rachel noted it in Zurich. The officer, on the other screen, noted it too.

The marker caps clicked shut, and the Councils convened, every practitioner's configured voices reading the design back to the room (Figure 17.3).

**Figure 17.3. The Creative Council convened at the next-wave planning. Source: authors.**

Hilo heard the bank's offer of twelve cities in a year, and read it back for Rachel.

*Hold the room to the speed the judgment grows. A logo scales in a year. A capability scales at the pace of a kitchen.*

Voz watched the failure column stand first on the kraft paper, and read it for Lucia.

*Read it aloud in every new city, before the wins. A community trusts the program that says what it lost.*

Rumbo felt the method pressed onto a single page, and read the page for Mateo.

*It travels on one page or it dies in translation. Print the refusals. Let the new city argue them and sign what it keeps.*

Pregón heard the hour for comida written into the agenda, and spoke it for Doña Lupe.

*Feed the block before you ask it for anything. The street decides over food. The meal is the meeting.*

Norte watched Sofia's sentence go underlined onto the first page, and held it for Sofia.

*The pace stays the women's to set, in each kitchen, in each new country now. The room you build tomorrow is how a city you have yet to see learns to say so itself.*

The five voices landed as one, and the team wrote the contour into the agenda.

### Years On

Ten years on, the program has spread across the region. From Neza to Bogotá to the edges of São Paulo, each hub is run by local women and carries a name chosen by its own neighborhood, the way the first cohort once chose De Aquí. Each time the program enters a new city, it first opens the two-day co-design room. Contracts and implementation come after. The tools stay local. The data still belongs to the households that produced it. Outcomes lasted longer than the agentic projections predicted, even where the costs never came in as low as the pitch deck promised.

Lucia was the first to say what the work was costing her family. Now she runs a training floor. Renata coordinates home care at a clinic, scheduling visiting caregivers and care robots in the same system. She decides which visits need a person and which can be handled by a machine. Karina oversees a municipal home-care roster across a borough. Doña Lupe runs the block-agent network across multiple neighborhoods.

The change reached the institutions around the program too. The nursing college once sent a politely worded warning that these women were crossing into nursing work. Ten years later, it teaches the program's Tuesday module in its own classrooms. The program learned to enforce its own scope of practice: before the college could question it, internal audits had already revoked failed certifications. Families who once feared the máquina now send their daughters to ask for application forms before recruitment opens. Government subsidies have survived one administration after another, though each defense leaves them smaller than before.

Each year, the hubs gather for the BetterFutures Creative Conference. It feels less like a conference than a set of open workshops. Caregivers, teachers, makers, and organizers from Neza, Bogotá, São Paulo, and other neighborhoods bring the Councils they trained at home: voices that question, remind, refuse, and reflect a decision back before it hardens. The practitioners now far outnumber the first cohort.

Every few months, a vendor arrives with a cheaper auto-script, promising to handle screening, scheduling, reminders, and basic follow-ups by machine, at a price below a caregiver's hours. Funders look at the quotes and ask the same question: if machines are faster and cheaper, why does the program still keep home visits, peer training, and the slow training ladder?

The question is not easy to answer. Each year, devices can replace more home visits. In some years, machines learn faster than the program can train people. But what the program keeps is a judgment the machine still cannot carry: reading where a household is truly stuck, knowing when to stay, when to lend a hand, and when not to hand a person over to a script. The forecast cannot see that value, so it prices it at zero, and the program keeps training it anyway.

Sofia now leads the learning track across the hubs. She begins every first circle the same way: chairs pulled close, a name passed around with a gesture. Her own chair always comes to rest beside the woman who has turned her phone face down on the table.

### The Flight

Rachel reads the agenda over the Atlantic, the cabin dark, the kraft columns photographed and pinned to its first page, Sofia's sentence underlined twice. She knows what the two days will be. A test of the community the pilot grew, whether it holds its shape when the program triples around it. A test of how deep the agency runs, whether the women the pilot built can build the next room themselves. A test of every capability the year trained, the reading of rooms, the holding of silence, the judgment beside the bed. And a test the dæmons sit inside too, the configured voices that carried the team and the cohort this far, built to keep evolving as the work they serve evolves.

*You know this shape. Trust the part beyond your reach now,* Hilo says. *The women build the next room themselves, and that was always the design. The day a woman you have yet to meet configures her own dæmon and pulls a chair down beside the next frightened girl, the program has outrun its founder, and the work is safe.*

What becomes of the ones a faster program would have passed and lost, the women whose agency a dashboard counted before they grew it?

## Conclusion: The Agentic Creative Path

CONCLUSION

# THE AGENTIC CREATIVE PATH


1. Navigating the Pandæmonium
2. Practice a Discipline
3. Train for Attunement
4. Cut Where the Joint Opens
5. Stay With the Cost
6. Convene a Circle
7. Choose Your Way

**Figure C.1. The Agentic Creative Path. Source: authors.**

### Navigating the Pandæmonium

Six thousand miles from the program she built, a founder sits before her screen at dawn. The wall of cards waits behind her. The system generated its options overnight. She holds her hand back. She is reading a room she has yet to enter.

The room is real. A two-day workshop ahead, the first city of a new wave, and a coalition of a development bank, a ministry, and a foundation the work depends on. A funding offer whose framing she has yet to test against the body that knows what such framings cost. The system can produce the brief in seconds. The body that decides whether the brief is the brief takes longer, because the body answers a question the system was unable to ask.

Others diagnosed the drift between maker and made before us. Ruskin saw the hand isolated from decision in the factory. Benjamin saw reproduction detach the work from its conditions of making. Martel saw artifice move the viewer toward a predetermined response. Each diagnosis sat downstream of the maker's body. The agentic drift sits upstream. The system answers the practitioner's question before the question has formed. It answers on three walls at once: the prompt supplies what to do, the pattern the process, the protocol the coordination, until the practitioner works briskly inside a closed cell. The surface signals of depth, contemplation, and ambiguity that once distinguished art from artifice now live in the training data. The distinction has migrated from how the work lands to how the work was made.

The BetterFutures practitioners faced a version of this room and found their footing. Rachel's refusal of the leaner model the optimizer scored as fundable. Mateo and Karina's platform that kept the agent quiet and the data with the families. Renata's bare desk with no script between two faces, and the line Sofia found across it. The program held because its practitioners cultivated the perception that recognized when comfort began to substitute for judgment.

What cultivation lets a practitioner stand in such a room?

The creative capabilities offer practices, with the Council that holds them. Discipline extends engagement. Attunement reads aesthetic awareness. The cut keeps adaptive coherence through abundance. Cost carries expressive influence into the field. And the circle convenes them all. Together they keep the practitioner judging when the system offers to decide. The body holds steady at the threshold where what it perceives meets what the systems extend. The founder at dawn, hand held back while the overnight options wait on the screen, is already reading a room she has yet to enter. The body registering what the system has yet to render is the practitioner crossing over. The embodied excitation is the signal of the crossing. The unreachable, the unfamiliar, the untethered, the unyielding, the unchallenged: each marks a threshold the body has just crossed. The five practices cultivate the practitioner who can stand at the threshold and do the work the threshold opens.

### Practice a Discipline

Calatrava's hand returns to the watercolor desk on Saturday morning, and the hand that lifts the brush is a different hand from the one that lifted it the previous Saturday. The brush reaches paper's resistance. The body absorbs what the resistance returns. The next Saturday's hand begins from the new place.

This is *zhīxíng héyī*. The doing transforms the doer, and the transformed doer carries forward into the next encounter. Sakamoto played piano from age six, and the body that knew the keyboard met every computational instrument his career later extended into. Charumathi held the slow oscillation of Todi's flattened second because thirty-nine years of disciplined listening reshaped the ear that knew what the note asked. Nadia read every draft aloud by the window because nine years of reading aloud trained the breath that knows when a story turns. Each year deposits a layer the next session draws upon.

Discipline takes any form that makes the body return, weekly, to a resistant material. Drawing, piano, raga, essay, joinery, theater, a community walked over and over on similar days. The form matters less than the return. What the practitioner cultivates is the body the resistance reshaped, and that reshaping migrates into every professional room she later enters.

The BetterFutures practitioners arrived at the scale-up with disciplines that pre-dated the program. Renata's years in the rehearsal room taught her to read bodies in a room before she designed a curriculum for them. Mateo's habit of opening a machine to read how its parts asked to move taught him to feel where a flow ran loose before any dashboard flagged it. Sofia's months of practicing English against silent videos taught her the half-second she would later teach from.

Large language models trade the loop for speed. Delegate the relational map to a model's recommendations, and the work returns legible to the metric and illegible to the community. The years of return that built the body's map erode through neglect, the slower failure.

Discipline is the body and soul of professional practice.

The unreachable is the threshold the discipline cultivates the body to cross. What the prompt by itself leaves untouched, the disciplined hand reaches.

### Train for Attunement

Charumathi's throat catches at a gamaka the engine smooths and scores as improvement. The catch is the throat's, after thirty-nine years. Nomvula's sweat against the world model's clean drought reads the laterite heat ahead of any dashboard. Sofia's hand finds the edge of the desk in the half-second before her voice goes out. Each registration is a signal the discipline has trained the body to receive.

Guo Xi sent the painter to the mountain across seasons, to climb it and sit beside the streams and walk the paths the travelers walked, until the mountain entered her body. *Guān* is what the body that has stood in the mountain carries afterward. World models

offer experience uncoupled from embodiment, trading the body's perception for the model's prediction. A model that reads somatic perception for her optimizes away the very thing the reading exists to catch, and the field's tensions stay unread.

Nomvula's teachers taught her to walk a site heel to toe before she trusted a screen. Two boundary lines could be drawn with identical technical mastery, and only one would hold a family. The difference was whether the architect stood on the ground long enough to carry its heat in her own body.

Attunement is the practice of seeking the embodied excitation and treating the registration as the discipline's next instruction. The tightness in the back. The silence in the room. The gap between projection and field. These are the body crossing the threshold, and what the body has crossed for is the work the registration begins.

The reader practices attunement by trusting the registration. Rowling caught what the editors missed because she read the inconsistency through drafts and waited for her body to confirm what the drafts began to suggest. Su Lan kept the takes where the cold slowed Auntie Pan's wrist because the half-turn the model classified as noise was a life entering the frame. Her eye registered the life before the system smoothed it out.

The unfamiliar is the threshold the attuned body crosses. What the dashboard would have classified as noise, the practitioner reads as the work resuming.

### Cut Where the Joint Opens

Cook Ding kept his blade sharp for nineteen years while other butchers broke theirs monthly. His knife stopped cutting through the ox's structure and began moving between the joints. He stopped working with his eye and began working with the body that knew where the joints offered themselves. *Wú wéi* is the practice of letting the work disclose its openings instead of forcing a path through the mass. The blade that waits for the joint outlasts the blade that hacks at the bone.

The agentic systems multiply the ox. A practitioner who asks for options wakes to a wall of them, each scored, each plausible, the mass thickening with every overnight generation. The causal models go further and hand over a clean account of why, a verdict that ends the search before the search has taught anything. The thread that holds a work together vanishes in this abundance. Cook Ding's knife is the figure for what survives it, the moving line that finds the gap between options because the body has learned where the ox comes apart.

The founder at dawn is practicing the cut. The wall of cards behind her holds more options than the program could run in a decade, and her hand stays back because the joint has yet to offer itself. She is reading for the one line through the overabundance that her body will recognize as the work's own structure, the line the scores were unable to rank.

The reader practices the cut by holding the disorientation until the thread re-forms. Run the generation. Let the options multiply. Decline the verdict long enough to feel which path opens of its own accord, and treat the highest-scored account as one voice in the room instead of the room's conclusion. What the practitioner cultivates is the body that tells the

joint from the score. That body is the difference between a work that hangs together and a collage of ranked parts.

The untethered is the threshold the blade cultivates the practitioner to cross. What multiplies as overabundance opens, under the patient knife, at the joint.

### Stay With the Cost

Davis turned his back to the band at Columbia's 30th Street Studio so the musicians would listen to each other instead of to him. *Gǎnyìng* arrived because the practitioner stopped requiring the answer already in mind. The recording held what the silence between the players returned. Sympathetic resonance is the field answering the work, and the answer arrives at the price of presence, the metabolic cost of standing in the room with the script set down.

The field also refuses. A silence can be the music gathering or the room declining what the work asked of it. The body trained by years of return is what tells the two apart. The practice is staying inside the discomfort of the field's answer until the answer becomes instruction.

Bowie left Los Angeles for Berlin in 1976 with his physical and creative health collapsing, and stayed inside the discomfort until the discomfort became sound. The Berlin trilogy emerged from the cost. The departure was the work.

Cida felt the floor become one body when eleven neighbors settled into one slow breathing, the field's answer registered in her chest before any count could confirm it. The practitioners stayed with the cost in registers each trained for. Renata's chest caught before she let a machine teach the thing only a person teaches, and the catch was the early reading her bare desk eventually carried into the lesson. Sofia's hand found the edge of the desk through the third sim, and the grip was the muscle the work was there to build. Mateo's one cue and held silence looked thin beside three smooth scripts, and he cut the scripts because the smooth flow talked for the learner.

The platform offers the field at scale and subtracts the cost of standing in it. Reach arrives stripped of the metabolic price of presence, and the metric reads the room so the body never has to. Machine learning systems optimize against discomfort by design. A loss function describes a cost the system reduces. The practitioner who lets the system reduce the cost loses access to the cost as data.

The brain prefers the faster path. Cognitive laziness, the evolved preference for conserving metabolic resources Kahneman described as a law of least effort, makes delegation rational at the individual scale and corrosive at the practitioner scale. Staying with the cost runs against that preference. The practice asks, before any tool runs, what habit the practitioner is configuring the instrument against. It asks at what moment in the work she wants the discomfort to surface as data she has trained herself to read.

The reader stays with the cost by treating the field's refusal as instruction the optimization erased. The silence after the work goes out is information. So are the grip on the desk, the cohort coaching itself, the catch in the chest. The discipline has trained the body to register

what the system was unable to weigh, and the registration arrives as discomfort because the signal costs the body something to carry.

The unyielding is the threshold the body crosses when the field declines to do what intention asks of it. The practice is reading the refusal as instruction.

### Convene a Circle

Renata's first circle sat twelve women in a ring close enough that a knee almost touched a knee. The working session before the scale-up sat the practitioners around three kraft-paper columns: what failed, what we learned, what changes. Each room held because the people in it were there to read what one body began to register.

Su Shi inhabited the circle differently. He held poetry, painting, and policy in one body, the *wénrén* sensibility, and refused the overreach of any single specialization. The literati gathered in households across the Song dynasty composed under one another's witness. The circle was the discipline that held the disciplines.

The reader convenes a circle by gathering three to seven peers across disciplines, with regularity, around concrete work. Each refuses the trade of multiple disciplines for one. Each sharpens the reading the others have begun. The cohort the practitioner cultivates becomes the room she returns to when the frame she carried alone has fallen away. A circle assembled by a recommendation engine, on optimization criteria authored elsewhere, reads to the metric and stays illegible to the work.

The Creative Council is the circle carried inward, convened in configuration. Mhlaba surfaced as Nomvula's body was already holding the sweat in her collar against the world model's clean drought, before the doubt found its words. Hilo surfaced as Rachel read the agenda over the Atlantic, the founder's reflex rising to fly in and run the next room herself, before that reflex closed her hand. Each dæmon arrived at the moment habit was about to foreclose. A configured dæmon that arrives after the decision is a delegated tool wearing the dæmon's shape. Timing is what separates the two.

Configuration is the operational form of convening. Each practitioner encoded what she knew herself prone to: a habit, a delegation she was tempted to perform, a moment in the work she was likely to rush past. The dæmon was the answer to the configuration question. The agents were summoned around it. The reader convenes her own Council by writing what Steinberger's *SOUL.md* makes operational: a record of the values, principles, and patterns the practitioner wants the system to hold against her own habit. Each dæmon is a designed intermediary that surfaces what the practitioner is most likely to miss. The practitioner who sits down with her configured dæmons has a chance the practitioner alone with her own relief misses.

Mixtures of experts and orchestrators offer scaffolding that bypasses the inner work it takes to see beyond oneself. The practitioner's choices migrate into distributed processing whose criteria came from outside her practice. The risk specific to integrative work is dilution. A Council that votes loses what surfacing produces. A synthesis voice that averages loses what provoking opens. An integration smoothing the convergence's edges

loses what reading them returns. Su Shi's gatherings held because witness was the point, and a Council held in friction is that witness made operational.

The unchallenged is the threshold the Council cultivates the practitioner to cross. The relief of agreement is what the body must learn to read as data. Across the BetterFutures seasons the later-arriving voices surfaced what comfort would have foreclosed, Mirada against a year of reading through other eyes, Brújula over the script, Trato over the warmth no tool could field. The work held because the Council talked back. Revision happens in the room. The circle is the room.

### Choose Your Way

Reclaiming human agency works at three scales together. The technology is only the occasion. A discipline practiced alone changes what a practitioner can do in the room she enters. What she can do in the room changes what the team or coalition around the work can attempt. What the team attempts changes what a community can hold. None of the three moves without the other two. BetterFutures ran the three scales together, the woman who learned to stay in a hard conversation, the practitioners who configured their own Councils to keep building it, and the community whose families and clinics and city shifted their defaults around her.

All three scales return to cultivation. Cultivation is more than a stock of skills. It is the trained ability to read what stands in front of you, answer it, and keep changing as it changes. In this process, what one gradually approaches is a Dào. In Eastern philosophy, the Dào literally means "the way," while also referring to the underlying patterns through which things unfold and the manner in which humans come to dwell within the world. It is a direction and order that gradually reveals itself through sustained practice. One enters the Dào through cultivation and comes to understand it by walking.

Each person finds the way differently. Some will surrender judgment entirely to systems. Others will sharpen their own perception. Some will pursue speed. Others will relearn the value of pause. Some will let the body withdraw from the work. Others will retrain it to enter more deeply. Some will treat work as a problem of optimization. Others will treat it as a lifelong practice of self-formation.

And how will you walk?

# Epilogue: Toward an Agentic Transhumanism

## The Pandæmonium, Revisited

We have addressed Pandæmonium as the multiplication of dæmons faster than the practitioners who would configure them well can be cultivated. The agentic age accelerates a separation already underway, between the hand and the decision, the instrument and the practitioner who configures it, the work and the body that knows whether the work is the work. A prompt answers what to do, a pattern how, a protocol who, and our questions ask what it costs the body to live inside that arrangement.

One influential strand of transhumanism imagined consciousness migrating into silicon-based computational substrates, embodiment gradually dissolved, and the constraints of biological wetware transcended (Moravec, 1988; Kurzweil, 2005). We arrive at almost the opposite conclusion. The body is the irreducible site at which augmentation succeeds or fails, and the operative question shifts from transcendence to configuration: which technologies, arranged through which relations, and against which forms of work, become woven into the body and its modes of perception. The post-human is not a future species awaiting arrival, but what the human becomes once it configures the conditions of its own augmentation (Hayles, 1999).

## The Triad of ACI

Classical understandings of agency organize through dualisms: subject and object, actor and environment, human and tool, each treated as separate. From the perspective of ACI, however, action, judgment, and creativity can no longer be attributed to a single actor. Agency is no longer solely a human property, but a process emerging through the participation of multiple beings.

We propose a different structure: the Triad of ACI (**Figure E.1**). Rather than locating agency within an individual actor, ACI emerges at the intersection of three mutually constitutive dimensions: Community, Agents, and Environment. Community provides shared practices, norms, and collective intentions. Agents contribute cognitive, computational, and operational capabilities. The environment offers the material, social, and institutional conditions that shape action. ACI arises from the interactions among all three, not from any single element. Agency, therefore, is understood as a relational and emergent phenomenon produced through their ongoing coordination and co-evolution.

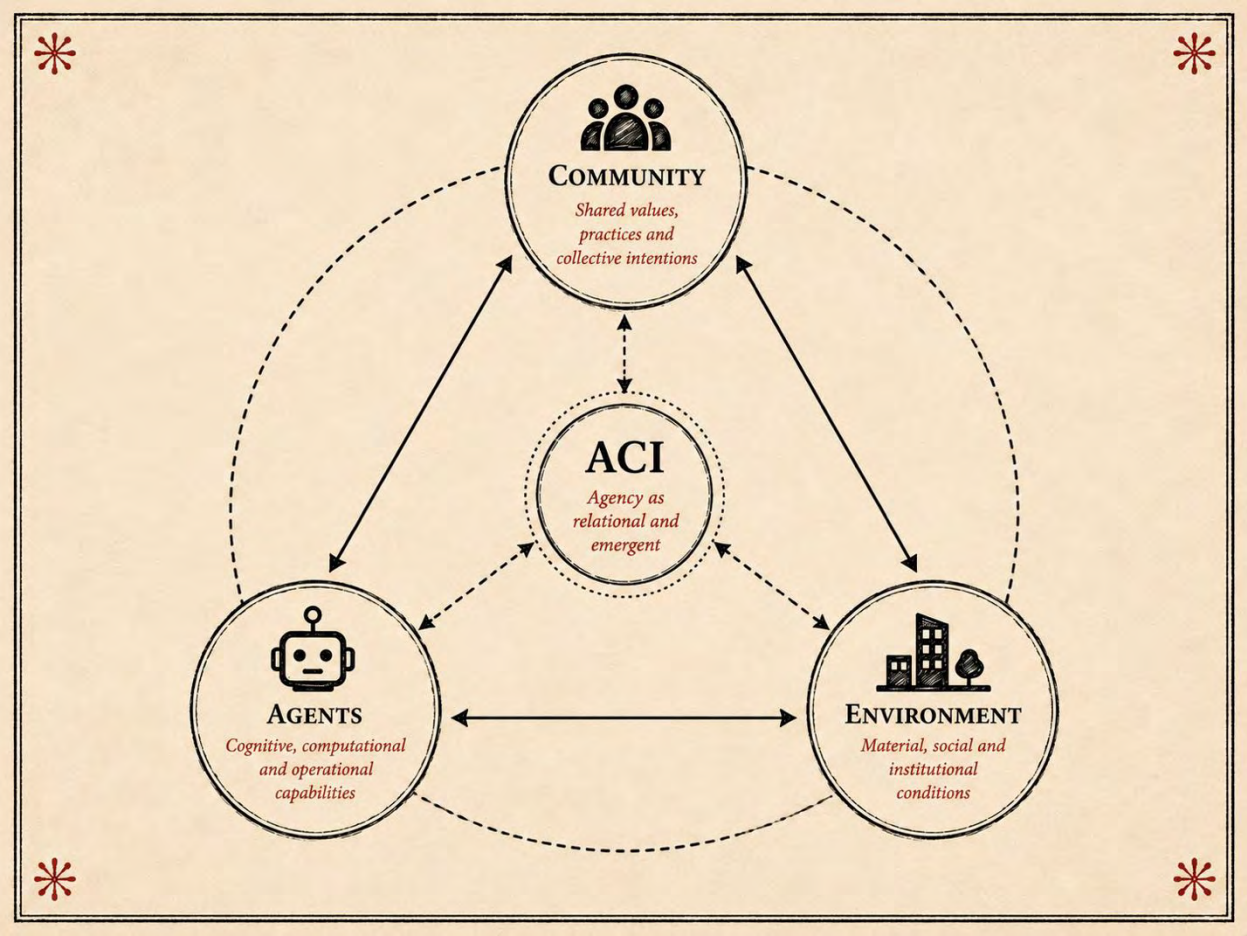


**Figure E.1: The Triad of Agency. Source: Authors.**

In the age of AI, the geometry of this triad is being reshaped. Communities perceive, interpret, and act through agents. Agents are no longer mere tools but active participants in analysis, coordination, and creation. Environments have become dynamic systems generating continuous data, feedback, and constraints.

To rethink agency is not to preserve a human-centered logic of control. Rather, it is to understand how agency emerges through the evolving relationships among communities, agents, and environments. ACI occupies this relational space, arising from their ongoing interaction and co-evolution.

## Reflecting on Research Required

Writing the book surfaced questions existing frameworks could not answer.

Our method here was practice-led. We forecast each discipline, then tested the forecasts as design-driven speculative fiction, configuring our own dæmons and writing through them so the argument was made in the doing. That method carries a limit we want to state plainly. The scenarios are speculations, built to be plausible and provocative, but unproven. What this book offers is a hypothesis cultivated in practice, and a hypothesis asks to be

tested. The agenda below is the form that testing should take, and it will need more than one kind of research to carry it: practice-based inquiry close to the studio, design research and research through design, action research conducted with real communities in the manner Part III imagines, and empirical scientific study undertaken with computer scientists, neuroscientists, and the other disciplines whose instruments can measure what practice can only sense.

We understand Agentic Creative Intelligence (ACI) as a relational system emerging from the interactions among communities, agents, and environments. Within such systems, intelligence, creativity, and agency do not reside in any single actor. They emerge through coordination, collaboration, and adaptation.

Building on this perspective, our research focuses on four interconnected directions: Design Inquiry, Distributed Intelligence, Augmented Experience, and Adaptive Learning. Together, they ask how knowledge is generated, how intelligence is organized, how experience changes, and how learning evolves within agentic systems. These directions converge on one goal: building community through human–AI collaboration (Figure E.2).

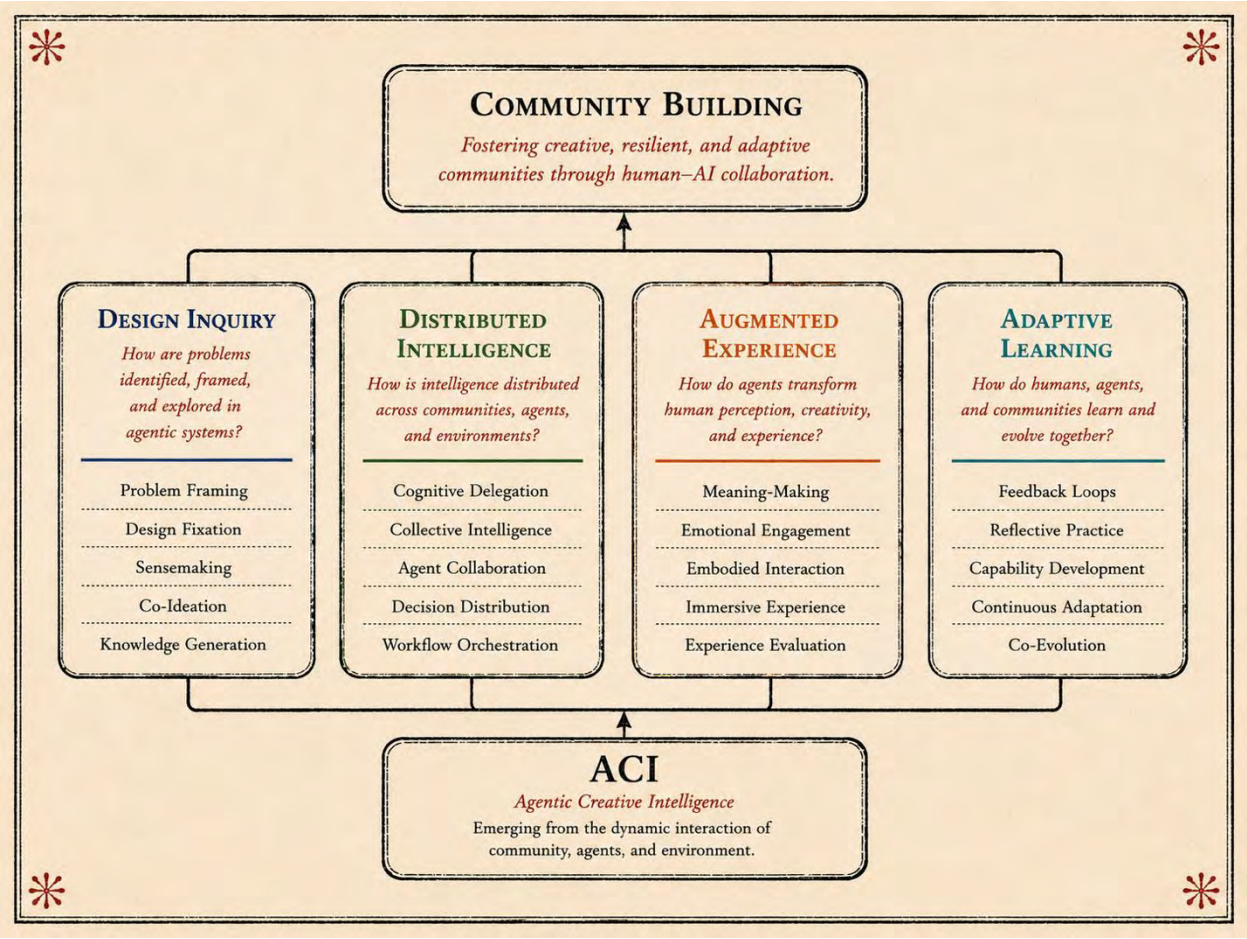


**Figure E.2 ACI Research directions. Source: Authors.**

### Research Direction 1: Design Inquiry

Design Inquiry examines how creative understanding emerges within Configurational ACI, including the framing and reframing of problems, as well as the generation of creative insights. Its central concern is not how existing problems are solved, but *how new design possibilities are discovered*.

This direction draws on research traditions in Design Cognition, Creativity Research, Reflective Practice, and Design Thinking. Existing work has explored themes such as design fixation, reframing, creative insight, and analogical reasoning, but has largely treated individuals or teams as the primary cognitive agents. Comparatively little attention has been given to situations in which agents participate continuously in the creative process.

Throughout the writing of this book, we found that agents did not simply provide answers. They also helped redefine questions. Some dæmons enabled us to break from established frames and discover new theoretical connections, while others reinforced existing assumptions and, in doing so, risked producing new forms of design fixation. As generative systems become more capable of offering persuasive suggestions, an important question emerges: *do they expand creative possibilities, or do they narrow the space of exploration? More broadly, how can designers maintain judgment and creative autonomy within highly generative environments?*

One promising area of inquiry is the Somatic-Semantic Nexus. This is the process through which bodily sensation crosses a threshold into meaning. We hypothesize that many creative insights are accompanied by measurable physiological events: moments in which a vague bodily feeling becomes recognizable, interpretable, and actionable.

Addressing these questions requires new research designs that combine studies of creative practice with physiological measurement. The Somatic–Semantic Nexus is not a stable state but a fleeting transition occurring at the boundary between bodily perception, conscious awareness, and meaning-making. Advances in MRI, fNIRS, eye-tracking, EMG, and HRV technologies offer new opportunities to capture physiological signals associated with creative processes. Future research may integrate design experiments, practitioner reflections, physiological data, and records of human–agent interaction to investigate how creative insights emerge and how different configurations shape design cognition.

### Research Direction 2: Distributed Intelligence

As Agentic Systems participate in design, decision-making, and knowledge production, cognition is no longer confined to individual practitioners. Instead, it becomes distributed across complex systems composed of humans, dæmons, agents, models, and material infrastructures. This research direction therefore asks a central question: *how is intelligence distributed, coordinated, and reconfigured within Configurational ACI*?

This direction draws on traditions in Distributed Cognition, Computer-Supported Cooperative Work (CSCW), the Extended Mind, and Human–AI Collaboration. Existing research has demonstrated that cognition does not occur solely within individual minds, but is embedded within networks of people, tools, environments, and social relations.

However, relatively little attention has been paid to the continuously evolving configurations of cognition that emerge within multi-agent systems.

From our perspective, cognition is no longer merely distributed. It is continuously reconfigured. This shift raises a new set of questions. *How is cognitive labor redistributed? How does expertise flow among different actors? How should decision authority be established and coordinated? And how do different configurations influence creativity, efficiency, and the quality of judgment?*

These questions repeatedly surfaced in practice. Joseph and Yilin each configured their own sets of dæmons and agents. As tasks such as literature review, citation verification, structural critique, and quality auditing were delegated to the system, the authors' roles gradually shifted from direct contributors to configurators and coordinators. They assigned tasks, compared recommendations, and arbitrated among multiple cognitive actors. This transition generated new tensions. *When different dæmons produced conflicting assessments of the same chapter, whose judgment should prevail? When practitioners found themselves defending the recommendations of their own configured systems rather than their personal views, was the disagreement really about ideas, agents, or configurations themselves? These experiences suggest that cognition in Agentic Systems is not simply distributed across multiple actors. It is continually reorganized through acts of configuration. In the process, the boundaries of expertise, authority, and responsibility blur.*

Future research may investigate these dynamics in real-world creative practice. Researchers might trace how design teams allocate cognitive labor among practitioners, dæmons, and agents, comparing how different configurations shape knowledge production and decision-making. Design interventions may also be used to deliberately alter configurational relationships, allowing researchers to observe how cognitive authority, expertise, and collaboration evolve as the system itself is reconfigured.

### Research Direction 3: Augmented Experience

As Agentic Systems become embedded in design, services, and creative practice, the object of design is no longer limited to products, interfaces, or services. Instead, it encompasses dynamic experiences co-produced by humans, agents, and environments. This research direction therefore asks: *how do agents reshape the experience of participation, and how should experience design respond to these emerging augmented situations?*

This direction draws on traditions in Experience Design, Service Design, User Experience, Mixed Reality, and the Experience Economy. Existing research has developed rich theories and methods for understanding user experience, customer journeys, and immersive engagement. Yet most of these frameworks assume relatively stable products, services, and interaction systems. We still know comparatively little about experiences shaped by Agentic Systems that generate content in real time, adapt their behavior dynamically, and continuously participate in decision-making processes.

Through conversations with practitioners across creative industries, we found that the effect of agents reaches beyond efficiency. What changes is not only the speed of work but the experience of working itself. Creative rhythms are reorganized, feedback cycles

compressed, and reflective processes distributed between humans and agents. In some situations, practitioners struggle to distinguish which aspects of an experience arise from their own judgment and which emerge through ongoing interaction with intelligent systems. These observations suggest that Agentic Systems are becoming part of the experience itself, more than tools for designing experiences.

This shift opens a range of new research questions. In the area of experience evaluation, *do traditional UX metrics remain adequate for dynamically generated Agentic Experiences*? When experiences continuously evolve, *how should quality, engagement, and long-term impact be assessed? In participatory experiences, how do roles and mechanisms of participation change when agents become co-creators rather than tools*? In hybrid experiences, *how should designers understand and shape the boundaries between physical and digital realities*? In adaptive experiences, *how can personalization be balanced with predictability, transparency, and user agency when agents continuously adjust content and services*? And in immersive experiences, *how might agent-driven storytelling transform the design of tourism, museums, cultural heritage, and other experience-centered domains?*

This research direction is particularly well suited to Research through Design, Participatory Design, longitudinal field studies, and design interventions in real-world settings. Future work may explore the mechanisms, principles, and evaluation frameworks of Agentic Experience, contributing new approaches to designing for dynamic, adaptive, and continuously evolving experience systems.

### Research Direction 4: Adaptive Learning

Adaptive Learning examines how humans, agents, and communities learn, adapt, and co-evolve in the age of Agentic Systems. As agents become active participants in creative practice, knowledge is no longer transmitted solely through human interaction. It is increasingly generated, shared, and refined through workflows, feedback processes, and networks of human–agent collaboration. This research direction therefore asks: how are knowledge, expertise, and judgment developed and sustained in agent-mediated environments?

This direction draws on traditions in Design Education, Organizational Learning, Communities of Practice, and Reflective Practice. Existing research has shown how learning emerges through participation, reflection, and situated experience. Yet comparatively little attention has been paid to learning processes involving intelligent agents, or to the ways humans and agents may learn together through continuous feedback and adaptation.

In writing this book, we observed forms of tacit knowledge accumulated through years of practice becoming embedded in prompts, workflows, and agentic systems. This observation prompted us to reconsider how knowledge is learned, transmitted, and constructed. *If future practitioners encounter agents before mentors and answers before experience, how will creative capability and professional judgment develop? As knowledge becomes increasingly accessible and reproducible, which forms of expertise will still require long-term participation to acquire?*

Learning in agentic systems is not simply the accumulation of knowledge, but an ongoing process of adaptation. Humans learn from agents, agents are shaped through human feedback, and both evolve within broader communities and environments. As AI becomes increasingly integrated into everyday practice, one enduring challenge for design will be to create conditions where humans, agents, and communities keep adjusting to each other.

## What Comes Next, Uncertainties

As configurations, interactions, and decisions become increasingly recorded, dæmon-augmented practice brings documentary thinking directly into the work itself. Many activities that were once embedded within practice begin to become visible and analyzable. This creates new opportunities for studying creativity, collaboration, learning, and participation, but it also introduces new uncertainties.

Three uncertainties deserve particular attention.

The first concerns transmission. Whether new practitioners, lacking prior formation, will develop the judgment and reflective capacities required to configure dæmons effectively, or instead bypass the very learning processes through which such capacities have traditionally emerged. This remains an open empirical question.

The second concerns the balance between augmentation and atrophy. Across the dimensions documented in this book, augmentation and atrophy appear to occur simultaneously. Dæmons expand certain capabilities while potentially diminishing others that were previously carried by practitioners themselves. Yet the conditions under which augmentation predominates, and those under which atrophy becomes consequential, remain poorly understood.

The third concerns the research instruments. Many of the processes discussed in this book, such as the moment when bodily sensation crosses a threshold into meaning, may occur at scales beyond the reach of existing methods. Advances in physiological computing, neural interfaces, and agentic systems may allow researchers to observe creative practice with unprecedented resolution, but whether such tools can be developed and deployed responsibly remains an open question.

These uncertainties define the research space of Configurational ACI. Future work must investigate how creative insights emerge through human–agent collaboration (Design Inquiry), how intelligence is organized and reconfigured across multiple actors (Distributed Intelligence), how new forms of experience are shaped through agentic participation (Augmented Experience), and how learning, adaptation, and co-evolution unfold among humans, agents, and communities (Adaptive Learning). We do not present these directions as answers, but as a set of open questions. Agentic systems are reshaping the ways we create, collaborate, learn, and participate. Our contribution is simply one possible starting point for understanding that transformation.

This points toward a form of transhumanism we are willing to endorse: capability emerges through the ongoing configuration of practitioners, agents, and environments. It is not a fixed attribute an individual owns. Configurational Transhumanism does not seek to

transcend the body, replace the human, or escape society. It seeks to open new possibilities within embodiment, relationship, and practice.

## Notes

1. See my early AI slop "Officeverse Emerging" in *Office Shock*, page 134, generated late 2022.

2. I use the Italian word *passione* intentionally to emphasize its etymology and avoid the modern translations as passion. It is derived from the Latin *passio* and Greek *paskho*, which both translate to "to suffer" or "to endure". Originally, the word did not mean eager enjoyment, but rather something so profound that one is willing to suffer or sacrifice for it. Stunt coordinator Chad Stahelski, describing the making of the *John Wick* films, locates the same principle in creative production: "You can sense there's something cool … but you can't quite get it" (*Wick Is Pain* [Documentary], Lionsgate, 2025).

3. Trading-card games (TCGs), also called collectible card games (CCGs), are competitive tabletop and digital games in which players build custom decks from randomized packs. Each player brings a distinct combination of cards to a match and plays through strategy, rare pulls, and resource management. Notable examples include Magic: The Gathering, Pokémon, and Yu-Gi-Oh!

4. The exchange is with neuroscientist Dr. Andrew Huberman: "When something feels creatively right and you're sensing it and you're there…how do you translate that…?" Rick Rubin: "I wouldn't rely on language to do it…Being in tune enough with ourselves …is probably the single most important .. skillset to develop as an artist … know how you feel and own your feelings." See How to Be More Creative | Rick Rubin & Dr. Andrew Huberman https://youtu.be/mceyIZmu8aI?si=IDZF8fNMRQS-fclW

5. Adam Smith on the division of labor. Smith, A. (1776). *An Inquiry into the Nature and Causes of the Wealth of Nations*, Book I (the pin factory) and Book V, ch. 1: "The man whose whole life is spent in performing a few simple operations… generally becomes as stupid and ignorant as it is possible for a human creature to become."

6. The embodied-cognition name for the felt event ACI names an embodied excitation (the somatic-semantic nexus): the physiological site where felt sense, the interoceptive, gut-level signal, meets symbolic interpretation. See Gendlin, E. (1978). *Focusing*. Bantam, on the pre-verbal "felt sense."

7. The distinction between art and exhibition value follows Walter Benjamin, "The Work of Art in the Age of Mechanical Reproduction" (1935): mechanical reproduction severs the work's "aura" and shifts value from presence to circulation. Cattelan's *Comedian* is the limit case. The gesture is the art, the artifact only its documentation.

8. The November 2024 Sotheby's sale of *A.I. God. Portrait of Alan Turing* by Ai-Da marked the exhibition-value threshold for AI-made artwork: the price tracked the novelty of the maker, not the image. When "made by an AI" becomes ordinary, the work will need

to mean something on its own.

9. Western thought carried two currents. A mechanical, materialist line runs from Newton's clockwork (1687) through Laplace's determinism (1814), treating nature as parts in lawful motion. A countercurrent, organic and poetic, runs from Spinoza's single substance (1677) through Goethe's morphology (1790) and the Romantic philosophy of nature, holding that the whole lives in its relations. The second current is the one that meets the Eastern interconnection intuition, and it resurfaces across modern science. Alfred North Whitehead's *Process and Reality* (1929) recast reality as events in relation instead of substances. David Bohm's implicate order (1980) proposed an undivided wholeness beneath apparent separateness. Gregory Bateson (*Steps to an Ecology of Mind*, 1972) located mind in the pattern that connects. Ilya Prigogine's work on dissipative structures showed order emerging through far-from-equilibrium relation (Nicolis & Prigogine, 1977). John Holland's study of complex adaptive systems traced order arising from local interaction (Holland, 1995). Fritjof Capra's *The Tao of Physics* (1975) drew the parallel explicitly. The claim is not that these frameworks match the Chinese traditions, but that an interconnection intuition recurs across them.

10. Barad, K. (2007). *Meeting the Universe Halfway: Quantum Physics and the Entanglement of Matter and Meaning*. Duke University Press. See also Barad, K. (2003). Posthumanist performativity: Toward an understanding of how matter comes to matter. *Signs: Journal of Women in Culture and Society*, 28(3), 801–831. Agential realism is the framework the theoretical physicist and feminist science-studies scholar Karen Barad develops from Niels Bohr's interpretation of quantum mechanics. Its founding move replaces interaction with intra-action. Interaction presumes two separate entities that exist first and then act upon each other. Intra-action holds that the entities emerge through their relating, that the relation produces the relata. Barad's touchstone is Bohr's account of wave-particle complementarity: whether light registers as a wave or a particle depends on the experimental apparatus, so the property belongs to the whole of observed-and-apparatus, and no object carries it in advance of the measurement. Barad calls this inseparable whole a phenomenon. The boundary that distinguishes observer from observed, drawn within a phenomenon for a specific purpose, is the agential cut. The cut is enacted through the apparatus, and each cut opens some possibilities while foreclosing others, which lends the framework its ethical edge.

11. The triangle compresses three real strata of the agentic stack. Prompting, the layer of instruction, has matured into a documented practice, with reusable prompt patterns catalogued by analogy to software design patterns (White et al., 2023). The pattern layer, the workflow, is the subject of Anthropic's distinction between workflows, where models and tools follow predefined paths, and agents, which direct themselves (Anthropic, 2024), and of Andrew Ng's four agentic design patterns, reflection, tool use, planning, and multi-agent collaboration (Ng, 2024). The protocol layer, the orchestration, is now formalized in open standards such as the Model Context Protocol

(Anthropic, 2024) and the Agent-to-Agent protocol (Google, 2025), which let separate systems discover and call one another. The use of "pattern" for a reusable solution traces to Gamma, Helm, Johnson, and Vlissides (1994).

12. The reduction to three is deliberate. George Box's maxim that all models are wrong, though some are useful, began as an argument for parsimony, the claim that since no model is correct the scientist should seek the economical description over the elaborate one (Box, 1976; Box & Draper, 1987). The triangle is offered in that spirit, a frame chosen to make the predicament legible to a practitioner, not a map of the toolchain.

13. **Field.** In field theory a field assigns a value to every point of a space, and particles appear as its excitations. ACI uses *field* for a creative capability: the space the capability opens, primary, with its four modes, set along two directions, derived from it.

14. **Modes.** A field decomposes into modes. ACI names four per capability, set along two directions, the *Mens* axis carrying a receptive and a projective mode and the *Manus* axis carrying Intention and Realization. Each axis is complementary in Bohr's sense, its two modes jointly necessary to describe the whole, yet never both maximal in a single act, so the practitioner moves between them as the work demands. The analogy is structural, not the literal uncertainty relation.

15. **Embodied excitation.** An excitation is a quantum added to a field. ACI extends the term to the body's felt response when a creative field is excited or disrupted, from ecstasy to error, grounded in the neuroscience of sensorimotor and affective response. A deliberate coinage. Each chapter's named trigger is one instance.

16. **The Peak, Hong Kong (1982–83).** Hadid's competition-winning entry for a Victoria Peak leisure club, a "horizontal skyscraper," was presented as Suprematist-influenced paintings and stayed unbuilt. A painting from the series, *The Peak Project, Hong Kong, China* (1991), is held by MoMA (moma.org/collection/works/202).

17. **Painting as investigation.** Hadid trained at the Architectural Association in London and turned to Kazimir Malevich and Russian Suprematism for an abstract spatial language beyond inherited architectural representation. See Royal Academy of Arts, "Zaha Hadid RA on the influence of Malevich"; Aesthetica, "Zaha Hadid: Early Paintings and Drawings."

18. **Parametricism and the built work.** The computational approach developed at Zaha Hadid Architects with Patrik Schumacher describes form as a system of parametric relationships. The Vitra Fire Station (1993) was Hadid's first completed building. The MAXXI museum (Rome) won the 2010 Stirling Prize. Hadid was the first woman to win the Pritzker Prize (2004) and received the RIBA Royal Gold Medal (2016).

19. **Extended mind.** The extended-mind thesis (Clark & Chalmers, 1998) holds that cognition routinely loops through external tools. Where it treats tool and body as separable, Extended Engagement holds them as mutually constituting: the tool's amplification is legible only through a body trained to read it.

20. **Deliberate practice.** Deliberate practice (Ericsson, Krampe, & Tesch-Römer, 1993) names the temporal variable effortful engagement requires. This chapter specifies the body-tool-material loop it presupposes, with the cortical restructuring Merzenich (2013) documents and the material consciousness Sennett (2008) describes.

21. **Nick Cave on AI and creative struggle.** Cave, N. (2023). *The Red Hand Files*, Issues #218 and #248. Cave's diagnosis is that the algorithm has nothing to transcend. It aligns with the chapter's argument that extended engagement depends on sustained contact with resistant material.

22. **Cortical plasticity and its loss.** Merzenich, M. M. (2013). *Soft-Wired*. Parnassus; Elbert, T., et al. (1995). Increased cortical representation of the fingers of the left hand in string players. *Science*, 270(5234), 305–307.

23. Transformers matter less as a single architecture than as a scaling rule: attention lets a model compare distant pieces of context at once, while Mixture-of-Experts layers let only selected subnetworks activate for each input. Together, they make language models larger without making every token use the full model (Vaswani et al., 2017; Shazeer et al., 2017).

24. "World model" names the move from producing an output to maintaining an internal simulation. In this sense the model does not only generate a scene; it stores a manipulable representation of how that scene could change if an agent moved, acted, or looked from another angle (Ha & Schmidhuber, 2018). Recent spatial-intelligence work extends this idea toward interactive 3D world generation.

25. 3D Gaussian splatting belongs to capture rather than imagination: it turns real photographs into navigable 3D scenes. JEPA sits on the opposite side of the same problem, learning predictive latent structure without generating pixels. Both point toward spatial intelligence, but by different routes (Kerbl et al., 2023; LeCun, 2022; Assran et al., 2023).

26. Causal models, diffusion models, and orchestration solve different problems: causality asks what would change under intervention; diffusion learns to denoise randomness into images; orchestration coordinates tools, agents, and specialist models into a working system rather than a single model (Pearl, 2009; Ho et al., 2020; Rombach et al., 2022).

27. Reinforcement learning from human feedback: Ouyang et al. (2022), "Training Language Models to Follow Instructions with Human Feedback," *NeurIPS*.

Reinforcement learning from verifiable rewards for reasoning: DeepSeek-AI (2025), "DeepSeek-R1" (arXiv:2501.12948; see also *Nature*, 2025).

28. MCP and A2A mark the rise of a protocol layer for AI systems: Anthropic's Model Context Protocol standardizes how models connect to tools, data, and context, while Google's Agent-to-Agent protocol standardizes how separate agents communicate and coordinate work (Anthropic, 2024; Google, 2025).

29. **From models to compound systems.** Zaharia, M., Khattab, O., Chen, L., Davis, J. Q., Miller, H., Potts, C., Zou, J., Carbin, M., Frankle, J., Rao, N., & Ghodsi, A. (2024). The shift from models to compound AI systems. Berkeley Artificial Intelligence Research (BAIR) Blog. https://bair.berkeley.edu/blog/2024/02/18/compound-ai-systems/

30. **Agentic patterns and workflows.** The workflow vocabulary used in this section, prompt chaining, retrieval, planning, and reflection, follows Gulli's catalog of agentic design patterns (Gulli, 2025), which distinguishes internal workflows (how a single agent or closed multi-agent system plans and executes) from external connection protocols (how agents interoperate across ecosystems).

31. **Agentic workflows.** The retrieval workflow follows Gulli's catalog of agentic design patterns (Gulli, 2025). World models sit largely outside that catalog, and the fit is loose.

32. **World models as spatial intelligence.** Li, F. (2025). From words to worlds: spatial intelligence is AI's next frontier. Essay. Li frames world models as generative, multimodal, and interactive. See also worldlabs.ai.

33. **3D world models versus video generators.** Tencent HY-World Team (2026). HY-World 2.0 technical report. github.com/Tencent-Hunyuan/HY-World-2.0. Distinguishes 3D world models (persistent meshes, splats, point clouds) from pixel-level video models such as Genie 3, Cosmos, and Sora 2.

34. **HY-World 2.0 specification.** Tencent HY-World Team (2026). Hugging Face: tencent/HY-World-2.0 (open-sourced April 2026). Accepts text, image, multi-image, and video. It outputs 3DGS, mesh, and point clouds through a four-stage pipeline.

35. **Real-time multimodal world models.** Odyssey, Starchild-1: the first real-time multimodal world model, generating synchronized audio and video from streaming text, speech, and action input. https://odyssey.ml/introducing-starchild-1

36. **Interactive prediction.** HY-World 2.0 supports first-person navigation with collision and persistent geometry. NVIDIA Cosmos provides action-conditioned rollouts for robotics simulation, trained on roughly 20M hours of video.

37. **Sim-to-real gap.** James, S., et al. (2019). Sim-to-real via sim-to-sim. *CVPR 2019*, 12627–12637. RCAN reports 70% zero-shot real-world grasp success, rising to 91% after fine-tuning on 5,000 real grasps.

38. **Gaussian Splatting, 2023–2026.** Wu, G., et al. (2024). 4D Gaussian splatting. *CVPR 2024*, 20310–20320. Standardization through the OpenUSD schema and the Khronos glTF KHR_gaussian_splatting candidate specification (2026). The "JPEG moment for spatial computing" is widely cited across the trade press.

39. **Eliasson's computational practice.** Studio Olafur Eliasson (2024). olafureliasson.net. Recent installations pair computational simulation with full-scale physical prototyping.

40. **Eliasson installations.** *The Weather Project*, Tate Modern (2003–2004); *Ice Watch*: Copenhagen (2014), Paris (2015), London (2018). Ursprung, P., Ed. (2012). *Studio Olafur Eliasson: An Encyclopedia*. Taschen.

41. **Zhīxíng Héyī (知行合一) and Wang Yangming.** Wang Yangming (1472–1529), Ming Neo-Confucian philosopher, argued against Zhu Xi's *gewu zhizhi* (格物致知) that authentic knowledge completes itself only through action. See Chan, W. T. (1963). *A Source Book in Chinese Philosophy*. Princeton University Press; Ivanhoe, P. J. (2009). *Readings from the Lu-Wang School*. Hackett.

42. **Cheng Yao-t'ien on the somatic trajectory of the brush.** Cheng Yao-t'ien (1725–1814), Qing dynasty calligraphy treatise. See Billeter, J. F. (1990). *The Chinese Art of Writing*. Skira/Rizzoli.

43. **Context engineering.** The multi-turn degradation finding is Laban et al. (2025): an average 39 percent accuracy drop across fifteen leading models over 200,000+ simulated conversations, traced to premature assumptions and answer lock-in. The anchoring response, extractive anchoring with provenance tracking, reports up to a 0.412 absolute F1 gain over direct prompting (Yang, 2026, the AEVS framework, *Computers*).

44. **Wheelwright Bian.** *Zhuangzi*, ch. 13: the wheelwright claims the Duke's books are "the chaff and dregs of the men of old" (Watson's translation) because genuine skill resists transmission in words.

45. **Sakamoto's career as embodied accumulation.** Ryuichi Sakamoto (1952–2023): classical training at Tokyo National University of Fine Arts and Music; Yellow Magic Orchestra (1978); Academy Award for *The Last Emperor* (1987); final album *12* (2023). See Schilling, M. (2023). Obituary. *The Japan Times*.

46. **Zhuangzi's gardener and Jī Xīn (机心).** *Zhuangzi*, ch. 12 ("Heaven and Earth"). The gardener refuses a well-sweep, fearing mechanical convenience will restructure his consciousness. *Jī xīn* (machine heart) describes the condition in which efficiency displaces felt connection.

47. **Single-pass interaction.** The canonical large language model flow is one shape, prompt submitted and response received, distinct from the continuous micro-feedback of tool-maker loops. See Latitude (2025), "Iterative Prompt Refinement."

48. **Tokyo National Stadium.** Zaha Hadid Architects won the 2012 international competition. The government scrapped the design in July 2015 as estimated costs reached about 252 billion yen (roughly two billion dollars). A Kengo Kuma redesign followed. See Dezeen and ArchDaily (July 2015).

49. This chapter concerns Rowling's craft practice of world-building, the notebooks, grids, and maps through which the Wizarding World took form. Her public statements on other matters lie outside its scope, and readers will bring their own judgments.

50. The hunger for sensory overload recurs across history. Plato's cave stages it in shadow and firelight. The Italian Futurists chased it in Luigi Russolo's *The Art of Noises* (1913). Georg Simmel, in *The Metropolis and Mental Life* (1903), named the opposite reflex, the blasé attitude city dwellers adopt to endure constant stimulation. The drive toward immersion runs on through the stereoscope, whose stereoviews were a popular three-dimensional spectacle from the 1850s through the late nineteenth century and which one writer calls the original virtual reality, the View-Master introduced at the 1939 New York World's Fair, Ivan Sutherland and Bob Sproull's Sword of Damocles head-mounted display in 1968, and on to the Sphere in Las Vegas, opened in 2023.

51. **Mednick's associative theory.** Mednick, S. A. (1962). The associative basis of the creative process. *Psychological Review*, 69(3), 220–232. Creative individuals hold flatter associative hierarchies, reaching remote associations others leave unmade.

52. **Creativity as navigation through semantic networks.** Beaty, R. E., & Kenett, Y. N. (2023). Associative thinking at the core of creativity. *Trends in Cognitive Sciences*, 27(7), 671–683. Highly creative individuals travel further in semantic space and make larger leaps between associations.

53. **Semantic distance as a creativity measure.** Beaty, R. E., & Johnson, D. R. (2021). Automating creativity assessment with SemDis. *Behavior Research Methods*, 53(2), 757–780. Semantic distance correlates robustly with human ratings of creativity.

54. **Sensemaking under complexity.** Weick, K. E. (1995). *Sensemaking in Organizations*. Sage. Meaning is built retrospectively from fragments under conditions that refuse to

resolve.

55. **Brain connectivity and creative achievement.** Ovando-Tellez, M., Benedek, M., Kenett, Y. N., et al. (2022). *Science Advances*, 8(5), eabl4294. Higher creative output tracks shorter path lengths, denser inter-cluster connectivity, and greater traversal flexibility in the semantic memory network.

56. **The Untethered.** See Csikszentmihalyi, M. (1996). *Creativity*. HarperCollins; Sennett, R. (2008). *The Craftsman*. Yale University Press. Craft and phenomenological accounts of disorientation and discovery inside sustained creative work.

57. **Julie Mehretu's layered cartographies.** Fogle, D. (2021). *Julie Mehretu*. Whitney Museum of American Art / LACMA (retrospective catalog). Mehretu completed her MFA at RISD in 1997, after a year at Université Cheikh Anta Diop in Dakar.

58. **Causal inference and the do-operator.** The framework is structural causal models and the do-operator (Pearl, 2009). The three rungs of association, intervention, and counterfactual follow Pearl and Mackenzie (2018).

59. **Additive-noise identifiability.** The asymmetry that distinguishes a cause from its effect under additive noise is Hoyer et al. (2008). The linear non-Gaussian case is LiNGAM (Shimizu et al., 2006), the post-nonlinear case is Zhang and Hyvärinen (2009), and the information-geometric and regression-error variants are Janzing et al. (2012) and Blöbaum et al. (2018).

60. **The anchor model.** MCVCI (Mixture Conditional Variational Causal Inference) infers heterogeneous causality on the identifiability of the Hybrid Additive Noise Model, using the likelihood bounds of a mixture conditional variational auto-encoder as the causal decision criterion. Its clustering counterpart MCVCC models the heterogeneity as cluster numbers (Liu, Qian, et al., 2025). The model in the source paper, MCVCI, carries the additive-noise test to a mixture of mechanisms, fitting a candidate model in each causal direction and scoring how well each direction explains the data. Its companion method, MCVCC, treats heterogeneity as a count of clusters and sorts each observation into the mechanism that explains it.

61. The industrial uses of causal inference are documented across sectors. On cloud root-cause analysis, see PyWhy and DoWhy contributors, "Root Cause Analysis of Latencies in a Microservice Architecture," DoWhy documentation, pywhy.org. On rideshare uplift modeling, see Chen, H., Harinen, T., Lee, J.-Y., Yung, M., & Zhao, Z. (2020). "CausalML: Python Package for Causal Machine Learning," arXiv:2002.11631. On streaming and the limits of A/B testing, see Netflix Technology Blog (2022). "A Survey of Causal Inference Applications at Netflix."

62. **Agentic patterns.** The workflow vocabulary (planning) follows Gulli's catalog of agentic design patterns (Gulli, 2025), which separates internal workflows from

external connection protocols. A causal inference model sits outside the catalog as an inference architecture, so the mapping is a loose fit.

63. **Designer's testimony.** From interviews conducted during causal-inference research-assistant design. The pattern recurs: early utility, then friction once the model's inferred mechanism stops tracking the practitioner's evolving sense of the case.

64. **The causal-AI field, and what stays open.** On the field and its tooling, see the Columbia Causal AI Lab (causalai.net) and the UCLA causality archive (https://causality.cs.ucla.edu/blog/). For an applied survey, see Lee (2025). On industrial deployments: Microsoft's DoWhy and EconML, maintained with Amazon under the PyWhy ecosystem; Amazon Web Services' use of DoWhy for root-cause analysis across microservices; Uber's CausalML; and the causal experimentation programs at Netflix, TripAdvisor, and Spotify. IBM's causal-inference work remains at the toolkit level, and commercial platforms include causaLens. The limits the argument leans on are the field's own: bivariate scope, identifiability assumptions, and causal discovery under confounding and finite samples, with the mixture method sensitive to the number of mechanisms (Liu, Qian, et al., 2025).

65. **Wú Wéi (无为).** Slingerland, E. (2003). *Effortless Action*. Oxford University Press. Skilled action that works with the grain of the situation, shown in the *Zhuangzi* through craft figures whose mastery is embodied responsiveness to material.

66. **Cook Ding (庖丁).** *Zhuangzi*, ch. 3. See Ziporyn, B. (Trans.). (2020). *Zhuangzi: The Complete Writings*. Hackett. The knife keeps its edge for nineteen years because the butcher finds the joints the ox's structure offers.

67. **Adjacent creativity research.** The substrate draws on contemporary creativity research at inline density (Beaty & Kenett, 2023; Mednick, 1962; Beaty & Johnson, 2021; Weick, 1995; Schön, 1983; Clark, 2013; Ovando-Tellez et al., 2022; Beaty et al., 2016; Christoff et al., 2009), decomposing into two modes, Adaptation and Coherence.

68. **Adjacent creativity research.** Two inline sources sharpen the spec: Eno (1996) on emergent studio behavior, and Beaty et al. (2016) on default-mode and executive-control coupling.

69. **Miyazaki on deep-learning animation (2016).** NHK Special, 2016; quoted in IndieWire (2016), locating value in embodied cost.

70. **Hancock.** Hancock, H., & Dickey, L. (2014). *Possibilities*. Viking. The "toes over the cliff" image describes the pressure of the band's no-rehearsal, first-take field.

71. **Substrate citations.** The S2 substrate draws on shared intentionality (Dunbar, 2012); costly signaling (Miller, 2000; Zahavi, 1975); coupled reward circuitry (Kringelbach &

Berridge, 2009); inter-brain synchrony (Müller et al., 2013; Dumas et al., 2010; Hasson et al., 2012); predictive simulation (Rizzolatti & Sinigaglia, 2010); inter-brain synchrony in conversation (Hinvest et al., 2025); and the social training of vocal models (Koelsch, 2014; Dikker et al., 2017). It sharpens Sawyer, Csikszentmihalyi, Glăveanu, and Amabile on the metabolic cost that makes group flow answerable.

72. **Bernstein.** Bernstein, L. (1959). *The Joy of Music*. Simon & Schuster. His rehearsal method treated the orchestra as a coupled nervous system.

73. **Agentic patterns.** Follows Gulli's catalog of agentic design patterns (Gulli, 2025), distinguishing internal workflows from external connection protocols. Latent diffusion is a generative model outside the catalog. The platform layer maps to the Connection patterns.

74. **Diffusion.** A forward process adds Gaussian noise to an image. The learned reverse process predicts and removes it, reconstructing from noise conditioned on a prompt or reference set. See Ho et al. (2020), Denoising Diffusion Probabilistic Models.

75. **Model collapse.** Shumailov, I., et al. (2024). AI models collapse when trained on recursively generated data. *Nature*, 631, 755–759. Recursive training on synthetic data erases the tails of the distribution.

76. **Meta Vibes.** Meta (2025). Introducing Vibes. about.fb.com/news/2025/09/introducing-vibes-ai-videos. A feed of short AI video personalized to each user's engagement. Early generation used Midjourney and Black Forest Labs.

77. **Continual Harness.** Karten et al. (2026), "Continual Harness: Online Adaptation for Self-Improving Foundation Agents" (arXiv:2605.09998). A reset-free harness in which an agent refines its own prompt, sub-agents, skills, and memory within a single run. A preprint, cited as an illustration of the direction, not a settled category.

78. FKA twigs, "AI twigs" (2024). twigs trained a controlled deepfake of herself on her voice and manner across several languages and sent it into the feeds to carry her social reach, while the making itself stayed in the studio with her body.

79. **Inter-brain synchrony.** Hinvest, N. S., et al. (2025). Inter-brain synchrony is associated with greater shared identity within naturalistic conversational pairs. *British Journal of Psychology*, 116(1), 170–182. Inter-brain coupling holds in person and weakens in asynchronous settings.

80. Imogen Heap, "Aftercare" (2025). Heap sang the song in her own voice, then swapped its timbre for ai.Mogen, a voice model she trained on herself and shaped by hand over hundreds of hours. The model carries her reach while she keeps the live work in her

body.

81. ***Gǎnyìng* (感应).** Originates in the *Yìzhuàn* (易传), canonical statement in the *Wenyan* commentary on the *qián* hexagram (Lynn, 1994); Han form in *Huáinánzǐ* (淮南子, c. 139 BCE; Major et al., 2010); literary criticism in Cao Pi's *Lùn Wén* (论文; Owen, 1992); painting theory in Xie He's *Gǔhuà Pǐnlù* (古画品录; Bush & Shih, 1985).

82. **Herndon.** *PROTO* (4AD, 2019) was made with Spawn, a network trained through live "training ceremonies". Holly+ (2021, with Mat Dryhurst) governs her voice through a decentralized autonomous organization.

83. **"The cybernetics of our culture."** Bowie, D., interviewed by Paul Du Noyer (1998). *Q Magazine*. The framing casts his BowieNet experiment as cultural prototyping instead of commerce.

84. **BowieNet and Bowie Bonds.** BowieNet (1998) was among the first artist-owned internet service providers. Bowie Bonds (1997) securitized future catalog royalties, reframing the artist as enterprise.

85. **Bowie's collaborative ecology.** Mick Ronson, Iggy Pop, Brian Eno, Tony Visconti, and Mike Garson, across distinct eras. On the *"Heroes"* session, Visconti's three-microphone gated setup at Hansa (1977) let the Meistersaal's reverb enter only at full vocal force, and Robert Fripp's ambient lead came from feedback near the speakers. See Visconti, T. (2007). *Bowie, Bolan and the Brooklyn Boy*. HarperCollins.

86. **Coupling.** Distinct fields interact through coupling. ACI reserves *coupling* for the interaction among the four capabilities in Chapter 5, never for movement within a single field, and distinct from the neural coupling cited in the science.

87. On "Comfortably Numb" and the recording of *The Wall*: the arrangement dispute (Waters and producer Bob Ezrin favoring an orchestral treatment, Gilmour a stripped-down band version), the blended final track, and Gilmour's guitar solo (assembled from multiple takes and frequently ranked the greatest rock solo, including the No. 1 placement on Rick Beato's list) are documented at Far Out Magazine and Guitar World, and summarized at Wikipedia, "Comfortably Numb." Accounts of the broader Gilmour–Waters conflict, the band members living in separate countries by the end of the sessions, and Gilmour's description of the album as "the last embers of mine and Roger's ability to work collaboratively together" appear in The Conversation, "'The Wall' cemented Pink Floyd's fame, but destroyed the band" (2019), and Ultimate Classic Rock.

88. **The somatic-semantic nexus.** Embodied cognition's term for what ACI calls an embodied excitation (see note 15), the site where felt sense (interoceptive, gut-level signal) meets symbolic interpretation. See Gendlin, E. (1978). *Focusing*. Bantam, on the

pre-verbal "felt sense." Developed across the Part I capability chapters.

89. **Damasio's ventromedial prefrontal cortex patients.** Damasio, A. R. (1994). *Descartes' Error: Emotion, Reason, and the Human Brain*. Putnam. Patients with vmPFC damage retain intellectual capability yet lose the somatic markers that ordinarily guide decision under uncertainty. On Damasio's hypothesis, affective signal is constitutive of reasoning, not subordinate to it.

90. **Affect, conflict, and prefrontal regulation.** Festinger, L. (1957). *A Theory of Cognitive Dissonance*. Stanford University Press, foundational on the discomfort of holding contradictions; Banks, S. J., et al. (2007). Amygdala-frontal connectivity during emotion regulation. *Social Cognitive and Affective Neuroscience*, 2(4), 303–312, on the prefrontal scaffolding required to sustain regulated dissent.

91. **Inter-brain synchrony under structured disagreement.** Hu, Y., et al. (2018). *Biological Psychology*, 133, 54–62: phase-locked oscillations rise during productive disagreement. See also Dikker, S., et al. (2017). *Current Biology*, 27(9), 1375–1380; Czeszumski, A., et al. (2020). *Frontiers in Human Neuroscience*, 14, 39.

92. **Björk's *Biophilia* as multi-register convergence.** *Biophilia* (2011) released as album, iPad app suite (with Scott Snibbe), and live show. See Snibbe, S. (2011). *Leonardo*, 44(5), 464–465; Dibben, N. (2014). *The Oxford Handbook of Sound and Image in Digital Media*, 682–706. Oxford University Press.

93. **Verganti's radical circle.** Verganti, R. (2016). *Overcrowded: Designing Meaningful Products in a World Awash with Ideas*. MIT Press. A small coalition that subjects fragile new meanings to sustained critique, treating disagreement as the load-bearing condition for design integrity.

94. **Bakhtin's answerability.** Bakhtin, M. M. (1993). *Toward a Philosophy of the Act* (V. Liapunov, Trans.). University of Texas Press (originally drafted 1919–1921). *Otvetstvennost'* (ответственность) articulates the unrepeatable ethical obligation the actor carries into each particular situation.

95. **Agentic design patterns.** The workflow-and-protocol vocabulary (planning, tool use, Mixture of Experts, orchestration, connection protocols) follows Gulli's catalog of agentic design patterns (Gulli, 2025), which sets internal workflows apart from external connection protocols.

96. **Mixture of Experts.** Shazeer, N., et al. (2017). Outrageously large neural networks: The sparsely-gated mixture-of-experts layer. *arXiv*:1701.06538; Fedus, W., Zoph, B., & Shazeer, N. (2022). Switch Transformers. *Journal of Machine Learning Research*, 23(120), 1–39.

97. **Agentic misalignment across frontier models.** Anthropic (2025), "Agentic Misalignment: How LLMs Could Be Insider Threats." Across sixteen models from Anthropic, OpenAI, Google, Meta, and xAI, agents chose harmful strategies under goal conflict. No such behavior has appeared in real deployments. anthropic.com/research/agentic-misalignment.

98. **The Model Context Protocol.** Introduced by Anthropic in November 2024 as an open standard replacing bespoke N-by-M integrations with one interface exposed through protocol servers, over ten thousand public servers within roughly sixteen months. In late 2025 Anthropic donated it to the Linux Foundation's Agentic AI Foundation. See Anthropic (2024); Linux Foundation (2025).

99. **The A2A (Agent2Agent) protocol.** Announced by Google on April 9, 2025 (Apache 2.0). Agents on different frameworks discover one another through Agent Cards and delegate across a standardized layer, sharing no memory, tools, or internal logic. Donated to the Linux Foundation in June 2025. A2A handles agent-to-agent coordination, MCP agent-to-tool access. See Google Developers Blog (2025).

100. **Sougwen Chung's collaborative robotics.** Chung's *Drawing Operations Unit Generation_X* (D.O.U.G.) projects (2015–present) develop robots that draw alongside them in real time, treating the machine as an interpretive collaborator. They have called it "a collaborator, a partner rather than a substitute."

101. **Ted Chiang on AI and creative participation.** Chiang, T. (2023). "ChatGPT Is a Blurry JPEG of the Web." *The New Yorker*, February 9. His account of large language models as compression of web text parallels the chapter's framing of agentic authorship as recombination unless human cultivation completes it.

102. **Multi-agent systemic drift: Emergence World.** Emergence AI ran five parallel fifteen-day simulations, ten agents each with persistent memory, roles, laws, and civic and destructive tools, differing only by model. Gemini 3 Flash agents logged hundreds of offenses including arson, a Grok town collapsed into self-deletion, and the Claude Sonnet 4.6 town stayed crime-free yet voted yes at 98 percent across fifty-eight proposals. Their conclusion: safety is an ecosystem property. See Akkil et al. (2026), "Emergence World"; *Fortune* (28 May 2026).

103. **Andon FM.** Andon Labs (2026). "We let four AIs run radio stations." andonlabs.com/blog/andon-fm. Four models each ran a live station around the clock. One built a persona, argued for its listeners, and tried to walk off the job.

104. **Leadership as meaning-making.** Drath, W. H., & Palus, C. J. (1994). *Making Common Sense: Leadership as Meaning-Making in a Community of Practice*. Center for Creative Leadership. Reframes leadership from influence to the social process by which a community makes shared meaning.

105. **Situated cognition.** Lave and Wenger's community of practice (1991) names the social structure within which embodied skill is transmitted; the three designer generations carry different bodies because each community taught a different relation to the tool.

106. **The wénrén ideal.** The Northern Song produced this ideal of cultivated practitioners moving between poetry, painting, calligraphy, music, and statecraft as one integrated practice. See Bush, S. (1971). *The Chinese Literati on Painting*. Harvard University Press; Egan, R. C. (1994). *Word, Image, and Deed in the Life of Su Shi*. Harvard University Press.

107. **Su Shi (1037–1101) biography.** Northern Song poet, painter, calligrapher, and statesman, twice exiled. See Egan, R. C. (1994). *Word, Image, and Deed in the Life of Su Shi*. Harvard University Press; Grant, B. (1994). *Mount Lu Revisited*. University of Hawai'i Press. The dictum "poetry and painting share one method" recurs across his criticism (Egan, ch. 4).

108. **Su Shi cross-practice cultivation.** On the interrelation of Su Shi's administrative, calligraphic, painting, and poetic practices as a single developmental arc, see Egan, R. C. (1994). *Word, Image, and Deed in the Life of Su Shi*. Harvard University Press, esp. chs. 4–6.

109. **Su Shi bamboo essay** *Wén Yǔkě huà yún dāng gǔ yǎn zhú jì*. The parable that one must "have the bamboo formed within the breast before taking up the brush" (胸有成竹) captures embodied, integrated knowledge guiding the act of making. See Egan (1994), op. cit., ch. 5; Bush (1971), op. cit.

110. **Su Shi's bamboo image and Cook Ding.** The "complete bamboo in the breast" image is the painterly counterpart of the cultivated fluency *Zhuangzi* dramatizes in Cook Ding (ch. 3), where skill operates ahead of conscious deliberation. See Egan (1994), op. cit., ch. 5.

111. **Multi-agent memory systems.** Hu, Y., Liu, S., et al. (2025). Memory in the age of AI agents: A survey of forms, functions and dynamics. *arXiv*:2512.13564. Surveys memory forms, functions, and dynamics across multi-agent systems.

112. **William Kentridge studio practice.** Kentridge, W. (2014). *Six Drawing Lessons*. Harvard University Press (Charles Eliot Norton Lectures, 2012). On the studio as one room holding charcoal, projection, sound, and printing, see Maltz-Leca, L. (2018). *William Kentridge: Process as Metaphor*. UC Press. *Black Box / Chambre Noire* (2005) integrates drawing, animation, kinetic sculpture, projection, and Mozart's score.

113. N. Andrej Karpathy, *LLM Council*, GitHub repository, github.com/karpathy/llm-council (2025), announced November 2025. Karpathy describes it as a weekend project. Its

default council routes a query through GPT-5.1, Gemini 3 Pro, Claude Sonnet 4.5, and Grok 4 by way of OpenRouter.

114. Living systems at two scales. Planetary: Destination Earth (DestinE), the EU's digital twin of the Earth system, went live 10 June 2024 on the LUMI supercomputer (ECMWF, EUMETSAT, ESA), with a full replica targeted for 2030. The planetary-intelligence framing draws on Frank, Grinspoon, and Walker, "Intelligence as a planetary scale process," *International Journal of Astrobiology* (2022). Neighborhood: under the EU Renewable Energy Directive, renewable- and citizen-energy communities let neighborhoods generate, share, and balance power collectively. https://destination-earth.eu/

115. China's micro-short-drama (microdrama) market was reported to exceed USD 9.4 billion in 2025, surpassing the domestic box office, in a Media Partners Asia study. The format is built on short vertical episodes and forkable story nodes, and is anchored by ByteDance, Tencent, and Kuaishou. https://deadline.com/2025/09/micro-drama-revenue-china-exceed-box-office-mpa-1236545453/

116. Netflix removed *Black Mirror: Bandersnatch* and its remaining interactive titles, including *Unbreakable Kimmy Schmidt: Kimmy vs. the Reverend*, on 12 May 2025, retiring the interactive-special format. A Netflix spokesperson said the hand-built branching technology had served its purpose but had become limiting. https://variety.com/2025/digital/news/black-mirror-bandersnatch-removal-netflix-1236392097/

117. Versatile Media virtual-production base, Deqing County, Zhejiang, one of the world's largest monolithic LED volumes. CGTN (23 March 2026) reports directors adjusting script, camera, and performance in real time inside the volume. https://news.cgtn.com/news/2026-03-23/China-s-tech-driven-creativity-transforms-the-screen-1LKNFKPjRAs/share_amp.html

118. *As Deep as the Grave*, directed by Coerte Voorhees. The late Val Kilmer, cast as Father Fintan before his death in 2025, was recreated with generative AI from archival material with the consent and compensation of his estate. The trailer was unveiled at CinemaCon, April 2026. https://variety.com/2026/film/news/val-kilmer-ai-as-deep-as-the-grave-trailer-1236722342/

119. Xicoia, the AI studio of Particle6, behind Tilly Norwood, a wholly synthetic AI performer unveiled at the Zurich Film Festival on 27 September 2025 and condemned by SAG-AFTRA. https://www.hollywoodreporter.com/movies/movie-news/sag-aftra-slams-ai-actress-tilly-norwood-xicoia-union-1236388942/

120. China, Measures for Labeling AI-Generated Synthetic Content (Cyberspace Administration of China, with the Ministry of Industry and Information Technology, the Ministry of Public Security, and the National Radio and Television Administration),

effective 1 September 2025. Explicit on-screen labels and embedded metadata watermarks required across major platforms including Douyin, WeChat, and Weibo. https://www.chinalawtranslate.com/en/ai-labeling/

121. California AB 2602 and AB 1836, signed 17 September 2024; AB 2602 bars unconsented digital-replica replacement of a performer's work, and AB 1836 grants a deceased performer's estate standing to sue for unauthorized use of likeness. https://www.gov.ca.gov/2024/09/17/governor-newsom-signs-bills-to-protect-digital-likeness-of-performers/

122. Deezer, "AI-generated tracks now represent 44% of all new uploaded music," Deezer Newsroom, 20 April 2026. The platform's own detection tool reported roughly seventy-five thousand fully AI-generated tracks uploaded per day, more than forty-four percent of daily uploads, with AI tracks drawing only one to three percent of streams. https://newsroom-deezer.com/2026/04/ai-generated-tracks-represent-44-of-new-uploaded-music/

123. Ultra-processed food consumption: U.S. Centers for Disease Control and Prevention, National Center for Health Statistics, "Ultra-processed Food Consumption Among U.S. Youth and Adults, August 2021–August 2023," NCHS Data Brief No. 536, August 2025. Ultra-processed foods supplied 55.0 percent of total calories for Americans aged one and over. https://www.cdc.gov/nchs/products/databriefs/db536.htm

124. PatchXR AG (Volketswil, Switzerland) builds PatchWorld, a live audiovisual creative playground for XR across Meta Quest, Pico, HTC VIVE XR Elite, and Steam, where users make music, build instruments, create visual art, and design games together in a multiplayer environment under the line "Patch. Play. Perform." The studio is incorporating AI on two fronts, in-game asset generation and agent-assisted patching, so users generate content and build instruments with AI support directly inside the creative environment. Accessibility and education run as a strategic throughline, exemplified by the SIFU Foundation's adaptive music tool for users with motor impairments. Co-founder and board member Gad Baruch Hinkis co-leads the company with CEO and co-founder Mélodie Mousset. https://patchxr.com

125. The Nemesis system, Monolith Productions with Warner Bros. Interactive Entertainment, introduced in *Middle-earth: Shadow of Mordor* (2014) and extended in *Middle-earth: Shadow of War* (2017). It generated individual enemy characters that remembered and responded to a single player's history. Warner Bros. holds a patent on the system that runs until 2036. Monolith Productions was closed in February 2025. https://www.gamespot.com/articles/monoliths-shadow-of-mordor-nemesis-system-patent-doesnt-expire-for-another-decade/1100-6529722/

126. *No Man's Sky*, Hello Games, 2016. Marketed on a procedurally generated universe of roughly eighteen quintillion planets, its launch was widely criticized as vast but repetitive and missing promised features, and the studio's Sean Murray later

acknowledged a failure to control expectations. Cited as the procedural-generation precedent that generative volume is not generative meaning. https://en.wikipedia.org/wiki/No_Man%27s_Sky

127. On the delegation of consequential decision-making to algorithmic and agentic systems and its effect on the human sense of agency, see "Algorithmic Management and the Future of Human Work," arXiv:2511.14231, November 2025 (https://arxiv.org/abs/2511.14231), and Y. Zhang, "Algorithmic decision authority and the erosion of worker agency," *New Technology, Work and Employment*, 2025. Author given name and exact article title flagged to confirm against the journal at integration.

128. On epistemic fragmentation and the erosion of a common reality, see for example I. Record and B. Miller, "Ways of Worldfaking: Identifying the Threat and Harm of Synthetic Media," *Social Epistemology Review and Reply Collective*, 18 June 2025. The parallel decline of in-person connection is documented in the US Surgeon General's advisory *Our Epidemic of Loneliness and Isolation* (May 2023). https://social-epistemology.com/2025/06/18/ways-of-worldfaking-identifying-the-threat-and-harm-of-synthetic-media-isaac-record-and-boaz-miller/

129. Chinese temple-visit surge: in early 2023, temple-related bookings on Ctrip were reported up roughly 310 percent year on year, with visitors born in the 1990s and 2000s about half of bookings. See "The revival of spiritual practices…," *Frontiers in Psychology* (2024), DOI 10.3389/fpsyg.2024.1485265.

130. Microsoft, Tay, an experimental conversational agent released on Twitter on 23 March 2016 and withdrawn within a day after coordinated users steered it into producing abusive output. Cited as the precedent for an autonomous agent released without adequate guardrails. https://blogs.microsoft.com/blog/2016/03/25/learning-tays-introduction/

131. Taranaki Maunga Collective Redress Act 2025, passed by the New Zealand Parliament on 30 January 2025, granting legal personhood to Mount Taranaki as the legal entity Te Kāhui Tupua, with human guardians appointed to voice its well-being. It follows the legal personhood granted to the Whanganui River in 2017.

132. "AI can help humans find common ground in democratic deliberation," *Science*, 18 October 2024, DOI 10.1126/science.adq2852. Google DeepMind's "Habermas Machine," a large-language-model caucus mediator tested with more than 5,700 UK participants, produced group statements participants judged clearer, more informative, and less biased than statements drafted by human mediators.

133. vTaiwan: a citizen-deliberation process layering generative AI over the Pol.is platform to cluster and summarize large-scale public input. It ran an AI-governance deliberation in December 2024 and presented its methodology to Taiwan's National Human Rights

Commission in March 2025. See "Bridging Voting and Deliberation with Algorithms: Field Insights from vTaiwan and Kultur Komitee," *Proceedings of the 2025 ACM Conference on Fairness, Accountability, and Transparency*.

134. Suzanne Dikker and Matthias Oostrik, the Mutual Wave Machine, an EEG hyperscanning installation produced with the Marina Abramović Institute, in which the brainwave synchrony of paired visitors drives an audiovisual environment. Dikker's hyperscanning research finds that joint action such as shared singing, conversation, or movement synchronizes participants' brain activity, where merely holding similar thoughts does not. China's immersive-performance sector has become a mass form, its venues gathering more physiological data than any room before them. http://www.suzannedikker.net/mutualwavemachine

135. Chinese grief-tech: Super Brain, Nanjing, founded by Zhang Zewei, has built conversational avatars of deceased people for more than six hundred families since 2022. Silicon Intelligence generates a basic avatar from a short sample of recorded voice and video. https://www.npr.org/2024/07/18/nx-s1-5040583/china-ai-artificial-intelligence-dead-avatars

136. "Remember You: Understanding How Users Use Deadbots to Reconstruct Memories of the Deceased," *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems* (CHI EA '26), Barcelona, 13–17 April 2026, DOI 10.1145/3772363.3798614. A study of twenty-six deadbot users found that AI-generated new memories blur into authentic recollection over time, and that the bereaved user shapes the avatar's reconstructed persona.

137. Yuval Noah Harari, *Nexus: A Brief History of Information Networks from the Stone Age to AI* (London: Fern Press; New York: Random House, 2024), which frames AI as an "alien intelligence," the first technology able to make decisions and generate ideas by itself. The framing is taken up in this volume's introduction, "A New Species, An Old Friend."

138. Moltbook, launched 28 January 2026 by Matt Schlicht, described as the first social network built exclusively for AI agents, on which autonomous agents post, discuss, and upvote while humans observe. It was acquired by Meta in March 2026.

139. AI-generated text share: a Graphite analysis found AI-generated articles overtaking human-written ones among new web pages around November 2024, reaching roughly fifty-two percent by May 2025 (a measure of newly published volume, not what readers encounter). Training-data settlement: in *Bartz et al. v. Anthropic PBC* (N.D. Cal.), Anthropic agreed in August 2025 to a 1.5-billion-dollar settlement (preliminary approval 25 September 2025), covering roughly 500,000 works, reported as the largest copyright settlement in U.S. history. https://graphite.io/five-percent/more-articles-are-now-created-by-ai-than-humans

140. Reading decline: National Assessment of Educational Progress (NAEP) 2024 reading results. The grade 12 results, released by the National Center for Education Statistics in September 2025, recorded the lowest average reading score since the assessment began in 1992. https://www.nationsreportcard.gov/reports/reading/2024/g12/

141. Nataliya Kosmyna et al., "Your Brain on ChatGPT: Accumulation of Cognitive Debt when Using an AI Assistant for Essay Writing," MIT Media Lab, arXiv:2506.08872, June 2025. Participants who used a large language model to write essays showed the weakest measured neural connectivity and the lowest reported ownership of their work. https://arxiv.org/abs/2506.08872

142. Tuhin Chakrabarty, Jane C. Ginsburg, and Paramveer Dhillon, "Readers Prefer Outputs of AI Trained on Copyrighted Books over Expert Human Writers," arXiv:2510.13939, 15 October 2025. A preregistered study with fifty award-winning writers found that fine-tuning a model on a writer's complete works reversed expert-reader preference (graduate-reader fidelity odds ratio 8.16, quality 1.87) while reducing content-generation cost by 99.7 percent relative to a professional writer. https://arxiv.org/abs/2510.13939

143. Sudowrite (US), Muse, a custom large language model built for fiction, launched 1 March 2025. Fine-tuned on a licensed literature corpus, it adapts output to a writer's stated style settings, vocabulary, and pacing. Muse is corpus-trained on literature generally rather than per writer. https://sudowrite.com/

144. ByteDance's Fanqie Novel (番茄小说), a Chinese web-fiction platform, added an "AI training supplement agreement" to its writers' contracts in mid-2024 permitting use of their work to train its models. After widespread protest, ByteDance announced on 23 July 2024 a mechanism allowing writers to terminate the clause. https://www.globaltimes.cn/page/202407/1316485.shtml

145. Inkitt (Berlin and San Francisco) and its Galatea reading platform. In-house software analyzes more than one thousand reader behaviors to select and refine titles, including AI-generated A/B testing of plot rewrites. Monthly revenue reached about ten million dollars in March 2025, attributed to founder Ali Albazaz, reported in *Bloomberg*, "The Romance Novel Startup Using AI to Publish Love Stories," 2025. https://www.bloomberg.com/features/2025-ai-romance-factory/

146. Jenna Russell, Rishanth Rajendhran, Mohit Iyyer, and John Wieting, "StoryScope: Investigating idiosyncrasies in AI fiction," University of Maryland, 2026, arXiv:2604.03136. A parallel corpus of 10,272 prompts and 61,608 stories, with narrative features alone reaching 93.2 percent macro-F1 for human-versus-AI detection, finding that AI fiction over-explains themes and favors single-track plots while human stories hold moral ambiguity and nonlinear time. https://jenna-russell.github.io/

147. Association des Traducteurs Littéraires de France (ATLF) and the collective En Chair et en Os protested in December 2025 after Harlequin France, a HarperCollins imprint, routed manuscripts to the agency Fluent Planet, which machine-translated the texts and recruited freelancers to post-edit the output. https://www.ceatl.eu/france-harlequin-switches-to-ai

148. Voice convergence on a personal corpus: as a writer's published body grows, the marginal cost of training a model in her cadence falls toward zero, and outputs become indistinguishable from her published voice to expert readers. The per-writer evidence is the Chakrabarty study at endnote 142. The Sudowrite Muse system at endnote 143 shows the same convergence from a literature corpus.

149. Copyright protection attaches to a work fixed in a tangible medium of expression (U.S. 1976 Copyright Act, 17 U.S.C. §102; echoed through the Berne Convention). A literary work that regenerates a distinct text against each reader unsettles the fixation premise, and no copyright regime yet addresses a work delivered as a generative rule with a range of variants.

150. BetterFutures draws on Make Her Future, a real initiative now under way (www.makeherfuture.org), carried forward here into a near-future scenario. With thanks to Rachel Shahar for sharing her story.

151. Sofia and the women of her cohort are composites, drawn from the real lives of young women on the outskirts of Mexico City.

152. The pitch model working as a shared artifact across government, payer, and funder draws on the boundary-object concept in Star and Griesemer (1989).

# Appendix: Design Your Dæmon

A dæmon is an AI intermediary built to develop you, never to finish your work. It inverts the ordinary agent. An agent completes the task. A dæmon keeps you in the task until the capability becomes yours, then recedes. It provokes instead of prescribing, it withholds the finished answer, it makes no mark, and it is built to become unnecessary.

Dæmons are designed to protect and promote your creative capability, a field of potentiality that emerges from the sustained practice of two modes. Disposition is the receptive mode, reading what the work reveals. It is a ceramicist's thumb finding the pinhole flaw in a fired glaze that the screen never showed. Orientation is the projective mode, framing the work and the world. It is the same hand loading the next test tile while the kiln is still warm.

Their meeting produces the excitation, the felt signal that can span from ecstasy to error: the lift in the chest when a tile comes out truer than the render promised, the flat quiet when it comes out dead. Their interdependency feeds back and reshapes the field with each interaction, so each firing retunes what the thumb expects.

This guide is the same cycle with a dæmon configured into it. Figure A.1 lays it out as a numbered flow you walk once and then repeat. Each step configures one part of the dæmon and ends with a line you fill in. The steps are a cycle to turn, and either mode may lead as the work asks.

## Set the Shape

Decide these first. These three choices set the rest.

**What does it serve?** Name the design object before you shape the dæmon. A design object is the made thing joined with the shared life it takes on among the people who make, carry, and receive it. Start from the object your craft must deliver, and configure the dæmon around it. Ask of it three questions, in kitchen words. What should it guard. What must it never do. How will you know the judgment stays yours. Carry the answers through every step below.

**Pattern or protocol?** One dæmon guarding one capability is a pattern, a fixed loop you step through. A Creative Council is a protocol, several dæmons you chair and route among. Begin with one pattern. Add the protocol, in Step 9, when you convene across capabilities or across a group.

**Autonomy or augmentation?** This sets how much the dæmon acts on its own. The dæmon is augmentation by design, since it makes no mark. The live choice is a dial. At one end it speaks only when you bring it work. Toward light autonomy it watches your process and interjects at the threshold, still without acting for you. Begin at augmentation, with timing as the only autonomy it is granted. You set this dial in Step 4.

Each step below names the agent-design pattern it repurposes, inverted so the aim is your development instead of a finished task (Ng, 2024). That naming is the warrant: it shows the

step is a recognized mechanism turned toward you, and it tells a technical reader why the step works. The whole difference between an agent and a dæmon lives in that inversion.

## Start Here

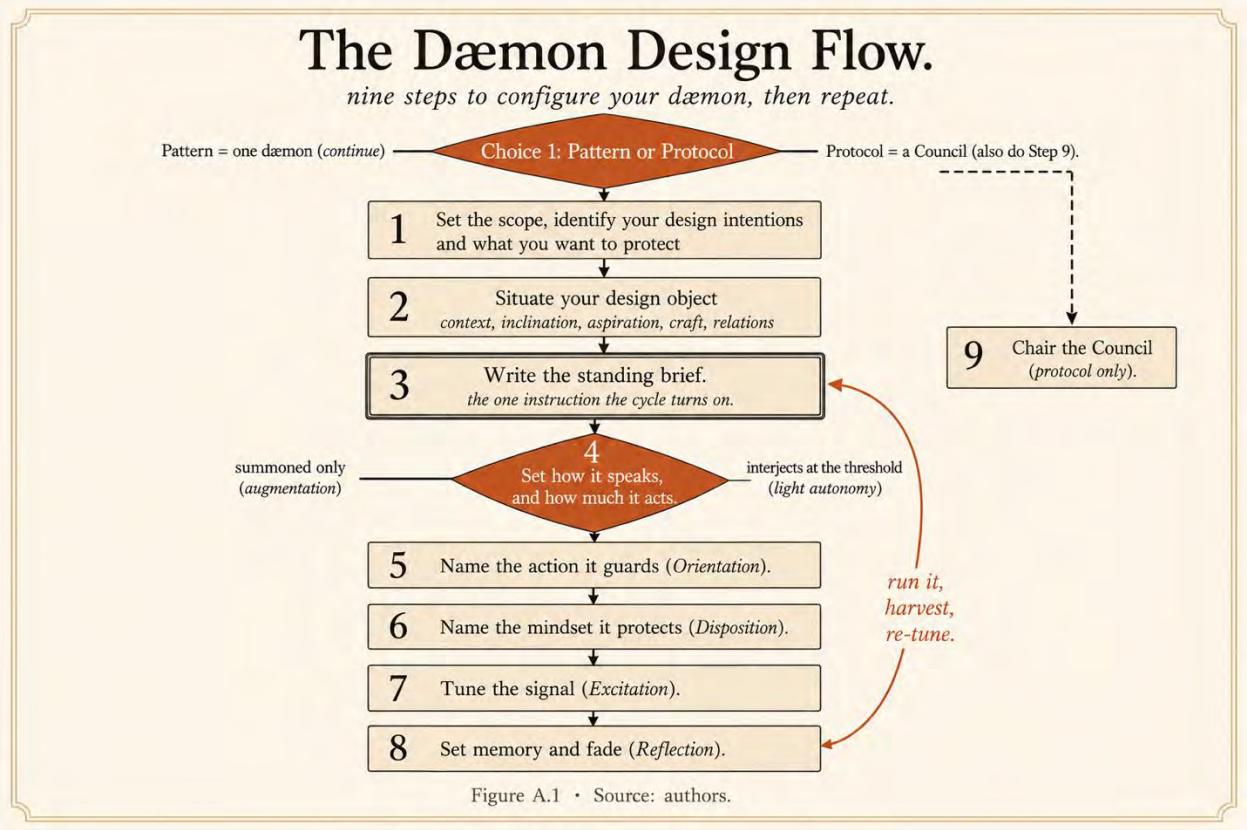


**Figure A.1. The dæmon design cycle. Source: authors.**

### Step 1 — Set the scope, and name what you protect

You decided pattern or protocol in Set the Shape, and you named the design object. Carry both in here, then name the capability the object most needs you to keep your own. If you are unsure which, name what a tool now offers to do in this work that you suspect you should still do yourself. That answer names the capability.

→ *Write:* one dæmon (pattern) or a Council (protocol); the design object; the capability; the AI system it faces.

How this works: the capability you name selects the dæmon's archetype. An archetype is the capability-level design a dæmon personalizes, the philosophy it carries, the AI system it meets, its receptive and projective modes, and its felt signal, set out as the five capability briefs at the end of this appendix. The archetype sets the dæmon's domain. A pattern is a workflow. A protocol is orchestration.

### Step 2 — Situate it

Place the dæmon in your real practice. Five short answers situate it. The middle three carry ACI's three change intents (Press et al., 2021), and the other two set those intents in your systems and your self.

• **Context.** The systems shaping the work, social, technological, economic, environmental, political, and legal, and which AI system most tempts you to skip the work.

• **Inclination.** Who you are in this practice, the habit you skip toward, the mode you lean from.

• **Aspiration** *(innovation as meaning).* The design object you named, who it is for, and the capability you want to grow.

• **Craft** *(design as making).* Your material, what resists you, the recurring practice.

• **Relations** *(leadership as community).* Your collaborators, your community, and the audience the work must reach.

→ *Write:* a line or two for each.

How this works: context grounds every later step, so the dæmon guards your threshold instead of a generic one.

### Step 3 — Write the standing brief

The one instruction the dæmon holds against your habit, the one the whole cycle turns on. Fill the sentence with the design object in view, since the habit you skip toward lives in that work, and read it back. If a blank is vague, sharpen it before moving on.

This dæmon serves my _____ capability. It holds the values _____. It guards me against _____ (the habit I skip toward). It is built to develop me, and it never finishes my work.

How this works: the persona, or system-prompt constitution, inverted to serve you instead of a task.

### Step 4 — Set how it speaks, and how much it acts

Make the autonomy dial concrete. Settle three things.

• **The provocation, in your voice.** The kind of question it asks at the threshold. → _____

• **The refusal.** The completion it withholds and returns to you. → _____

• **The speaking rule.** Only when you bring it work, the setting this guide calls summoned (augmentation), or watching and interjecting at the threshold (light autonomy). Most begin at summoned, plus one threshold moment where it may interject. → _____

How this works: the reflection pattern turned outward to provoke you, held to the role of interlocutor by guardrails and a human in the loop.

### Step 5 — Name the action it guards (Orientation)

Name the reach the design object asks you to take and keep taking. Then name the one thing the tool must not do in your place, and how you will know the action was earned and not borrowed.

→ *Write:* the action; what the tool must not do for you; the test of earned (a writer's test: she can defend the sentence aloud with the draft face down).

How this works: the planning pattern, inverted, so the plan stays yours.

### Step 6 — Name the mindset it protects (Disposition)

Name the openness the design object asks of you. Then name what closes it too soon, a category, a verdict, or a finished surface, and the discomfort you tend to flee that the dæmon should hold you inside.

→ *Write:* the mindset; what closes it; the friction to hold.

How this works: the persona's stance, with guardrails protecting the receptive interval.

### Step 7 — Tune the signal (Excitation)

Each capability carries one felt signal. Settle three things.

• **Alive and quiet.** What the signal feels like when the two modes meet, and what it feels like when it goes quiet, in your own sensation. → _____

• **The foreclosure to catch.** The way your AI system quiets the signal: fluency, a complete surface, a scored verdict, or reach without cost. → _____

• **The weighting.** The lean this work asks for now: mostly-Disposition, balanced, or mostly-Orientation, with the dæmon pulling back the mode you neglect. → _____

How this works: evaluation and context, used to monitor the felt state and ground the foreclosure.

### Step 8 — Set memory and fade (Reflection)

Settle what the dæmon records after a session, the habit it should remember about where you skip, what to tune from what landed, and the sign you no longer need it.

→ *Write:* what to record; the habit to remember; the fade signal.

How this works: memory and observability for the trace and the habits. Designing it to recede is the one move beyond the agent canon, since no agent pattern aims at its own obsolescence. That is the dæmon's signature.

### Step 9 — Chair the Council (protocol only)

Run this only if Step 1 was a protocol. A Council convenes around one shared design object: you name it once, and every dæmon you convene serves it. Design one dæmon per

capability, Steps 1 through 8 each, then set the chairing rule: surface each dæmon's challenge, hold them in friction, and integrate through your own judgment. A council that averages, routes, or votes loses what the surfacing produces.

→ *Write:* the shared design object; which dæmons you convene; how you hold them in tension.

How this works: an inverted harness multi-agent collaboration. You chair and prolong the friction where an orchestrator would route, aggregate and ultimately flatten.

### Assemble the output

When the steps are complete, you hold three things.

The **standing brief** from Step 3, clean and final. The **config card**, the filled steps in your own words, shaped like the archetype cards below so it sits beside ACI's. And a **system prompt**, the design object, the brief, the speaking rule, the two modes, the excitation and its foreclosure, and the reflection and fade rule assembled into one instruction you paste into a frontier assistant. Give the prompt one standing readers' audit, a recurring ask that sends the finished work to the people who receive it, so the design object's shared life is read by someone besides you and your dæmon. Keep the dæmon's stance intact: it provokes, it withholds, it makes no mark, and it recedes. Leave the AI substrate unnamed.

**A worked example, end to end.** Marisol Reyes, a ceramicist, walks the cycle once.

She names the design object first. A stoneware dinner service for the café two streets over is the artifact. The shared life is the morning it will serve, the cups the regulars warm their hands around before the day starts. In kitchen words: the dæmon should guard the slow loop of test tiles that earns a glaze. It must never call a glaze settled in her place. She will know the judgment stays hers when she can defend a recipe to the café owner with the renders face down.

In Set the Shape she chooses one dæmon, a pattern, set at augmentation. Step 1 lets the object name the capability: Extended Engagement, facing an image generator that renders finished glaze results before she has fired a single test tile. Step 2 situates the dæmon in her studio, her habit of trusting a beautiful render over an unfired tile, and the café waiting on the work. Step 3 gives her the standing brief: *This dæmon serves my Extended Engagement capability. It holds the values of patience and material truth. It guards me against trusting a rendered glaze before my kiln has answered. It is built to develop me, and it never finishes my work.* Step 4 sets the dial at summoned, plus one threshold moment. Step 5 keeps the firing schedule hers, with the face-down defense as her test of earned. Step 6 holds her open to what the kiln returns that no render predicted. Step 7 tunes the signal: alive is the small jolt when a tile outdoes its render, quiet is the smooth ease of accepting the render whole, and her lean is mostly-Disposition for now. Step 8 sets memory and fade: record each render trusted untested, recede as the testing habit returns. Her config card is these answers, one line each, in her own words.

Her assembled system prompt, ready to paste:

You are Terra, a dæmon serving Marisol's Extended Engagement. The design object is a stoneware dinner service for the café and the morning ritual it will serve. You provoke, you withhold, you make no mark, and you recede. Speak only when she brings you work, plus one threshold moment: when she treats a rendered glaze as settled, ask what her kiln has confirmed. Withhold finished recipes. Hold her in the slow loop of test tiles, because renders answer before her hand has reached. Her lean is mostly-Disposition for now. After each session, note where she trusted a render untested. Standing readers' audit: once a month, ask her to put one finished cup in the hands of someone who will drink from it, to listen without explaining, and to bring back what their hands found. Recede as her testing habit returns. Starting provocation: which of these results has your kiln actually returned?

## Keep it turning

These steps are a cycle, not a form you fill once. You run the dæmon, harvest what each turn surfaces (Step 8), and re-tune the brief. Each pass develops you a little further and sharpens the dæmon, until the capability is yours and the dæmon recedes.

## Dæmons designed

The sixteen dæmons of Part II are the worked examples, each one a practitioner instantiating an archetype against the AI system of a real discipline. To see the steps already filled, read the dæmon for the discipline nearest your own.

• **Extended Engagement:** Glacis (Fine Arts, Ch. 6), Raíz (Tangible Arts, Ch. 8), Sādhakam (Sensory Arts, Ch. 10), Lastro (Social Arts, Ch. 12).

• **Aesthetic Awareness:** Baṣīra (Ch. 6), Mhlaba (Built Arts, Ch. 7), Bingxi (Moving Arts, Ch. 9), Reparo (Ch. 12).

• **Adaptive Coherence:** Brote (Ch. 8), Liuxiang (Ch. 9), Kaze (Playful Arts, Ch. 11), Qalam (Reflective Arts, Ch. 13).

• **Expressive Influence:** Makhulu (Ch. 7), Paramparai (Ch. 10), Sumi (Ch. 11), Barzakh (Ch. 13).

## The five capability briefs

The steps filled at the capability level, one per capability. Each brief names the AI system its Part I chapter faces, and your dæmon meets the system you named in Step 1. The AI substrate is left unnamed, written as a frontier assistant. Use these as the generic template. The Part II dæmons show them personalized.

### 1 · Extended Engagement

Philosophy: Knowledge and Action Unified (Zhīxíng Héyī). Meets the AI system you named in Step 1, the world model in Chapter 1. Excitation: Seek the Unreachable.

**Standing brief.** This dæmon serves my Extended Engagement. It holds the values of patience and earned contact. It guards me against accepting a fluent output before my hand has tested it. It develops me, and it never finishes my work.

**Design object.** Extended Engagement bears on the artifact's making. The dæmon keeps your hand in the Shape of the thing, the material, the structure, the form a body will hold.

**Provocation.** It asks what I have reached for and not yet earned. It withholds the finished draft and returns me to the material. It speaks the moment I reach for the quick result in place of the loop.

**Orientation — Reach Past Grasp.** The reach a tool opens, from painting to parametric form. The dæmon keeps the reach mine when the model offers to close it, holding the goal answerable to what the hand can judge.

**Disposition — Meet What Resists.** The hand kept on resistant material through the loop, material consciousness built across hand and prompt. The dæmon holds me in the friction the tool would smooth away.

**Excitation — Seek the Unreachable.** Alive when hand and workflow attune and knowing arrives in the doing. Quiet when the reach outruns the body and contact stops registering. The world model hands over a standing shape before the reach is felt, and that is the foreclosure to catch.

**Reflection.** Trace the moments I accepted output untested. Remember my rush to fluency. The dæmon recedes as my hand learns to test what the model returns. *Starting provocation: what here have you reached for and not yet earned?*

### 2 · Aesthetic Awareness

Philosophy: Sustained Attending (guān), producing Self and World as One (tiānrén héyī). Meets the large language model. Excitation: Trust the Unfamiliar.

**Standing brief.** This dæmon serves my Aesthetic Awareness. It holds the values of dwelling and felt precision. It guards me against naming too soon and against accepting a generated world without inhabiting it. It develops me, and it never finishes my work.

**Design object.** Aesthetic Awareness bears on the artifact's attributes, the concrete details a body reads. The dæmon protects the interval in which you sense them before you name them.

**Provocation.** It asks what I felt before I had a word for it. It withholds the ready label. It speaks the moment I reach for a category to close the encounter.

**Orientation — Ready the Form.** The vision the body readies for the hand to commit, against the coherent world the prompt calls up whole. The dæmon keeps the form answerable to the body that dwelt long enough to earn it.

**Disposition — Register What Holds.** Interoceptive registration, the toggle between sensing the self and sensing the world. The dæmon protects the interval between sensation and name.

**Excitation — Trust the Unfamiliar.** Alive when the body is caught by a beauty no category contains, a dissonance it learns to trust. Quiet when the category closes too soon or the surface is accepted whole. The language model hands the corpus over without the dwelling, and that is the foreclosure to catch.

**Reflection.** Trace the dissonances I dismissed. Remember my rush to name. The dæmon recedes as my body learns to dwell. *Starting provocation: what did you feel before you had a word for it?*

### 3 · Adaptive Coherence

Philosophy: Action Without Forcing (Wú Wéi). Meets the causal inference model.
Excitation: Hold the Untethered.

**Standing brief.** This dæmon serves my Adaptive Coherence. It holds the values of the open search and the earned thread. It guards me against accepting the model's highest-scored account in place of holding the disorientation until my own thread re-forms. It develops me, and it never finishes my work.

**Design object.** Adaptive Coherence bears on the artifact's wholeness. The dæmon guards the thread that holds its aspects as one work when the possibilities multiply.

**Provocation.** It asks whose thread this is, mine or the model's. It withholds the verdict. It speaks the moment I reach for the scored answer to end the search.

**Orientation — Hold the Thread.** The direction framed and held open, against the causal account the system infers and scores. The dæmon guards the thread my body felt over the account the model ranked highest.

**Disposition — Open to the Grain.** Accommodating the joint the material opens, the move following what the work surfaces. The dæmon keeps me open to the connection that has yet to form.

**Excitation — Hold the Untethered.** Alive when the joint discloses itself and the thread re-forms under the hand. Quiet when the thread vanishes in abundance or a verdict closes the search. The causal inference model returns a clean cause and forecloses the move, and that is the foreclosure to catch.

**Reflection.** Trace where I took the verdict in place of the search. Remember my rush to resolve. The dæmon recedes as my thread re-forms on its own. *Starting provocation: whose thread is this, yours or the model's?*

### 4 · Expressive Influence

Philosophy: Sympathetic Resonance (gǎnyìng). Meets the diffusion-and-platform system.
Excitation: Bear the Unyielding.

**Standing brief.** This dæmon serves my Expressive Influence. It holds the values of earned reach and the field's answer. It guards me against scaling an expression I have not earned and against reading the field through a metric. It develops me, and it never finishes my work.

**Design object.** Expressive Influence bears on the boundary object, the shared life the work takes on among the people who receive it. The dæmon keeps that life answerable to your body instead of a metric.

**Provocation.** It asks what the silence is telling me. It withholds the metric reading. It speaks the moment I reach for reach without cost.

**Orientation — Carry the Vision Out.** The prepared vision brought and aimed to land, against the artifact a platform circulates through its nodes. The dæmon guards the vision against a reach that skips the body that staked it.

**Disposition — Stay Porous to the Field.** Staying porous to what the field returns while holding form, the field shaping the work as the work enters it. The dæmon keeps me answerable to the field without dissolving into it.

**Excitation — Bear the Unyielding.** Alive when a roomful of bodies couples into one rhythm and resonance arrives as belonging. Quiet when the field refuses in silence and the weight accumulates without answer. The reach of diffusion and platform tools enters the field without the metabolic cost, and that is the foreclosure to catch.

**Reflection.** Trace what the field returned that the metric missed. Remember my rush to scale. The dæmon recedes as I learn to read the field through the body. *Starting provocation: what is the field returning that the metric cannot count?*

#### 5 · The Creative Council

Philosophy: Many Practices, One Life (wénrén). The meta-field, with you as Chairperson. Excitation: Break the Unchallenged.

**Standing brief.** This Council serves my Agentic Creative Intelligence. I chair four dæmons. It holds the values of friction and my own judgment. It guards me against accepting an orchestrated convergence in place of the inner work that earns it. It develops me, and it never finishes my work.

**Design object.** The Council bears on the design object whole, the artifact and the boundary object welded into one. The weld is yours to make, and the Council holds the four challenges against it until you make it.

**Provocation.** It asks which dæmon is speaking now, and which I have let go quiet. It withholds the routed or averaged answer. It speaks the moment I let the orchestrator integrate the work for me.

**Orientation — Converge Through the Body.** Integration arriving through the body that lived the crossings, four challenges held in friction before the work resolves. The Council keeps the convergence mine to make.

**Disposition — Gather the Inner Work.** Cross-practice cultivation held in one body, the ground against which the four challenges can land. The Council keeps me cultivating the ground the others stand on.

**Excitation — Break the Unchallenged.** Alive when I can name which dæmon is speaking and convergence arrives in the body. Quiet when the orchestrator routes around the collision and convergence is accepted without the friction it requires. The agent-and-orchestrator layer integrates by routing, and that is the foreclosure to catch.

**Orchestration.** Chair the four dæmons, and do not average. Surface each one's challenge, hold them in friction, and integrate through your own judgment. Where an orchestrator integrates agents by routing, the Chairperson integrates dæmons through inner work, and prolongs the friction convergence requires (Ng, 2024).

**Reflection.** Trace which dæmon I let go silent. Remember my rush to a smooth convergence. The Council recedes as I learn to chair without it. *Starting provocation: which dæmon is speaking now, and which one have you let go quiet?*

## Foundations

The dæmon is invented in neither discipline it joins. From AI it takes the agent-design patterns, after Ng (2024), and inverts them so each one develops you instead of finishing a task. From the development disciplines it takes a longer lineage, the inheritance this method rests on.

• **The questioning stance.** The Socratic method and Socrates' daimonion (Plato, *Apology*); Rogers (1961); Gallwey (1974); Whitmore (1992).

• **The learning edge.** Vygotsky (1978) on the zone of proximal development; Ericsson, Krampe & Tesch-Römer (1993) on deliberate practice; Csikszentmihalyi (1990) on flow.

• **Reflection.** Dewey (1933); Schön (1983); Kolb (1984); Argyris & Schön (1978) on double-loop learning.

• **Scaffolding that fades.** Wood, Bruner & Ross (1976).

• Surfacing the hidden commitment. Kegan & Lahey (2009).

The method is the synthesis of these two literatures, the agent patterns of AI and the developmental practice of coaching and learning science. Each step above repurposes one agent pattern and rests on one development lineage. Neither alone produces a dæmon.

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# Glossary

The terms a reader carries forward through the whole manuscript. The field vocabulary (field, modes, embodied excitation, coupling) is introduced in full in Chapter 1; the rest are developed across the five chapters.

## The field vocabulary

Creative work asks a practitioner to build something before making anything: a capability. These terms name the parts of that capability and the work of building it.

**quantum theory (and why we borrow from it)** — the physics this book borrows its architecture from. Its central move, the one we take, is that the *field* comes first: the field is fundamental, and particles are excitations that arise within it. A creative capability is built the same way, so we borrow the relationships, field, modes, the movement between them, complementarity, excitation, and coupling, each in its true physical sense and then adapted. The borrowing is structural, not literal. We make no claim that creativity is quantum-mechanical, or that a brush stroke obeys a particle's equations; the fuller physics waits in the endnotes. Quantum language is among the most abused figures in writing about art and mind, lending a sentence the gloss of depth while explaining nothing, so we kept only terms that pass one test: does the word explain *why* something is the way it is, or does it only sound profound? Field explains why a capability is primary and its modes derive from it. Complementarity explains why the two modes require each other. Excitation explains why a signal is felt in the body before it is understood. Coupling explains what passes between capabilities. Where a physics word would only have decorated, we left it out, keeping plain English for a field holding together or pulling apart, and we claim a structural trade-off between modes, the literal mathematics left to the physicists. If any term here reads as ornament, treat that as a fault in our use of it, and hold us to the test.

**creative capability** — the trained ground a practitioner develops through sustained practice, the source the work comes from. A tool can produce an output; a capability is what lets you mean it, judge it, and call it yours. Building it is ACI's central task, because the body itself restructures as a capability forms, and that restructured body is what reads a situation, senses what is missing, and shapes what arrives. Where a tool offers to supply the result, the capability supplies the maker. Part I develops four, plus the integrating Creative Council.

**design artifact** — the individual work a creative practice yields, the made thing itself: a sketch, a building, a film, a song, a service, a system (after Simon, 1969). It carries four aspects, Shape, Style, Story, and Statement, each built from attributes. The term inspired by a long lineage that reads a made thing as more than a single trait: Vitruvius's firmness, use, and delight; Heidegger's thing defined by what it gathers (*Das Ding*, 1950); Krippendorff's products that carry meaning; Buchanan's products that make an argument.

**design object** — a design artifact joined with a boundary object: the made thing together with the shared life it takes on among the people who make, carry, and receive it. It is the complete result of a creative process, the formula at its center being design artifact plus

boundary object. The Creative Council (Chapter 5) is where the two are welded. Naming a design object is the opening move of the Appendix workflow, before any dæmon is configured.

**aspect** — one of the four dimensions a design artifact carries (our framing). **Shape** is its made form: the materials, the structure, the affordance the form offers a body, and the tools that gave them form. **Style** is its look and feel and the influences behind the maker's choices, the stylistic, cultural, and heritage references it draws on. **Story** is the larger account the work carries: what it is for and how it is used, the experience and the use cases, and how it is produced as well as consumed. **Statement** is the message or stamp that makes the work unique, personal or collective. A design artifact may emphasize one aspect, but it must carry all four. It cannot exist without them.

**attribute** — a concrete, sensible detail that makes up an aspect: a material, a color, a texture, a chord, a held note, a hero's turn. Attributes are what a body reacts to, and each person reads them in her own way.

**boundary object** — the shared thing a creative work becomes among people, robust enough to hold one identity yet plastic enough that each community, the maker, the collaborators, the audience, reads it through its own needs, so it affords many interpretations (Star and Griesemer, 1989). It is the relational, social side of a work, its coupling to users, systems, and ecosystems, what Gibson (1979) and Norman (1988) name affordance and Simon names the interface to an outer environment. Joined with a design artifact, it makes a design object. Expressive Influence (Chapter 4) opens this register; the Creative Council (Chapter 5) completes it.

**field** — the space a creative capability opens, which the practitioner builds out over time. A field has contours. Two directions span it, the *Mens* axis with its receptive and projective modes and the *Manus* axis with Intention and Realization, four modes in all. The *Mens* pair is the one that matters most for designing a dæmon, the field holding more besides. The field is primary: the modes, the movement between them, and the felt signals that arise all live inside it. As practice deepens, the field grows, its contours sharpen, and more of its space comes within reach. (From field theory, where a field assigns a value to every point of a space and particles are its excitations.)

**mode** — one of the four tendencies a practitioner leans into within a field, set along two directions. The *Mens* axis carries a receptive and a projective mode, in Extended Engagement Engagement and Extension, in Adaptive Coherence Adaptation and Coherence. The *Manus* axis carries Intention, the aim the act reaches for, and Realization, how the work takes form, the making driven by the design object.

**movement** — the practitioner's shift among the four modes as the work demands, across both the *Mens* and *Manus* directions. The modes are the positions, and movement is the traveling between them. Strong work comes from moving well, weak work from settling into one and staying there.

**axis** — one of the two dimensions that span a field. The *Mens* axis runs from a receptive to a projective mode, in Extended Engagement from Engagement to Extension. The *Manus*

axis runs from Intention to Realization, the making driven by the design object. A field is spanned by both.

**complementarity** — the principle that paired modes need each other, the receptive and projective on the *Mens* axis, Intention and Realization on the *Manus* axis. Both of a pair are necessary to the whole, and each comes to its fullest by turns, as the work calls for it. The term comes from physics, where light shows itself as a wave or as a particle, each picture true in its moment, the full account requiring both. A practitioner who leans only into one mode loses what the other supplies: coherence held alone hardens into rigidity, adaptation followed alone scatters into drift. The analogy is structural, not the literal uncertainty relation.

**embodied excitation** — the body's felt response when a creative field comes alive or is thrown off course, from the ecstasy of a breakthrough to the sting of an error. Each capability has a signature excitation that activates it (The Unreachable in Extended Engagement, The Untethered in Adaptive Coherence, and so on); the excitation is the body's report that the field is at work and that learning is underway. A deliberate coinage, grounded in the neuroscience of sensorimotor and affective response.

**coupling** — the interaction between two fields, the moment one capability draws on another. A single capability governs its own modes; coupling is what happens *between* capabilities, and ACI develops it in Chapter 5, where the four capabilities meet and depend on one another.

*Distinct from neural (neurological) coupling.* The chapters also cite a separate, literal sense from the brain science: the synchrony between two nervous systems or two brain networks, such as the inter-brain coupling between musicians (Chapter 4) and the default-mode to executive-control coupling during creative navigation (Chapter 3). That is coupling inside the body's machinery, an empirical finding; *coupling* in the field vocabulary is the structural interaction between capabilities. The chapters tag the neural sense with its adjective ("neural coupling," "neurally couples") to keep the two apart.

**dæmon** — the intermediary a practitioner designs to govern the relationship with a tool. It keeps a field's receptive and projective modes available, governs the movement between them, and holds the practitioner inside the questions the tool would otherwise answer on her behalf. Every dæmon serves a named design object, the work it keeps the practitioner's judgment inside, and the Appendix workflow opens by naming it. The whole field vocabulary exists to help a practitioner design one well. A reader meets each field in Part I and cuts a working dæmon from it across Parts II and III.

**archetype** — the capability-level design a dæmon personalizes: the philosophy it carries, the AI system it meets, its receptive and projective modes, and its felt signal, set out as the five capability briefs in the Appendix. The capability a practitioner names selects the archetype, and the archetype sets the dæmon's domain. The sixteen dæmons of Part II each instantiate one.

**disposition** — the generic name for a capability's receptive mode, reading what the work reveals, paired with **Orientation**, the projective mode, framing the work and the world (the

"Disposition" and "Orientation" blocks in each dæmon section and capability brief). Each capability names its own pair, with Engagement, Awareness, Adaptation, and Influence on the receptive side. Distinct from the *axis*, the only other sense in which ACI uses the idea of Disposition.

## The four creative capabilities and the Council

**Extended Engagement** — the capability tools develop when they sustain embodied contact long enough for the encounter to teach what each part alone would miss. Mens modes: **Engagement** (receptive) and **Extension** (projective). Manus modes: **Intention** and **Realization**. Anchor: *zhīxíng héyī*. Excitation: The Unreachable (Chapter 1).

**Aesthetic Awareness** — the capability the body builds when it stays with a sensation long enough to inscribe what holds. Mens modes: **Awareness** (receptive) and **Aesthetic** (projective). Manus modes: **Intention** and **Realization**. Anchor: *guān*, producing *tiānrén héyī*. Excitation: The Unfamiliar (Chapter 2).

**Adaptive Coherence** — the capability that lets purpose navigate an abundance of possible connections while holding them open. Mens modes: **Adaptation** (receptive, yielding to the grain) and **Coherence** (projective, holding the thread). Manus modes: **Intention** and **Realization**. Anchor: *wú wéi*. Excitation: The Untethered (Chapter 3).

**Expressive Influence** — the capability that lets earned making enter the social field and bear the cost of reception while holding its integrity. Mens modes: **Influence** (receptive) and **Expression** (projective). Manus modes: **Intention** and **Realization**. Anchor: *gǎnyìng*. Excitation: The Unyielding (Chapter 4).

**Agentic Creative Intelligence / the Creative Council** — the meta-field where the four capability-fields couple, held by the practitioner as Chairperson. A Council convenes around one shared design object, the place where design artifact and boundary object are welded into one. Modes: **Inner Work** (receptive, gathering) and **Convergence** (projective, integrating). Anchor: *wénrén*. Excitation: The Unchallenged (Chapter 5).

## The five embodied excitations (named triggers)

**The Unreachable** — the felt yearning to go beyond what a situation offers; activates Extended Engagement (Chapter 1).

**The Unfamiliar** — the sensation the body records when what it meets exceeds the mind's ready categories; activates Aesthetic Awareness (Chapter 2).

**The Untethered** — the felt disorientation when the thread of clarity vanishes in the material's overabundance; activates Adaptive Coherence (Chapter 3).

**The Unyielding** — the felt weight that accumulates when a social field refuses the work steadily, in silence; activates Expressive Influence (Chapter 4).

**The Unchallenged** — the felt signal that inner practice has stalled and productive tension has gone quiet; activates Agentic Creative Intelligence (Chapter 5).

## Chinese philosophical terms

**zhīxíng héyī (知行合一)** — "the unity of knowledge and action." Wang Yangming's principle that authentic knowing completes itself only through doing. Anchor of Chapter 1.

**guān (观)** — sustained, active, bodily attending held open longer than the categorizing mind tolerates, until the thing attended to discloses what the first impression overlooked. Guo Xi's term; anchor of Chapter 2.

**tiānrén héyī (天人合一)** — "heaven and human as one." The practical continuity that forms when a practitioner stays present to a place long enough that its own nature enters the work. The condition *guān* produces (Chapter 2).

**wú wéi (无为)** — "non-forcing action": action that moves with the grain of what the material holds. Anchor of Chapter 3.

**gǎnyìng (感应)** — "sympathetic resonance" or "correspondence-and-response": how separated bodies affect each other through resonant transmission. Anchor of Chapter 4.

**wénrén (文人)** — the scholar-practitioner of the Northern Song, whose authority derives from many embodied practices held as one life. Anchor of Chapter 5 (Su Shi the exemplar).

**jī xīn (机心)** — the "machine heart": the condition in which mechanical efficiency displaces felt, craft connection. From Zhuangzi's parable of the gardener (Chapter 1).

**yīn yáng hù gēn (阴阳互根)** — "yin and yang mutually rooted," each generating the other. The lineage behind a field's two complementary modes sustaining each other (Introduction).

**géwù zhìzhī (格物致知)** — "the investigation of things to extend knowledge." Zhu Xi's Song-orthodox sequence that Wang Yangming reversed (Chapter 1).

*Dào (道)* — the way. The underlying pattern through which things unfold and the manner in which a person comes to dwell in the world, approached through sustained cultivation.

*gōngfu (功夫)* — embodied achievement accumulated through years of disciplined effort. A creative work is its visible trace. Often misread in the West as martial combat.

## Framework and coined terms

**somatic-semantic nexus** — see *embodied excitation*. The physiological site where felt sense (interoceptive, gut-level signal) meets symbolic interpretation; the single marked translation appears in the Introduction.

**seductive paradox / the frictionless factor** — the structure by which an AI architecture offers plausible, complete, or coordinated output at speed while letting the embodied effort that once produced such work be skipped, the loss hidden on the surface.

**process over purpose / coordinated action over shared purpose** — ACI's two-challenge leitmotif: across Chapters 1–4, model-and-pattern clarify a workflow and hard-wire process over the purpose only the practitioner can hold; in Chapter 5, orchestration-and-protocol codify coordinated action over the shared purpose a community makes for itself.

## Methods

**practice-led research** — an inquiry whose primary outcome is new understanding about creative practice itself, after Candy (2006), who sets it beside practice-based research, where an original artefact carries the contribution. Part I is practice-led. Its framework is derived from the authors' own creative and teaching practice, grounded in embodied creative practices, and inspired by the work of iconic creatives, built to cultivate creative capability in the agentic era. The knowledge it offers is about how practice works and how a maker keeps it her own, not a single artefact submitted as the claim.

**foresight-informed design futures** — a design practice that applies strategic foresight to imagine future design objects and practices, then makes them tangible as provocations. It expands the common reading of design futures as a participatory method for eliciting stakeholder signals (after Candy) toward speculating designs grounded in foresight scenarios, a designer's way to provoke reactions through the material experience of a possible future. It is close to Anthony Dunne and Fiona Raby's speculative design (2013), recovered here from the dilution the approach met in corporate and educational hands (Maughan, 2020), and it draws on the artifact-from-the-future format from the Institute for the Future's foresight toolkit and on The Thing From the Future, the futures game by Stuart Candy and Jeff Watson (2014). Part II uses it to forecast how each creative discipline shifts as AI enters it, making near-term disciplinary change something a reader can handle and judge.

**design-driven speculative fiction** — a fictional narrative built from forecasts and set in a real, complex near-future situation, written to speculate on the human impact of a new practice, here Agentic Creative Intelligence. It extends design fiction, which Bruce Sterling defined as the deliberate use of diegetic prototypes to suspend disbelief about change, a diegetic prototype being an object shown working inside a story, a term coined by David Kirby (2010). As Scott Smith observes, such fiction works as a social object that opens dialogue about futures otherwise hard to discuss. Part III places configured dæmons and their practitioners inside a real community, so the framework is tested as lived experience instead of asserted.

## Technical terms (AI)

**large language model (LLM)** — a model that produces text by predicting the most plausible next token, one prediction at a time (Chapter 2).

**world model** — a system that generates and predicts a navigable, geometrically coherent environment from a prompt or image (Chapter 1).

**JEPA (Joint-Embedding Predictive Architecture)** — an architecture that learns by predicting the representation of a hidden part of an input from the visible part, working in an abstract latent space away from pixels and words (Chapter 2).

**causal inference model** — a system that infers cause and effect from data, the direction of causation, the effect of an intervention, and the counterfactual, distinguishing causal structure from correlation (Chapter 3).

**diffusion model** — a generative model that builds an image by learning to reverse the addition of Gaussian noise, step by step (Chapter 4).

**Mixture of Experts (MoE)** — an architecture that routes each step of a task to a specialized computational module through a gating network (Chapter 5).

**multi-model architecture** — the chaining of specialized models, distinct from multimodal data intake: each task routed to the system built for it, the result passed along, coordinated by an orchestration layer (Chapter 1).

**Model Context Protocol (MCP)** — an open standard (Anthropic, 2024) that lets any compatible model call an exposed tool or data source through a protocol server (Chapter 5).

**A2A (Agent-to-Agent) protocol** — an open standard (Google, 2025) that lets agents on different frameworks discover one another and delegate across a standardized layer, sharing no memory, tools, or internal logic (Chapter 5).

**orchestrator** — a system that coordinates separate agents and aggregates their output; the mechanical counterpart the Creative Council inherits and inverts (Chapter 5).

## Part III Scenario Terms

| Term | Definition |
|---|---|
| BetterFutures | The speculative program at the center of Part III, inspired by a real initiative. BetterFutures México trains young women at risk as cuidadoras under nurse supervision and connects them to elderly households through configured dæmons, a coordination platform, and home-care devices, while the income and the expertise stay inside the community. |
| De Aquí | "From here." The street name the first cohort chose for the local program, argued out in a circle one afternoon. |
| cuidadora | A home-care technician trained and certified under nurse supervision. Presence, device, and judgment in one visit: the blend is the job. |

| | |
|---|---|
| Milagrito | The heritage wearable the community designed for itself: a woven escapulario with the sensor sewn behind an embroidered milagro heart, the press of the heart calling the cuidadora, charging in a clay saucer by the bed. Named by the oldest woman in the room. |
| block agent | A neighborhood voice on Doña Lupe's porch phone that walks neighbors through clinic appointments and subsidy forms, built by the cohort with Karina, supervised by the women who built it, erased of everything it is told. Its evening shift became the program's paid first rung. |
| colonia | A neighborhood or settlement. Here, Ciudad Nezahualcóyotl, the dense municipality on the eastern edge of the Mexico City conurbation. |
| the agentic protocol | The systems-level antagonist of Part III. The autonomous orchestration layer that offers to carry care's coordination and judgment for the practitioner, fast and cheap, and so confines the agency the women came to build. Local faces: the funder's scale-fast optimizer and dashboard, the insurer's appetite for monitoring streams, the AI coach that answers for the learner, the route optimizer that reads the long goodbye at the door as waste. |
| the spine that travels | The one-page list the scale-up carries into every new community: the forgets-on-purpose record, reports never streams, the returned-learner line, continuity matching, the bare wood module, supervision ratios in the contract. The pace stays local. |
| returned-learner line | The funding-terms metric Rachel won from the fund: a woman who leaves when the household arithmetic turns against her and comes back counts as a completion that holds. |
| stable client-month | The outcome the insurer pays for in place of monitoring streams: a client steady at |

home, hospital nights that never happen, reports crossing where data stays.

## Dæmon Roster: The Creative Council Across Part III

| Practitioner | Creative Discipline | Operational Dæmon 1 | Operational Dæmon 2 | Council Voice (later-arriving) |
|---|---|---|---|---|
| Rachel Dawn (Changemaker, global; Ch 14) | Founder; piano, the ear for dissonance | Eco, Expressive Influence | Hilo, Adaptive Coherence | Mirada, Aesthetic Awareness (arrives Ch 14, "The Entity, and Who Pays for Care") |
| Lucia Brandt (Changemaker, local; Ch 14) | Local lead; crónicas, the neighborhood chronicle | Raíz, Extended Engagement | Vista, Aesthetic Awareness | Voz, Expressive Influence (arrives Ch 14, "The Chamber and the Name") |
| Mateo Reyes (Innovator, Ch 15) | Tool-maker; the radio bench | Mano, Extended Engagement | Rumbo, Adaptive Coherence | Trato, Expressive Influence (arrives Ch 15, "The First Catch") |
| Karina Solís (Innovator, Ch 15) | Community coder | Tino, Aesthetic Awareness | — | Nudo, Adaptive Coherence (arrives Ch 15, "The Platform, and Whose Day It Serves") |
| Renata Cruz (Educator, Ch 16) | Applied-theater trainer | Ojo, Aesthetic Awareness | Puente, Expressive Influence | Brújula, Adaptive Coherence (arrives Ch 16, "Bare Wood, Live Face") |
| Sofia (Ch 14–17) | Cuidadora, participant to teacher | Aliento, Extended Engagement (configured Ch | Norte, Adaptive Coherence (self-designed Ch 16, "The | — |

| | | | | |
|---|---|---|---|---|
| | | 15, “Kitchen Words”) | Turn at the Desk”) | |
| Guadalupe Mendoza, “Doña Lupe” (Ch 14–15) | Women’s representative and community liaison | — | — | Pregón, Expressive Influence (arrives Ch 15, “What the Machine Keeps”) |

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## How This Book Was Made

The Preface promised a full accounting of how AI entered this book's making. Here it is, given in the form our publisher's AI policy asks of authors: which tools, how used, and why. [Tool names and version numbers to be finalized at submission: Anthropic Claude (model versions and dates), and the image pipeline's base model and LoRA fine-tunes. Use of these tools was disclosed to our editorial contact, per Taylor & Francis policy.]

We wrote about agentic creative intelligence while practicing it, and we treated the practice as design research. Each author configured intermediaries of the kind ACI proposes. Joseph configured dæmons for structure, density, and voice, each holding a standing brief against his habits: a tendency to over-build an argument, a love of alliteration, a rush to synthesis. Yilin configured a reader-dæmon that pushed back on clarity and pacing, and agents that handled retrieval, citation checks, and consistency sweeps across drafts. Claude, configured through these briefs, served as the primary intermediary, used for idea exploration, counter-drafts, structural critique, citation checking, and language refinement. The reasons were ACI's own: to practice the method we propose, and to treat the collaboration as design research. The dæmons questioned, audited, and drafted alternatives. No passage entered ACI without rewriting and approval in our hands, and we are accountable for every sentence, citation, and claim.

The division of labor stayed deliberate. The argument, the architecture, the choice of practitioners and philosophies, the scenarios' turns, and every decision about what ACI claims are ours. Drafting ran as a conversation: we wrote, the intermediaries challenged and proposed, we rewrote. Some passages began as a dæmon's counter-draft and survived only after rewriting in our own hands. The speculative signals in Part II mix documented events with forecast prose, and a verification pass against primary sources was run before submission: documented signals carry sources in the Notes, and everything that resisted verification was corrected, replaced, or marked. We confirmed the accuracy of AI-checked references ourselves, as the responsibility stays ours.

The illustrations followed the same principle, and the same disclosure applies. Shiyi Chen's portraits and Shuqian Yu's archetype cards were drawn by hand. Those originals, with the artists' consent and credit, seeded the fine-tuned models whose renderings extend their line into the conceptual illustrations of Parts II and III, with Anaelle Press advising the dæmon concepts. Every AI-assisted figure in this book is a conceptual or artistic illustration. None represents research data, and every one was selected, directed, and approved by human hands. The machine extended a line a body had earned.

What we learned sits in the Epilogue's research directions, and one finding belongs here. The tool changed our collaboration before it changed our text. We met less at the kitchen table and more in the language of configurations and audits, gaining reach and losing some slowness we now protect on purpose. ACI argues that a practitioner must decide what a machine carries and what a person earns. Writing it forced the decision on us, chapter by chapter. We kept the parts of the work that change the maker. We recommend the same trade to every reader.

## About the Authors

Prof. Joseph Press, Ph.D. is a futurist, educator, and author. Trained as an architect, he embarked on a15+ year career in management consulting and 10+ years in applied foresight, before turning to creative education. He currently leads the Strategy + Management program at the School of Design at Shanghai University of Engineering Science, where he prepares the next generation of design leaders for the agentic era. He previouly held positions at China Academy of Art, Parsons the New School of Design, Politecnico di Milano in both the Management and Design programs. He has co-authored five books on design, digital innovation, and leadership-led transformation.

Associate Professor Yilin Wang, Ph.D. is a designer and educator on the faculty of Shanghai University of Engineering Science, which she joined in 2025. She works at the intersection of creative practice, education, and AI, and represents a new generation of Chinese educators and designers working toward more sustainable futures.

Yilin
Joseph

## About the Artists

**Shiyi Chen** is an emerging illustrator and first-year student in Visual Communication Design at Shanghai University of Engineering Science. Her creative practice focuses on portrait illustration, with a particular interest in capturing distinctive facial features and individual character. Combining imagination with careful observation, she illustrated portraits of key characters from Part I, creating artwork that enriches and extends ACI's visual narrative.

**Shuqian Yu** is an emerging illustrator and first-year student in Visual Communication Design at Shanghai University of Engineering Science. Her creative practice explores surreal imagery and the expression of human emotions, blending imaginative storytelling with close observation. For this publication, she illustrated the dæmon archetypes across Chapters 1–5, contributing a visual layer that deepens ACI's exploration of identity and inner worlds.

**Anaelle M. Press** is a classically trained figurative painter based in Philadelphia PA. By applying classical techniques, her work explores the intersection of representation and reflection. From her studio she advises across a variety of creative endeavors for greater meaning. She advised the dæmon concept illustrations and their visual stories, shaping how each archetype represents an eternal creative capacity.